Chapter 2

Maxwell Operators and Radiation Operators

2.0 Introduction

Antenna theory, inverse scattering, near-field metrology, and radiation synthesis all reduce, at fixed frequency, to two engineering questions that recur on every range and in every synthesis code [6]: *What field can this source produce?* and *What source produced this field?* The first question governs forward design—a horn mouth, patch feed, or array excitation is specified and the far-zone pattern on a large sphere Γ_R is sought [2].

The second governs diagnosis and inversion—probes on a scanner record tangential fields and one asks which impressed currents could have generated the observed radiation [17]. Maxwell's equations answer both questions in principle; the purpose of Chapter 2 is to organize those answers so that laboratory measurements, reactive storage, and non-radiating degrees of freedom are not conflated [1]. The two questions are not symmetric in practice. Forward design begins with a declared source class and asks what radiative image it radiates; inversion begins with a measured field sample and asks which source configurations remain compatible with that sample. Chapter 2 keeps both questions inside one operator vocabulary that separates the full Maxwell solution from the smaller radiative image a range can report.

Maxwell's equations fixed the local balances from which every antenna measurement still begins [1]. Poynting's identification of directed energy flux made the wattmeter on an enclosing sphere a primary instrument rather than a quantity inferred only after secondary calculation [5].

Stratton's treatment of the radiation condition and Green-function propagation turned open-region horn and scatterer problems into the boundary-value laboratory practice that compact ranges still emulate [2]. Chu later sharpened the same finite-support picture into a counting question: electrical size fixes how much angular structure a bounded radiator can radiate before bandwidth and stored energy dominate [7].

Harrington's time-harmonic field theory and method-of-moments habit carried those limits into the discrete source-to-pattern maps laboratories use today [6]. Chapter 2 continues that measurement line by asking which Maxwell states bounded impressed currents radiate after outgoing selection, which remain in reactive storage, and which finite instruments can resolve on a declared far shell [2]. The governing thesis, carried forward from Volume I, is that *finite geometry limits realizable, accessible, and observable angular electromagnetic complexity* [18]. Forward design, inverse scattering, and near-field metrology differ in hardware, yet they share the same historical obligation—declare support, enforce outgoing radiation, then report watts and patterns only within the rank the outline

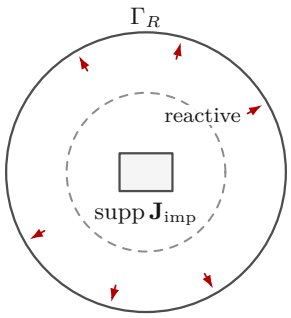


Figure 2.1: Finite impressed support, reactive near shell, and far-zone observation surface Γ_R on which outgoing Poynting flux defines radiative content recoverable on $\mathcal{F}_{\text{rad}}$.

and probe count permit [6].

The governing idea of 2 is *radiation as accessibility*: which Maxwell states generated by bounded sources remain observable after outgoing selection on a declared far-zone shell Γ_R [1]. The radiation operator reports what finite sources radiate once Sommerfeld outgoing radiation condition is enforced; it does not create angular structure [1].

At fixed frequency, Maxwell's equations define a source-to-field operator $\mathcal{S}_\omega$—the harmonic *solution map* that sends impressed current $\mathbf{J}_{\text{imp}}$ to the field pair $(\mathbf{E}, \mathbf{H})$ consistent with sources and boundary termination—and, after outgoing selection, a radiation map $\mathcal{R}_\omega = \Pi_{\text{rad}} \circ \mathcal{S}_\omega$, where Π_{rad} is the outgoing selector that retains only far-zone transverse content [1]. The selector Π_{rad} acts on $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]$ to remove evanescent and standing reactive content; it is not a second Maxwell solver.

Green's retarded potentials and Sommerfeld's radiation condition made the exterior problem precise long before modern mathematical theory was applied to antennas and scattering [2]. Hertz [19] demonstrated in the laboratory that Maxwell's equations predict propagating electromagnetic waves.

The scientific question is classical: which field content generated by finite impressed currents $\mathbf{J}_{\text{imp}} \in \mathcal{J}_\Omega$ is radiated to the far zone, which remains in reactive storage, and which lies in the null space of $\mathcal{R}_\omega$? Chapter 1 fixed Maxwell consistency, conservation laws, gauge limits on local readouts, and the measurement sequence; here those results become the radiation-selection maps [2]. No new electromagnetic law is introduced; the maps restate consequences of Maxwell theory together with outgoing selection and finite instrumentation [1].

Radiative probe rank. Far-zone observability is a composed map: radiation selection, geometric regime, and finite measurement rank act in sequence. Evanescent and reactive content may dominate near scans without enlarging the angular rank-N recovery of radiation on $\mathcal{F}_{\text{rad}}$ [2].

Figure 2.1 orders the physical sequence from finite sources downward, not from angular labels upward. Impressed support and boundary data fix the Maxwell problem; outgoing selection on the exterior branch isolates the radiative image; harmonic coefficients are flux-normalized coordinates on that image, not independent radiative degrees of freedom [6].

1 established admissibility; 2 establishes radiation through $\mathcal{R}_\omega$—which Maxwell-generated content survives outgoing radiation condition and reaches the far zone [1]. Maxwell field specification fixes what can exist on a declared domain; radiation fixes what leaves that domain under outgoing selection [1]. A cavity mode and a compact-range pattern may share the same frequency yet differ completely in radiation; the distinction is not visible in a single near-field snapshot [6].

Consider a shorted coaxial stub and an open-ended dipole at 1 GHz in the same anechoic cell. Only the radiating structure supports watt completion on Γ_R once outgoing Sommerfeld condition is enforced; the stub stores reactive energy without net radiation [2]. That contrast is the operational

content of the operator split developed below: reactive storage is diagnosed on $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$, while far-zone reporting attaches to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$. In laboratory measurements the distinction is immediately visible. A network analyzer may report substantial stored energy and strong local fields in both structures, yet only the dipole produces sustained outward power flow and a measurable far-field pattern [6]. Radiation operators express this experimentally familiar distinction [1].

Source specification, Maxwell completion, radiation selection, observation, and inference form one connected argument, not a sequence of unrelated topics [1]. Not every Maxwell field radiates; not every radiating field is observable; not every observable field identifies its source [2]. The three subsections of section 2.0 develop that sequence in order: Subsection 2.0.1 names the maps and their domains; Subsection 2.0.2 packages Maxwell's equations as the operator whose inversion produces $\mathcal{S}_\omega$; Subsection 2.0.3 records continuity with the $\mathcal{D}_J = 1$ and measurement criteria of 1 [2]. section 2.1 then supplies the function spaces on which those maps act with mathematical precision.

2.0.1 Radiation as a source-to-field operator

section 2.0 posed radiation as a question of accessibility rather than of field existence alone. The distinction between field existence and radiation through $\mathcal{R}_\omega$ is central to radiation theory [2]. Maxwell's equations determine which fields may arise from a specified source distribution, whereas the radiation map $\mathcal{R}_\omega$ determines which portion of those fields survives as observable outgoing content [1]. From an antenna engineer's perspective, $\mathcal{S}_\omega$ answers what the structure *stores and circulates* locally, while $\mathcal{R}_\omega$ answers what outward-propagating flux and angular structure survive once the reactive shell is left behind [6]. Port reflection $|S_{11}|$, tangential field samples on Γ_R, and a far-zone pattern cut are three coordinate readings on one composed operator: impressed data, Maxwell closure, outgoing radiation, finite observation [1, 2]. Misplacing readout before outgoing selection—reporting reactive storage as radiated power—remains the common compact-range failure mode [5, 3].

Laboratory measurements presuppose an excitation model even when it is left implicit: a port drive, a probe sample, or a termination fixes how energy is partitioned between storage and escape [2]. Impressed data from section 1.1.2 map through Maxwell field specification to $(\mathbf{E}, \mathbf{H})$, and only then is the outward-propagating fraction identified as the carrier of far-zone power and angular flux (Eq. (1.5); sections 1.1.4–1.1.5). At fixed ω the maps are the harmonic solution operator $\mathcal{S}_\omega$ and radiation operator $\mathcal{R}_\omega$ under the $\exp(+\mathrm{j}\omega t)$ convention of Chapter 1.

Let $\Omega \subset \mathbb{R}^3$ be a piecewise smooth domain with boundary $\Gamma = \partial\Omega$. Phasor fields $\mathbf{E}(\mathbf{r})$ and $\mathbf{H}(\mathbf{r})$ are suppressed in time unless causality is restored below. Impressed electric current $\mathbf{J}_{\mathrm{imp}}$, charge density ρ, and the data tuple $\mathfrak{D}$ of Eq. (1.17) are not redefined; they are the inputs on which the maps act. Finite-source admissibility is assumed in the sense of the 1 $\mathcal{D}_J = 1$ conditions (Eq. (1.160)): only configurations with $\mathcal{D}_J = 1$ enter the domain of the radiation maps developed here [2].

Eq. (2.2) displays the operator composition at fixed frequency: impressed current $\mathbf{J}_{\mathrm{imp}}$ maps through $\mathcal{S}_\omega$ to the full Maxwell pair $(\mathbf{E}, \mathbf{H})$; outgoing selection Π_{rad} retains only the transverse radiative content that carries far-zone Poynting flux [1, 5].

Function spaces are declared before the maps. Let $\mathcal{F}_\Omega$ be the space of phasor pairs $(\mathbf{E}, \mathbf{H})$ on Ω whose components have L^2-integrable curls and divergences in the sense required by Eq. (1.7), and whose boundary traces lie in the domain of $\mathcal{B}_\Gamma$. Let $\mathcal{J}_\Omega \subset L^2(\Omega)^3$ be the impressed-current space compatible with finite-energy class (section 1.0.4), charge continuity (Eq. (1.8)), and the same boundary class. Refined Sobolev and distributional extensions of these spaces are supplied in section 2.1.1; here the declaration is minimal but precise enough to state the maps [2]. $\mathcal{J}_\Omega$ houses impressed currents; $\mathcal{F}_{\mathrm{rad}}$ houses satisfying the Sommerfeld radiation condition far-zone field

pairs radiated by $\mathcal{R}_\omega$. The harmonic Maxwell *solution map* is the first quantitative object in the radiation sequence: it records, for each admissible impressed current $\mathbf{J}_{\text{imp}} \in \mathcal{J}_\Omega$, the Maxwell field pair that the governing equations and boundary data jointly permit. Physically, this map is the harmonic solution of Maxwell's equations on a declared domain; its image is a *complete* field state, not yet a radiative image [2]. On a closed surface enclosing the source, both reactive circulation and outward-propagating flux appear in $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]$ until outgoing selection is applied [6]. The solution map sends impressed data to a field state:

$$\mathcal{S}_\omega : \mathcal{J}_\Omega \longrightarrow \mathcal{F}_\Omega, \qquad \mathbf{J}_{\text{imp}} \longmapsto (\mathbf{E}, \mathbf{H}), \tag{2.1}$$

defined by $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}] = (\mathbf{E}, \mathbf{H})$ satisfying the Maxwell operator residuals of Eq. (1.7) with sources $\mathfrak{D}$ and boundary data $\mathbf{g}_\Gamma$ [2]. Eq. (2.1) is the 2 statement of the initial-boundary value problem proven in section 1.1.1; it does not alter the law, only the input–output viewpoint. Injectivity fails in general: distinct $\mathbf{J}_{\text{imp}}$ can produce identical $(\mathbf{E}, \mathbf{H})$ on observational subdomains, and equivalent source distributions related by divergence-free reparametrizations inside Ω can leave exterior patterns unchanged when $\mathcal{B}_\Gamma$ permits reactive storage (section 1.2.5). Those equivalences are not nuisances to be removed by definition; they foreshadow null-space and realizability analyses in sections 2.7 and 4. Radiation accounting therefore never begins from a guessed modal field; it begins from $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]$ with explicit completion. The full state typically contains both stored and radiated electromagnetic energy. Reactive fields exchange power with sources and terminations at distances small compared with k_0^{-1}; radiative fields carry net outward flux at infinity in open configurations. Let $\mathcal{F}_{\text{rad}} \subset \mathcal{F}_\Omega$ denote pairs whose far-zone asymptotics satisfy outgoing selection (made analytic in section 2.4). Radiation accounting therefore requires a second map that discards non-outgoing storage before pattern or transport claims are made. The outgoing selector $\Pi_{\text{rad}} : \mathcal{F}_\Omega \to \mathcal{F}_{\text{rad}}$ performs that discard: it accepts a completed Maxwell pair and returns only the component whose leading far-zone asymptotics satisfy Sommerfeld outgoing radiation condition [2]. From a measurement standpoint, Π_{rad} is the mathematical counterpart of moving a probe beyond the reactive shell and reporting only the transverse outgoing field that persists with $1/r$ decay [6]. Composing source excitation with that selector defines the *radiation operator*

$$\mathcal{R}_\omega : \mathcal{J}_\Omega \longrightarrow \mathcal{F}_{\text{rad}}, \qquad \mathcal{R}_\omega[\mathbf{J}_{\text{imp}}] = \Pi_{\text{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]. \tag{2.2}$$

Eq. (2.2) composes source excitation with outgoing selection: $\mathcal{R}_\omega$ accepts the same impressed currents as $\mathcal{S}_\omega$ but returns only the radiative image $\Pi_{\text{rad}}(\mathbf{E}, \mathbf{H}) \in \mathcal{F}_{\text{rad}}$, where $\mathcal{F}_{\text{rad}}$ is the subspace of field pairs whose leading far-zone behavior is outgoing and transverse. Factoring $\mathcal{S}_\omega$ and Π_{rad} separates retarded generation from outgoing radiation so each map can be diagnosed before harmonic labeling. The composition law records that radiation is never computed independently of Maxwell completion:

$$\mathcal{R}_\omega = \Pi_{\text{rad}} \circ \mathcal{S}_\omega, \qquad \mathcal{D}_J = 1 \Longrightarrow \mathcal{S}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_\Omega, \quad \mathcal{R}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_{\text{rad}}. \tag{2.3}$$

where $\mathcal{S}_\omega$ is the harmonic Maxwell solution map, Π_{rad} the outgoing selector, $\mathcal{R}_\omega$ the radiation map, $\mathcal{J}_\Omega$ the impressed-current space, $\mathcal{F}_\Omega$ the admissible field space, $\mathcal{F}_{\text{rad}}$ the radiative subspace, and $\mathcal{D}_J = 1$ the condition for outward angular-momentum flux inherited from 1 [2]. Eq. (2.3) composes $\mathcal{S}_\omega$ with outgoing selection Π_{rad} on $\mathcal{J}_\Omega$ [2]. The order of this composition is the physical point of the display. $\mathcal{S}_\omega$ answers the first Maxwell question by producing the full admissible field generated by the declared source data. Π_{rad} answers the second question by retaining only the part of that field that qualifies as outward outgoing radiation. The two maps therefore express successive scientific judgments on one source configuration, not two interchangeable notations for the same

field report. A compact-range report that collapses the two maps into one claim cannot say whether a discrepancy arose in full-wave solution, in outgoing selection, or in harmonic listing. On a ball B_a with $k_0 a = 0.5$, reactive-to-radiative field-energy ratios on S_R with $R = 2a$ typically exceed unity by an order of magnitude; applying pattern or transport metrics to $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ without Π_{rad} therefore overstates radiation by roughly $10\times$ in magnitude bookkeeping before phase errors accumulate [14]. Superposition of radiative images is legitimate only after each constituent satisfies Eq. (2.8) and outgoing radiation condition [6]. Until Subsection 2.2.3 fixes $\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}$, treat Π_{rad} as an abstract outgoing selector [2]. Angular-flux transport from sections 1.2.4–1.2.5 and structured-beam interpretations in section 1.6.2 attach to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$; coefficients read from reactive components of $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ describe near-zone storage, not radiated flux [13]. When Green operators are available (section 2.3), $\mathcal{R}_\omega$ admits a retarded integral representation that makes the propagation step explicit. The integral representation below is the frequency-domain statement of what a method-of-moments code evaluates when it integrates the dyadic Green function against impressed currents: each source element at $\mathbf{r}'$ contributes to the observation point $\mathbf{r}$ with phase and amplitude fixed by outgoing propagation [2]. Physically, the integral superposes the radiative images of all current elements; only the portion of $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ that Π_{rad} retains enters the integrand in the completed theory developed in section 2.3. For $\mathbf{r} \notin \Omega_s$ with $\operatorname{supp} \mathbf{J}_{\mathrm{imp}} = \Omega_s \Subset \Omega$,

$$\mathbf{E}_{\mathrm{rad}}(\mathbf{r}) = -\mathrm{j}\omega\mu_0 \int_{\Omega_s} \boldsymbol{\Gamma}(\mathbf{r}, \mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3 r', \tag{2.4}$$

In Eq. (2.4), $\mathbf{E}_{\mathrm{rad}}(\mathbf{r})$ is the radiative electric phasor at $\mathbf{r}$ ($\mathrm{V\,m^{-1}}$); $\boldsymbol{\Gamma}(\mathbf{r}, \mathbf{r}')$ is the outgoing dyadic Green tensor ($\mathrm{m^{-1}}$) that propagates unit current dipoles from $\mathbf{r}'$ to $\mathbf{r}$; $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$ is the vacuum permeability; $\mathbf{r}'$ ranges over the source volume Ω_s; and the volume element $d^3 r'$ carries SI units $\mathrm{m^3}$. The factor $-\mathrm{j}\omega\mu_0$ arises from the harmonic Ampère law under the locked phasor convention. Here $\boldsymbol{\Gamma}$ is the outgoing dyadic Green function [2]. Eq. (2.4) is a frequency-domain representation of the radiative field generated by a compact source distribution through the outgoing dyadic Green function [2]. It should be read as a source-to-observer conversation carried by retarded propagation. The integration variable $\mathbf{r}'$ runs over the compact region where the impressed current is physically supplied, while $\mathbf{r}$ names the exterior point at which one asks for the radiative consequence of that source. The dyadic kernel therefore does not introduce a new physical layer beyond Maxwell theory; it makes explicit how source generation, outgoing selection, and observable field report remain one continuous chain. Every application of this representation assumes the $\mathcal{D}_J = 1$ and boundary conditions stated above. Unless noted otherwise, $\omega > 0$ is fixed, materials are linear isotropic passive regions, and Ω is either bounded with declared $\mathcal{B}_\Gamma$ or an exterior domain with compact $\operatorname{supp} \mathbf{J}_{\mathrm{imp}}$ and outgoing radiation condition at infinity. Maps are defined only on $\mathcal{J}_\Omega$ satisfying Eq. (1.8); generalized sources appear in section 2.3.3. Maxwell linearity on Ω induces linearity of both maps on fixed boundary data: for $\mathbf{J}_{\mathrm{imp}}^{(1)}, \mathbf{J}_{\mathrm{imp}}^{(2)} \in \mathcal{J}_\Omega$ and scalars a_1, a_2,

$$\mathcal{R}_\omega\left[a_1 \mathbf{J}_{\mathrm{imp}}^{(1)} + a_2 \mathbf{J}_{\mathrm{imp}}^{(2)}\right] = a_1 \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(1)}] + a_2 \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(2)}], \tag{2.5}$$

In Eq. (2.5), $a_1, a_2 \in \mathbb{C}$ are complex amplitudes and $\mathbf{J}_{\mathrm{imp}}^{(1)}, \mathbf{J}_{\mathrm{imp}}^{(2)} \in \mathcal{J}_\Omega$ are admissible impressed currents; $\mathcal{R}_\omega$ is the radiation map; and linearity holds when boundary data $\mathbf{g}_\Gamma$ and the material layout are held fixed [1]. Composite-source analysis decomposes $\mathbf{J}_{\mathrm{imp}}$ into constituents, maps each through $\mathcal{R}_\omega$, and superposes radiative outputs—a discipline that prevents reactive interference from masquerading as transport (section 1.6.4). Write $\operatorname{ran} \mathcal{S}_\omega \subset \mathcal{F}_\Omega$ for all field pairs reachable from finite sources and $\operatorname{ran} \mathcal{R}_\omega \subset \mathcal{F}_{\mathrm{rad}}$ for far-zone radiative outputs. Strict inclusion $\operatorname{ran} \mathcal{R}_\omega \subsetneq \operatorname{ran} \mathcal{S}_\omega$ occurs whenever reactive storage is nontrivial: distinct near-zone states can share identical radiative

images (section 1.2.5). Pattern measurements probe ran $\mathcal{R}_\omega$ and cannot invert $\mathcal{S}_\omega$ without additional near-zone data—a limitation revisited in section 2.4.4. Radiative power is metered with inherited Poynting machinery. The practical question is how much time-averaged power crosses a closed surface that encloses the source and lies beyond the reactive shell; that flux is the scalar quantity laboratories report as radiated power. On a sphere $S_R = \{|\mathbf{r}| = R\}$ enclosing $\operatorname{supp}\mathbf{J}_{\text{imp}}$, with outward unit vector $\hat{\mathbf{r}}$ and complex Poynting vector $\mathbf{S} = \frac{1}{2}\mathbf{E} \times \mathbf{H}^*$ from section 1.1.4,

$$P_{\text{rad}}(R) = \operatorname{Re} \oint_{S_R} \mathbf{S} \cdot \hat{\mathbf{r}}\, dS. \tag{2.6}$$

In Eq. (2.6), $P_{\text{rad}}(R)$ is the time-averaged outward power (W) through the sphere of radius R (m); $\mathbf{S}$ is the complex Poynting vector ($\mathrm{W\,m^{-2}}$) built from the radiative field pair; $\hat{\mathbf{r}} = \mathbf{r}/|\mathbf{r}|$ is the outward unit normal on S_R; and dS is the area element ($\mathrm{m^2}$). The real part $\operatorname{Re}\{\cdot\}$ extracts the time-averaged power flux carried by the phasor fields under the $\exp(+\mathrm{j}\omega t)$ convention. Eq. (2.6) inherits the Poynting law of section 1.1.4. For $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}]$, $P_{\text{rad}}(R)$ becomes R-independent beyond the reactive shell [5]. Angular-momentum budgets from section 1.1.5 attach to the same radiative component.

Spatial nesting seen by the maps: $\mathcal{S}_\omega$ populates all regions; Π_{rad} passes only the outgoing transverse content compatible with section 1.2.5.

The solution map energizes reactive and radiative regions alike; the radiation map discards non-outgoing storage before pattern or transport claims are made. Misidentifying reactive circulation as radiated angular structure is a recurring error in high-gradient near-zone beams [13]. Finite measurement channels (section 1.5) implement linear functionals on field space. Composing a declared far-zone observable with $\mathcal{R}_\omega$ yields a linear functional of $\mathbf{J}_{\text{imp}}$; composing the same observable with $\mathcal{S}_\omega$ without selection generally includes reactive contamination. No new measurement axioms are introduced—only the correct map for the declared observable class [2]. Non-radiating currents illustrate kernel visibility. If $\mathbf{J}_{\text{imp}}^{(\text{NR})} \in \mathcal{J}_\Omega$ has vanishing far-zone transverse fields while maintaining near-zone storage (section 1.2.5), then

$$\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}^{(\text{NR})}] = \mathbf{0} \quad \text{in } \mathcal{F}_{\text{rad}}, \tag{2.7}$$

In Eq. (2.7), $\mathbf{J}_{\text{imp}}^{(\text{NR})}$ is an admissible impressed current ($\mathrm{A\,m^{-2}}$) that satisfies $\mathcal{D}_J = 1$ yet carries no outgoing transverse radiation; $\mathcal{F}_{\text{rad}}$ is the far-zone radiative field space selected by Π_{rad}; and $\ker\mathcal{R}_\omega$ denotes the set of currents for which the radiation map vanishes on that space, while $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}^{(\text{NR})}] \neq \mathbf{0}$ in general. Eq. (2.7) targets the null space analyzed in section 2.7.1; ignoring $\ker\mathcal{R}_\omega$ invites spurious modal assignments to invisible sources. Membership in $\mathcal{J}_\Omega$ requires more than square integrability: the Maxwell-closed restriction below states, in compact form, which impressed currents may enter the radiation chain without contradicting charge continuity or boundary compatibility. $\mathcal{D}_J$ is not a mathematical regularity condition alone: it encodes the laboratory requirement that impressed currents be charge-consistent and boundary-compatible before Maxwell field specification is attempted [1]. Violating the restriction produces formal solutions that no passive feed can realize [6]:

$$\mathcal{D}_J[\mathbf{J}_{\text{imp}}] = 1 \iff \Big(\mathbf{J}_{\text{imp}} \in L^2(\Omega)^3,\ \nabla\cdot\mathbf{J}_{\text{imp}} = -\mathrm{j}\omega\rho,\ \mathbf{J}_{\text{imp}} \text{ satisfies BC compatibility on } \Gamma\Big), \tag{2.8}$$

In Eq. (2.0.1), $\mathcal{D}_J$ is the source Maxwell-closed restriction (dimensionless) inherited from Eq. (1.160); $\mathbf{J}_{\text{imp}}$ is the impressed electric current density ($\mathrm{A\,m^{-2}}$) on compact $\Omega \subset \mathbb{R}^3$; ρ is the impressed charge density ($\mathrm{C\,m^{-3}}$) linked by $\nabla\cdot\mathbf{J}_{\text{imp}} = -\mathrm{j}\omega\rho$; and $\Gamma = \partial\Omega$ is the boundary on which tangential traces

must match the declared boundary class. with $\mathcal{D}_J$ as in Eq. (1.160). Violating $\nabla \cdot \mathbf{J}_{\text{imp}} = -\mathrm{j}\omega\rho$ contradicts Eq. (1.8) before Maxwell field specification is attempted. Boundary compatibility ensures induced surface equivalents do not contradict Eq. (1.18) [2]. Causality constrains harmonic representations. A frequency-domain phasor is physically meaningful only when it can be traced to a retarded time-domain evolution that respects finite propagation speed. With time-domain current $j(\mathbf{r}, t)$ and retarded dyadic impulse $\mathcal{G}(\mathbf{r}, \mathbf{r}', t - t')$,

$$\mathbf{E}(\mathbf{r}, t) = -\mu_0 \frac{\partial}{\partial t} \int \mathcal{G}(\mathbf{r}, \mathbf{r}', t - t') \cdot j(\mathbf{r}', t') \, d^3 r' \, dt', \tag{2.9}$$

integrated only over $t' \leq t$ [2]. In Eq. (2.9), $\mathbf{E}(\mathbf{r}, t)$ is the real time-domain electric field ($\mathrm{V\,m^{-1}}$); $j(\mathbf{r}', t')$ is the impressed current density ($\mathrm{A\,m^{-2}}$) at source point $\mathbf{r}'$ and retarded time t'; $\mathcal{G}(\mathbf{r}, \mathbf{r}', t - t')$ is the causal dyadic Green impulse response ($\mathrm{m^{-1}\,s^{-1}}$) that vanishes for $t - t' < 0$; and the temporal integral enforces that the field at $(\mathbf{r}, t)$ depends only on sources in the past light cone. The steady-state operator $\mathcal{R}_\omega$ may therefore be viewed as the frequency-domain specialization of the retarded causal evolution [2]. The equivalence between the time-domain and harmonic descriptions follows under the standard assumptions of linearity, stationarity, and finite-energy excitation [1]. Vacuum scales from Eq. (1.12) fix the carrier on which selection acts: $k_0 = \omega\sqrt{\mu_0\varepsilon_0}$, $\eta_0 = \sqrt{\mu_0/\varepsilon_0}$. Outgoing components retained by Π_{rad} carry $\exp(-\mathrm{j}k_0 r)$ phase fronts at leading order; components whose phase does not lock to outgoing $k_0 r$ remain in $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]$ but outside $\mathcal{F}_{\text{rad}}$. Power metered by Eq. (2.6) stabilizes only on the latter [2]. Distance labels become meaningful through operator action. For compact $\operatorname{supp} \mathbf{J}_{\text{imp}}$, $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]$ contains $1/r^2$ and $1/r^3$ reactive terms; $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}]$ retains leading $1/r$ transverse fields with kr phase in the far zone (section 1.2.5). Asymptotic partitions are sharpened in section 2.5.2. Observability enters as functional composition rather than as a new postulate. Let $\mathcal{O}_{\text{far}}$ denote a linear functional that extracts a declared far-zone scalar or vector observable—pattern voltage, axial ratio proxy, or a single spherical-harmonic coefficient—from a field pair in $\mathcal{F}_{\text{rad}}$. The measured quantity associated with impressed current $\mathbf{J}_{\text{imp}}$ is the scalar $\mathcal{O}_{\text{far}}\big[\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}]\big]$, not $\mathcal{O}_{\text{far}}\big[\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]\big]$ unless the observable has been shown to vanish on reactive content. Because both $\mathcal{R}_\omega$ and $\mathcal{O}_{\text{far}}$ are linear on their domains, the laboratory reading is a linear functional $\Lambda_{\text{rad}} = \mathcal{O}_{\text{far}} \circ \mathcal{R}_\omega$ on $\mathcal{J}_\Omega$. That composition expresses in map language the measurement limitations in section 1.5, finite apertures and finite modal sets appear as rank-deficient $\mathcal{O}_{\text{far}}$, not as modifications of Maxwell law [2]. Angular-transport claims require further that Λ_{rad} be compatible with the flux budgets of section 1.1.5 when the observable is interpreted as momentum or power radiation. The outgoing selector Π_{rad} is idempotent on $\mathcal{F}_{\text{rad}}$: applying selection twice leaves a radiative state unchanged. On the larger space $\mathcal{F}_\Omega$, Π_{rad} is a projection only after the Sommerfeld-class uniqueness of section 2.4.3 is imposed; without that completion, multiple outgoing decompositions can coexist for the same truncated near-zone data. The idempotence condition is therefore a *joint* property of field space and boundary class, not an algebraic trick independent of physics [6]. Numerical methods that window fields in space and call the result "radiative" without outgoing radiation condition are approximating Π_{rad} by a non-idempotent mask; such outputs must not be fed into transport criteria without an explicit error budget. The composition law used downstream is

$$\mathcal{R}_\omega = \Pi_{\text{rad}} \circ \mathcal{S}_\omega, \qquad \mathcal{D}_J = 1 \Longrightarrow \mathcal{S}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_\Omega, \quad \mathcal{R}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_{\text{rad}}. \tag{2.10}$$

In Eq. (2.10), $\mathcal{S}_\omega$ is the Maxwell solution map from admissible sources to field pairs, Π_{rad} is the outgoing selector that retains far-zone transverse radiation, $\mathcal{R}_\omega = \Pi_{\text{rad}} \circ \mathcal{S}_\omega$ is the radiation map, $\mathcal{D}_J = 1$ denotes source Maxwell-closed from Eq. (2.8), and $\mathcal{J}_\Omega$, $\mathcal{F}_\Omega$, $\mathcal{F}_{\text{rad}}$ are the source and field spaces declared in Subsection 2.0.1. Green kernels realize $\mathcal{S}_\omega$; Sommerfeld conditions define Π_{rad}; spectral theory in section 2.6 studies eigenstructure on spaces refined in section 2.1 [6]. Nothing

in Eqs. (2.1)–(2.10) re-proves results listed in section 1.7.1; the Maxwell, charge-continuity, stress-flux, and gauge relations established in Chapter 1 are used here by citation without re-derivation. Subsection 2.0.1 has named $\mathcal{S}_\omega$, Π_{rad}, and $\mathcal{R}_\omega$, introduced their Green and Poynting realizations, and connected finite measurement rank to composition with far-zone observables. Subsection 2.0.2 identifies the Maxwell operator whose inversion, under declared boundary completion, produces $\mathcal{S}_\omega$.

Radiation on Γ_R therefore remains the laboratory requirement: watt must close through outgoing flux before angular labels are compared [2].

2.0.2 Maxwell equations in operator form

Subsection 2.0.1 named the solution map $\mathcal{S}_\omega$ and the outgoing selector Π_{rad}; the display below identifies the Maxwell operator whose inversion, under declared boundary completion, produces that map—the same inversion Maxwell theory prescribes once impressed currents, frequency, and boundary data on Γ are declared [2, 3]. Maxwell's retarded radiation to Γ_R remains the governing constraint [1]. The governing question is which field pair $(\mathbf{E}, \mathbf{H})$ is consistent with impressed currents $\mathbf{J}_{\mathrm{imp}}$ and with the termination class $\mathcal{B}_\Gamma$ on $\Gamma = \partial\Omega$—the same question radiation theory answers before any far-zone pattern is extracted [6]. Harmonic Maxwell theory on Ω is packaged as a first-order residual system on a six-component state, augmented by boundary traces through the operator Eq. (1.18) imported from section 1.1.2. Local curl and divergence balances remain those of Eq. (1.7); the Maxwell radiation sequence developed here does not alter those local balances [2]. First-order Maxwell theory couples electric and magnetic fields through circulation; packaging both components in a single state vector is the standard device that permits resolvent notation and variational formulations without abandoning the physical coupling [2]. $\mathbf{u}_\omega$ is not an arbitrary six-vector: each admissible state satisfies Gauss constraints and carries tangential traces compatible with the boundary class declared with the impressed data tuple $\mathfrak{D}$. From an antenna engineer's perspective, $\mathbf{u}_\omega$ is the coupled $(\mathbf{E}, \mathbf{H})$ phasor pair at fixed ω on the declared domain—the local field state before far-zone reduction yields patterns or port quantities [6]. Collect phasor fields into the Maxwell state

$$\mathbf{u}_\omega = \begin{bmatrix} \mathbf{E} \\ \mathbf{H} \end{bmatrix} \in \mathcal{F}_\Omega, \tag{2.11}$$

where $\mathbf{u}_\omega$ denotes the six-component Maxwell state at fixed ω, with $\mathbf{E}$ and $\mathbf{H}$ carrying the SI units noted above and lying in the admissible field space $\mathcal{F}_\Omega$ declared in Subsection 2.0.1. The pair $(\mathbf{E}, \mathbf{H})$ is constrained jointly: divergence rows of Eq. (1.7) must hold, and tangential traces on $\Gamma = \partial\Omega$ must lie in the domain of the boundary class declared with $\mathfrak{D}$. Treating $\mathbf{u}_\omega$ as a single unknown is what permits resolvent notation, variational forms, and the spectral compression developed in section 2.6 [6]. Every boundary-value problem begins by distinguishing prescribed quantities from unknown quantities. In Maxwell theory the impressed currents, charges, and boundary data constitute the prescribed excitation, whereas the electromagnetic field remains the unknown state to be determined [1]. The operator formulation merely records this distinction in a compact mathematical form suitable for analysis [6]. Impressed data enter through a forcing vector aligned with the Maxwell residual rows. The forcing packages charge and current drive on the rows where impressed sources appear, leaving divergence constraints to be satisfied by the field unknown. For the tuple Eq. (1.17),

$$\mathbf{s}_\omega(\mathfrak{D}) = \begin{bmatrix} \mathbf{0} \\ -\mathbf{J}_{\mathrm{imp}} \\ -\rho \\ \mathbf{0} \end{bmatrix}, \tag{2.12}$$

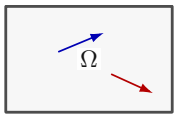


Figure 2.2: Volume Ω with interior $(\mathbf{E}, \mathbf{H})$ and boundary $\Gamma = \partial\Omega$: tangential traces on Γ close the harmonic problem jointly with Maxwell balance in Ω.

In Eq. (2.12), $\mathbf{s}_\omega(\mathfrak{D})$ is the Maxwell forcing six-vector (SI units per row) assembled from the data tuple $\mathfrak{D}$; $\mathbf{J}_{\mathrm{imp}}$ enters the Ampère row ($\mathrm{A\,m^{-2}}$) and ρ the Gauss row ($\mathrm{C\,m^{-3}}$); the remaining rows are zero for the impressed-current class treated here; row order matches Eq. (1.7). Equation (2.12) records drive on the Ampère and Gauss rows only for the impressed-current class treated here; sheet and magnetic excitations extend the vector through Eq. (1.19) when trace spaces are enriched in section 2.1.1 [2]. The forcing is *data*, not a field variable: it specifies how $\mathcal{S}_\omega$ is sourced while $\mathbf{u}_\omega$ remains the unknown. Operator language separates *states* $(\mathbf{E}, \mathbf{H})$ from *data* $(\mathbf{J}_{\mathrm{imp}}, \rho, \mathfrak{D})$: the feed specifies what is driven, the solver returns what Maxwell allows, and radiation operators report what is radiated [1]. Treating a tangential-field template as independent forcing without a matching current repeats the realizability failure already flagged in Chapter 1 [2]. Let $\mathcal{M}_\omega$ denote the harmonic Maxwell operator of Eq. (1.7). The volume subsystem reads

$$\mathcal{M}_\omega[\mathbf{u}_\omega] = \mathbf{s}_\omega(\mathfrak{D}), \tag{2.13}$$

and is satisfied when each residual row vanishes. Linearity in $\mathbf{u}_\omega$ follows from linearity of Eq. (1.5); spatially varying $\varepsilon(\mathbf{r})$ and $\mu(\mathbf{r})$ modify coefficients inside $\mathcal{M}_\omega$ without changing the first-order block shape (section 2.2.2) [2]. In uniform vacuum regions, $\mathcal{M}_\omega$ has translation-invariant symbol, the property exploited by dyadic Green kernels in section 2.3. Figure 2.2 maps the coupled channels of Eqs. (2.13) and (1.18). A solver that enforces only volume rows produces a formal solution compatible with Maxwell *sources* but not necessarily with termination physics; conversely, boundary data without volume balance violate conservation. Subsection 2.0.1 applies $\mathcal{R}_\omega$ only after both channels are closed [6]. Tangential traces are extracted by the restriction

$$\gamma_\Gamma : \mathcal{F}_\Omega \to \mathcal{T}_\Gamma, \qquad \gamma_\Gamma[\mathbf{u}_\omega] = \begin{bmatrix} \hat{\mathbf{n}} \times \mathbf{E}|_\Gamma \\ \hat{\mathbf{n}} \times \mathbf{H}|_\Gamma \end{bmatrix}, \tag{2.14}$$

with outward unit normal $\hat{\mathbf{n}}$ on Γ and trace space $\mathcal{T}_\Gamma$ induced by interior states. The map γ_Γ is not defined on arbitrary L^2 six-tuples; Sobolev and distribution refinements in section 2.1.1 specify its domain [2]. The closed operator stacks volume and boundary residuals:

$$\mathcal{L}_\omega[\mathbf{u}_\omega] = \begin{bmatrix} \mathcal{M}_\omega[\mathbf{u}_\omega] \\ \mathcal{B}_\Gamma[\gamma_\Gamma[\mathbf{u}_\omega]] \end{bmatrix}. \tag{2.15}$$

A physically closed harmonic problem requires both volume Maxwell balance and boundary termination to hold simultaneously; neither channel alone suffices [2]. The composite operator $\mathcal{L}_\omega$ encodes that joint requirement: its first block tests whether $(\mathbf{E}, \mathbf{H})$ satisfies the curl–curl structure with impressed sources removed to the forcing vector, while its second block tests whether tangential traces on Γ match the declared termination (PEC, impedance, or outgoing) [6]. Experimentally, a solver that minimizes volume residuals while ignoring port or chamber walls produces a formal Maxwell solution that may disagree with what a network analyzer reports on the same structure [2]. The harmonic boundary-value problem is to find $\mathbf{u}_\omega \in \mathcal{F}_\Omega$ such that

$$\mathcal{L}_\omega[\mathbf{u}_\omega] = \begin{bmatrix} \mathbf{s}_\omega(\mathfrak{D}) \\ \mathbf{g}_\Gamma \end{bmatrix}. \tag{2.16}$$

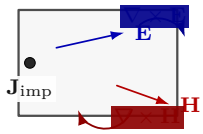


Figure 2.3: First-order Maxwell coupling in $\mathcal{M}_\omega$: $\nabla \times \mathbf{E}$ sources $\mathbf{H}$, $\nabla \times \mathbf{H}$ closes on $\mathbf{E}$ with impressed current $\mathbf{J}_{\mathrm{imp}}$ as external drive.

Perfect conductors impose $\hat{\mathbf{n}} \times \mathbf{E} = \mathbf{0}$; radiation and Leontovich completions encode outgoing or dissipative relations on $\gamma_\Gamma[\mathbf{u}_\omega]$. The operator $\mathcal{L}_\omega$ is the Maxwell balance operator for this coupled system [6]. Hence $\mathcal{L}_\omega$ enforces Maxwell balance with impressed drive and boundary termination; without outgoing selection it does not distinguish radiate from storage, and every near-field probe reading $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ presupposes its inversion on the declared domain [2]. Eq. (2.1) is the resolvent relation

$$\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{u}_\omega \quad \text{with} \quad \mathcal{L}_\omega[\mathbf{u}_\omega] = \begin{bmatrix} \mathbf{s}_\omega(\mathbf{J}_{\mathrm{imp}}, \rho) \\ \mathbf{g}_\Gamma \end{bmatrix}, \tag{2.17}$$

where ρ is tied to $\mathbf{J}_{\mathrm{imp}}$ through Eq. (1.8) when only electric impressed currents drive the system. Invertibility of $\mathcal{L}_\omega$ is not universal: existence and uniqueness depend on $\mathcal{B}_\Gamma$ and on whether outgoing selection is imposed (section 2.4.3 for exterior domains; section 2.6.2 for cavities) [2]. $\mathcal{S}_\omega$ is therefore a *physical branch* of $\mathcal{L}_\omega^{-1}$, not an abstract algebraic inverse on all of $\mathcal{F}_\Omega$. Figure 2.3 maps the schematic structure: Circulation coupling in : each curl row feeds the partner field; appears as external drive on the magnetic ....

Single-field Helmholtz reductions equivalent to Eqs. (1.13)–(1.15) are used later when a scalar symbol simplifies kernels; the elimination belongs to 1 and is not repeated. The electric-wave symbol used in 2 is recorded as

$$\mathcal{W}_{E,\omega}\mathbf{E} \equiv \nabla \times \left(\mu_0^{-1}\nabla \times \mathbf{E}\right) - \omega^2 \varepsilon_0 \mathbf{E} = -\mathrm{j}\omega\mu_0 \mathbf{J}_{\mathrm{imp}} + \mu_0^{-1}\nabla \times \mathbf{J}_{\mathrm{eff}}^{(\mathrm{mag})}, \tag{2.18}$$

with $\mathbf{J}_{\mathrm{eff}}^{(\mathrm{mag})}$ collecting equivalent magnetic drive when present [2]. Subsection 2.2.2 develops the magnetic counterpart and material-weighted generalizations. Variational structure is inherited from section 1.1.4. For test state $\mathbf{v}_\omega = (\mathbf{E}', \mathbf{H}')$,

$$\mathfrak{a}_\omega(\mathbf{u}_\omega, \mathbf{v}_\omega) = \int_\Omega \left[\mu_0\, \mathbf{H}\cdot\nabla \times \mathbf{E}' - \varepsilon_0\, \mathbf{E}\cdot\nabla \times \mathbf{H}'\right] d^3r. \tag{2.19}$$

Integration by parts connects Eq. (2.19) to surface fluxes and to the weak formulation

$$\mathfrak{a}_\omega(\mathbf{u}_\omega, \mathbf{v}_\omega) = \mathfrak{f}_\omega(\mathbf{v}_\omega) \quad \forall\, \mathbf{v}_\omega \in \mathcal{F}_\Omega^0, \tag{2.20}$$

where $\mathcal{F}_\Omega^0$ carries homogeneous boundary traces compatible with $\mathcal{B}_\Gamma$ and $\mathfrak{f}_\omega$ is induced by $\mathbf{s}_\omega(\mathfrak{D})$ [5]. Equation (2.20) is the variational face of Eq. (2.16); finite-element discretizations favor this route when material coefficients are piecewise constant [6]. Well-posedness of Eq. (2.20) is not automatic on unbounded domains or on multiply connected cavities. On a bounded region with passive $\mathcal{B}_\Gamma$ and homogeneous traces in $\mathcal{F}_\Omega^0$, coercivity fails only along divergence-free cavity modes that satisfy both curl-curl balance and homogeneous tangential data—precisely the null space that section 2.1.4 isolates before imposing gauge or polarization fixes [6]. For exterior problems, coercivity is recovered only after restricting to the outgoing subspace selected by Sommerfeld-type completion; without that restriction, the bilinear form is indefinite and the weak problem admits non-physical growing solutions. The practical rule for antenna analysis is therefore threefold: declare $\mathcal{D}(\mathcal{L}_\omega)$, specify $\mathcal{B}_\Gamma$ on every connected piece of Γ, and confirm that the impressed tuple lies in the compatible

range of Eq. (2.23). Numerical codes that enforce volume Maxwell rows on truncated domains but leave exterior faces open implement an *implicit* $\mathcal{B}_\Gamma$; unless that implicit completion matches the laboratory termination, $\mathcal{S}_\omega$ in the code is a different operator from the one measured on the bench [6]. Domain specification cannot remain at C^∞ when sources are distributional. The minimal energy class declared here is

$$\mathcal{D}(\mathcal{L}_\omega) \subset \{\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H}) : \mathbf{E}, \mathbf{H} \in L^2(\Omega)^3,\ \nabla \times \mathbf{E}, \nabla \times \mathbf{H} \in L^2(\Omega)^3\}, \tag{2.21}$$

where $\mathcal{L}_\omega$ is the composite Maxwell operator on the six-component state $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H})$, $\mathcal{D}(\mathcal{L}_\omega)$ its domain of square-integrable curl fields compatible with $\mathcal{B}_\Gamma$, and Ω the analysis region [2]. section 2.1.1 refines embeddings and compactness; evaluating Eq. (2.13) outside $\mathcal{D}(\mathcal{L}_\omega)$ generates spurious boundary terms that corrupt flux identities [2]. Piecewise smooth $\varepsilon(\mathbf{r})$ and $\mu(\mathbf{r})$ modify coefficients inside $\mathcal{M}_\omega$ while preserving the block template of Eq. (1.7). Interface jumps satisfy Eq. (1.19); multi-region Green operators in section 2.3.5 realize $\mathcal{L}_\omega^{-1}$ on composite domains [2]. The coupling structure described above extends naturally to practical terminal conditions. Finite feed structures and external circuits frequently impose impedance relations between tangential field components on the boundary [6]:

$$\mathcal{Z}_\Gamma \begin{bmatrix} \hat{\mathbf{n}} \times \mathbf{E}|_\Gamma \\ \hat{\mathbf{n}} \times \mathbf{H}|_\Gamma \end{bmatrix} = \mathbf{0}, \tag{2.22}$$

In Eq. (2.22), $\mathcal{Z}_\Gamma$ is a linear boundary impedance operator (SI units absorbed in its definition) acting on the tangential Maxwell traces $\hat{\mathbf{n}} \times \mathbf{E}|_\Gamma$ and $\hat{\mathbf{n}} \times \mathbf{H}|_\Gamma$ on $\Gamma = \partial\Omega$; admissible states satisfy the terminal or absorbing relation $\mathcal{Z}_\Gamma[\text{traces}] = \mathbf{0}$. encoding Leontovich, absorbing, or mode-matched completions [6]. When Eq. (2.22) replaces explicit $\mathbf{g}_\Gamma$ in Eq. (2.16), admissible states satisfy the impedance manifold on Γ. Exterior outgoing problems correspond to a limit in which $\mathcal{Z}_\Gamma$ selects the Sommerfeld branch of section 2.4.3; conflating a finite-chamber termination with that limit without normalization corrupts far-zone patterns [2]. Forcing must lie in the range implied by charge conservation. Divergence of the Ampère row together with Eq. (1.8) yields

$$\nabla \cdot \mathbf{s}_{\omega,2} + \mathrm{j}\omega\, \mathbf{s}_{\omega,3} = 0 \tag{2.23}$$

on the second and third blocks of Eq. (2.12). Inconsistent $(\mathbf{J}_{\text{imp}}, \rho)$ pairs lie outside physically realizable drive; numerical solvers may still return vectors that violate Eq. (2.23) [2]. A compact current filament with vanishing ρ automatically satisfies Eq. (2.23) because $\mathbf{s}_{\omega,3} = 0$. When a volumetric ρ is prescribed independently of $\mathbf{J}_{\text{imp}}$, the pair must be adjusted so that their divergence balance matches the Ampère–Gauss rows; otherwise the closed problem is mathematically posed but physically inconsistent. Feed models that inject $\mathbf{J}_{\text{imp}}$ without updating ρ are common in method-of-moments implementations; they remain valid only when the discretization enforces continuity on the impressed grid or when the problem is magnetostatic in the relevant band [6]. Linking to Subsection 2.0.1, radiation acts on closed solutions:

$$\mathcal{R}_\omega = \Pi_{\text{rad}} \circ \mathcal{S}_\omega = \Pi_{\text{rad}} \circ \mathcal{L}_\omega^{-1} \circ \begin{bmatrix} \mathbf{s}_\omega(\cdot) \\ \mathbf{g}_\Gamma \end{bmatrix}, \tag{2.24}$$

with $\mathcal{L}_\omega^{-1}$ understood on the outgoing-admissible branch fixed by $\mathcal{B}_\Gamma$ and with Π_{rad} from Eq. (2.2) [2]. Completion must precede radiative selection: applying Π_{rad} to non-closed states mixes reactive content into far-zone observables [6]. The energy form supports an integration-by-parts identity

$$\mathfrak{a}_\omega(\mathbf{u}_\omega, \mathbf{v}_\omega) = \langle \mathcal{M}_\omega[\mathbf{u}_\omega], \mathbf{v}_\omega \rangle_\Omega + \int_\Gamma \big(\hat{\mathbf{n}} \times \mathbf{E} \cdot \mathbf{H}' - \hat{\mathbf{n}} \times \mathbf{E}' \cdot \mathbf{H}\big) dS, \tag{2.25}$$

where $\langle\cdot,\cdot\rangle_\Omega$ is the componentwise L^2 pairing [2]. Self-adjoint truncations on bounded domains require that $\mathcal{B}_\Gamma$ cancel the boundary integrand on admissible traces; criteria are supplied in section 2.1.4. Equation (2.25) is the adjoint pairing on which those arguments rest. Fredholm structure note: for bounded Ω with passive $\mathcal{B}_\Gamma$, $\mathcal{L}_\omega$ is a Fredholm operator on $\mathcal{D}(\mathcal{L}_\omega)$ modulo cavity modes, with index tied to topology and boundary class. Exterior domains replace discrete spectra with continuous radiation modes, and $\mathcal{L}_\omega^{-1}$ becomes a resolvent on the outgoing subspace rather than an ordinary inverse. section 2.6.4 develops the spectral/Fredholm foundation; here the point is that invertibility questions for $\mathcal{S}_\omega$ are questions about $\mathcal{L}_\omega$ on the *declared* domain, not about formal elimination of $\mathbf{H}$ alone [6]. Linearity of $\mathcal{L}_\omega$ in $\mathbf{u}_\omega$ implies that superposition holds for fixed boundary class: if $\mathcal{L}_\omega[\mathbf{u}_\omega^{(a)}] = [\mathbf{s}^{(a)}, \mathbf{g}^{(a)}]$ and $\mathcal{L}_\omega[\mathbf{u}_\omega^{(b)}] = [\mathbf{s}^{(b)}, \mathbf{g}^{(b)}]$ with identical $\mathcal{B}_\Gamma$, then $\mathcal{L}_\omega[\mathbf{u}_\omega^{(a)} + \mathbf{u}_\omega^{(b)}] = [\mathbf{s}^{(a)} + \mathbf{s}^{(b)}, \mathbf{g}^{(a)} + \mathbf{g}^{(b)}]$. Changing $\mathcal{B}_\Gamma$ while holding $\mathbf{J}_{\mathrm{imp}}$ fixed breaks this principle because boundary rows are part of the operator, not of the drive. Laboratory comparisons that swap terminations without re-solving the coupled system therefore compare different $\mathcal{L}_\omega$, not different "modes" of the same map [2]. The radiation operator $\mathcal{R}_\omega$ inherits the same sensitivity: composition Eq. (2.24) is linear in impressed data only after $\mathcal{B}_\Gamma$ and the outgoing branch are held fixed [2]. Subsection 2.0.2 has packaged harmonic Maxwell theory as the composite operator $\mathcal{L}_\omega$, recorded its resolvent relation to $\mathcal{S}_\omega$, and linked radiation composition Eq. (2.24) to operator inversion on the outgoing-admissible branch. Subsection 2.0.3 records how those maps inherit the admissibility, regime, and measurement criteria established in 1.

2.0.3 Admissibility and Maxwell radiation sequence

A coax-fed pyramidal horn over a ground plane defines the impressed-current and boundary-data tuple technicians declare before Maxwell radiation is computed [2]. Maxwell's retarded radiation to Γ_R remains the governing constraint [1]. Subsections 2.0.1–2.0.2 introduced the radiation maps and Maxwell radiation sequence; the display below invokes 1 $\mathcal{D}_J = 1$ [2]. Chapter 2 applies the foundations established in 1 to radiation problems [1]. The Maxwell-closed criteria, conservation laws, and measurement principles developed in 1 remain in force; radiation operators connect admissible sources to observable radiative fields [6]. The operators $\mathcal{S}_\omega$, $\mathcal{R}_\omega$, and $\mathcal{L}_\omega$ are applied on admissible inputs without reopening the 1 proofs [2]. Impressed data enter the solution domain only when the 1 source conditions hold; radiative reports attach to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$, not to the full solution pair by default [2]. The continuity relations below restate, in operator language, the 1 regime and measurement criteria under which angular-flux integrals attach to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ rather than to the unreconstructed solution pair [2]. Physically, this means that near-field reactive content diagnosed on $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ must not be reported as radiated radiative flux unless the regime restriction $\mathcal{D}_{J,\mathrm{reg}}$ from 1 is satisfied on the measurement shell [6].

$$\text{Under Chapter 1 admissibility,}\quad \mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega \ \wedge\ \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \in \mathcal{F}_\Omega. \tag{2.26}$$

$$\mathcal{D}_{J,\mathrm{reg}} = 1 \implies \text{angular-flux reports attach to } \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}],\ \text{not to } \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ by default,} \tag{2.27}$$

$$\Pi(\theta,\phi;r) = \Pi_{\mathrm{a}}(\theta,\phi;r) + \Pi_{\mathrm{re}}(\theta,\phi;r), \qquad \Pi_{\mathrm{a}} = \mathrm{Re}\{\mathbf{S}\cdot\hat{\mathbf{n}}\}, \tag{2.28}$$

with reactive sector Π_{re} the remainder on the declared control surface.

$$Q_{\mathrm{re}}(r) = \mathrm{Im}\oint_{S_r} \mathbf{S}\cdot\hat{\mathbf{r}}\,dS, \qquad P_{\mathrm{rad}}(r) = \mathrm{Re}\oint_{S_r} \mathbf{S}\cdot\hat{\mathbf{r}}\,dS, \tag{2.29}$$

$$\chi(r) = \frac{|Q_{\mathrm{re}}(r)|}{P_{\mathrm{rad}}(r) + \epsilon_P}, \qquad \epsilon_P > 0. \tag{2.30}$$

$$P_{\mathrm{rad}}(r) > 0 \ \wedge \ \chi(r) \le \chi_{\mathrm{th}}, \tag{2.31}$$

$$r_c : \ \chi(r_c) = \chi_{\mathrm{th}}, \tag{2.32}$$

$$\mathcal{D}_{J,\mathrm{reg}}(r) = 1 \iff \Big(\chi(r) \le \chi_{\mathrm{th}} \ \wedge \ \mathcal{D}_J(r) = 1\Big), \tag{2.33}$$

$$\oint_{S_r} \mathrm{Re}\{\mathbf{S}_R \cdot \hat{\mathbf{r}}\}\, dS = \oint_{S_r} \mathrm{Re}\{\mathbf{S} \cdot \hat{\mathbf{r}}\}\, dS, \tag{2.34}$$

$$\mathcal{I}[\mathcal{P}_R \mathcal{F}] = \mathcal{I}[\mathcal{F}], \tag{2.35}$$

for scalar observables $\mathcal{I}$ under passive frame rotation $\mathcal{P}_R$.

$$\mathcal{R}_{\mathcal{F}} = \left\| \mathcal{R}_\omega[\mathcal{T}_{R,\mathbf{a}} \mathbf{J}_{\mathrm{imp}}] - \mathcal{T}_{R,\mathbf{a}}\, \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \right\|, \tag{2.36}$$

$$\mathbf{J}_{\mathrm{int}} = \mathbf{J}_{\mathrm{em}} - (\mathbf{r} - \mathbf{r}_c) \times \mathbf{P}_{\mathrm{tot}}, \tag{2.37}$$

$$\mathbf{E}_\perp = \mathbf{E} - (\hat{\mathbf{r}} \cdot \mathbf{E})\hat{\mathbf{r}}, \qquad \mathbf{H}_\perp = \mathbf{H} - (\hat{\mathbf{r}} \cdot \mathbf{H})\hat{\mathbf{r}}, \tag{2.38}$$

$$S_0 = |E_x|^2 + |E_y|^2, \quad S_1 = |E_x|^2 - |E_y|^2, \quad S_2 = 2\,\mathrm{Re}(E_x E_y^*), \quad S_3 = 2\,\mathrm{Im}(E_x E_y^*), \tag{2.39}$$

$$\Omega_P = \hat{\mathbf{n}} \cdot (\nabla \times \mathbf{P}), \qquad \mathbf{P} = \mathrm{Re}\{\mathbf{S}\}, \tag{2.40}$$

where $\mathcal{D}_{J,\mathrm{reg}}$ is the regime flag from Eq. (2.33). The continuity composition law composes inherited 1 constraints with radiation selection and finite observation:

$$\mathfrak{A}_{\mathrm{ch2}}^{(\mathrm{report})} = \mathfrak{A}_{\mathrm{meas}} \circ \mathfrak{A}_{\mathrm{g}} \circ \mathfrak{A}_{\mathrm{cov}} \circ \mathfrak{A}_{\mathrm{tr}} \circ \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega, \tag{2.41}$$

$$\Lambda_{\mathrm{lab}} = \mathcal{O}_{\mathrm{meas}} \circ \mathcal{O}_{\mathrm{far}} \circ \mathcal{R}_\omega, \qquad \mathcal{Q} \in \{\mathrm{direct},\ \mathrm{surrogate},\ \mathrm{inference}\}, \tag{2.42}$$

$$\mathcal{R}_\omega[\mathcal{T}_{R,\mathbf{a}} \mathbf{J}_{\mathrm{imp}}] = \mathcal{T}_{R,\mathbf{a}}\, \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \quad \text{on admissible data}, \tag{2.43}$$

with residual Eq. (2.36) serving as the numerical diagnostic when discretization breaks Eq. (2.43) [2]. Function spaces, Green functions, and radiation operators exist because Maxwell field specification requires them—not as parallel formulations [1]. section 2.0 has established the radiation maps, Maxwell radiation sequence, and 1 continuity; section 2.1 now supplies the Hilbert and Sobolev apparatus through which those maps acquire precise domains, traces, and adjoints [2].

2.1 Function Spaces and Maxwell Operators

Every practical electromagnetic system occupies finite space, is driven by finite sources, and is observed through finite apertures [2]. These physical limitations restrict what fields can be generated, measured, and transported. Before the Sobolev symbols $\mathcal{H}_{\nabla\times}(\Omega)$ and $L^2(\Omega)$ appear, ask what electromagnetic object lives here: not an abstract Hilbert vector, but a horn aperture, a slot on a ground plane, a finite feed current, or a near-field scan surface whose tangential $\mathbf{E}$ must be the trace of a volume field [2].

A horn feed launches aperture currents whose rim singularity is visible in near-zone scans; a patch or slot on a ground plane carries sheet-like $\mathbf{J}_{\mathrm{imp}}$ in method-of-moments assembly; a finite-element port model enforces tangential $\hat{\mathbf{n}} \times \mathbf{E}$ on conductors while impressing currents drawn from $\mathcal{J}_\Omega$. Each construction asks whether a field pair $(\mathbf{E}, \mathbf{H})$ can satisfy Maxwell's equations and match what probes record on $\Gamma = \partial\Omega$.

The solution map $\mathcal{S}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_\Omega$, the composite Maxwell operator $\mathcal{L}_\omega$ on $\mathcal{D}(\mathcal{L}_\omega)$, and the radiation map $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ were named in section 2.0.1; here $\mathcal{F}_\Omega$ is the admissible field space and Π_{rad} the outgoing projection that qualifies far-zone content [6]. The adjoint $\mathcal{R}_\omega^*$ is the reception

map dual to transmission: reciprocity identities pair excitation with measurement functionals on the same radiation operator. Rank deficiency is an algebraic null of $\mathcal{R}_\omega$; spectral compression is the gradual singular-value decay of the same map on finite support—related, but not identical. Function spaces provide the mathematical language through which those restrictions are expressed. They do not replace Maxwell's equations; rather, they specify the class of fields and sources to which Maxwell's equations may be consistently applied [1]. Self-adjoint completion is invoked from standard functional analysis; this section records electromagnetic consequences rather than reproving operator theory.

Every space symbol enters in physical sequence before formal sequence. First comes the source or field one is trying to represent on declared hardware; next comes the regularity required for Maxwell closure, trace evaluation, and reciprocity pairing; only then does the Hilbert or Sobolev notation become useful. This order matters because Function spaces here are not mathematical ornament. It introduces them so that later claims about radiation, null spaces, compactness, and inversion rest on fields and sources that could actually belong to the laboratory situations the chapter has named.

section 2.1 now supplies the Hilbert and Sobolev apparatus—domain, range, adjoint, boundedness, and completion—through which finite source extent and finite observation apertures appear as accessibility constraints on Maxwell maps rather than as after-the-fact modeling choices [2]. Near, intermediate, and far regimes are separated by $kr \sim 1$ and $kr \gg 1$ scales; those transitions are asymptotic order markers, not sharp physical discontinuities. Spectral decomposition enters only after outgoing selection; consequences of compact geometry are developed once in section 2.6 and invoked thereafter by reference [7].

The finite-energy criterion of Eq. (1.3) and admissible state class Eq. (1.4) from Subsection 1.0.4 remain the physical floor. Phasor reduction discards transient envelope and broadband spectral content; it is valid only after a single-frequency laboratory band has been declared. Causality is imposed on the retarded operator $\mathcal{S}_\omega$, not merely on time-domain field samples: advanced Green branches violate the laboratory excitation order. The dyadic Green singularity is the self-field of the impressed current at coincident points—a near-zone reactive effect, not a far-zone radiative channel; self-term regularization is both a numerical necessity and a modeling choice that specifies how finite feed extent replaces the point singularity.

Far-field reduction discards evanescent and reactive sheets that cannot be recovered from Γ_R data alone. The operator norm bounds worst-case radiated energy per unit source norm—a capacity measure for how strongly compact support can radiate to $\mathcal{F}_{\mathrm{rad}}$. Unstable inversion means distinct source distributions produce indistinguishable far-zone data—the laboratory consequence is non-unique source reconstruction. The Green operator is the integral map on $\mathcal{J}_\Omega$; the Green function is its kernel—two views of the same retarded propagation law. The vector Sommerfeld condition adds polarization and divergence structure to the scalar outgoing selector; it is not a mere notational generalization. Uniqueness eliminates the ambiguity of adding incoming or non-radiating complements that leave far-zone probes unchanged yet alter near-zone storage.

Subsection 1.1.2 fixed source tuples, boundary operators, and charge continuity; every space declaration below must remain compatible with Eq. (2.26) and must not reopen 1 proofs [2]. The guiding constraint from Subsection 2.0.3 persists without restatement here. Chapter 2 radiation analysis joins Chapter 1 physical admissibility—charge continuity, finite energy, and gauge-fixed transport—with radiation-operator admissibility and outgoing selection; failure at either level voids spectral claims.

Subsection 2.1.1 defines source, field, and dual spaces for phasor Maxwell theory, including distributional sources and trace spaces on $\Gamma = \partial\Omega$, importing finite-energy charge-continuous constraints from section 1.0.4 and source–boundary coupling from section 1.1.2. The singular-value decay rate is set primarily by electrical size $k_0 D_s$ and support outline through the lateral spectrum

of $\mathbf{J}_{\mathrm{imp}}$. Rotational symmetry makes $\mathcal{R}_\omega$ commute with angular momentum, so distinct harmonic labels can share the same singular value—geometric degeneracy, not measurement redundancy.

Subsection 2.1.2 treats linear operators on those spaces: domain, range, graph norm, and adjoint pairing compatible with Eq. (2.25). Operator reciprocity reduces to measurable two-port symmetry: interchanged excitation and observation ports yield the same coupled watt reading when media are reciprocal. Mode orthogonality is stated with respect to the energy inner product on the declared domain; far-zone power orthogonality follows only after outgoing selection.

Subsection 2.1.3 develops boundedness and compactness criteria, rank deficiency of observation maps, and the spectral compression viewpoint developed for radiation operators in section 2.6. Fredholm structure is cited from standard compact-operator theory; only the radiation consequences are developed here. Angular-spectrum selection is generally lossy: evanescent content suppressed at the measurement plane cannot be recovered from far-zone probes alone. Non-reconstructable content refers to modal coefficients below the noise floor of the measurement functional, not to a second physical field copy.

Subsection 2.1.4 addresses self-adjoint realizations, Friedrichs and other completions, and the $\mathcal{D}_J = 1$ constraints under which truncated Maxwell operators inherit coercivity or passivity on bounded domains [6].

Several constructions deliberately remain outside section 2.1. Dyadic Green kernels and retarded inverses belong to section 2.3; Sommerfeld outgoing selection and exterior uniqueness to section 2.4; radiation integrals and flux operators at the radiative boundary to section 2.5. Plane-wave decomposition is established in Chapter 3 §3.8; here it is used as the angular-spectrum chart for radiation fields. Suppressing evanescent modes sets a practical sub-wavelength resolution ceiling on any probe grid confined to a single measurement plane.

Chapter 3 Hilbert-space machinery uses §2.6 spectral results as soft prerequisites for normalization and completeness statements. section 2.1 prepares the domains on which those maps act. Attempting to evaluate $\mathcal{L}_\omega^{-1}$ through a Green function before trace spaces and dual pairings are fixed produces kernel formulas that ignore boundary rows and corrupt flux identities [2]. Conversely, staying at the L^2-level of Eq. (2.21) without Sobolev refinement is adequate for uniqueness statements on smooth domains but insufficient for sheet sources, impedance boundaries, and finite-element discretizations where $\nabla\times$ and $\hat{\mathbf{n}}\times$ must be closed simultaneously [6].

The energy bilinear form Eq. (2.19) and weak formulation Eq. (2.20) from Subsection 2.0.2 anchor the Hilbert-space choices. section 2.1 identifies the minimal completion in which $\mathfrak{a}_\omega$ is continuous and coercive modulo cavity modes, and in which the forcing functional $\mathfrak{f}_\omega$ induced by $\mathfrak{s}_\omega(\mathfrak{D})$ is bounded on test states with homogeneous traces. Well-posedness of the weak problem on bounded domains then follows from Lax–Milgram arguments once null spaces aligned with divergence-free resonances are quotiented; the quotient construction is owned by Subsection 2.1.4 and feeds the cavity-mode analysis of section 2.6.4 [2]. Exterior domains require outgoing subspaces rather than full Hilbert self-adjointness; that distinction is preserved throughout so that interior cavity spectra are not conflated with continuous radiation spectra [6]. Rank deficiency enters at two levels that section 2.1 must keep separate. Instrument maps—finite apertures, modal truncations, probe coupling—are finite-rank operators on field space even when $\mathcal{L}_\omega$ is injective on the physical branch. Equivalent source distributions in $\ker \mathcal{R}_\omega$ produce identical far-zone patterns while differing in reactive near fields; that null-space structure is analyzed in section 2.7, but the compact embedding lemmas used there are developed in Subsection 2.1.3. Measurement surrogates from section 1.5 therefore appear as rank-deficient compositions on the right of Eq. (2.41), not as modifications of Maxwell law [2]. Method-of-moments and finite-element discretizations are valid only when the discrete spaces inherit the continuity and trace structure defined in Subsections 2.1.1–2.1.2. section 2.1 states the continuous targets those discretizations approximate; it does not substitute for mesh

convergence analysis [6]. Material inhomogeneity and interface conditions from Eq. (1.19) enter through piecewise constant coefficients in $\mathcal{M}_\omega$ without changing the space template: fields remain in product Sobolev spaces with jump conditions enforced on traces. Multi-region formulations in section 2.3.5 rely on the interface trace classes introduced in Subsection 2.1.1. Perfect conductors impose homogeneous tangential $\mathbf{E}$; impedance and Leontovich boundaries modify the boundary pairing in Eq. (2.25) and are treated as closure operators in Subsection 2.1.4 [2]. Subsection 2.1.4 completes the function-space foundation begun here with self-adjoint truncations and Maxwell-closed completions that connect directly to the spectral/Fredholm program of section 2.6.4. The subsection sequence begins with the space definitions that make $\mathcal{D}(\mathcal{L}_\omega)$ in Eq. (2.21) precise enough for the Green, Physically, radiation, and spectral sections that follow.

The measurable consequence is therefore a directional or rank limit on radiation, not a calibration correction: recalibration cannot remove obstructions imposed by $\ker \mathbf{\Gamma}_\omega$ [6].

2.1.1 Spaces for sources, fields, and distributions

Function spaces describe admissible currents and fields before any antenna synthesis [2]. Compact support and finite power are laboratory facts, not optional regularity [1]. The physical problem is source realizability on finite hardware: every antenna measurement begins with currents and boundary data that occupy bounded regions, carry finite power, and meet conductors where tangential $\mathbf{E}$ or $\mathbf{H}$ is measured [2]. A horn feed, patch edge, or waveguide flange excites currents that are not smooth functions on all of $\mathbb{R}^3$: sheet sources, rim currents, and port terminals appear as distributions whose support is bounded yet whose derivatives fail at edges [2]. Without the Sobolev and trace spaces declared here, one obtains Green formulas that ignore boundary rows, mis-state flux through $\Gamma = \partial\Omega$, and corrupt near-field transforms on spherical scanners [6]. Sobolev and trace classes encode the physical constraints imposed by finite power, bounded supports, conducting boundaries, and realizable feed structures [6]. Synthesis that ignores these classes may produce mathematically sparse currents that no laboratory feed can realize [17].

A PEC-backed slot fed by $\mathbf{J}_{\text{imp}} \in L^2(\Omega_s)$ and an idealized $1/r$ current template illustrate the admissibility distinction at fixed frequency. Only the compact L^2 source class supports finite radiated power; divergent templates violate the finite-energy $\mathcal{D}_J = 1$ conditions inherited from Chapter 1 [2]. Compact support and finite power are laboratory facts, not optional regularity [1]. Method-of-moments and finite-element antenna models represent impressed currents on surface patches and volume filaments, then test Maxwell's equations with localized basis functions [6]. That procedure requires square-integrable $\nabla\times$ fields, well-defined tangential traces on conductors and apertures, and distributional extensions when sources collapse to sheets—precisely the analytic gaps that naive L^2 current models leave open. Subsections 2.0.1–2.0.2 declared minimal spaces $\mathcal{F}_\Omega$ and $\mathcal{J}_\Omega$ adequate to *name* the maps $\mathcal{S}_\omega$, $\mathcal{R}_\omega$, and $\mathcal{L}_\omega$; Subsection 2.1.1 fixes the Sobolev, trace, and distributional structure those declarations presuppose. Finite-energy Maxwell-closed from Subsection 1.0.4 and the source–boundary tuple of Subsection 1.1.2 remain authoritative; embeds them in spaces where $\nabla\times$, $\nabla\cdot$, and tangential restriction are continuous maps with explicit domains [2]. Nothing here re-proves charge continuity, interface jumps, or outer-boundary classes already fixed in 1. Let $\Omega \subset \mathbb{R}^3$ be a bounded Lipschitz domain or a finite union of Lipschitz patches $\{\Omega_a\}$ with outward boundary $\Gamma = \partial\Omega$. Phasor fields use the $\exp(+\mathrm{j}\omega t)$ convention throughout. In the display below, $L^2(\Omega)$ denotes square-integrable field energy on Ω; $\mathcal{H}_{\nabla\times}(\Omega)$ and $\mathcal{H}_{\text{div}}(\Omega)$ are the minimal vector Sobolev classes in which curl-coupled Maxwell balances close weakly; $\mathcal{J}_\Omega$ collects impressed currents of compact support and finite power; $\mathcal{F}_\Omega$ is the admissible Maxwell field class; and $\mathcal{S}_\omega$, $\mathcal{R}_\omega$, and $\mathcal{L}_\omega$ are the solution, radiation, and composite operators declared in section 2.0.1 [2]. Square-integrability alone is too weak for curl-coupled Maxwell theory: one needs control of

both the field and its curl so that circulation balances are meaningful in the L^2 sense. An engineer measuring tangential $\mathbf{E}$ on a patch boundary is, in effect, asking whether the recorded trace is the restriction of a field with square-integrable curl; a method-of-moments code that tests $\nabla \times \mathbf{E}$ against rooftop functions presupposes the same control. The space $\mathcal{H}_{\nabla\times}(\Omega)$ names precisely that charge-continuous class: elements of $L^2(\Omega)^3$ whose curls also lie in $L^2(\Omega)^3$, so Faraday and Ampère balances can be posed weakly without assuming pointwise smoothness at horn rims or slot edges [2]. Note that $\mathcal{H}_{\nabla\times}(\Omega)$ is stronger than L^2 alone yet weaker than requiring classical C^1 fields everywhere; it is the minimal completion in which curl-coupled Maxwell operators close on bounded Ω [6]. The companion $\mathcal{H}_{\mathrm{div}}(\Omega)$ plays the parallel role for Gauss constraints when divergence must be controlled in the L^2 sense. With that physical reading fixed, the formal definitions are: Single-vector Sobolev classes encode the minimal regularity that curl-coupled Maxwell theory requires on bounded domains. Membership in $\mathcal{H}_{\nabla\times}(\Omega)$ controls both field amplitude and circulation in the L^2 sense, whereas plain L^2 control would not guarantee that Faraday and Ampère balances can be posed weakly at horn rims or slot edges [2]. From a measurement standpoint, these spaces establish that tangential fields recorded on Γ are traces of physically admissible volume states, not arbitrary boundary templates [6]. Single-vector Sobolev classes are

$$\mathcal{H}_{\nabla\times}(\Omega) = \left\{ \mathbf{u} \in L^2(\Omega)^3 : \nabla \times \mathbf{u} \in L^2(\Omega)^3 \right\}, \tag{2.44}$$

Watt on Γ_R changes only through the outgoing field pair already constrained by Maxwell structure on the selected branch [2].

$$\mathcal{H}_{\mathrm{div}}(\Omega) = \left\{ \mathbf{u} \in L^2(\Omega)^3 : \nabla \cdot \mathbf{u} \in L^2(\Omega) \right\}, \tag{2.45}$$

where $\nabla\cdot\mathbf{u}$ is the divergence of $\mathbf{u}$ (m^{-2} for current-like fields, $\mathrm{C\,m}^{-3}$ when $\mathbf{u}$ represents a displacement-related quantity), required to be square integrable so that Gauss-type constraints can be enforced weakly. each equipped with the graph norm $\|\mathbf{u}\|^2_{\mathcal{H}_{\nabla\times}} = \|\mathbf{u}\|^2_{L^2} + \|\nabla \times \mathbf{u}\|^2_{L^2}$ and the analogous $\mathcal{H}_{\mathrm{div}}$ norm [2]. Equations (2.44)–(2.45) are the vector analogues of the finite-energy floor Eq. (1.3): they permit Maxwell curl-coupling without assuming pointwise smoothness of $\mathbf{E}$ or $\mathbf{H}$. The consequence for antenna modeling is direct: horn rims, slot edges, and patch boundaries may host singularities, yet the impressed current and recovered field remain admissible as long as they lie in the declared Sobolev classes [6]. For piecewise constant ε, μ, fields are understood patchwise in Eqs. (2.44)–(2.45) with interface traces constrained by Eq. (1.19) [6]. The Maxwell state space is the product

$$\mathcal{U}_\omega = \mathcal{H}_{\nabla\times}(\Omega) \times \mathcal{H}_{\nabla\times}(\Omega), \qquad \mathbf{u}_\omega = (\mathbf{E}, \mathbf{H}) \in \mathcal{U}_\omega. \tag{2.46}$$

Eq. (2.46) refines the six-component packaging Eq. (2.11): both components carry square-integrable curls, matching the first-order structure of Eq. (1.7). A weighted energy norm compatible with Eq. (2.19) is

$$\|\mathbf{u}_\omega\|^2_{\mathcal{U}} = \|\mathbf{E}\|^2_{L^2} + \|\mathbf{H}\|^2_{L^2} + \alpha \|\nabla \times \mathbf{E}\|^2_{L^2} + \beta \|\nabla \times \mathbf{H}\|^2_{L^2}, \tag{2.47}$$

with positive weights α, β fixed by constitutive scaling in each patch. Norm (2.47) controls the bilinear form $\mathfrak{a}_\omega$ of Eq. (2.20) on bounded domains and makes $\mathcal{U}_\omega$ a Hilbert space [2].

Embedding ladder for Maxwell fields: $\mathcal{H}_{\nabla\times}$ and $\mathcal{H}_{\mathrm{div}}$ sit above L^2; tangential traces map $\mathcal{H}_{\nabla\times}$ fields to boundary data compatible with Eq. (1.18).

Divergence control enters through source spaces and through Gauss constraints rather than through requiring both components to lie in $\mathcal{H}_{\mathrm{div}}$; forcing $\mathbf{E} \in \mathcal{H}_{\mathrm{div}}$ and $\mathbf{H} \in \mathcal{H}_{\mathrm{div}}$ simultaneously is stronger than Maxwell's first-order system requires and is reserved for scalar-potential reductions deferred to later chapters [2]. Impressed sources refine the declaration of Subsection 2.0.1. The

admissible impressed-current space collects currents of finite support and finite power that satisfy the 1 $\mathcal{D}_J = 1$ restriction. Let $\Omega_s \subseteq \Omega$ denote compact source support. The admissible impressed current space is

$$\mathcal{J}_\Omega = \{\mathbf{J}_{\mathrm{imp}} \in \mathcal{H}_{\nabla\times}(\Omega) : \operatorname{supp}\mathbf{J}_{\mathrm{imp}} \subseteq \Omega_s,\ \mathcal{D}_J[\mathbf{J}_{\mathrm{imp}}] = 1\}, \tag{2.48}$$

where $\mathcal{D}_J$ is the criterion of Eq. (2.0.1) inherited from Eq. (1.160) and Eq. (1.22) [2]. Equation (2.48) strengthens square integrability to $\mathcal{H}_{\nabla\times}$ regularity so that $\nabla\times\mathbf{J}_{\mathrm{imp}}$ is defined in L^2 whenever numerical or analytical measurement sequences differentiate sources. Charge-neutral subclasses used in radiation null-space analysis satisfy

$$\mathcal{J}_\Omega^{(0)} = \{\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega : \nabla\cdot\mathbf{J}_{\mathrm{imp}} = 0\}, \tag{2.49}$$

with $\rho = 0$ implied on Ω_s. Eq. (2.49) is the subspace on which divergence-free reparametrizations act without altering $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ in simply connected regions; null-space structure is developed in section 2.7.1 [6]. Boundary data from the tuple Eq. (1.17) live in trace spaces rather than in volume Sobolev classes. Define

$$\mathcal{T}_\Gamma = H^{-1/2}(\Gamma)^3 \times H^{-1/2}(\Gamma)^3, \tag{2.50}$$

as the product space for tangential electric and magnetic traces induced by $\mathcal{H}_{\nabla\times}(\Omega)$ fields on Lipschitz Γ [2]. The tangential trace operators extend Eq. (2.14):

$$\gamma_{\tau,E} : \mathcal{H}_{\nabla\times}(\Omega) \to H^{-1/2}(\Gamma)^3, \quad \gamma_{\tau,E}[\mathbf{E}] = \hat{\mathbf{n}}\times\mathbf{E}|_\Gamma, \tag{2.51}$$

The display fixes flux through Γ_R: no coefficient rearrangement can create radiation where Π_{rad} has removed incoming content [5].

$$\gamma_{\tau,H} : \mathcal{H}_{\nabla\times}(\Omega) \to H^{-1/2}(\Gamma)^3, \quad \gamma_{\tau,H}[\mathbf{H}] = \hat{\mathbf{n}}\times\mathbf{H}|_\Gamma, \tag{2.52}$$

with outward unit normal $\hat{\mathbf{n}}$. Equations (2.51)–(2.52) make Eq. (1.18) an operator between declared Banach spaces: $\mathcal{B}_\Gamma : \mathcal{T}_\Gamma \to \mathcal{Y}_\Gamma$ for data space $\mathcal{Y}_\Gamma$ fixed by the boundary class of Eq. (1.20) or outgoing traces of Eq. (1.21) [2].

Figure 2.2 maps the trace–data chain for boundary completion—volume fields map to $\mathcal{T}_\Gamma$; the boundary operator from Eq. (1.18) selects admissible data. Surjectivity of γ_τ onto compatible trace subclasses fails globally—only fields in specified $\mathcal{H}_{\nabla\times}$ subspaces with prescribed divergence pairings map to physically realizable $\mathbf{g}_\Gamma$ [2]. Subsection 2.1.4 identifies self-adjoint boundary realizations; here the point is that $\mathcal{L}_\omega$ of Eq. (2.15) acts on $\mathcal{U}_\omega$ only when $\gamma_\tau[\mathbf{u}_\omega]$ lies in the domain of $\mathcal{B}_\Gamma$. Sheet currents and interface channels from Eq. (1.19) require distributional extensions. Let $\mathcal{D}'(\Omega)^3$ denote temporal-harmonic distributional three-vector fields. Impressed volume data remain $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$, while surface equivalents $\mathbf{J}_{s,ab}$ on Σ_{ab} embed as

$$\langle \mathbf{J}_s, \boldsymbol{\phi}\rangle = \sum_{ab} \int_{\Sigma_{ab}} \mathbf{J}_{s,ab}\cdot\boldsymbol{\phi}\, dS, \qquad \boldsymbol{\phi} \in C_c^\infty(\Omega)^3, \tag{2.53}$$

linking the data tuple Eq. (1.17) to a continuous linear functional on smooth test fields [2]. Equation (2.53) does not alter the jump laws of Eq. (1.19); it specifies the functional-analytic class in which those jumps appear in weak formulations. Interface trace spaces on Σ_{ab} are denoted

$$\mathcal{T}_{\Sigma_{ab}} = H^{-1/2}(\Sigma_{ab})^3, \tag{2.54}$$

with tangential continuity of $\mathbf{E}$ and jump of $\hat{\mathbf{n}}\times\mathbf{H}$ encoded as constraints on γ_τ taken from each patch [6]. Dual pairings for source and test fields use the L^2 inner product

$$\langle \mathbf{J}_{\mathrm{imp}}, \boldsymbol{\psi}\rangle_\Omega = \int_\Omega \mathbf{J}_{\mathrm{imp}}\cdot\boldsymbol{\psi}^*\, d^3r, \qquad \boldsymbol{\psi} \in \mathcal{H}_{\nabla\times}(\Omega), \tag{2.55}$$

which induces a bounded functional on test states when $\mathbf{J}_{\mathrm{imp}} \in L^2(\Omega)^3$ and extends to Eq. (2.53) for sheet data. Forcing in Eq. (2.12) enters the weak form Eq. (2.20) through $\mathfrak{f}_\omega(\mathbf{v}_\omega) = \langle \mathbf{s}_\omega, \mathbf{v}_\omega \rangle$ once $\mathbf{v}_\omega \in \mathcal{U}_\omega$ carries homogeneous boundary traces [2]. The symbol $\mathcal{F}_\Omega$ names field pairs $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H})$ that are not merely square integrable but carry controlled curls and admissible tangential traces on Γ; every output of $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on the physical branch must lie in this sharpened class [6]. The admissible Maxwell field space compatible with 1 is

$$\mathcal{F}_\Omega = \left\{ \mathbf{u}_\omega = (\mathbf{E}, \mathbf{H}) \in \mathcal{U}_\omega : \gamma_\tau[\mathbf{u}_\omega] \in \mathcal{D}(\mathcal{B}_\Gamma),\ (\mathbf{E}, \mathbf{H}) \text{ satisfy patchwise Gauss constraints} \right\}, \quad (2.56)$$

where $\mathcal{D}(\mathcal{B}_\Gamma)$ is the domain of boundary data compatible with Eq. (1.23) imported from section 1.1.2. Eq. (2.56) refines the minimal declaration of Subsection 2.0.1, same symbol $\mathcal{F}_\Omega$, sharpened topology. Membership implies Eq. (1.3) but adds curl and trace control needed for integration by parts in Eq. (2.25) [2]. The composite operator $\mathcal{L}_\omega$ couples volume residuals with boundary rows; its domain $\mathcal{D}(\mathcal{L}_\omega)$ is the closed Maxwell solution class on which both $\mathcal{M}_\omega$ and $\mathcal{B}_\Gamma$ are simultaneously evaluable in the declared data spaces [6]. The closed Maxwell solution domain becomes

$$\mathcal{D}(\mathcal{L}_\omega) = \left\{ \mathbf{u}_\omega \in \mathcal{F}_\Omega : \mathcal{M}_\omega[\mathbf{u}_\omega] \in L^2(\Omega)^4 \right\}, \quad (2.57)$$

which specializes Eq. (2.21) to Lipschitz domains with patchwise smooth coefficients [2]. On $\mathcal{D}(\mathcal{L}_\omega)$, the volume residuals of Eq. (2.13) are square integrable, and the boundary row of Eq. (2.15) is evaluated in $\mathcal{T}_\Gamma$. Exterior domains append outgoing subspaces rather than changing Eqs. (2.44)–(2.56); that selection is function-space compatible with weighted norms that decay at infinity and is detailed in section 2.4.3 [6]. Free-charge density paired with $\mathbf{J}_{\mathrm{imp}}$ in Eq. (1.17) is taken in the scalar class $\rho \in L^2(\Omega)$ whenever volumetric charge is prescribed, with distributional interpretation $\nabla \cdot \mathbf{J}_{\mathrm{imp}} + \mathrm{j}\omega\rho = 0$ understood in $\mathcal{D}'(\Omega)$ as in Eq. (1.22). Scalar L^2 control is sufficient because Gauss laws in Eq. (1.7) involve $\nabla\cdot$ on $\varepsilon_0\mathbf{E}$ and $\nabla\cdot$ on $\mu_0\mathbf{H}$, not gradients of ρ. When ρ is absent and $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega^{(0)}$, the electric Gauss row reduces to a divergence constraint on $\mathbf{E}$ enforced weakly through test functions in $\mathcal{H}_{\nabla\times}(\Omega)$ [2]. Approximation properties matter for numerical projection. Smooth compactly supported fields are dense in $\mathcal{H}_{\nabla\times}(\Omega)$ and in $\mathcal{U}_\omega$ equipped with norm (2.47); every admissible state is a limit of C_c^∞ test pairs modified to satisfy boundary class on Γ [2]. Density underpins Galerkin convergence: edge-element spaces approximate $\mathcal{F}_\Omega$ when their degrees of freedom match tangential trace continuity on Γ_{PEC}, where

$$\mathcal{U}_\omega^{\mathrm{PEC}} = \{ (\mathbf{E}, \mathbf{H}) \in \mathcal{U}_\omega : \hat{\mathbf{n}} \times \mathbf{E}|_{\Gamma_{\mathrm{PEC}}} = \mathbf{0} \}, \quad (2.58)$$

encodes perfect-conductor tangential data from Eq. (1.20) without assuming $\mathbf{E} = \mathbf{0}$ in Ω [6]. Integration by parts on $\mathcal{U}_\omega^{\mathrm{PEC}}$ yields the boundary-term cancellation used in Eq. (2.25) when $\mathcal{B}_\Gamma$ selects homogeneous tangential electric trace. Compact embeddings used later for Fredholm arguments are recorded as

$$\mathcal{H}_{\nabla\times}(\Omega) \hookrightarrow L^2(\Omega)^3 \hookleftarrow H^{-1/2}(\Gamma)^3, \quad (2.59)$$

with the first inclusion compact on bounded Ω and the trace map continuous but not compact [2]. Eq. (2.59) explains why interior cavity spectra are discrete while boundary observation maps are rank deficient: compact embeddings produce accumulating eigenvalues in volume, whereas non-compact traces map infinite-dimensional field spaces to finite observation vectors [6]. The quantitative singular-value decay of radiation maps on compact support is developed in section 2.6 and cited thereafter [2]. Subsection 2.1.3 develops the consequences for radiation accessibility and finite observation. Subsection 2.1.1 has refined $\mathcal{F}_\Omega$, $\mathcal{J}_\Omega$, and $\mathcal{D}(\mathcal{L}_\omega)$ from minimal declarations to Sobolev, trace, and distributional classes compatible with 1 charge-continuous. The consequence

for measurement is immediate: near-field scanners and method-of-moments codes inherit these spaces as the continuous targets their discretizations must approximate. Subsection 2.1.2 treats the Maxwell maps as linear operators with explicit domains, ranges, and adjoints on those spaces [2]. As a worked example, suppose $\mathbf{g}_\Gamma$ prescribes tangential $\mathbf{E}$ on Γ with $H^{1/2}$ regularity but is not the trace of any $\mathcal{H}_{\nabla\times}$ field in Ω. Then no $\mathbf{u}_\omega \in \mathcal{F}_\Omega$ satisfies $\gamma_{\tau,E}[\mathbf{E}] = \mathbf{g}_\Gamma$, regardless of how accurately a volume solver minimizes Maxwell residuals. The failure is a trace-space mismatch, not a Green-function error; correcting it requires either lifting $\mathbf{g}_\Gamma$ into $\mathcal{T}_\Gamma$ as an image under γ_τ or relaxing the boundary class through impedance completion in Subsection 2.1.4 [2].

The physical limit to retain is therefore radiative rank on the declared support, not harmonic listing width alone [17].

2.1.2 Linear operators: domain, range, and adjoint

The physical problem is operator accountability: every radiation operator presupposes that each map in the source-to-field composition accepts only admissible excitations and returns only field pairs compatible with Maxwell closure and outgoing selection [2]. Every linear map—$\mathcal{S}_\omega$, $\mathcal{L}_\omega$, $\mathcal{R}_\omega$, and finite-rank measurement surrogates—requires explicit domain, range, and adjoint pairings [2]. Recall that $\mathcal{S}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_\Omega$ is the harmonic solution map, $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ selects outgoing radiative content, and $\mathcal{L}_\omega$ packages volume and boundary residuals jointly [6]. Reciprocity is a symmetry principle of accessibility: network analyzers swap transmit and receive ports, and the adjoint is the mathematical shadow of that hardware symmetry when sources remain passive [6]. Treating "adjoint" as jargon detached from range calibration misses this link [17]. When a synthesis residual persists under port reversal, the fault often lies in domain mismatch between trial currents and measured tangential data [3]. Continuity and divergence constraints tie impressed currents to charge conservation on real feeds [1]. When those constraints are violated, numerical solvers may still return finite vectors, but the corresponding fields exhibit unphysical reactive blow-up that no anechoic termination can absorb. Domain, range, and adjoint language records which excitations the operator accepts and which field pairs are returned [2]. Subsection 2.1.1 fixed the Hilbert and trace spaces on which Maxwell maps act. Subsection 2.1.2 treats those maps as linear operators with explicit domains, ranges, and adjoints [2]. The volume operator $\mathcal{M}_\omega$, boundary operator $\mathcal{B}_\Gamma$, composite $\mathcal{L}_\omega$, and solution map $\mathcal{S}_\omega$ introduced in Subsections 2.0.1–2.0.2 are not re-defined; their Maxwell field structure is recorded here as the definitive construction of domain–range–adjoint theory. Weak and energy forms of Eqs. (2.20) and (2.25) supply the duality pairings below and are cited, [2]. Let $\mathcal{X}$ and $\mathcal{Y}$ be Hilbert spaces. The linear map representing $T : \mathcal{D}(T) \to \mathcal{Y}$ is specified by its domain $\mathcal{D}(T) \subseteq \mathcal{X}$ and range $\mathcal{R}(T) = T\mathcal{D}(T) \subseteq \mathcal{Y}$. For Maxwell operators, $\mathcal{X} = \mathcal{U}_\omega$ from Eq. (2.46) and $\mathcal{Y}$ is a product of volume and boundary data spaces. When a finite-element solver reports residual norms, it is effectively measuring a graph norm: how large the field is in $\mathcal{U}_\omega$ together with how large Maxwell's residuals become when that field is inserted into $\mathcal{L}_\omega$. Engineers should read the graph norm as a joint flux measurement record of field energy and equation balance, not as a single L^2 amplitude [6]. A trial vector that is small in L^2 but drives large curl residuals cannot lie in $\mathcal{D}(\mathcal{L}_\omega)$ on the physical branch; conversely, a state with controlled graph norm is the continuous target that Galerkin discretizations on edge elements approximate [2]. The graph norm

$$\|\mathbf{u}\|^2_{\mathrm{gr}(T)} = \|\mathbf{u}\|^2_{\mathcal{X}} + \|T\mathbf{u}\|^2_{\mathcal{Y}}, \qquad \mathbf{u} \in \mathcal{D}(T), \tag{2.60}$$

turns $\mathcal{D}(T)$ into a Hilbert space when T is closed [2]. Eq. (2.60) is the diagnostic used in Subsection 2.1.4 to construct self-adjoint completions; here it organizes domain control for $\mathcal{L}_\omega$. Each Maxwell map acts from a declared domain of admissible states into a data space of residuals or traces; that domain–range discipline applies to each Maxwell map once $\mathcal{X}$ and $\mathcal{Y}$ are chosen. Confusing

$\mathcal{D}(T)$ with all of $\mathcal{X}$ is a common source of spurious boundary terms in flux identities [2]. The volume Maxwell operator $\mathcal{M}_\omega$ packages the four local Maxwell balance rows that a harmonic solver tests row by row; it returns residuals in $\mathcal{Y}_{\mathrm{vol}}$, not the field itself, and therefore acts only on states whose curls are controlled enough that those residuals lie in L^2 [2]. When a finite-element log reports "Maxwell residual," the engineer is observing an approximate norm of $\mathcal{M}_\omega[\mathbf{u}_\omega]$; convergence to zero on the physical branch is the numerical signature of membership in $\mathcal{D}(\mathcal{L}_\omega)$ [6]. The volume Maxwell operator of Eq. (2.13) acts as

$$\mathcal{M}_\omega : \mathcal{D}(\mathcal{M}_\omega) \longrightarrow \mathcal{Y}_{\mathrm{vol}}, \qquad \mathcal{Y}_{\mathrm{vol}} = L^2(\Omega)^3 \times L^2(\Omega)^3 \times L^2(\Omega) \times L^2(\Omega)^3, \tag{2.61}$$

with $\mathcal{D}(\mathcal{M}_\omega) \subset \mathcal{F}_\Omega$ the subspace on which all four residual rows of Eq. (1.7) lie in $\mathcal{Y}_{\mathrm{vol}}$ [2]. Operand interpretation: $\mathcal{M}_\omega$ evaluates curl-coupled Maxwell residuals pointwise in the L^2 sense; physically, it returns whether a candidate $(\mathbf{E}, \mathbf{H})$ satisfies local Maxwell balance with impressed data removed to the forcing vector Eq. (2.12). The map is linear in $\mathbf{u}_\omega$ and bounded on bounded subsets of $\mathcal{D}(\mathcal{M}_\omega)$ but is not bounded on all of $\mathcal{U}_\omega$ because curl norms are not controlled by L^2 norms alone [6]. Tangential traces form bounded boundary operators on $\mathcal{H}_{\nabla\times}(\Omega)$:

$$\gamma_\tau : \mathcal{H}_{\nabla\times}(\Omega) \longrightarrow H^{-1/2}(\Gamma)^3, \qquad \gamma_\tau[\mathbf{u}] = \hat{\mathbf{n}} \times \mathbf{u}|_\Gamma, \tag{2.62}$$

with $\|\gamma_\tau\|_{\mathrm{op}} < \infty$ on Lipschitz Γ [2]. Eq. (2.62) specializes Eqs. (2.51)–(2.52) to a single template. The boundary operator of Eq. (1.18) is

$$\mathcal{B}_\Gamma : \mathcal{D}(\mathcal{B}_\Gamma) \longrightarrow \mathcal{Y}_\Gamma, \qquad \mathcal{D}(\mathcal{B}_\Gamma) \subseteq \mathcal{T}_\Gamma, \tag{2.63}$$

where $\mathcal{T}_\Gamma$ and $\mathcal{Y}_\Gamma$ are from Subsection 2.1.1. Impedance and outgoing classes from Eqs. (1.20)–(1.21) restrict $\mathcal{D}(\mathcal{B}_\Gamma)$ to admissible trace pairs compatible with Eq. (1.23) [2]. The composite operator of Eq. (2.15) is

$$\mathcal{L}_\omega : \mathcal{D}(\mathcal{L}_\omega) \longrightarrow \mathcal{Y}_{\mathrm{vol}} \times \mathcal{Y}_\Gamma, \tag{2.64}$$

with $\mathcal{D}(\mathcal{L}_\omega)$ given by Eq. (2.57). Range membership is *coupled*: a state cannot satisfy volume rows while violating boundary rows in the product sense of Eq. (2.16). The graph norm (2.60) on $\mathcal{L}_\omega$ controls both residuals simultaneously and is the natural norm for convergence tests in numerical Maxwell solvers [6]. The harmonic solution map is an operator on source space:

$$\mathcal{S}_\omega : \mathcal{D}(\mathcal{S}_\omega) \longrightarrow \mathcal{F}_\Omega, \qquad \mathcal{D}(\mathcal{S}_\omega) = \mathcal{J}_\Omega, \tag{2.65}$$

defined by Eq. (2.17) as $\mathcal{L}_\omega[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]] = [\mathbf{s}_\omega(\mathbf{J}_{\mathrm{imp}}), \mathbf{g}_\Gamma]$. Well-posedness requires invertibility of $\mathcal{L}_\omega$ on the physical branch selected by $\mathcal{B}_\Gamma$; then

$$\mathcal{S}_\omega = \mathcal{L}_\omega^{-1} \circ (\mathbf{s}_\omega(\cdot), \mathbf{g}_\Gamma), \tag{2.66}$$

with $\mathcal{L}_\omega^{-1}$ understood as the outgoing or cavity-truncated resolvent of sections 2.4.3 and 2.6.4, not as an algebraic inverse on all of $\mathcal{U}_\omega$ [2]. Radiation and observation operators compose accordingly:

$$\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega, \qquad \Lambda_{\mathrm{rad}} = \mathcal{O}_{\mathrm{far}} \circ \mathcal{R}_\omega, \tag{2.67}$$

with Π_{rad} from Eq. (2.2) and $\mathcal{O}_{\mathrm{far}}$ from Subsection 2.0.1 [6].

Excitation-to-flux sequence on declared domains: $\mathcal{S}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_\Omega$, $\mathcal{R}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_{\mathrm{rad}}$.

(2.10) in domain–range language. Each arrow is meaningful only on the declared domain; extending $\mathcal{S}_\omega$ to non-admissible sources produces elements of $\mathcal{U}_\omega$ that need not lie in $\mathcal{F}_\Omega$ [2]. Adjoints

are defined through duality pairings compatible with Eq. (2.55). For the volume operator, the formal adjoint $\mathcal{M}_\omega^\dagger$ satisfies

$$\langle \mathcal{M}_\omega[\mathbf{u}_\omega], \mathbf{v}_\omega \rangle_{\mathrm{vol}} = \langle \mathbf{u}_\omega, \mathcal{M}_\omega^\dagger[\mathbf{v}_\omega] \rangle_{\mathcal{U}} + \mathcal{B}_{\mathrm{surf}}(\mathbf{u}_\omega, \mathbf{v}_\omega), \tag{2.68}$$

where $\langle \cdot, \cdot \rangle_{\mathrm{vol}}$ is the componentwise L^2 pairing on $\mathcal{Y}_{\mathrm{vol}}$, $\langle \cdot, \cdot \rangle_{\mathcal{U}}$ is induced by Eq. (2.47), and $\mathcal{B}_{\mathrm{surf}}$ collects boundary terms from integration by parts [2]. On $\mathcal{D}(\mathcal{M}_\omega) \cap \mathcal{U}_\omega^{\mathrm{PEC}}$ from Eq. (2.58), Eq. (2.25) identifies $\mathcal{B}_{\mathrm{surf}}$ with the tangential trace pairing of Eq. (2.25); that identity is imported, not re-proved [2]. The composite adjoint satisfies the block structure

$$\mathcal{L}_\omega^\dagger = \begin{bmatrix} \mathcal{M}_\omega^\dagger & 0 \\ 0 & \mathcal{B}_\Gamma^\dagger \end{bmatrix} \quad \text{on compatible pairs,} \tag{2.69}$$

with cross-coupling entering only through shared trace data when boundary and volume domains are not synchronized. Self-adjoint *realizations* of truncated Maxwell problems require choosing $\mathcal{B}_\Gamma$ so that $\mathcal{B}_{\mathrm{surf}}$ cancels on $\mathcal{D}(\mathcal{L}_\omega)$; Forcing functionals induced by impressed data are bounded linear operators on test spaces. The map

$$\mathfrak{f}_\omega : \mathcal{U}_\omega^0 \longrightarrow \mathbb{C}, \qquad \mathfrak{f}_\omega[\mathbf{v}_\omega] = \langle \mathbf{s}_\omega(\mathfrak{D}), \mathbf{v}_\omega \rangle_{\mathrm{vol}}, \tag{2.70}$$

is continuous when $\mathbf{s}_\omega \in \mathcal{Y}_{\mathrm{vol}}$ and $\mathbf{v}_\omega \in \mathcal{U}_\omega^0$ carries homogeneous boundary traces compatible with $\mathcal{B}_\Gamma$ [2]. Equation (2.70) is the operator reading of the right-hand side of Eq. (2.20). Source-to-field operators inherit continuity on bounded subsets: if $\|\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{J}} \le C$ and $\mathcal{L}_\omega$ is boundedly invertible on a fixed branch, then

$$\|\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]\|_{\mathcal{U}} \le C_{\mathrm{res}} \, \|\mathbf{s}_\omega(\mathbf{J}_{\mathrm{imp}})\|_{\mathrm{vol}}, \tag{2.71}$$

with $C_{\mathrm{res}} = \|\mathcal{L}_\omega^{-1}\|$ on the selected subspace [6]. Inequality (2.71) is the operator form of solver stability under source perturbations. Kernel and range spaces are recorded for later null-space analysis:

$$\ker \mathcal{L}_\omega = \{ \mathbf{u}_\omega \in \mathcal{D}(\mathcal{L}_\omega) : \mathcal{L}_\omega[\mathbf{u}_\omega] = \mathbf{0} \}, \tag{2.72}$$

radiation on the radiative shell therefore tracks $\mathbf{E} \times \mathbf{H}$ flux on the same outgoing pair used in the derivation [2].

$$\mathcal{R}(\mathcal{L}_\omega) = \left\{ [\mathbf{s}, \mathbf{g}]^{\mathsf{T}} \in \mathcal{Y}_{\mathrm{vol}} \times \mathcal{Y}_\Gamma : \exists \, \mathbf{u}_\omega \in \mathcal{D}(\mathcal{L}_\omega) : \ \mathcal{L}_\omega[\mathbf{u}_\omega] = [\mathbf{s}, \mathbf{g}]^{\mathsf{T}} \right\}. \tag{2.73}$$

On bounded domains with passive $\mathcal{B}_\Gamma$, $\ker \mathcal{L}_\omega$ contains cavity modes; on exterior domains with outgoing radiation condition, $\ker \mathcal{L}_\omega$ is trivial on the outgoing branch [2]. section 2.7.1 analyzes $\ker \mathcal{R}_\omega$; Subsection 2.1.3 treats compactness and Fredholm index on $\mathcal{R}(\mathcal{L}_\omega)$. Extension operators lift boundary data into volume fields. A right inverse $E_\Gamma : \mathcal{Y}_\Gamma \to \mathcal{H}_{\nabla\times}(\Omega)$ satisfying $\gamma_\tau \circ E_\Gamma = \mathrm{id}$ on compatible subclasses exists only for $\mathbf{g}_\Gamma$ in the range of γ_τ; otherwise no $\mathbf{u}_\omega \in \mathcal{F}_\Omega$ meets the boundary row [2]. Extension operators enter Calderón preconditioners and boundary-element formulations developed in later sections; here they clarify that $\mathcal{B}_\Gamma$ acts on traces, not on arbitrary L^2 boundary vectors [6]. As a worked example, fix $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ and two distinct outgoing radiation conditions $\mathcal{B}_\Gamma^{(a)}$ and $\mathcal{B}_\Gamma^{(b)}$ on the same exterior domain. The operators $\mathcal{L}_\omega^{(a)}$ and $\mathcal{L}_\omega^{(b)}$ have the same volume domain but different boundary domains; Eq. (2.66) yields two resolvents $\mathcal{S}_\omega^{(a)} \neq \mathcal{S}_\omega^{(b)}$ in general. Comparing analysis to measurement without matching $\mathcal{D}(\mathcal{B}_\Gamma)$ is therefore a comparison of different operators, not of different discretizations of one operator [2]. Subsection 2.1.2 has packaged domain, range, graph norm, adjoint pairings, and the composition $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ on declared spaces. Subsection 2.1.3 records boundedness, compact embeddings, and the rank Physically, deficiency that finite instruments impose on $\mathcal{F}_{\mathrm{rad}}$ [2].

2.1.3 Boundedness, compactness, rank deficiency, spectral compression

Spectral compression on compact support limits how sharply a finite aperture can beam [14]. Sidelobe floors on a range trace the same law [2]. Maxwell's retarded radiation to Γ_R remains the governing constraint [1]. The physical problem is observability under finite geometry: a compact source, a bounded domain, and a finite aperture cannot couple to infinitely many independent field degrees of freedom [3]. Subsections 2.1.1–2.1.2 fixed spaces and operator domains. Subsection 2.1.3 records boundedness and compactness properties, rank Compactness encodes the finite-geometry fact that bounded support cannot sustain arbitrarily many independent radiative channels: singular values of $\mathcal{R}_\omega$ must decay on declared support. deficiency of observation maps, and the viewpoint from which section 2.6 develops spectral compression for radiation operators [2]. The physical content is familiar on a compact range: finite sources and finite probes cannot couple to infinitely many independent far-zone degrees of freedom, so accessible rank is limited by geometry and observation extent [3]. Fredholm index computation and complete spectral decomposition remain in sections 2.6.4 and 2.6.2; here the constructive development establishes the lemmas on boundedness, compactness, and rank deficiency those sections invoke by reference [2]. In the display below, $\mathcal{F}_{\mathrm{rad}} \subset \mathcal{F}_\Omega$ denotes the outgoing radiative subspace selected by Π_{rad}; $\mathcal{S}_\omega$ and $\mathcal{R}_\omega$ are the solution and radiation maps on $\mathcal{J}_\Omega$ [6]. The linear map representing $T : \mathcal{X} \to \mathcal{Y}$ between Hilbert spaces is *bounded* if

$$\|T\|_{\mathrm{op}} = \sup_{\|\mathbf{x}\|_{\mathcal{X}}=1} \|T\mathbf{x}\|_{\mathcal{Y}} < \infty. \tag{2.74}$$

Boundedness is equivalent to continuity at the origin and is the default hypothesis for composition with measurement readouts [2]. On the Maxwell chain, operators split into bounded and unbounded classes. Trace and observation maps are bounded on declared spaces. The tangential trace Eq. (2.62) satisfies $\|\gamma_\tau\|_{\mathrm{op}} < \infty$ from $\mathcal{H}_{\nabla\times}(\Omega)$ to $H^{-1/2}(\Gamma)^3$ [2]. Far-zone readouts from Subsection 2.0.1 define bounded functionals when restricted to $\mathcal{F}_{\mathrm{rad}}$:

$$\|\mathcal{O}_{\mathrm{far}}\|_{\mathrm{op}} = \sup_{\|\mathbf{u}\|_{\mathcal{U}}=1} \left|\mathcal{O}_{\mathrm{far}}[\mathbf{u}]\right| < \infty \quad \text{on } \mathcal{F}_{\mathrm{rad}}. \tag{2.75}$$

Laboratory maps $\mathcal{O}_{\mathrm{meas}} \circ \mathcal{O}_{\mathrm{far}}$ in Eq. (2.42) are bounded on the same subspace because composition of bounded operators is bounded [6]. Volume and composite Maxwell operators are unbounded on full $\mathcal{U}_\omega$. The map $\mathcal{M}_\omega$ of Eq. (2.61) has $\|\mathcal{M}_\omega\|_{\mathrm{op}} = \infty$ when $\mathcal{D}(\mathcal{M}_\omega) = \mathcal{U}_\omega$ because controlling $\|\mathcal{M}_\omega[\mathbf{u}]\|_{\mathrm{vol}}$ would require uniform curl bounds beyond L^2 control [2]. Restriction to $\mathcal{D}(\mathcal{L}_\omega)$ with graph norm Eq. (2.60) restores boundedness of the *resolvent* $\mathcal{L}_\omega^{-1}$ on the physical branch where invertibility holds:

$$\|\mathcal{L}_\omega^{-1}\|_{\mathrm{op}} < \infty \quad \Longleftrightarrow \quad \mathcal{L}_\omega \text{ bijective on } \mathcal{D}(\mathcal{L}_\omega)/\ker \mathcal{L}_\omega. \tag{2.76}$$

Eq. (2.76) specializes Eq. (2.71); exterior outgoing problems use resolvent bounds on weighted subspaces rather than on all of $\mathcal{U}_\omega$ [6].

Compact embedding $\mathcal{H}_{\nabla\times}(\Omega) \hookrightarrow L^2(\Omega)^3$ on bounded Ω; trace map is bounded but not compact (Eq. (2.59)).

(2.59). Rellich-type compactness for $\mathcal{H}_{\nabla\times}(\Omega) \hookrightarrow L^2(\Omega)^3$ on bounded domains implies that bounded sequences in $\mathcal{H}_{\nabla\times}$ have L^2-convergent subsequences [2]. Finite geometry cannot radiate arbitrarily fine angular complexity; propagation therefore acts as a compression mechanism on electromagnetic structure. An operator $K : \mathcal{X} \to \mathcal{Y}$ is *compact* if bounded sets in $\mathcal{X}$ map to relatively compact sets in $\mathcal{Y}$:

$$K \text{ compact} \iff K(B_{\mathcal{X}}(0,1)) \text{ has compact completion in } \mathcal{Y}. \tag{2.77}$$

Compact perturbations of identity produce Fredholm structure used in section 2.6.4; Subsection 2.1.3 records the embedding lemmas without computing index [6]. Radiation and Green maps on fixed bounded domains often decompose as

$$\mathcal{S}_\omega = G_{\text{sing}} + K_{\text{cpt}}, \tag{2.78}$$

with G_{sing} capturing singular near-field kernels and K_{cpt} compact on $\mathcal{J}_\Omega$ when sources have compact support and Ω is bounded [2]. Equation (2.78) is the operator template for section 2.3; kernel construction is not repeated here. The split separates non-compact singular behavior at coincident points from compact smoothing at finite separation [6]. Rank deficiency enters through finite observation. An aperture-limited measurement map $\mathcal{O}_{\text{meas}} : \mathcal{F}_{\text{rad}} \to \mathbb{C}^N$ has finite-dimensional range with $N < \infty$:

$$\mathcal{O}_{\text{meas}} = \sum_{n=1}^{N} \langle \cdot, \mathbf{f}_n \rangle_{\mathcal{U}} \, \mathbf{e}_n, \tag{2.79}$$

for dual vectors $\mathbf{f}_n$ and coordinate directions $\mathbf{e}_n$. Therefore $\operatorname{rank}(\mathcal{O}_{\text{meas}}) \le N$ and

$$\ker \mathcal{O}_{\text{meas}} = \{\mathbf{u} \in \mathcal{F}_{\text{rad}} : \mathcal{O}_{\text{meas}}[\mathbf{u}] = \mathbf{0}\} \tag{2.80}$$

is infinite-dimensional whenever $\mathcal{F}_{\text{rad}}$ is infinite-dimensional [2]. Equation (2.80) is the operator form of the measurement instrument rank limits in section 1.5, finite modal sets cannot resolve all angular content. Rank deficiency is structural, not a discretization artifact alone; the spectral compression and singular-value decay law for radiation maps on compact support are owned by section 2.6 [2].

(2.67). Angular-spectrum reports that omit unresolved harmonics lie in the range of $\mathcal{O}_{\text{meas}}$ but not in its injective sector [2]. Subsection 2.1.3 does not classify report types; section 1.5 remains authoritative. Singular values quantify rank deficiency. For compact $K : \mathcal{X} \to \mathcal{Y}$, the singular value sequence $\{\sigma_n(K)\}_{n \ge 1}$ decreases to zero, and

$$\|K - R_N K\|_{\text{op}} = \inf_{\operatorname{rank} R_N \le N} \|K - R_N\|_{\text{op}} \sim \sigma_{N+1}(K), \tag{2.81}$$

where R_N ranges over rank-N operators [2]. For radiation maps truncated to N spherical or plane-wave modes, Eq. (2.81) controls how much field content lies outside the retained modal subspace; explicit VSH eigenbases are deferred to section 2.6.2 and Chapter 3 (*Angular Momentum Operators and SO(3)*) [6]. *Spectral compression* names the reduction of infinite-dimensional radiation dynamics to a finite or countable spectral model on the physical branch. Let $\{\phi_n\}$ be a spectral family in $\mathcal{F}_\Omega$ and $\{P_n\}$ the associated one-dimensional projectors. A truncated radiation operator

$$\mathcal{R}_\omega^{(N)} = \sum_{n=1}^{N} \lambda_n \, P_n \circ \mathcal{S}_\omega \tag{2.82}$$

retains only modes with eigenvalues λ_n above a declared threshold [2]. Equation (2.82) is the abstract template for section 2.6; concrete eigenfunctions are not constructed here. On bounded cavities, compact embeddings yield accumulating eigenvalues $\lambda_n \to \infty$ for interior resonances; on exterior domains, continuous spectra require outgoing projectors rather than discrete sums [2]. Compression error is bounded in operator norm when K is compact:

$$\|\mathcal{R}_\omega - \mathcal{R}_\omega^{(N)}\|_{\text{op}} \le C \, \sigma_{N+1}(K_{\text{cpt}}), \tag{2.83}$$

with K_{cpt} from Eq. (2.78) and C independent of impressed data on bounded $\mathcal{J}_\Omega$ [6]. Physically, Eq. (2.83) states that truncating modal content beyond the $(N+1)$st singular value cannot be

recovered by any rank-N surrogate operator; the quantitative decay law is as in section 2.6 [2]. Rank deficiency of $\mathcal{R}_\omega$ itself is distinct from observation rank deficiency. Non-radiating sources satisfy $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}] = \mathbf{0}$ while $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}] \neq \mathbf{0}$ (Eq. (2.7)); section 2.7.1 develops that null space. Here the point is compositional:

$$\operatorname{rank}(\mathcal{O}_{\text{meas}}\mathcal{R}_\omega) \leq \min\left\{\operatorname{rank}(\mathcal{O}_{\text{meas}}), \dim \mathcal{R}(\mathcal{R}_\omega)\right\}, \tag{2.84}$$

In Eq. (2.84), $\mathcal{O}_{\text{meas}}$ is a finite-rank linear measurement functional (pattern probe bank, modal fitter, or spherical-harmonic extractor) composed with $\mathcal{R}_\omega$; $\operatorname{rank}(\cdot)$ is the dimension of the accessible report subspace; and $\mathcal{R}(\mathcal{R}_\omega)$ is the range of the radiation map on $\mathcal{J}_\Omega$. So measurement rank caps reported angular content even when $\mathcal{R}_\omega$ is injective on admissible sources [2]. Both deficits must appear in uncertainty budgets for angular-transport claims imported from section 1.2.4. Fredholm structure note: on bounded Ω with passive $\mathcal{B}_\Gamma$, $\mathcal{L}_\omega$ is Fredholm on $\mathcal{D}(\mathcal{L}_\omega)$ modulo cavity kernels when written as

$$\mathcal{L}_\omega = \mathcal{L}_0 + K, \qquad K \text{ compact on graph domain}, \tag{2.85}$$

with $\mathcal{L}_0$ an invertible reference operator on the quotient by $\ker \mathcal{L}_\omega$ [2]. Index and solvability criteria are developed in section 2.6.4; Eq. (2.85) supplies the split Subsection 2.1.2 stated the introductory relation in Subsection 2.0.2. As a worked example, a spherical near-field scanner records $N = 50$ tangential field samples on a closed surface. The map $\mathcal{O}_{\text{meas}}$ has rank at most 50, so angular harmonics beyond the resolved set lie in $\ker \mathcal{O}_{\text{meas}}$. Applying Eq. (2.81) to the compact smoothing operator from sample interpolation shows that recovered far-zone coefficients error is controlled by σ_{51}, not by Maxwell residual alone [2]. Increasing frequency without increasing N lowers σ_{N+1} slowly for fixed aperture, worsening compression error even when $\mathcal{L}_\omega[\mathbf{u}_\omega] = \mathbf{0}$ in the volume. As a worked example, in a closed cavity, compact embedding yields discrete resonances and Eq. (2.82) sums over cavity modes. In an exterior problem with outgoing radiation condition, the same formula requires a continuous spectral integral rather than a finite sum; using cavity compression on an open antenna model conflates discrete storage modes with continuous radiation modes [2]. section 2.4.3 selects the outgoing branch; Subsection 2.1.3 only records that compression templates differ by domain class [2]. Finite-geometry interpretation (compactness and rank). Compact embeddings and finite-rank observation maps express one enduring electromagnetic truth: finite geometry destroys information. A bounded source support Ω_s cannot excite arbitrarily fine angular structure at every observation point; a finite aperture integrates over unresolved directions; a finite modal set truncates accessible rank. Inaccessible angular structure is therefore not a numerical inconvenience but a consequence of Maxwell propagation from compact sources through outgoing selection to finite instruments. flux report is fundamentally limited, and the mathematics of compactness makes that limit quantifiable through decaying singular values and infinite-dimensional observation null spaces [2]. Subsection 2.1.3 has established boundedness classes, compact embeddings, and finite-rank observation structure. The quantitative singular-value decay for radiation maps on compact support is developed in section 2.6; here only the structural lemmas are recorded. Subsection 2.1.4 completes the function-space foundation with self-adjoint realizations and admissibility completions on bounded domains [2].

2.1.4 Self-adjointness, completions, and admissibility

Self-adjoint extensions of Maxwell operators are not formal conveniences—they decide which boundary traces belong to the same radiative inner product. Before radiation is read, one must decide which completion matches outgoing laboratories before spectral partitions are read [2]. The physical problem is passive realizability on bounded hardware: cavity walls, matched terminations,

and dissipative linings must not permit unbounded storage of reactive energy under admissible excitation [2]. Friedrichs and other completions extend domains just enough that $\nabla\times$ and $\hat{\mathbf{n}}\times$ are simultaneously closed on the same space—precisely what finite-element antenna models require when sheet sources sit on interior interfaces [6]. Improper domain extension predicts unphysical growth of reactive storage—a numerical symptom that must be corrected before radiation maps are trusted [1]. Completion choices are radiation-physics choices about which reactive storage remains inside Ω and which energy is radiated [16].

Subsections 2.1.1–2.1.3 supplied spaces, solution domains, compactness, and spectral compression templates. Subsection 2.1.4 identifies self-adjoint and coercive realizations of truncated Maxwell operators on bounded domains, records completion and quotient constructions that remove cavity kernels, and states the $\mathcal{D}_J = 1$ embedding that links Chapter 1 finite-energy criteria to the solution domains declared here [2]. Weak and energy forms Eqs. (2.19)–(2.20) and the adjoint pairing Eq. (2.25) are presupposed; they are [2]. A densely defined operator T on Hilbert space $\mathcal{H}$ is *symmetric* if $\langle T\mathbf{u}, \mathbf{v}\rangle = \langle \mathbf{u}, T\mathbf{v}\rangle$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{D}(T)$, and *self-adjoint* if $\mathcal{D}(T^\dagger) = \mathcal{D}(T)$ with equality of adjoints [2]. Maxwell operators are symmetric on suitable domains because Eq. (2.25) cancels boundary pairings when $\mathcal{B}_\Gamma$ is passive and homogeneous on test states. Self-adjointness requires matching domain and range of adjoints; that match fails on exterior domains without restricting to outgoing subspaces [6]. On bounded Ω with passive $\mathcal{B}_\Gamma$, the sesquilinear form $\mathfrak{a}_\omega$ of Eq. (2.19) is coercive on the quotient by cavity modes:

$$\mathfrak{a}_\omega(\mathbf{u}, \mathbf{u}) \geq \alpha \, \|\mathbf{u}\|_{\mathcal{U}}^2, \qquad \mathbf{u} \in \mathcal{U}_\omega^0 / \ker \mathcal{L}_\omega, \tag{2.86}$$

for some $\alpha > 0$ when $\ker \mathcal{L}_\omega$ is finite-dimensional and consists of divergence-free resonances compatible with boundary class [2]. Eq. (2.86) is the Lax–Milgram hypothesis that makes Eq. (2.20) well posed on the quotient. Coercivity fails on the unquotiented space because cavity modes satisfy $\mathfrak{a}_\omega(\mathbf{u}, \mathbf{u}) = 0$ while $\|\mathbf{u}\|_{\mathcal{U}} > 0$.

Cavity-mode quotient: coercivity and uniqueness hold on the quotient space after removing $\ker \mathcal{L}_\omega$.

Physical excitation at a cavity frequency requires impedance or source coupling that breaks pure standing-wave balance; Subsection 2.1.4 records the linear-algebraic quotient; section 2.6.4 develops spectral consequences [2]. Impedance and Leontovich completions from Eq. (2.22) yield self-adjoint boundary realizations when surface impedance is passive:

$$\mathrm{Re}\{Z_s\} \geq 0 \text{ on } \Gamma_Z \quad \Longrightarrow \quad \mathcal{B}_\Gamma \text{ symmetric on } \mathcal{T}_\Gamma. \tag{2.87}$$

Passive $\mathrm{Re}\{Z_s\} \geq 0$ dissipates power through the boundary rather than injecting net energy; active or negative-real parts require revisiting Maxwell-closed as noted in section 1.1.2 [2]. Perfect conductors select Eq. (2.58); impedance boundaries enlarge $\mathcal{D}(\mathcal{B}_\Gamma)$ while preserving symmetry when Eq. (2.87) holds [6]. When $\mathcal{B}_\Gamma$ is symmetric but the Maxwell operator is only densely defined, the Friedrichs extension supplies a self-adjoint realization:

$$\mathcal{L}_\omega^{\mathrm{F}} = \overline{\mathcal{L}_\omega}\big|_{\mathcal{D}(\mathfrak{a}_\omega)}, \tag{2.88}$$

the completion of $\mathcal{L}_\omega$ on the energy domain of $\mathfrak{a}_\omega$ [2]. Equation (2.88) is the locked definition for for self-adjoint Maxwell truncations on bounded domains. Exterior outgoing problems use a different completion template: restrict to the outgoing subspace selected in section 2.4.3 rather than applying Eq. (2.88) to all of $\mathcal{U}_\omega$ [6]. Well-posedness on the quotient combines coercivity and bounded forcing:

$$\exists!\, \mathbf{u}_\omega \in \mathcal{U}_\omega^0 / \ker \mathcal{L}_\omega : \mathfrak{a}_\omega(\mathbf{u}_\omega, \mathbf{v}) = \mathfrak{f}_\omega(\mathbf{v}) \quad \forall\, \mathbf{v} \in \mathcal{U}_\omega^0 / \ker \mathcal{L}_\omega, \tag{2.89}$$

with $\mathfrak{f}_\omega$ from Eq. (2.70). Stability follows from Eqs. (2.86) and (2.71) [2]. Resolvent bound Eq. (2.76) holds on the quotient when $\mathcal{L}_\omega^{\mathrm{F}}$ is bijective there.

Self-adjoint completion on bounded domains: passive $\mathcal{B}_\Gamma$ supports coercive $\mathfrak{a}_\omega$ and stable $\mathcal{S}_\omega$ on the cavity quotient.

Changing $\mathcal{B}_\Gamma$ while leaving $\mathfrak{a}_\omega$ formally unchanged can destroy coercivity or symmetry; that sensitivity was already visible in the resolvent-branch illustration of Subsection 2.1.2 [2]. charge-continuous from Subsection 1.0.4 embeds into solution domains. The 1 finite-energy class Eq. (1.4) requires $(\mathbf{E}, \mathbf{H}) \in L^2(\Omega)^3 \times L^2(\Omega)^3$ with compatible boundary and source data [2]. The Chapter 2 admissibility embedding is

$$\mathcal{X}_{\mathrm{ch1}}(\Omega) \cap \mathcal{U}_\omega \subseteq \mathcal{F}_\Omega \subseteq \mathcal{D}(\mathcal{L}_\omega), \tag{2.90}$$

with strict inclusions in general because $\mathcal{H}_{\nabla\times}$ control and trace membership strengthen L^2 Maxwell-closed without replacing Eq. (1.160) [2]. Equation (2.90) specializes Eq. (2.26): when Chapter 1 admissibility holds, impressed data lie in $\mathcal{J}_\Omega$ and closed solutions lie in $\mathcal{F}_\Omega$. Operator Maxwell-closed adds graph-norm control:

$$\mathbf{u}_\omega \in \mathcal{F}_\Omega \implies \|\mathbf{u}_\omega\|_{\mathrm{gr}(\mathcal{L}_\omega)} < \infty \text{ when } \mathcal{L}_\omega[\mathbf{u}_\omega] \text{ is defined.} \tag{2.91}$$

Inequality (2.1.4) is the quantitative form of "physically realizable" field pairs for variational solvers [6]. Source-side Maxwell-closed remains coupled to Eq. (2.48) and Eq. (1.22) by reference. Completion operators must not introduce incompatible traces:

$$\mathcal{B}_\Gamma \text{ admissible} \iff \mathcal{R}(\gamma_\tau|_{\mathcal{F}_\Omega}) \supseteq \mathcal{D}(\mathcal{B}_\Gamma). \tag{2.92}$$

Failure of Eq. (2.92) means prescribed $\mathbf{g}_\Gamma$ is not the trace of any $\mathcal{H}_{\nabla\times}$ field, reproducing the trace-mismatch illustration of Subsection 2.1.1 [2]. Exterior domains: outgoing radiation condition from section 2.4.3 defines a *physically self-adjoint* branch rather than a Hilbert self-adjoint operator on all of $\mathcal{U}_\omega$:

$$\mathcal{U}_\omega = \mathcal{U}_{\mathrm{out}} \oplus \mathcal{U}_{\mathrm{re}}, \qquad \mathcal{L}_\omega \text{ coercive on } \mathcal{U}_{\mathrm{out}}, \tag{2.93}$$

with reactive storage content in $\mathcal{U}_{\mathrm{re}}$ controlled by regime criteria of section 1.2.5 rather than by full coercivity on $\mathcal{U}_\omega$ [2]. Conflating Eq. (2.88) with exterior radiation problems produces incorrect cavity-like spectra in open regions. With Subsections 2.1.1–2.1.4 complete, section 2.1 has moved from minimal declarations in section 2.0 to a closed operator calculus on Sobolev spaces: traces, adjoints, compactness, and self-adjoint truncations compatible with 1 charge-continuous [2]. section 2.2 applies those spaces to time–frequency equivalences; sections 2.3–2.4 inherit $\mathcal{D}(\mathcal{L}_\omega)$ and $\mathcal{F}_\Omega$ as fixed input classes. 4 states which operator outputs survive finite source support and manufacturable geometry [17].

2.2 Time- and Frequency-Domain Maxwell Operators

Function spaces and Maxwell operators in section 2.1 fix admissible sources and field pairs before time-harmonic reduction. section 2.2 carries the same physics into frequency-domain maps used on every fixed-frequency range [2]. The laboratory never measures abstract Hilbert norms; it measures phasor fields at declared ω [1]. An oscilloscope trace records $\mathbf{E}(\mathbf{r}, t)$ and $\mathbf{H}(\mathbf{r}, t)$ as functions of time; a vector network analyzer records complex amplitudes at each stepped tone. Both instruments interrogate the same structure, but only the time-domain picture makes retarded support visible before any phasor reduction is declared [6].

Fixed-ω phasors are frequency slices of retarded evolution, valid when excitation and media are narrowband on the declared interval [2]. Each harmonic operator $\mathcal{L}_\omega$ is a labeled cross-section of a

single causal evolution law, not an independent postulate. A vector network analyzer sweep therefore samples one harmonic operator family at each tone—not a separate physics from time-domain causality [1]; every compact-range calibration and harmonic finite-difference cell presupposes that reduction. section 2.2 develops the time–frequency equivalences that connect harmonic operators to causal evolution, Helmholtz symbols, and propagating-versus-evanescent splits under the $\exp(+\mathrm{j}\omega t)$ convention locked in 1. It records when that reduction is valid, how retarded time-domain maps relate to $\mathcal{R}_\omega$, and which spectral components carry far-zone transport content importable from section 1.2.5. Harmonic Maxwell theory in section 2.0 treated $\omega > 0$ as a fixed parameter; time-domain formulations restore the full operator pencil $\partial/\partial t$ and clarify why frequency-domain solvers implement causal limits when inversion is performed correctly. Subsection 2.2.1 is the account within Chapter 2 of time-domain evolution; Subsection 2.2.2 packages frequency-domain Maxwell and curl–curl Helmholtz operators without re-displaying Eq. (1.5); Subsection 2.2.3 proves equivalence between retarded Green propagation and $\mathcal{S}_\omega$ stated in Eq. (2.9); Subsection 2.2.3 applies the reactive-radiative partition of section 1.2.5 to propagating, evanescent, and non-propagating subspaces in operator form [2].

Several related constructions are deferred deliberately. In particular, dyadic Green-function development and inversion theory are postponed until section 2.3, where the harmonic operators introduced here acquire explicit integral representations [2]. section 2.2 does not re-open function-space declarations from section 2.1 or local Maxwell derivations from 1.

Helmholtz and curl–curl reductions developed in Subsection 2.2.2 are single-field tools: they simplify kernel construction but do not replace the first-order $\mathcal{L}_\omega$ packaging of Eq. (2.15) when boundary rows and impedance closures must be enforced jointly [6]. Within the locked harmonic band, $\mathcal{S}_\omega$ is the ω-slice of a causal time-domain map whenever impressed data and $\mathcal{B}_\Gamma$ are held fixed across the transform [2]. Finite-difference time-domain schemes approximate Subsection 2.2.1; frequency-domain finite-element solvers approximate Subsection 2.2.2 on the domains of Eq. (2.57). Modal and plane-wave decompositions in Subsection 2.2.3 filter which components of $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ may enter $\mathcal{R}_\omega$ under Eq. (2.27) [6].

Self-adjoint completion therefore fixes the inner product on which radiation orthogonality is measured: the wrong extension mislabels watt overlap as gauge drift. Retardation and causal propagation follow on the same domain (Subsection 2.3.4).

2.2.1 Time-domain evolution

The physical problem is causal propagation: a switched feed, a pulsed radar, or a transient immunity test excites fields whose support in $(\mathbf{r}, t)$ must respect finite signal speed [1]. Time-harmonic phasors are how network analyzers, compact ranges, and chamber receivers report steady-state radiation at each frequency step [2]. The $\exp(+\mathrm{j}\omega t)$ convention fixed in Chapter 1 aligns Poynting flux with wattmeter readings and VNA phase references [5]. Phasor analysis is best viewed as a steady-state reduction of a fundamentally causal theory rather than as the primary formulation of electromagnetic dynamics [6]. Even small phase errors can rotate synthesized patterns when near-field data are back-propagated [3]. The time-domain picture below states the causal support that every VNA tone must inherit when $\mathcal{L}_\omega$ is interpreted as a frequency slice rather than an independent postulate [2]. Subsection 2.2.1 provides the time-domain foundation underlying every harmonic formulation used later in the chapter. It introduces the first-order Maxwell evolution system, the associated evolution operator, and the causal support restrictions that all physically admissible frequency-domain solutions must inherit [2]. Time-domain evolution specifies retarded trajectories: the field observed at $(\mathbf{r}, t)$ depends only on source events contained within the corresponding past light cone [2]. Any numerical or analytical approximation that violates this support structure

introduces nonphysical precursors and loses its connection to Maxwell causality [6]. Local Maxwell balances remain those of Eq. (1.27); they are cited. Function spaces from section 2.1 enter through time-dependent embeddings developed below; Sobolev and trace declarations are not repeated [2]. Let $\Omega \subset \mathbb{R}^3$ be a Lipschitz domain with boundary $\Gamma = \partial\Omega$. Real time-domain fields $\mathbf{E}(\mathbf{r}, t)$, $\mathbf{H}(\mathbf{r}, t)$ use SI units and the same constitutive class as 1. The evolution problem is posed on trajectories of the electromagnetic state rather than on isolated snapshots at a single frequency. A pulsed or switched excitation on a feed line makes the distinction visible: the field at $(\mathbf{r}, t)$ cannot respond until source events inside the backward light cone have occurred. Time-domain evolution is therefore the laboratory picture behind every harmonic slice $\mathcal{S}_\omega$ used later in the chapter [2]. The state vector $\mathbf{U}(t)$ packages the coupled $(\mathbf{E}, \mathbf{H})$ fields at frozen time t into a single object on which the Maxwell generator acts; this packaging mirrors the phasor state $\mathbf{u}_\omega$ of Eq. (2.11), with time retained rather than suppressed [6]. Collect the electromagnetic state at time t as

$$\mathbf{U}(t) = \begin{bmatrix} \mathbf{E}(\cdot, t) \\ \mathbf{H}(\cdot, t) \end{bmatrix} \in \mathcal{V}(\Omega), \tag{2.94}$$

where $\mathcal{V}(\Omega)$ is the time-dependent product space of square-integrable field pairs compatible with Eq. (1.4) at each frozen time slice [2]. Eq. (2.94) is the time-domain analogue of Eq. (2.11); phasor symbols are reserved for the $\exp(+\mathrm{j}\omega t)$ reduction in Subsection 2.2.2. The time-domain Maxwell system is the causal law from which every harmonic operator in section 2.2 is obtained by phasor substitution. Physically, Eqs. (2.95) state that changing $\mathbf{E}$ in space drives $\mathbf{H}$ in time through Faraday's law, and changing $\mathbf{H}$ in space drives $\mathbf{E}$ in time through Ampère's law with impressed current as external drive [1]. From an antenna engineer's perspective, this is the system that finite-difference time-domain codes march forward in time; the harmonic solvers of Subsection 2.2.2 are steady-state limits of the same law when all data are single-frequency [6]. The time-domain Maxwell system imported from Eq. (1.27) reads in vacuum constitutive form as

$$\begin{aligned} \nabla \times \mathbf{E}(\mathbf{r}, t) &= -\mu_0\, \partial_t \mathbf{H}(\mathbf{r}, t), \\ \nabla \times \mathbf{H}(\mathbf{r}, t) &= \mathbf{j}_{\mathrm{imp}}(\mathbf{r}, t) + \varepsilon_0\, \partial_t \mathbf{E}(\mathbf{r}, t), \\ \nabla \cdot \mathbf{E}(\mathbf{r}, t) &= \frac{\rho(\mathbf{r}, t)}{\varepsilon_0}, \qquad \nabla \cdot \mathbf{H}(\mathbf{r}, t) = 0, \end{aligned} \tag{2.95}$$

with impressed current density $\mathbf{j}_{\mathrm{imp}}(\mathbf{r}, t)$ and charge density $\rho(\mathbf{r}, t)$. Eq. (2.95) packages local dynamics as a first-order system in $(\mathbf{E}, \mathbf{H})$ and ∂_t; the hyperbolic coupling is the time-domain face of the curl structure in Eq. (1.7) [2]. Charge continuity requires

$$\nabla \cdot \mathbf{j}_{\mathrm{imp}}(\mathbf{r}, t) + \partial_t \rho(\mathbf{r}, t) = 0 \tag{2.96}$$

in the sense of distributions when ρ is present [2]. Pairs $(\mathbf{j}_{\mathrm{imp}}, \rho)$ violating Eq. (2.96) lie outside the time-domain admissible class parallel to Eq. (2.0.1).

Boundary data enter through the same $\mathcal{B}_\Gamma$ of Eq. (1.18), now evaluated on time-dependent traces $\gamma_\tau[\mathbf{U}(t)] \in \mathcal{T}_\Gamma$ for each $t \geq 0$ [2]. The closed time-domain problem is to find $\mathbf{U}(t)$ such that

$$\begin{cases} \partial_t \mathbf{U} = \mathcal{A}\,\mathbf{U} + \mathbf{F}(t), & \mathbf{r} \in \Omega,\ t > 0, \\ \mathbf{U}(0) = \mathbf{U}_0, & \mathbf{r} \in \Omega, \\ \mathcal{B}_\Gamma[\mathbf{U}(t)] = \mathbf{g}_\Gamma(t), & \mathbf{r} \in \Gamma,\ t \geq 0, \end{cases} \tag{2.97}$$

where $\mathcal{A}$ is the Maxwell generator induced by Eq. (2.95) on $\mathcal{D}(\mathcal{A}) \subset \mathcal{V}(\Omega)$ and $\mathbf{F}(t)$ collects impressed-source rows [2]. Equation (2.97) is the time-domain counterpart of Eq. (2.16); initial data and

impressed sources drive the semigroup $\mathcal{T}(t)$ to field state $\mathbf{U}(t)$ under boundary class $\mathcal{B}_\Gamma$. Uniqueness on a fixed branch is inherited from section 1.1.3 when data classes coincide with the harmonic limit [2]. Define the evolution operator (semigroup) $\mathcal{T}(t)$ by

$$\mathcal{T}(t) : \mathbf{U}_0 \longmapsto \mathbf{U}(t), \qquad \mathcal{T}(0) = \mathrm{id}, \qquad \mathcal{T}(t+s) = \mathcal{T}(t)\mathcal{T}(s), \tag{2.98}$$

when $\mathbf{F} \equiv \mathbf{0}$ and $\mathcal{B}_\Gamma$ is homogeneous. With forcing,

$$\mathbf{U}(t) = \mathcal{T}(t)\mathbf{U}_0 \int_0^t \mathcal{T}(t-s)\,\mathbf{F}(s)\,ds, \tag{2.99}$$

the Duhamel representation on $\mathcal{D}(\mathcal{A})$ [2]. Eq. (2.99) makes causality explicit: $\mathbf{U}(t)$ depends only on $\mathbf{F}(s)$ for $s \le t$. The time-domain *solution map* sends impressed data and initial state to trajectories:

$$\mathcal{S}_t : (\mathbf{U}_0, \mathbf{j}_{\mathrm{imp}}) \longmapsto \{\mathbf{U}(t)\}_{t\ge 0}. \tag{2.100}$$

Harmonic $\mathcal{S}_\omega$ of Eq. (2.1) is a single-frequency slice of Eq. (2.100) when all data are proportional to $\exp(+\mathrm{j}\omega t)$ and $\mathcal{B}_\Gamma$ is fixed [2]; the equivalence is proved in Subsection 2.2.3. Retarded representation specializes Eq. (2.99) when Green kernels exist:

$$\mathbf{E}(\mathbf{r},t) = \mathbf{E}_{\mathrm{hom}}(\mathbf{r},t) - \mu_0\,\partial_t \int_0^t \int_\Omega \mathcal{G}(\mathbf{r},\mathbf{r}',t-t')\cdot\mathbf{j}_{\mathrm{imp}}(\mathbf{r}',t')\,d^3r'\,dt', \tag{2.101}$$

with $\mathcal{G}(\mathbf{r},\mathbf{r}',t-t') = \mathbf{0}$ for $t - t' < 0$ [2]. Eq. (2.101) refines the Eq. (2.9) without constructing $\mathcal{G}$ here; kernel development is in section 2.3. The support condition

$$\mathrm{supp}_{(\mathbf{r},t)}\,\mathbf{E} \subseteq \big\{(\mathbf{r},t) : t \ge t_{\mathrm{ret}}(\mathbf{r})\big\} \tag{2.102}$$

expresses finite propagation speed $c_0 = 1/\sqrt{\mu_0\varepsilon_0}$ on vacuum regions [2]. Figure 2.4 schematizes Eq. (2.102). Energy balance in time domain imports the Poynting and storage laws of section 1.1.4.

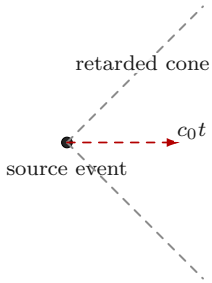


Figure 2.4: Causal support: field at $(\mathbf{r},t)$ depends on sources inside the backward light cone; retarded integration uses $t' \le t$ only.

With volume energy density

$$u(\mathbf{r},t) = \frac{\varepsilon_0}{2}\|\mathbf{E}(\mathbf{r},t)\|^2 + \frac{\mu_0}{2}\|\mathbf{H}(\mathbf{r},t)\|^2, \tag{2.103}$$

integration over a control volume $V \subset \Omega$ yields

$$\frac{d}{dt}\int_V u\,dV + \int_{\partial V} \mathbf{S}\cdot\hat{\mathbf{n}}\,dS = -\int_V \mathbf{E}\cdot\mathbf{j}_{\mathrm{imp}}\,dV, \tag{2.104}$$

with $\mathbf{S} = \mathbf{E}\times\mathbf{H}$ [2]. Equation (2.104) is cited; it supplies the physical reading of Eq. (2.99): impressed work on the right drives stored and radiated energy on the left. Reactive exchange

with boundaries appears in the surface term and motivates the regime split of Subsection 2.2.3 [6]. Harmonic operators follow from the substitution $\mathbf{E}(\mathbf{r},t) = \mathrm{Re}\{\underline{\mathbf{E}}(\mathbf{r})\mathrm{e}^{+\mathrm{j}\omega t}\}$ with $\underline{\mathbf{H}}$ analogous, yielding

$$\partial_t \longrightarrow +\mathrm{j}\omega, \qquad \mathcal{A} \longrightarrow \mathcal{M}_\omega, \qquad \mathcal{S}_t\big|_\omega \longrightarrow \mathcal{S}_\omega, \tag{2.105}$$

on admissible data of Eq. (2.90) [2]. Eq. (2.105) is symbolic; Subsection 2.2.3 states equivalence conditions under which frequency-domain solvers implement causal time-domain maps. Exterior domains require outgoing boundary conditions in time domain compatible with Eq. (1.21); those completions are sharpened in section 2.4.3 [2]. Time-domain Maxwell-closed couples Eq. (1.4) at each time slice to Eq. (2.96):

$$\mathbf{U}(t) \in \mathcal{F}_\Omega \text{ a.e. } t \ \wedge\ (\mathbf{j}_{\mathrm{imp}}, \rho) \text{ satisfy charge continuity } \implies \mathbf{U} \in \mathcal{D}(\mathcal{S}_t). \tag{2.106}$$

Finite-difference time-domain schemes discretize Eq. (2.95) on staggered grids; stability requires discrete analogues of Eq. (2.102) and consistent boundary enforcement of $\mathcal{B}_\Gamma$ [6]. Subsection 2.2.1 states the continuous targets those schemes approximate. Subsection 2.2.1 has posed the time-domain Maxwell system, the evolution semigroup $\mathcal{T}(t)$, and the retarded convolution form that enforces causal support. Subsection 2.2.2 carries the same physics into fixed-frequency Physically, operators and Helmholtz symbols on the spaces of section 2.1 [2].

The measurable consequence is therefore a directional or rank limit on radiation, not a calibration correction: recalibration cannot remove obstructions imposed by $\ker \mathbf{\Gamma}_\omega$ [6].

2.2.2 Frequency-domain Maxwell operators

The physical problem is steady-state interrogation at fixed frequency: every calibrated S-parameter, resonator Q measurement, harmonic finite-element solve, and compact-range pattern extraction assumes that excitation, media, boundary class, probe standoff, and wattmeter reference on Γ_R are frozen while ω is stepped [6, 3]. A vector network analyzer sweep is the canonical laboratory observation that precedes every harmonic operator declaration. The instrument steps ω through a calibrated band, records complex S-parameters at each tone, and treats each frequency as a steady-state slice of the same passive structure [6]. The composite operator $\mathcal{L}_\omega$ introduced in section 2.1 is precisely that slice: Maxwell's equations with $\partial_t \to +\mathrm{j}\omega$ on impressed data and boundary class held fixed. $\mathcal{L}_\omega$ is not an abstract Hilbert map alone but as the operator whose inversion reproduces the phasor fields a VNA implicitly assumes when it reports $S_{21}(\omega)$ after de-embedding [2]. Frequency-domain formulations dominate practical electromagnetics because measurements, simulations, and design specifications are usually organized around frequency. Network analyzers, resonator measurements, and antenna-pattern ranges all interrogate electromagnetic systems one frequency at a time [6]. The harmonic operators introduced below state that experimentally familiar viewpoint while remaining consistent with the causal evolution described in Subsection 2.2.1. Subsection 2.2.1 fixed time-domain evolution and Eq. (2.105). Subsection 2.2.2 is the account within Chapter 2 of harmonic Maxwell operators and curl–curl Helmholtz symbols on the spaces of section 2.1 [2]. Recall that $\mathcal{S}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_\Omega$ is the harmonic solution map, $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ selects outgoing radiative content, and $\mathcal{L}_\omega$ packages volume and boundary residuals jointly [6]. The first-order system Eq. (1.7) and composite $\mathcal{L}_\omega$ of Eq. (2.15) are presupposed; local Maxwell derivations from 1 are not repeated. The electric-wave operator Eq. (2.18) was recorded in Subsection 2.0.2; here it is embedded in domain declarations and extended by magnetic and material-weighted counterparts. Full vector-field separation is deferred to section 3.3 [2]. Under the locked $\exp(+\mathrm{j}\omega t)$ convention, differentiation with respect to time converts to multiplication by $+\mathrm{j}\omega$ on phasor quantities, and the second time derivative maps to $-\omega^2$. Every phasor on a network analyzer trace has already

undergone this substitution; it is not a notational convenience alone but the algebraic condition under which a time-domain IBVP and a frequency-domain BVP describe the same passive structure at fixed ω [2]. The substitution fails when excitation is broadband without superposition, when $\mathcal{B}_\Gamma$ varies in time, or when material response is not single-frequency on the declared band [6]. With those limits stated, the formal rule is: Time derivatives map to

$$\partial_t \longrightarrow +\mathrm{j}\omega, \qquad \frac{\partial^2}{\partial t^2} \longrightarrow -\omega^2, \tag{2.107}$$

on phasor fields $\mathbf{E}(\mathbf{r})$, $\mathbf{H}(\mathbf{r})$ and impressed data $\mathbf{J}_{\mathrm{imp}}(\mathbf{r})$, $\rho(\mathbf{r})$ [2]. In Eq. (2.107), ∂_t acting on a time-harmonic quantity is replaced by $+\mathrm{j}\omega$, where ω is in $\mathrm{rad\,s^{-1}}$; and ∂_t^2 is replaced by $-\omega^2$. This substitution is valid only when all time dependence is single-frequency or when a superposition of such terms is taken with consistent phasor definitions. Equation (2.107) specializes Eq. (2.105) and converts Eq. (1.27) to the harmonic BVP of Eq. (1.28) by reference. The harmonic residual collects the four Maxwell rows in frequency form. A finite-element or method-of-moments solve at fixed ω is, operationally, the search for $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H})$ that drives $\mathcal{M}_\omega[\mathbf{u}_\omega]$ to zero in the volume while satisfying the boundary row enforced by $\mathcal{B}_\Gamma$. The engineer observes residuals indirectly through solver convergence and port matching; the mathematician packages the same requirement as membership in $\mathcal{D}(\mathcal{L}_\omega)$ with $\mathcal{L}_\omega[\mathbf{u}_\omega]$ balanced against impressed data [6]. Note that Eq. (2.108) is not a new physics postulate but the frequency image of Eq. (2.95) once $\partial_t \to +\mathrm{j}\omega$ on admissible data [2]. The harmonic Maxwell residual operator $\mathcal{M}_\omega$ of Eq. (2.13) acts on $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H}) \in \mathcal{U}_\omega$:

$$\mathcal{M}_\omega[\mathbf{u}_\omega] = \begin{bmatrix} \nabla \times \mathbf{E} + \mathrm{j}\omega\mu_0 \mathbf{H} \\ \nabla \times \mathbf{H} - \mathrm{j}\omega\varepsilon_0 \mathbf{E} - \mathbf{J}_{\mathrm{imp}} \\ \nabla \cdot (\varepsilon_0 \mathbf{E}) - \rho \\ \nabla \cdot (\mu_0 \mathbf{H}) \end{bmatrix}, \tag{2.108}$$

row-aligned with Eq. (1.7) [2]. Each row of Eq. (2.108) is a residual that must vanish for a physically consistent harmonic state: the first row enforces Faraday's law with phasor substitution; the second enforces Ampère's law with impressed current $\mathbf{J}_{\mathrm{imp}}$; the third enforces Gauss's law for electric flux with charge density ρ; and the fourth enforces the magnetic Gauss law. Eq. (2.108) is the frequency-domain formulation of Eq. (2.95); completion and traces remain in $\mathcal{L}_\omega[\mathbf{u}_\omega] = [\mathbf{s}_\omega, \mathbf{g}_\Gamma]^{\mathsf{T}}$ from Eq. (2.16).

Frequency-domain reduction ladder: first-order $\mathcal{M}_\omega$ yields electric and magnetic Helmholtz operators when partner fields are eliminated under stated domain and gauge constraints.

Single-field Helmholtz forms equivalent to Eqs. (1.13)–(1.15) were established in 1 and are; elimination steps are not repeated [2]. The 2 electric-wave operator Eq. (2.18) is the vacuum electric Helmholtz map on $\mathcal{D}(\mathcal{W}_{E,\omega}) \subset \mathcal{H}_{\nabla\times}(\Omega)$:

$$\mathcal{D}(\mathcal{W}_{E,\omega}) = \{\mathbf{E} \in \mathcal{H}_{\nabla\times}(\Omega) : \mathcal{W}_{E,\omega}\mathbf{E} \in L^2(\Omega)^3\}. \tag{2.109}$$

The magnetic counterpart is

$$\mathcal{W}_{H,\omega}\mathbf{H} \equiv \nabla \times \left(\varepsilon_0^{-1} \nabla \times \mathbf{H}\right) - \omega^2 \mu_0 \mathbf{H} = \mathrm{j}\omega\mu_0 \mathbf{J}_{\mathrm{imp}} - \varepsilon_0^{-1} \nabla \times \mathbf{J}_{\mathrm{imp}}, \tag{2.110}$$

on $\mathcal{D}(\mathcal{W}_{H,\omega}) \subset \mathcal{H}_{\nabla\times}(\Omega)$ analogously [2]. Equations (2.18) and (2.110) are dual single-field symbols; using one without the other is valid only when the eliminated partner is recoverable from curl relations without ambiguity. In piecewise smooth media with $\varepsilon(\mathbf{r})$, $\mu(\mathbf{r})$,

$$\mathcal{W}_{E,\omega}^{(\varepsilon,\mu)}\mathbf{E} \equiv \nabla \times \left(\mu^{-1} \nabla \times \mathbf{E}\right) - \omega^2 \varepsilon \mathbf{E} = -\mathrm{j}\omega\mu \mathbf{J}_{\mathrm{imp}} + \mu^{-1} \nabla \times \mathbf{J}_{\mathrm{eff}}^{(\mathrm{mag})}, \tag{2.111}$$

Each relation above is a watt statement on $\mathcal{F}_{\mathrm{rad}}$: magnitude alone never substitutes for qualified flux on Γ_R [6].

$$\mathcal{W}_{H,\omega}^{(\varepsilon,\mu)}\mathbf{H} \equiv \nabla \times \left(\varepsilon^{-1}\nabla \times \mathbf{H}\right) - \omega^2\mu\mathbf{H} = \mathrm{j}\omega\mu\mathbf{J}_{\mathrm{imp}} - \varepsilon^{-1}\nabla \times \mathbf{J}_{\mathrm{imp}}, \tag{2.112}$$

with interface jumps on $\mathcal{D}(\mathcal{W}_{\cdot,\omega}^{(\varepsilon,\mu)})$ fixed by Eq. (1.19) [2]. Material weights modify symbols but not the first-order coupling pattern of Figure 2.3; multi-region Green operators in section 2.3.5 invert Eqs. (2.111)–(2.112) on composite domains [6]. The local Helmholtz symbol is

$$k(\mathbf{r}) = \omega\sqrt{\mu(\mathbf{r})\varepsilon(\mathbf{r})}, \qquad k_0 = \omega\sqrt{\mu_0\varepsilon_0}, \tag{2.113}$$

from Eq. (1.12) in uniform vacuum [2]. In source-free regions, Eqs. (1.14) apply with $k(\mathbf{r})$ replacing k_0. The symbol $k(\mathbf{r})$ labels propagating versus evanescent wavenumber locally; it does not replace Sommerfeld outgoing selection on open ranges [2]. Evanescent sheets with large $|\mathbf{E}|$ can satisfy the symbol while radiating no watts through Γ_R [7]. Single-field reduction is admissible only under explicit constraints:

$$\text{eliminate } \mathbf{H} \implies \mathbf{J}_{\mathrm{imp}} \in \mathcal{H}_{\nabla\times}(\Omega),\ \rho \text{ compatible},\ \mathcal{B}_\Gamma \text{ expressible in } \mathbf{E}\text{-traces alone}, \tag{2.114}$$

with the magnetic elimination requiring the parallel conditions on $\mathbf{H}$-traces [2]. Failure of Eq. (2.114) —for example impedance boundaries coupling both tangential components—forces retention of the first-order $\mathcal{L}_\omega$ system Eq. (2.15) [6]. Forcing maps between impressed data and Helmholtz right-hand sides are linear:

$$\mathbf{f}_{E,\omega} = -\mathrm{j}\omega\mu\mathbf{J}_{\mathrm{imp}} + \mu^{-1}\nabla \times \mathbf{J}_{\mathrm{eff}}^{(\mathrm{mag})}, \qquad \mathcal{W}_{E,\omega}^{(\varepsilon,\mu)}\mathbf{E} = \mathbf{f}_{E,\omega}, \tag{2.115}$$

with $\mathbf{f}_{E,\omega} \in L^2(\Omega)^3$ when $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ from Eq. (2.48) [2]. The harmonic solution operator on the electric Helmholtz branch is

$$\mathcal{G}_{E,\omega} : \mathbf{f}_{E,\omega} \longmapsto \mathbf{E}, \qquad \mathcal{W}_{E,\omega}^{(\varepsilon,\mu)}\mathcal{G}_{E,\omega} = \mathrm{id} \tag{2.116}$$

on the physical resolvent branch selected by $\mathcal{B}_\Gamma$; explicit $\mathcal{G}_{E,\omega}$ construction is developed in section 2.3 [6]. Composition with the full map yields

$$\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \left(\mathcal{G}_{E,\omega}\mathbf{f}_{E,\omega},\ -\frac{1}{\mathrm{j}\omega\mu}\nabla \times \mathbf{E}\right) \quad \text{when } \mathbf{H} \text{ recovered from Ampère row}, \tag{2.117}$$

linking Eq. (2.1) to Helmholtz inversion without re-displaying Eq. (2.17) [2]. Spectral viewpoint note: on bounded domains with discrete cavity spectra, $\mathcal{W}_{E,\omega}^{(\varepsilon,\mu)}$ admits eigenpairs $(\lambda_n, \mathbf{e}_n)$ accumulating at infinity because $\mathcal{H}_{\nabla\times}(\Omega) \hookrightarrow L^2(\Omega)^3$ compactly (Eq. (2.59)) [2]. On exterior domains, continuous spectrum replaces discrete eigenvalues; section 2.6.4 develops Fredholm and spectral consequences. Vector spherical harmonic separation of eigenmodes is deferred to section 3.3 [6].

First-order $\mathcal{L}_\omega$ retains coupled boundary rows; single-field $\mathcal{W}_{E,\omega}$ is valid only when $\mathcal{B}_\Gamma$ can be projected without losing impedance coupling.

Impedance and outgoing radiation conditions modify both volume and trace operators simultaneously [2]. Weak form of the electric Helmholtz operator uses the same sesquilinear structure as Eq. (2.20) restricted to $\mathbf{E}$-only test spaces when reduction is valid [2]. Coercivity on the quotient by cavity modes from Eq. (2.86) applies to $\mathcal{W}_{E,\omega}^{(\varepsilon,\mu)}$ on bounded domains with passive $\mathcal{B}_\Gamma$; Subsection 2.1.4 develops the self-adjoint realizations [6]. Subsection 2.2.2 has packaged harmonic Maxwell residuals, electric and magnetic Helmholtz operators, and the conditions under which single-field reduction is

admissible. Subsection 2.2.3 proves that $\mathcal{S}_\omega$ is the frequency-domain image of a causal time-domain map when impressed data, media, and boundary class are held Physically, fixed [2].

Angular-momentum transport therefore remains qualified on outgoing pairs $(\mathbf{E}, \mathbf{H})$: gauge-dependent density splits cannot be read as laboratory observables without the admissibility predicates from Chapter 1 [2].

2.2.3 Causality and time-frequency equivalence

The physical problem is retarded response: no field at $(\mathbf{r}, t)$ may depend on source events that have not yet had time to propagate from $\mathbf{r}'$ to $\mathbf{r}$ [1]. On the range, this is the difference between a measured impulse response that vanishes before the light-travel delay and an unphysical advanced template that leads the feed. A pulsed compact-range measurement makes the distinction visible: energy arrives only after transit time $|\mathbf{r} - \mathbf{r}'|/c_0$; a single-frequency phasor can hide violations that broadband synthesis would expose once frequency stepping and wattmeter closure on Γ_R are treated as laboratory prerequisites rather than coefficient-extraction conveniences [6, 3]. Causality is therefore the principle that links time-domain evolution to harmonic analysis before any Laplace or resolvent notation is introduced [2]. A time-domain operator is causal when no response can occur before the corresponding excitation [2]. In electromagnetics this requirement appears through retarded propagation: disturbances generated at one location influence another only after the finite transit time imposed by the speed of light [16]. Subsection 2.2.3 establishes that
$mathcalS_{omega}$ and $\mathcal{R}_\omega$ are the frequency-domain images of causal time-domain maps when impressed data, media, and boundary class are held fixed [2]. The retarded Eq. (2.9) and symbolic relation Eq. (2.105) are upgraded to an equivalence theorem; Green-function construction remains in section 2.3 [2]. Causal support. A time-domain operator $\mathcal{K}$ is *causal* if

$$\mathcal{K}[f](t) = 0 \text{ for } t < t_0 \quad \text{whenever } \operatorname{supp}_t f \subseteq [t_0, \infty), \tag{2.118}$$

for admissible test functions f [2]. Retarded propagation forbids response before transit time $|\mathbf{r} - \mathbf{r}'|/c$; frequency-domain Green functions used on ranges are analytic continuations of that support, while broadband pulses expose violations that single-frequency phasors can hide [16]. The retarded dyadic kernel in Eq. (2.101) satisfies

$$\mathcal{G}(\mathbf{r}, \mathbf{r}', t - t') = \mathbf{0} \quad \text{for } t - t' < 0, \tag{2.119}$$

which enforces Eq. (2.102) on $\mathbf{E}$ [2]. Advanced kernels with $\mathcal{G}(\mathbf{r}, \mathbf{r}', t - t') \neq \mathbf{0}$ for $t - t' < 0$ violate Eq. (2.118) and are excluded from the equivalence class treated here.

Let $\mathbf{j}_{\text{imp}}(\mathbf{r}, t)$ be an impressed current with compact support in $\Omega_s \times [0, \infty)$ and let $\mathbf{U}_0 = \mathbf{0}$ for simplicity. The Laplace transform in s with $\operatorname{Re}\{s\}$ sufficiently large is

$$\widetilde{\mathbf{j}}(\mathbf{r}, s) = \int_0^\infty \mathbf{j}_{\text{imp}}(\mathbf{r}, t)\, e^{-st}\, dt. \tag{2.120}$$

On the phasor axis $s = -\mathrm{j}\omega$ with $\omega > 0$,

$$\widetilde{\mathbf{j}}(\mathbf{r}, -\mathrm{j}\omega) = \int_0^\infty \mathbf{j}_{\text{imp}}(\mathbf{r}, t)\, e^{\mathrm{j}\omega t}\, dt, \tag{2.121}$$

which is the Fourier transform restricted to causal support [2]. For $\mathbf{j}_{\text{imp}}(\mathbf{r}, t) = \operatorname{Re}\{\underline{\mathbf{j}}(\mathbf{r}) e^{+\mathrm{j}\omega_0 t}\}$ with fixed $\omega_0 > 0$, Eq. (2.121) selects the phasor $\underline{\mathbf{j}}$ under the locked convention $\exp(+\mathrm{j}\omega t)$ [2]. Applying the Laplace transform to Eq. (2.97) with fixed $\mathcal{B}_\Gamma$ and homogeneous initial data yields, formally,

$$\widetilde{\mathbf{U}}(\mathbf{r}, s) = \mathcal{R}(s)\, \widetilde{\mathbf{F}}(s), \qquad \mathcal{R}(s) = \mathcal{L}_s^{-1}, \tag{2.122}$$

where $\mathcal{L}_s$ is the Laplace-domain Maxwell operator obtained from $\mathcal{A}$ by Eq. (2.107) with $\partial_t \to s$ [2]. On the axis $s = -\mathrm{j}\omega$, $\mathcal{L}_s = \mathcal{L}_\omega$ from Eq. (2.64), so

$$\mathcal{R}(-\mathrm{j}\omega) = \mathcal{L}_\omega^{-1} \quad \text{on the outgoing/resolvent branch,} \tag{2.123}$$

and the retarded convolution Eq. (2.101) is the inverse Laplace transform of Eq. (2.122) when $\mathcal{G}$ is the causal Green operator constructed in section 2.3 [2]. Eq. (2.123) is the operator form of Eq. (2.9): harmonic inversion is not an independent postulate but the $s = -\mathrm{j}\omega$ evaluation of a causal resolvent. The equivalence theorem requires explicit assumptions:

$$\begin{aligned}
&\text{(i) } \mathcal{B}_\Gamma \text{ and } \varepsilon, \mu \text{ fixed in time,}\\
&\text{(ii) } \mathbf{j}_{\mathrm{imp}} \text{ causal with } \operatorname{supp}_t \subseteq [0, \infty),\\
&\text{(iii) outgoing/resolvent branch selected for exterior domains,}\\
&\text{(iv) } \mathbf{U}_0 = \mathbf{0} \text{ or compatible with harmonic steady limit.}
\end{aligned} \tag{2.124}$$

Under Eqs. (2.124),

$$\mathcal{S}_\omega[\underline{\mathbf{j}}] = \widetilde{\mathbf{U}}(\mathbf{r}, s)\Big|_{s=-\mathrm{j}\omega} = \mathcal{L}_\omega^{-1}\widetilde{\mathbf{F}}(s)\Big|_{s=-\mathrm{j}\omega}, \tag{2.125}$$

where $\underline{\mathbf{j}}$ is the phasor of $\mathbf{j}_{\mathrm{imp}}(\mathbf{r}, t) = \mathrm{Re}\{\underline{\mathbf{j}}(\mathbf{r})e^{+\mathrm{j}\omega t}\}$ and $\widetilde{\mathbf{F}}$ is the Laplace forcing induced by $\widetilde{\mathbf{j}}$ [2]. Equation (2.125) is the formal statement that frequency-domain solvers implement causal time-domain maps when Eqs. (2.124) hold [6]. For single-frequency excitation with damping or outgoing radiation,

$$\lim_{t\to\infty} \mathbf{U}(t) = \mathrm{Re}\left\{\mathcal{S}_\omega[\underline{\mathbf{j}}]\, e^{+\mathrm{j}\omega t}\right\}, \tag{2.126}$$

which justifies steady-state phasor measurements after switch-on transients decay [2]. Because Π_{rad} is linear and commutes with harmonic extraction on the outgoing subspace,

$$\mathcal{R}_\omega[\underline{\mathbf{j}}] = \Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\underline{\mathbf{j}}], \tag{2.127}$$

with transport and declared size and distance regimes from Eq. (2.27) unchanged [2]. Frequency-domain codes that invert $\mathcal{L}_\omega$ without outgoing or passive completion implement a different resolvent branch than the causal retarded $\mathcal{G}$ of Eq. (2.101) [6]. Advanced Green functions satisfy

$$\mathcal{G}_{\mathrm{adv}}(\mathbf{r}, \mathbf{r}', t - t') \neq \mathbf{0} \text{ for some } t - t' < 0, \tag{2.128}$$

and cannot satisfy Eq. (2.125) for any causal $\mathbf{j}_{\mathrm{imp}}$ [2]. Broadband synthesis requires superposition over ω, and time-varying $\mathcal{B}_\Gamma(t)$ breaks assumption (i) in Eq. (2.124) [2]. Subsections 2.2.1–2.2.3 fixed evolution, harmonic operators, and causality. Subsection 2.2.3 develops spectral partitions into propagating, evanescent, and non-propagating subspaces in operator form [2]. Reactive versus radiative regime criteria from section 1.2.5 are invoked by reference; the outgoing selector Π_{rad} of Eq. (2.2) is refined spectrally, not replaced [5]. The physical problem is spectral transport: a near-field scanner records both traveling and decaying longitudinal content, yet only outgoing propagating modes may enter far-zone power reports [3]. Subsection 2.2.3 defines propagating, evanescent, and non-propagating spectral subspaces and refines $\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}$ so that only outgoing propagating content qualifies for far-zone transport reports [2]. Recall that $\mathcal{S}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_\Omega$ produces the full harmonic field pair, $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ retains only radiative content, and Π_{rad} acts after propagating selection [6]. A near-field scanner routinely measures both propagating and evanescent spectral content. The distinction becomes visible when reconstructed fields are propagated away from the measurement surface: propagating components persist, whereas evanescent

components decay exponentially with distance [3]. The spectral decomposition developed below states that experimentally familiar observation in spectral language. In a homogeneous region with symbol k_0 from Eq. (2.113), plane-wave trials decompose the field into lateral and longitudinal wavenumber components. The dispersion relation below is the local criterion that separates traveling from decaying longitudinal behavior: when lateral wavenumber $k_t = \sqrt{k_x^2 + k_y^2}$ exceeds k_0, no real propagating longitudinal root exists and the mode becomes evanescent [2]. Experimentally one would observe that near-field scan content with $k_t > k_0$ dies away within a penetration depth $\delta_{\mathrm{ev}} = 1/\alpha(k_t)$ as the field is propagated outward [3]. Plane-wave trials $\mathbf{u}_{\mathbf{k}}(\mathbf{r}) = \mathbf{u}_0\, e^{-\mathrm{j}\mathbf{k}\cdot\mathbf{r}}$ satisfy the Helmholtz condition when

$$|\mathbf{k}| = k_0, \qquad k_z^2 = k_0^2 - k_x^2 - k_y^2, \tag{2.129}$$

with lateral wavenumber $k_t = \sqrt{k_x^2 + k_y^2}$ [2]. Equation (2.129) splits spectral content by longitudinal behavior:

$$k_z = \begin{cases} \pm\sqrt{k_0^2 - k_t^2}, & k_t \le k_0 \text{ (propagating)}, \\ -\mathrm{j}\,\alpha(k_t), & k_t > k_0 \text{ (evanescent)},\ \alpha > 0, \end{cases} \tag{2.130}$$

where the evanescent root is chosen with decay in the $+z$ half-space [2]. For $k_t > k_0$ the decay rate is

$$\alpha(k_t) = \sqrt{k_t^2 - k_0^2}, \tag{2.131}$$

so a pure evanescent trial on $z > 0$ scales as $e^{-\alpha(k_t)z}$ with penetration depth $\delta_{\mathrm{ev}} = 1/\alpha(k_t)$ [2]. Propagating modes carry real $k_z = \pm\sqrt{k_0^2 - k_t^2}$ and oscillatory phase along the longitudinal axis; evanescent modes carry imaginary k_z and store reactive energy in the near zone without far-zone power radiation [6]. At fixed (k_x, k_y), the Helmholtz symbol $\mathcal{L}_\omega$ from Subsection 2.2.2 acts on each longitudinal branch as a scalar multiplier in the plane-wave basis, which is why spectral partitions commute with harmonic extraction once outgoing radiation condition is fixed on the propagating sheet [2]. Transverse polarization completes the mode description. For a trial electric field $\widetilde{\mathbf{E}}(k_x, k_y)\, e^{-\mathrm{j}(k_x x + k_y y + k_z z)}$, source-free admissibility requires $\mathbf{k} \cdot \widetilde{\mathbf{E}} = 0$ in the homogeneous region [2]. Decompose $\mathbf{k} = \mathbf{k}_t + k_z\hat{\mathbf{z}}$ with $\mathbf{k}_t = k_x\hat{\mathbf{x}} + k_y\hat{\mathbf{y}}$. Then

$$k_x\widetilde{E}_x + k_y\widetilde{E}_y + k_z\widetilde{E}_z = 0, \tag{2.132}$$

which leaves two independent transverse amplitudes on each branch of Eq. (2.130). On the propagating sheet, Eq. (2.132) supports traveling power along $\hat{\mathbf{z}}$ when outgoing k_z is selected; on the evanescent sheet the same constraint forces predominantly tangential storage with longitudinal E_z tied to $\alpha(k_t)$ [2]. Subsection 2.2.3 does not classify TE/TM guide nomenclature—that labeling is deferred to waveguide sections—but Eq. (2.132) is the local polarization criterion paired with the dispersion criterion of Eq. (2.129) [6].

Dispersion split Eq. (2.130): lateral wavenumber k_t selects propagating versus evanescent longitudinal behavior at fixed k_0.

Angular-spectrum representations in section 2.9 expand fields in these modes; here the subspaces are defined as solution domains [2]. Let $\mathcal{F}_{\mathrm{prop}} \subset \mathcal{F}_\Omega$ be the completion of finite superpositions of propagating plane waves with outgoing k_z selected by exterior completion, let $\mathcal{F}_{\mathrm{ev}} \subset \mathcal{F}_\Omega$ be the closure of evanescent superpositions, and let $\mathcal{F}_{\mathrm{non}} \subset \mathcal{F}_\Omega$ collect modes that are neither outgoing-propagating nor evanescent-decaying in the declared half-space—for example cutoff guide modes and purely reactive standing content tied to terminals [2]. Orthogonal spectral projectors (formal) satisfy

$$\Pi_{\mathrm{prop}} + \Pi_{\mathrm{ev}} + \Pi_{\mathrm{non}} = \mathrm{id} \quad \text{on } \widehat{\mathcal{F}}_\Omega, \tag{2.133}$$

with $\widehat{\mathcal{F}}_\Omega$ the spectral completion of admissible states [2]. Idempotency and mutual annihilation read

$$\Pi_{\mathrm{prop}}^2 = \Pi_{\mathrm{prop}}, \quad \Pi_{\mathrm{ev}}^2 = \Pi_{\mathrm{ev}}, \quad \Pi_{\mathrm{non}}^2 = \Pi_{\mathrm{non}}, \quad \Pi_{\mathrm{prop}}\Pi_{\mathrm{ev}} = \Pi_{\mathrm{ev}}\Pi_{\mathrm{prop}} = 0, \tag{2.134}$$

with analogous zero products for Π_{non} [2]. On $\mathcal{D}(\mathcal{L}_\omega)$ from Subsection 2.2.2, each projector block is an invariant subspace of the scalar Helmholtz symbol in the plane-wave basis, Eqs. (2.133)–(2.134) are *formal* on the spectral completion $\widehat{\mathcal{F}}_\Omega$: idempotency and orthogonality are justified in the rigged Hilbert space treatment of Chapter 3 (*Angular Momentum Operators and SO(3)*) Subsections 3.1.3–3.5.2 [2]. Until that completion, projector algebra is used only on admissible finite superpositions and on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, where outgoing selection is already enforced [6].

$$\mathcal{L}_\omega(\Pi_{\mathrm{prop}}\mathbf{u}) = \Pi_{\mathrm{prop}}(\mathcal{L}_\omega\mathbf{u}), \tag{2.135}$$

and similarly for Π_{ev} and Π_{non} when boundary closure does not mix sheets [6]. Equation (2.135) is the operator reason harmonic solvers can treat propagating and evanescent content separately before radiation selection; Sommerfeld outgoing radiation condition in section 2.4 acts only on Π_{prop} [2]. The decomposition

$$\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{prop}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] + \Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] + \Pi_{\mathrm{non}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \tag{2.136}$$

holds when the impressed spectrum excites all three classes. Only $\Pi_{\mathrm{prop}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is eligible for far-zone radiation content after outgoing radiation condition; evanescent and non-propagating components remain in reactive storage exchange [2]. Energy norms from Eq. (2.47) decompose orthogonally in the spectral basis when the impressed angular spectrum has sufficient decay in (k_x, k_y) [2].

$$\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}, \tag{2.137}$$

where $\mathcal{O}_{\mathrm{out}}$ enforces outgoing selection on propagating content compatible with Eq. (1.21) [2]. Eq. (2.137) refines Π_{rad}: radiative subspace $\mathcal{F}_{\mathrm{rad}}$ is a subset of $\Pi_{\mathrm{prop}}\mathcal{F}_\Omega$, not of all of $\mathcal{F}_\Omega$ [6]. Evanescent and non-propagating components lie in the null space of far-zone radiation:

$$\Pi_{\mathrm{rad}}\Pi_{\mathrm{ev}} = 0, \qquad \Pi_{\mathrm{rad}}\Pi_{\mathrm{non}} = 0, \tag{2.138}$$

while $\Pi_{\mathrm{rad}}\Pi_{\mathrm{prop}} = \Pi_{\mathrm{rad}}$ after outgoing closure [2]. Eqs. (2.138) and (2.137) restate, in projector form, the regime logic of section 1.2.5, reactive near fields may be large even when $\Pi_{\mathrm{rad}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0}$ [6]. Causality from Subsection 2.2.3 ensures retarded support but does not, by itself, promote evanescent content into $\mathcal{F}_{\mathrm{rad}}$. For all $r > r_{\min}(a, \lambda_0)$ and every transport, flux, near-to-far, and angular-spectrum statement in sections 2.5–2.10,

$$\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}} \quad \text{as in Eq. (2.137).}$$

Earlier uses of Π_{rad} in section 2.0 are to be read through this refinement [2]. Outward transport condition from section 1.2.5 enters as an operator criterion:

$$\mathcal{D}_{J,\mathrm{reg}} = 1 \implies \text{transport reports use } \Pi_{\mathrm{prop}}\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}], \text{ not } \Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}], \tag{2.139}$$

with $\mathcal{D}_{J,\mathrm{reg}}$ from Eq. (2.33) [2]. Equation (2.139) is the operator restatement of Eq. (2.27); regime indices $\chi(r)$ and χ_Ω diagnose when Eq. (2.139) fails on a measurement shell [2]. Waveguide and cutoff structures illustrate non-propagating subspaces. For a uniform guide with lateral eigenvalue κ_n, mode n propagates only when

$$k_0^2 > \kappa_n^2 \quad \Longleftrightarrow \quad \omega > \omega_{c,n} = \frac{c_0\kappa_n}{\sqrt{\varepsilon_r\mu_r}}, \tag{2.140}$$

with $\omega_{c,n}$ the cutoff angular frequency [2]. Below cutoff, the mode lies in $\mathcal{F}_{\mathrm{ev}}$ or $\mathcal{F}_{\mathrm{non}}$ depending on boundary termination; applying far-zone transport formulas to such fields violates Eq. (2.139) as in the TE_{10} illustration of section 1.1.2 [6]. In a rectangular guide of width a, lateral eigenvalues $\kappa_{m,n} = \sqrt{(m\pi/a)^2 + (n\pi/b)^2}$ replace k_t in Eq. (2.140); each mode index (m, n) switches from $\mathcal{F}_{\mathrm{non}}$ to $\mathcal{F}_{\mathrm{prop}}$ only when ω exceeds its cutoff [2]. Near cutoff the propagating group velocity vanishes and stored reactive energy dominates even though the mode has crossed into Π_{prop}; Eq. (2.31) remains the far-zone transport criterion, not Eq. (2.140) alone [6]. Angular-spectrum expansion: expand $\mathbf{E}(\mathbf{r})$ on $z = \text{const}$ as

$$\mathbf{E}(\mathbf{r}) = \iint \widetilde{\mathbf{E}}(k_x, k_y)\, e^{-\mathrm{j}(k_x x + k_y y + k_z(k_t) z)}\, dk_x\, dk_y, \tag{2.141}$$

with $k_z(k_t)$ from Eq. (2.130) [2]. Eq. (2.141) is the 2 spectral template for section 2.9; full radiation operator action on angular spectra is developed there. Subsection 2.2.3 only defines the propagating/evanescent partition induced by Eq. (2.129) [6]. Figure 2.5 separates propagating and



Figure 2.5: Spectral criterion: only Π_{prop} content passes to Π_{rad}; evanescent and non-propagating components remain reactive.

evanescent spectral content: only Π_{prop} contributions pass to Π_{rad} and carry far-zone Poynting flux; evanescent and non-propagating components remain in reactive storage within $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2, 5].

Causality from Subsection 2.2.3 does not automatically place evanescent energy in $\mathcal{F}_{\mathrm{rad}}$; spectral and declared size and distance regimes are additional [2]. Near-zone reactive fields decompose into large evanescent and non-propagating fractions even when $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is exact. As $r \to \infty$, Π_{prop} content dominates P_{rad} when Eq. (2.31) holds [2]. Operator norms on subspaces satisfy

$$\|\Pi_{\mathrm{ev}} \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]\|_{\mathcal{U}} \gg \|\Pi_{\mathrm{prop}} \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]\|_{\mathcal{U}} \quad \text{on near shells,} \tag{2.142}$$

while the reverse inequality governs far-zone radiative observables [6]. Subsection 2.2.3 does not compute shell integrals; it supplies the spectral vocabulary for section 2.5.2 and flux operators in section 2.5. With Subsections 2.2.1–2.2.3 complete, section 2.2 has connected time-domain evolution, harmonic Maxwell and Helmholtz operators, causal equivalence, and propagating–evanescent spectral split on the function spaces of section 2.1. Green-function inversion in section 2.3 realizes $\mathcal{L}_\omega^{-1}$ and $\mathcal{G}_{E,\omega}$ with retarded causality; Sommerfeld selection in section 2.4 fixes outgoing branches for Π_{prop} and Π_{rad}; radiation integrals in section 2.5 meter flux on the propagating subspace subject to Eq. (2.139) [2].

2.3 Green Functions and Electromagnetic Inversion

Time- and frequency-domain operators in section 2.2 specify how impressed currents enter Maxwell's equations; section 2.3 supplies the propagators that make that coupling computable—the computational heart of method-of-moments assembly and near-field scanning [2, 6]. Propagation must be understood before radiation is selected [16]. Maxwell's equations couple impressed currents to fields throughout a domain; in practice one needs the field at a specified observation point produced by sources elsewhere. The Green function is the impulse response that reduces that coupling to a single propagator integral rather than a fresh boundary-value solve at every new observation coordinate

[2]. Before dyadic kernels are tabulated, fix the lowest-dimensional laboratory calibration every range engineer uses: a short dipole or probe feed at $k_0 a \ll 1$ on a turntable, with watt closure on a sphere Γ_R and a transverse $\sin\theta$ pattern that is invariant under rotation [14]. That Hertzian test is the physical image of $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ when $\mathbf{J}_{\mathrm{imp}}$ collapses to a point dipole moment; Subsection 2.3.2 closes the calculation, but the range observation precedes the dyad formulas [2]. Two laboratory questions follow. Given currents on a feed or equivalent surface, how is the field computed at an observation point? Given field samples on a scan surface or range contour, how may the equivalent source be recovered? Method-of-moments assembly, near-field back-propagation, compact-range transformation, and microwave imaging differ in discretization yet implement the same propagation law [6]. Scalar, vector, and dyadic kernels, singular regularization, retarded causality, and inversion on composite domains are developed below; outgoing selection at infinity belongs to section 2.4. The radiation map below $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ realized through G and $\overline{\mathbf{G}}_E$ is the physical response of the medium to declared impressed currents; radiation through $\mathcal{R}_\omega$ and Π_{rad} is deferred until section 2.4 has fixed the outgoing branch on which those operators act [2].

Laboratory reports therefore require the same outgoing branch and gauge assumed in the derivation [3].

2.3.1 Scalar, vector, and dyadic Green functions

Physically, G and $\overline{\mathbf{G}}_E$ are the medium's response functions: each records how a unit impulse at $\mathbf{r}'$ launches scalar or vector fields to $\mathbf{r}$ under the same boundary completion as $\mathcal{L}_\omega$ [2]. Every compact-range measurement begins with a dipole elementary check: an electrically small feed must reproduce the transverse far-zone law before multipole catalogs are trusted on larger apertures [14]. That observation anchors the Green-function program developed here—first the scalar monopole response, then vector potentials, then the electric dyad that maps $\mathbf{J}_{\mathrm{imp}}$ to $\mathbf{E}$ directly [2]. The dyadic Green function is the electromagnetic impulse response of the governing Maxwell operator: the field at an observation point produced by an elemental current located elsewhere. In uniform media it is also the continuous limit of one radiating cell in a phased array and the local kernel that probe correction applies when near-field data are translated to a reference surface [2]. Admissible kernels are retarded—field response follows source excitation only after finite propagation delay—so localized impressed currents remain consistent with Maxwell causality [16]. Scalar Helmholtz functions describe monopole-like propagation; polarization coupling on vector feeds requires the full dyadic [15]. The *kernel formulas* developed below must be distinguished from and the *domain declarations* that make those formulas meaningful on a declared boundary-value problem [2]. Unless noted otherwise, $\omega > 0$ is fixed, $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ from Eq. (2.48), charge continuity Eq. (1.8) holds, and $\mathcal{B}_\Gamma$ is declared on $\Gamma = \partial\Omega$. In uniform vacuum regions the wavenumber is k_0 from Eq. (2.113). Kernels are stated first in operator form on $\mathcal{D}(\mathcal{W}_{E,\omega}) \subset \mathcal{H}_{\nabla\times}(\Omega)$ from Eq. (2.109); piecewise $k(\mathbf{r})$ extensions use the same template with patchwise coefficients [2]. The harmonic solution operator $\mathcal{G}_{E,\omega}$ introduced in Subsection 2.2.2 is realized below by convolution with G and $\overline{\mathbf{G}}_E$; observation coordinate $\mathbf{r}$ and source coordinate $\mathbf{r}'$ must remain distinct throughout every range transformation [6]. Imagine exciting a uniform medium with the smallest source one can conceive: a unit impulse concentrated at a single point $\mathbf{r}'$. The field that appears at every other location $\mathbf{r}$ is the scalar Green function $G(\mathbf{r}, \mathbf{r}')$—the impulse response of the Helmholtz operator at fixed frequency [2]. Laboratory engineers never place a mathematical δ-function on a bench, yet every extended feed, patch, or volume current is assembled from such elemental responses by superposition: each infinitesimal slice of $\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')$ radiates as if it were a point source scaled by the local source strength, and the total field is the sum of those contributions weighted by propagation from $\mathbf{r}'$ to $\mathbf{r}$ [6]. Green functions are therefore the natural language of propagation: they replace repeated boundary-value solves

with a single propagator integral that method-of-moments codes, near-field scanners, and imaging algorithms all discretize [2]. The physical significance of Eq. (2.143) is that it isolates the response of the medium to the smallest source that can be imagined—a unit point excitation. Once this elementary response is known, more complicated sources may be synthesized by superposition [14]. The scalar Green function $G(\mathbf{r}, \mathbf{r}')$ inverts the Helmholtz operator $(\nabla^2 + k_0^2)$ on scalar potentials; $\mathbf{r}$ denotes the observation point and $\mathbf{r}'$ the source point, with $R = |\mathbf{r} - \mathbf{r}'|$ the geometric separation in metres. On a compact range, the scalar kernel is the monopole reference against which probe calibration and layered-substrate corrections are measured [3]. It satisfies

$$(\nabla^2 + k_0^2)\, G(\mathbf{r}, \mathbf{r}') = -\delta(\mathbf{r} - \mathbf{r}'), \tag{2.143}$$

with δ the three-dimensional Dirac distribution and boundary conditions on Γ inherited from $\mathcal{B}_\Gamma$ [2]. Equation (2.143) is the local symbol of spectral inversion: for fixed $\mathbf{r}'$, $G(\cdot, \mathbf{r}')$ is the response to a unit point source at $\mathbf{r}'$ under the same completion as $\mathcal{L}_\omega$ [14]. Convolution with G therefore propagates an arbitrary scalar source to the field ϕ that satisfies the inhomogeneous Helmholtz equation on the same domain. The associated solution operator is

$$\phi(\mathbf{r}) = \int_\Omega G(\mathbf{r}, \mathbf{r}')\, q(\mathbf{r}')\, d^3 r', \qquad (\nabla^2 + k_0^2)\phi = q, \tag{2.144}$$

for scalar sources q in the dual class paired with ϕ [2]. Eq. (2.144) as a weighted accumulation of responses arriving from every source point inside the domain Ω. The Green function determines how strongly a source at $\mathbf{r}'$ influences the observation point $\mathbf{r}$, while $q(\mathbf{r}')$ specifies how much source strength is available at that location [2]. In plain language: $\phi(\mathbf{r})$ is the scalar field one would measure or compute at the observation point; $q(\mathbf{r}')$ is the source density distributed throughout the impressed region Ω; and $d^3 r'$ sums contributions from every infinitesimal source element [14]. Mathematically, $\phi(\mathbf{r})$ carries units that depend on the physical quantity represented (potential, pressure, or a component precursor to $\mathbf{E}$); $q(\mathbf{r}')$ is the corresponding source density per cubic metre; integration extends only over Ω, the region where sources are nonzero [2]. On exterior domains, G must satisfy outgoing decay compatible with section 2.4; on bounded domains, Dirichlet, Neumann, or impedance data on Γ replace the free-space radiation template [6]. Before examining Eq. (2.145) itself, it is useful to recall the familiar physical picture of a disturbance spreading outward from a localized source. An expanding spherical wavefront carries phase $e^{-\mathrm{j}k_0 R}$ and amplitude dilution proportional to $1/R$; the same geometric spreading responsible for the inverse-square law in power appears here as the inverse-distance amplitude factor [14]. Phase accumulates linearly with propagation distance R, and outgoing energy transport requires the retarded phase sign fixed by the locked phasor convention $\exp(+\mathrm{j}\omega t)$ [2]. In unbounded vacuum with outgoing selection at infinity, Sommerfeld radiation at large R singles out the spherical wave whose amplitude falls as $1/R$ and whose phase advances with propagation distance.

$$G_0(\mathbf{r}, \mathbf{r}') = \frac{e^{-\mathrm{j}k_0 R}}{4\pi R}, \qquad R = |\mathbf{r} - \mathbf{r}'|, \tag{2.145}$$

with $R > 0$ and the branch of k_0 fixed by the locked phasor axis [2]. Eq. (2.145) satisfies Eq. (2.143) for $\mathbf{r} \neq \mathbf{r}'$; the distributional definition at coincidence is completed in Subsection 2.3.3. Physically, Eq. (2.145) is the spherical outgoing wave produced by a unit monopole source at $\mathbf{r}'$; its phase gradient defines local propagation direction on the propagating sheet of Subsection 2.2.3 [2]. Green kernels realize the resolvent Eq. (2.66) developed in section 2.2, the scalar and dyadic definitions below implement $\mathcal{G}_{E,\omega}$ of Eq. (2.116) and feed the radiation integral Eq. (2.4) [2]. Electric-wave operators and material-weighted symbols from Subsection 2.2.2 are presupposed; gauge enters

only through the Lorenz reduction below. Coincidence limits belong to Subsection 2.3.3; outgoing completion at infinity to section 2.4. Vector potential and vector kernels. Direct field inversion from measured $\mathbf{E}$ alone is difficult because Maxwell's equations couple $\mathbf{E}$ and $\mathbf{H}$ through curls, material response, and boundary data; the map from sources to fields is nonlocal and orientation-dependent [2]. Vector potentials are introduced not as additional fields one would place on a probe calibration sheet but as mathematical intermediaries that simplify reconstruction. In Lorenz gauge, each Cartesian component of $\mathbf{A}$ satisfies the scalar Helmholtz template of Eq. (2.143) with source $-\mu_0\mathbf{J}_{\mathrm{imp}}$ [14]. The vector potential is not an additional physical field but as a mathematical intermediary that simplifies the reconstruction of electromagnetic quantities; gauge choices remove redundant degrees of freedom so that propagation integrals take a standard convolution form while what an engineer measures remains $\mathbf{E}$ and $\mathbf{H}$ [2]. Lorenz-gauge vector potentials $\mathbf{A}$ reduce vector Maxwell inversion to a vector Helmholtz system driven by $\mathbf{J}_{\mathrm{imp}}$. Under the gauge and continuity constraints fixed in Chapter 1, each Cartesian component of $\mathbf{A}$ satisfies Eq. (2.143) with k_0 and source $-\mu_0\mathbf{J}_{\mathrm{imp}}$ [2]. In Lorenz gauge the vector potential $\mathbf{A}(\mathbf{r})$ carries units of $\mathrm{T\,m}$ (equivalently $\mathrm{Wb\,m^{-1}}$ in SI); each Cartesian component satisfies the scalar Helmholtz template with source $-\mu_0\mathbf{J}_{\mathrm{imp}}$. The electric field phasor follows from $\mathbf{A}$ and its divergence through

$$\mathbf{E}(\mathbf{r}) = \mathrm{j}\omega\mu_0\,\mathbf{A}(\mathbf{r}) + \frac{1}{\mathrm{j}\omega\varepsilon_0}\,\nabla\,\nabla\cdot\mathbf{A}(\mathbf{r}), \tag{2.146}$$

with $\mathbf{E}$ in $\mathrm{V\,m^{-1}}$, ε_0 the vacuum permittivity ($\mathrm{F\,m^{-1}}$), and $\mathbf{H}$ obtained from $\nabla\times\mathbf{E}$ when the first-order system is required [6]. Eq. (2.146) is a vector reduction tool, not a substitute for the dyadic definition below when $\mathcal{B}_\Gamma$ couples $\mathbf{E}$ and $\mathbf{H}$ at boundaries: impedance and PEC closures are enforced on $\mathcal{L}_\omega$, not on $\mathbf{A}$ alone [2]. A vector Green function $\mathbf{G}^A(\mathbf{r},\mathbf{r}')$ may be assembled from three scalar kernels; the electric dyadic is preferred when $\mathbf{J}_{\mathrm{imp}}$ is the primary excitation because it maps $\mathbf{J}_{\mathrm{imp}}$ to $\mathbf{E}$ directly without intermediate gauge potentials [6]. The scalar Green function describes how a point disturbance propagates. Real electromagnetic sources, however, possess orientation and polarization. A current element directed along $\hat{\mathbf{x}}$ generally produces field components in multiple directions. The scalar kernel is therefore insufficient, and the propagation law must be upgraded to a tensorial form [2]. Let $\overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}')$ denote a rank-two tensor (dyad) acting on vector sources by dot product from the left; $\mathbf{r}$ is the observation coordinate and $\mathbf{r}'$ the source coordinate. The electric dyadic Green function answers, in dyadic form, the impulse-response question: what electric field $\mathbf{E}(\mathbf{r})$ is produced at $\mathbf{r}$ when a vector current impulse is placed at $\mathbf{r}'$? Method-of-moments assembly implements Eq. (2.148) by testing this kernel against patch basis functions; near-field scanners apply the same propagation law when translating probe-corrected samples to a reference surface [6]. Note that the dyad encodes polarization coupling absent from the scalar G: a transverse feed excites both radiative and reactive dyadic components, which is why the Hertzian dipole test at $k_0a \ll 1$ remains the lowest-dimensional calibration before aperture catalogs are trusted [14]. It inverts the electric-wave operator Eq. (2.18) in the sense

$$\mathcal{W}_{E,\omega}\,\overline{\mathbf{G}}_E(\cdot,\mathbf{r}') = \bar{\mathbf{I}}\,\delta(\mathbf{r}-\mathbf{r}'), \tag{2.147}$$

where $\bar{\mathbf{I}}$ is the identity dyad and the operator $\mathcal{W}_{E,\omega}$ acts on the $\mathbf{r}$-dependence of $\overline{\mathbf{G}}_E$ [2]. Equation (2.147) is the dyadic analogue of Eq. (2.143): each column of $\overline{\mathbf{G}}_E$ is the electric field produced by an impulsive vector source aligned with a Cartesian unit vector at $\mathbf{r}'$ [14]. Every method-of-moments code and every near-field scanner relies on the convolution structure below: given an impressed current distribution $\mathbf{J}_{\mathrm{imp}}$, the electric field at observation points follows by superposition through the dyadic propagator $\overline{\mathbf{G}}_E$. The physical measurement problem is forward prediction—what field does a known or assumed current produce?—which is the prerequisite for calibration, coupling

analysis, and array synthesis [6]. Inversion asks the inverse question from sparse $\mathbf{E}$ samples and is substantially harder even though forward propagation is linear and explicit [3]. Superposition over an extended impressed current $\mathbf{J}_{\mathrm{imp}}$ then yields the electric field through the dyadic convolution below; the prefactor $-\mathrm{j}\omega\mu_0$ ties the integral to the locked phasor convention $\exp(+\mathrm{j}\omega t)$. The electric field due to impressed current is

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0 \int_\Omega \overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r' \;+\; \mathbf{E}_{\mathrm{hom}}(\mathbf{r}), \tag{2.148}$$

The importance of Eq. (2.148) lies not in the integral itself but in the asymmetry between prediction and reconstruction. Given $\mathbf{J}_{\mathrm{imp}}$, the field follows directly. Given only sparse measurements of $\mathbf{E}$, however, recovering $\mathbf{J}_{\mathrm{imp}}$ becomes substantially more difficult [3]. Here $\mathbf{E}_{\mathrm{hom}}$ is the homogeneous solution fixed by $\mathcal{B}_\Gamma$ [2]. In Eq. (2.148), $\mathbf{E}(\mathbf{r})$ is the electric field phasor in $\mathrm{V\,m^{-1}}$, $\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')$ is the impressed current density in $\mathrm{A\,m^{-2}}$, μ_0 is the vacuum permeability ($\mathrm{H\,m^{-1}}$), and ω is the angular frequency in $\mathrm{rad\,s^{-1}}$; the volume integral extends over the source region Ω, and $\mathbf{E}_{\mathrm{hom}}$ collects excitations not attributable to $\mathbf{J}_{\mathrm{imp}}$, including incident fields and cavity modes fixed by $\mathcal{B}_\Gamma$. Eq. (2.148) is the constructive radiation map every method-of-moments impedance matrix discretizes: patch currents $\mathbf{J}_{\mathrm{imp}}$ tested against $\overline{\mathbf{G}}_E$ produce the $\mathbf{E}$ that near-field scanners back-propagate to equivalent surfaces [6]. The consequence for inversion is immediate: specifying $\mathbf{J}_{\mathrm{imp}}$ determines $\mathbf{E}$ through a single propagator integral, while recovering $\mathbf{J}_{\mathrm{imp}}$ from sparse $\mathbf{E}$ samples is ill-posed and requires the regularization program of Subsection 2.3.3 [2]. Eq. (2.148) is the formal realization of Eq. (2.4) with outgoing dyad $\mathbf{\Gamma} = \overline{\mathbf{G}}_E$ on the physical branch [6]. The prefactor $-\mathrm{j}\omega\mu_0$ matches the locked phasor convention and the forcing map Eq. (2.115). The operator $\mathcal{G}_{E,\omega}$ in Eq. (2.149) should be read as the machine that converts admissible forcing functions into electromagnetic responses: it accepts a current distribution in the forcing class and returns the corresponding field throughout the observation domain [6]. The integral operator $\mathcal{G}_{E,\omega}$ of Eq. (2.116) is realized by

$$(\mathcal{G}_{E,\omega}\mathbf{f})(\mathbf{r}) = \int_\Omega \overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}') \cdot \mathbf{f}(\mathbf{r}')\, d^3r', \tag{2.149}$$

for $\mathbf{f}$ in the forcing class of Eq. (2.115) [2]. When $\mathbf{f}_{E,\omega}$ is the electric Helmholtz drive induced by $\mathbf{J}_{\mathrm{imp}}$, Eqs. (2.148) and (2.149) coincide up to the linear map from $\mathbf{J}_{\mathrm{imp}}$ to $\mathbf{f}_{E,\omega}$ [6]. The solution map $\mathcal{S}_\omega$ is not merely an abstract mapping notation—it represents the complete electromagnetic experiment beginning with impressed current and ending with observable fields, including homogeneous background $\mathbf{E}^{\mathrm{hom}}$ and boundary data $\mathcal{B}_\Gamma$ when present [2]. The solution map on impressed currents is therefore

$$\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{u}_\omega \quad \text{with} \quad \mathbf{E} \text{ from Eq. (2.148)},\ \mathbf{H} \text{ from } \nabla\times\mathbf{E} \text{ when single-field reduction is valid}, \tag{2.150}$$

on domains satisfying Eq. (2.114); otherwise the first-order dyadic system of section 2.3.5 applies [2].

Eq. (2.148) is a Fredholm integral of the first kind when $\mathbf{r}$ is fixed and $\mathbf{J}_{\mathrm{imp}}$ is unknown, and a constructive radiation map when $\mathbf{J}_{\mathrm{imp}}$ is specified. The prefactor $-\mathrm{j}\omega\mu_0$ ties the kernel to the locked phasor axis $\exp(+\mathrm{j}\omega t)$; reversing the sign convention would flip incoming versus outgoing content without changing the local curl–curl symbol. In vacuum, for $k_0R = 3$ with $R = |\mathbf{r}-\mathbf{r}'|$, the transverse dyadic component of Eq. (2.151) contributes $\approx 70\%$ of $|\overline{\mathbf{G}}_{E,0}|$ in operator norm, while the R^{-2} and R^{-3} reactive terms still matter for $k_0a \gtrsim 1$ near-zone scans [2]. Boundary data $\mathcal{B}_\Gamma$ enter through the kernel that satisfies the same traces, not through a separate post-multiplication on $\mathbf{E}$ alone [6]. In uniform vacuum with outgoing condition at infinity, let $\mathbf{R} = \mathbf{r}-\mathbf{r}'$, $R = |\mathbf{R}|$, and $\widehat{\mathbf{R}} = \mathbf{R}/R$. When observation and source points are separated ($R > 0$), the outgoing vacuum dyad may be written explicitly in terms of these geometric quantities. The projector $\bar{\mathbf{I}} - \widehat{\mathbf{R}}\widehat{\mathbf{R}}$

represents transverse radiation: only field components perpendicular to the separation vector carry energy to infinity in the far zone [2]. The inverse-power corrections in $k_0 R$ represent stored reactive energy—magnetic and electric energy oscillating locally without net outward transport at short range. An engineer measuring fields close to an antenna would observe significant contributions from these reactive terms; far from the source they become progressively less important, leaving only the transverse radiative component [6]. The standard outgoing electric dyad is

$$\overline{\mathbf{G}}_{E,0}(\mathbf{r},\mathbf{r}') = \frac{e^{-\mathrm{j}k_0 R}}{4\pi R}\left[\bar{\mathbf{I}} - \hat{\mathbf{R}}\hat{\mathbf{R}} + \left(\frac{1}{\mathrm{j}k_0 R} - \frac{1}{k_0^2 R^2}\right)\left(\bar{\mathbf{I}} - 3\,\hat{\mathbf{R}}\hat{\mathbf{R}}\right)\right], \tag{2.151}$$

valid for $R > 0$ [2]. The $\bar{\mathbf{I}} - \hat{\mathbf{R}}\hat{\mathbf{R}}$ term is transverse to $\mathbf{R}$ and dominates in the radiative far zone; the R^{-2} and R^{-3} terms in the parentheses supply near-field reactive structure through inverse powers of $k_0 R$ and dominate on electrically small probes placed within a wavelength of the feed [2]. Eq. (2.151) satisfies Eq. (2.147) for $\mathbf{r} \neq \mathbf{r}'$; coincidence evaluation is deferred to Subsection 2.3.3. The same kernel appears in Eq. (2.4) once $\mathbf{r}$ lies outside $\operatorname{supp}\mathbf{J}_{\mathrm{imp}}$ and outgoing radiation condition is active [6]. The magnetic field may be reconstructed from the electric field through Maxwell curl relations when the observation region is source-free and simply connected: $\mathbf{H}$ is not an independent measurement channel in the Lorenz reduction but follows from the spatial derivatives of $\mathbf{E}$ [2]. This reconstruction is valid in homogeneous regions away from impressed currents and fails across material interfaces or inside the source support where curl relations must be supplemented by impressed-current terms [6]. The magnetic field may be generated by the electric dyad through Maxwell curl relations when $\mathbf{J}_{\mathrm{imp}}$ is divergence-free in the source region:

$$\mathbf{H}(\mathbf{r}) = \frac{1}{\mathrm{j}\omega\mu_0}\,\nabla \times \mathbf{E}(\mathbf{r}), \qquad \mathbf{E} \text{ from Eq. (2.148)}, \tag{2.152}$$

on simply connected source-free observation regions [2]. A magnetic dyadic $\overline{\mathbf{G}}_H(\mathbf{r},\mathbf{r}')$ $(\mathrm{A\,m^{-1}})$ inverts Eq. (2.110) analogously to Eq. (2.147), with operator $\mathcal{W}_{H,\omega}$ in place of $\mathcal{W}_{E,\omega}$: it is the machine that maps magnetic impressed sources to $\mathbf{H}$, while $\overline{\mathbf{G}}_H$ denotes the corresponding tensorial impulse response [6]. Duality $\varepsilon_0\mathbf{E} \leftrightarrow \mu_0\mathbf{H}$, $\mathbf{J}_{\mathrm{imp}} \leftrightarrow \mathbf{J}_{\mathrm{eff}}^{(\mathrm{mag})}$ maps $\overline{\mathbf{G}}_{E,0} \leftrightarrow \overline{\mathbf{G}}_{H,0}$ in vacuum without re-deriving the magnetic-wave symbol. Subsection 2.3.1 states $\overline{\mathbf{G}}_H$ only where magnetic impressed currents are primary; otherwise Eq. (2.152) avoids duplicate kernel tabulations [2]. The preceding formulas assume a uniform medium. Real structures seldom satisfy this assumption. Substrates, radomes, biological tissues, and layered materials alter propagation and therefore modify the Green response [15]. In piecewise media, local symbols $k(\mathbf{r})$, $\varepsilon(\mathbf{r})$, $\mu(\mathbf{r})$ modify Eq. (2.147) to

$$\mathcal{W}_{E,\omega}^{(\varepsilon,\mu)}\,\overline{\mathbf{G}}_E^{(\varepsilon,\mu)}(\cdot,\mathbf{r}') = \bar{\mathbf{I}}\,\delta(\mathbf{r}-\mathbf{r}'), \tag{2.153}$$

with $\mathcal{W}_{E,\omega}^{(\varepsilon,\mu)}$ from Eq. (2.111) [2]. Free-space Eq. (2.151) applies patchwise only when $k(\mathbf{r})$ is constant in each patch and interface jumps are enforced by continuity of tangential $\mathbf{E},\mathbf{H}$ across shared boundaries (Eq. (1.19)); multi-patch assembly is Subsection 2.3.5 [6]. Reciprocity expresses a fundamental symmetry of passive electromagnetic environments: interchanging transmitter and receiver does not alter the propagation law itself [2]. On passive media without magneto-electric coupling, reciprocity requires that interchanging observation and source points leaves the dyadic invariant up to transposition:

$$\overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}') = \overline{\mathbf{G}}_E^{\mathsf{T}}(\mathbf{r}',\mathbf{r}), \tag{2.154}$$

at fixed ω; the full proof is in section 2.8 [2]. Eqs. (2.148)–(2.150) hold when $\mathbf{J}_{\mathrm{imp}}$ lies in the domain of the forcing map, when $\mathcal{B}_\Gamma$ is time-independent and identical under phasor reduction, and when

the selected kernel branch matches the resolvent of Eq. (2.123). They fail for resonant cavities without quotienting, for incoming rather than outgoing exterior conditions, and for sources with $\nabla \cdot \mathbf{J}_{\mathrm{imp}} \neq 0$ unless charge density ρ is included in the complete source tuple of section 1.1.2 [2]. Near $\mathbf{r} = \mathbf{r}'$, Eqs. (2.145) and (2.151) are pointwise undefined; convolution meaning is distributional (Subsection 2.3.3). The scalar monopole kernel G supplies the lowest-dimensional propagation law; the electric dyad $\overline{\mathbf{G}}_E$ is the vector generalization demanded by polarization on real feeds. Subsection 2.3.2 closes the loop in the inverse measurement order: a measured dipole pattern on Γ_R precedes the dyadic tabulation, so the laboratory observation remains the anchor when the full tensor formulas are applied to extended apertures [14]. Scalar monopole law G, vector dyad $\overline{\mathbf{G}}_E$, and operator realization $\mathcal{G}_{E,\omega}$ are now fixed with $\mathbf{r}$ as observation and $\mathbf{r}'$ as source throughout; coincidence limits and Physically, retarded causality refine how those kernels may be convolved with extended $\mathbf{J}_{\mathrm{imp}}$ [2].

Angular-momentum transport therefore remains qualified on outgoing pairs $(\mathbf{E}, \mathbf{H})$: gauge-dependent density splits cannot be read as laboratory observables without the admissibility predicates from Chapter 1 [2].

2.3.2 Point electric dipole and the dyadic Green operator

Engineering reading (Hertzian reference sequence). *Physical setup:* a short wire or probe feed whose electrical size satisfies $k_0 a \ll 1$ on a turntable range. *Objective:* verify rank-one far-zone radiation before multipole catalogs are applied to larger apertures. *Expected intuition:* a transverse $\sin\theta$ pattern with a single dominant radiative channel. *Measurement:* watt closure on a sphere Γ_R and turntable invariance of the total radiated power [14]. *Misconception:* equating mesh unknowns in a numerical model with the radiative rank of the physical source [7].

On a turntable compact range, the dipole test is performed before any aperture catalog: rotate the electrically small feed, integrate $|\mathbf{E}|^2$ on Γ_R, and confirm that the total radiated power is invariant and that the far-zone pattern carries a single transverse lobe [3]. That watt closure is the laboratory observation that precedes the dyadic formulas of Subsection 2.3.1; when it fails, the fault lies in calibration, outgoing branch selection, or source admissibility—not in the multipole catalog applied later [2]. The Hertzian dipole is the canonical image of the dyadic Green operator: a localized source, a retarded kernel, a radiation map, and a far-zone angular spectrum linked in one closed calculation [2]. Short-wire or probe feeds at $k_0 a \ll 1$ on a turntable should reproduce transverse $\sin\theta$ radiation and turntable-invariant watt totals before multipole catalogs are applied to larger apertures [14]. Every compact feed should be checked against this lowest-dimensional outgoing test at low ka [7]. The range observation fixed above—watt closure on Γ_R, turntable invariance, and a single transverse lobe—is what a correct Green propagation must reproduce before any multipole catalog is trusted on larger structures [3]. Only after that calibration does it pay to write the dyadic convolution in closed form: replace the extended source $\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')$ by a rank-one impulse and read off the field pair that every method-of-moments code must match at electrically small $k_0 a$ [6]. The Hertzian dipole is chosen as the canonical calibration source because it is the smallest oriented radiator that still exercises the full dyadic structure: one moment vector, one propagation direction, and a transverse far-zone pattern that every radiation theory must reproduce [14]. Compact feeds on turntable ranges, probe references, and method-of-moments self-tests all reduce to this limit at $k_0 a \ll 1$; when a code or measurement sequence cannot match the dipole field pair, no larger aperture catalog built on the same Green operator should be trusted [3]. The current moment $\mathbf{p}_J = \int \mathbf{J}_{\mathrm{imp}}\, d^3 r'$ has units A m; under harmonic time dependence it is related to the electric dipole moment $\mathbf{p}_e = -\mathrm{j}\mathbf{p}_J/\omega$ (units C m) that enters standard far-zone formulas [14]. Substituting $\mathbf{J}_{\mathrm{imp}}(\mathbf{r}') = \mathbf{I}\,\delta(\mathbf{r}')$ into Eq. (2.148) collapses the volume integral to a single dyadic–vector product

and yields the closed-form field pair

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0\,\overline{\mathbf{G}}_{E,0}(\mathbf{r},\mathbf{0})\cdot\mathbf{p}_e, \qquad \mathbf{H}(\mathbf{r}) = \frac{1}{\mathrm{j}\omega\mu_0}\,\nabla\times\mathbf{E}(\mathbf{r}), \tag{2.155}$$

with $\mathbf{p}_e = -\mathrm{j}\mathbf{p}_J/\omega$ and outgoing dyad Eq. (2.151). This is $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ in closed form; $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is the same pair after outgoing selection on the exterior domain [2]. Physically, Eq. (2.155) states that every Cartesian component of $\mathbf{p}_e$ excites a column of $\overline{\mathbf{G}}_{E,0}$, so polarization coupling is already present in the rank-one source that calibrates compact feeds [2].

Signature chain for a Hertzian dipole: impressed moment convolves with the outgoing dyadic Green function, producing $(\mathbf{E},\mathbf{H})$; the radiation map extracts the far-zone amplitude and angular spectrum.

In the radiative far zone $k_0 r \gg 1$, with $r = |\mathbf{r}|$ the observation radius in metres, the reactive R^{-2} and R^{-3} terms in Eq. (2.151) become negligible and Eq. (2.155) reduces to Eq. (2.246) with

$$\mathbf{E}(\mathbf{r}) \approx -\mathrm{j}\omega\mu_0\,\frac{e^{-\mathrm{j}k_0 r}}{4\pi r}\,\overline{\mathbf{I}} - \widehat{\mathbf{r}}\widehat{\mathbf{r}}\cdot\mathbf{p}, \tag{2.156}$$

so the transverse radiative image is fixed by $\widehat{\mathbf{r}}\times(\widehat{\mathbf{r}}\times\mathbf{p})$. This expression reveals that only transverse components survive at large distance. The far field therefore forgets most local source details and retains only radiatively accessible information—precisely the content a compact-range pattern measurement can resolve on Γ_R [2]. Power through sphere S_r follows Eq. (2.6) with $P_{\mathrm{rad}} \propto k_0^4|\mathbf{p}|^2$ in the electrically small limit [2]. Experimentally, the angular spectrum is the decomposition of the far-zone field into plane waves propagating in different directions $\widehat{\mathbf{k}}$: each outgoing direction receives a distinct amplitude fixed by how the dipole moment projects transversely onto that direction [3]. Pattern measurements on a spherical scanner therefore sample this angular distribution; synthesis codes invert the same lateral Fourier image when reconstructing aperture fields from far-zone data [2]. The plane-wave angular spectrum of Subsection 2.9.1 reads

$$a(\widehat{\mathbf{k}}) = -\mathrm{j}k_0^2\sqrt{\mu_0/\varepsilon_0}\,(\overline{\mathbf{I}} - \widehat{\mathbf{k}}\widehat{\mathbf{k}})\cdot\mathbf{p}, \tag{2.157}$$

where $a(\widehat{\mathbf{k}})$ is the far-zone plane-wave amplitude ($\mathrm{V\,m^{-1}}$) associated with propagation direction $\widehat{\mathbf{k}}$ (dimensionless unit vector) and $\widehat{\mathbf{k}}$ parametrizes the angular-spectrum parametrization of Subsection 2.9.1. Eq. (2.157) is the lateral Fourier image of the same radiative channel: each outgoing direction receives transverse amplitude fixed by $(\overline{\mathbf{I}} - \widehat{\mathbf{k}}\widehat{\mathbf{k}})\cdot\mathbf{p}$. One source, one kernel, one radiation map, one far field, one angular spectrum—every abstract operator statement in Chapter 2 reduces to this geometry when $\mathbf{J}_{\mathrm{imp}} = \mathbf{p}\,\delta(\mathbf{r}')$ [2]. The dipole test on Γ_R is therefore not a numerical afterthought; it is the lowest-dimensional witness that G, $\overline{\mathbf{G}}_E$, $\mathcal{S}_\omega$, and Physically, $\mathcal{R}_\omega$ describe the same hardware [3].

2.3.3 Singularities, distributions, and regularization

Layered substrates in patch and handset designs modify Green propagation through reflection coefficients at material interfaces [2]. Printed antennas and handset boards propagate through layered media; the free-space dyadic is the reference impulse, not the laboratory default when a probe hovers above a dielectric stack [15]. Probe correction files must include substrate Green response, not free-space alone, because the governing question—how a point source on a feed line launches fields to a probe above the substrate—already presupposes outgoing propagation in each half-space [16]. Kernels transport impressed currents; they do not specify which currents the feed can realize [17]. Subsection 2.3.1 defined scalar and dyadic Green kernels that satisfy

Eqs. (2.143) and (2.147) for $\mathbf{r} \neq \mathbf{r}'$ and deferred coincidence evaluation to the present subsection [2]. Subsection 2.3.3 supplies the definitive account of distributional convolutions, principal-value and finite-part prescriptions, static–dynamic kernel splits, and self-term regularization that make Eq. (2.148) well-defined on $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ including sheet and filamentary support. The Sobolev and distributional source classes of Subsection 2.1.1 are presupposed through Eqs. (2.53) and (2.55); their functional-analytic construction is not repeated. Retarded time-domain support belongs to Subsection 2.3.4; outgoing branch selection at infinity belongs to section 2.4. Let $\Omega \Subset \mathbb{R}^3$ be Lipschitz with piecewise smooth internal interfaces, $\omega > 0$ fixed, and $\mathbf{J}_{\mathrm{imp}}$ in the refined source class Eq. (2.48) or its distributional extension Eq. (2.53). Kernels are the vacuum templates Eqs. (2.145) and (2.151) unless $\mathcal{B}_\Gamma$ or piecewise $k(\mathbf{r})$ modify them patchwise. Observation points $\mathbf{r}$ may approach $\operatorname{supp} \mathbf{J}_{\mathrm{imp}}$ but remain in the domain where $\mathbf{E} \in \mathcal{H}_{\nabla\times}(\Omega)$ is sought. For $\mathbf{r} \notin \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$, the convolution Eq. (2.148) is a classical integral. What happens when observation and source points become arbitrarily close? The free-space dyad develops a non-integrable leading singularity: field amplitudes grow as inverse powers of separation R, reflecting reactive storage and self-interaction between neighboring source elements [2]. These singularities do not imply infinite radiated power—they signal that pointwise source models and naive quadrature break down, while extended patches with finite extent carry finite reactive energy [14]. As $\mathbf{r} \to \mathbf{r}' \in \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$, the free-space dyad Eq. (2.151) develops a non-integrable leading singularity; the local expansion in powers of $k_0 R$ separates static storage from dynamic radiation and is the analytic starting point for every regularization scheme in the display below. With separation vector $\mathbf{R} = \mathbf{r} - \mathbf{r}'$, distance $R = |\mathbf{R}|$ in metres, observation point $\mathbf{r}$ and source point $\mathbf{r}'$, and unit direction $\widehat{\mathbf{R}} = \mathbf{R}/R$,

$$\overline{\mathbf{G}}_{E,0}(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi R^3}\, \overline{\mathbf{P}}_t(\widehat{\mathbf{R}}) \;+\; \frac{\mathrm{j}k_0}{4\pi R^2}\, \overline{\mathbf{Q}}(\widehat{\mathbf{R}}) \;+\; \frac{k_0^2}{4\pi R}\, \overline{\mathbf{S}}(\widehat{\mathbf{R}}) \;+\; \mathcal{O}(k_0^3), \tag{2.158}$$

where $\overline{\mathbf{P}}_t(\widehat{\mathbf{R}}) = \bar{\mathbf{I}} - 3\widehat{\mathbf{R}}\widehat{\mathbf{R}}$ is transverse to $\mathbf{R}$ and $\overline{\mathbf{Q}}$, $\overline{\mathbf{S}}$ collect the next two anisotropic terms in the locked phasor convention [6]. Eq. (2.158) matches the R^{-3}, R^{-2}, and R^{-1} pieces of Eq. (2.151) and separates reactive near storage from radiative R^{-1} decay. The R^{-3} term is non-integrable in a neighborhood of coincidence; convolution with $\mathbf{J}_{\mathrm{imp}}$ therefore requires a distributional or regularized definition, not pointwise Riemann integration [2]. The scalar kernel Eq. (2.145) carries the parallel ordering R^{-1}, R^0, … with R^{-1} non-integrable on neighborhoods of the source. In the electrostatic limit $k_0 \to 0$ at fixed separation $R > 0$ (metres), the dynamic terms in Eq. (2.158) vanish and only the R^{-3} dipolar projector survives; that static piece is extracted analytically in method-of-moments codes before the smooth remainder is integrated numerically. Eq. (2.158) reduces to

$$\overline{\mathbf{G}}^{(\mathrm{st})}_{E,0}(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi R^3}\Big(\bar{\mathbf{I}} - 3\,\widehat{\mathbf{R}}\widehat{\mathbf{R}}\Big), \tag{2.159}$$

the field of a unit dipole aligned with each Cartesian source direction [14]. Physically, Eq. (2.159) captures static-field dominance at small $k_0 R$: reactive storage and self-interaction between neighboring current elements overwhelm the radiative R^{-1} term until separation or frequency increases [2]. Eq. (2.159) is the leading singular part extracted in method-of-moments static–dynamic splits: dynamic corrections are $\mathcal{O}(k_0 R)$ relative to the static term when $k_0 R \ll 1$ on source patches [6]. For divergence-free $\mathbf{J}_{\mathrm{imp}}$, the $\nabla\nabla\cdot$ structure of Lorenz reduction Eq. (2.146) removes monopole artifacts; the dominant self-coupling on compact patches is dipolar and transverse to the separation vector between observation and source centroids [2].

Singular–regular split at coincidence: non-integrable G_{sing} is extracted analytically; remainder K_{cpt} is smooth for Galerkin or point evaluation away from the extracted neighborhood.

(2.78) at kernel level. The decomposition separates analytically known local physics from the smoother long-range interaction that can be handled numerically: $\overline{\mathbf{G}}^{(\mathrm{sing})}_E$ collects the non-integrable

near-zone coupling, while $\overline{\mathbf{G}}_E^{(\mathrm{reg})}$ is the remainder suitable for standard quadrature [6]. Write

$$\overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}') = \overline{\mathbf{G}}_E^{(\mathrm{sing})}(\mathbf{r},\mathbf{r}') + \overline{\mathbf{G}}_E^{(\mathrm{reg})}(\mathbf{r},\mathbf{r}'), \tag{2.160}$$

with $\overline{\mathbf{G}}_E^{(\mathrm{sing})}$ collecting the non-integrable static and leading dynamic singularities of Eq. (2.158) and $\overline{\mathbf{G}}_E^{(\mathrm{reg})}$ bounded on $\Omega\times\Omega$ [2]. The corresponding solution operator split is

$$\mathcal{S}_\omega = \mathcal{G}_{\mathrm{sing}} + \mathcal{K}_{\mathrm{cpt}}, \qquad \mathcal{G}_{\mathrm{sing}}[\mathbf{J}_{\mathrm{imp}}] = -\mathrm{j}\omega\mu_0 \int_\Omega \overline{\mathbf{G}}_E^{(\mathrm{sing})}(\mathbf{r},\mathbf{r}')\cdot\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\,d^3r', \tag{2.161}$$

with $\mathcal{K}_{\mathrm{cpt}}$ compact on bounded Ω when $\overline{\mathbf{G}}_E^{(\mathrm{reg})}$ is square-integrable in both variables [2]. Here $\mathcal{G}_{\mathrm{sing}}$ is the singular propagation operator carrying non-integrable near-zone coupling, $\mathcal{K}_{\mathrm{cpt}}$ is the compact remainder suitable for stable numerical quadrature, and $\overline{\mathbf{G}}_E^{(\mathrm{sing})}$ and $\overline{\mathbf{G}}_E^{(\mathrm{reg})}$ are the corresponding kernel split defined in Eq. (2.160) [6]. Eqs. (2.160) and (2.161) realize Eq. (2.78) for Green-based inversion; Fredholm consequences are developed in section 2.6.4 [6]. When $\mathbf{J}_{\mathrm{imp}}$ is a bounded function, Eq. (2.148) defines $\mathbf{E}\in L^2_{\mathrm{loc}}(\Omega)^3$ away from the source. For sheet currents Eq. (2.53), the convolution is defined by duality: for all smooth compactly supported test fields $\boldsymbol{\phi}$ (dimensionless weighting fields that probe how the electric field pairs against admissible test states),

$$\langle \mathbf{E},\boldsymbol{\phi}\rangle_\Omega = -\mathrm{j}\omega\mu_0 \sum_{ab}\int_{\Sigma_{ab}}\int_\Omega \mathbf{J}_{s,ab}(\mathbf{r}')\cdot\overline{\mathbf{G}}_E^{\mathsf{T}}(\mathbf{r}',\mathbf{r})\cdot\boldsymbol{\phi}(\mathbf{r})\,d^3r\,dS', \tag{2.162}$$

Here $\boldsymbol{\phi}$ is a smooth test field used to probe the electric field when pointwise evaluation fails; Σ_{ab} is a surface patch carrying sheet current $\mathbf{J}_{s,ab}$ ($\mathrm{A\,m^{-1}}$), the surface density of impressed current on patch (a,b); and $\mathbf{J}_{s,ab}(\mathbf{r}')$ is the physical current flowing on that patch at source point $\mathbf{r}'$ [2], with appropriate Fubini interpretation and outgoing radiation condition on exterior problems [2]. Eq. (2.162) is the weak form of Eq. (2.148) when pointwise evaluation fails; it pairs with Eq. (2.20) once $\boldsymbol{\phi}$ is replaced by admissible test states in $\mathcal{F}^0_\Omega$ [6]. Physically, sheet sources produce jump conditions on $\hat{\mathbf{n}}\times\mathbf{E}$ and $\hat{\mathbf{n}}\times\mathbf{H}$; the distributional definition ensures those jumps arise from the same kernel as volume currents, not from ad hoc post-factors [2]. For volume $\mathbf{J}_{\mathrm{imp}}$ and observation points $\mathbf{r}\notin\operatorname{supp}\mathbf{J}_{\mathrm{imp}}$ approaching the support, one may attempt to define the field as the symmetric limit of integrals that exclude a small ball about the observation point. The principal value is that symmetric exclusion limit: ordinary integration fails because the R^{-3} pole is not integrable over a neighborhood of coincidence, but symmetric cancellation about the observation point can still yield a finite field when the singularity is weak enough [14]. Define the principal-value convolution

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0\,\mathrm{p.v.}\int_\Omega \overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}')\cdot\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\,d^3r', \tag{2.163}$$

as the limit of integrals over $\Omega\setminus B_\varepsilon(\mathbf{r})$ as $\varepsilon\downarrow 0$ when that limit exists in $\mathcal{H}_{\nabla\times}(\Omega)$ [14]. For observation on the source support—for example $\mathbf{r}\in\Sigma_{ab}$ for a surface patch—the principal value generally diverges because of the R^{-3} term; the *Hadamard finite part* subtracts the singular expansion explicitly and retains the finite reactive coupling between extended patches. Finite-part subtraction works because the non-integrable piece is known analytically from Eq. (2.158) and can be removed before the remainder is integrated numerically—the engineering prescription MoM codes implement as static–dynamic splitting [6]:

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0 \lim_{\varepsilon\to 0}\left[\int_{\Omega\setminus B_\varepsilon(\mathbf{r})}\overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}')\cdot\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\,d^3r' \;+\; \int_{B_\varepsilon(\mathbf{r})\cap\Omega}\overline{\mathbf{G}}_E^{(\mathrm{sing})}(\mathbf{r},\mathbf{r}')\cdot\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\,d^3r'\right], \tag{2.164}$$

provided $\mathbf{J}_{\mathrm{imp}}$ is smooth enough on the patch that the limit exists [6]. Eq. (2.164) is the prescription for on-surface field evaluation; it differs from Eq. (2.163) by the explicit analytic subtraction of $\overline{\mathbf{G}}_E^{(\mathrm{sing})}$, not by numerical mesh refinement alone [2]. Mathematically, the finite part is the constant term in the Laurent expansion of the integral as $\varepsilon \to 0$; physically, it removes infinite self-energy of pointwise source models while retaining finite reactive coupling between extended patches [14]. Method-of-moments and boundary-element codes replace the continuum convolution Eq. (2.148) by discrete impedance entries obtained by testing the dyadic kernel against pairs of basis functions on source patches. In plain language: $\mathbf{f}_m$ and $\mathbf{f}_n$ are the mth and nth current basis functions on patches Ω_m and Ω_n; Ω_m and Ω_n are the surface or volume patches where those basis functions live; and Z_{mn} is the mutual impedance—the complex voltage response at patch m per unit current on patch n—that a numerical code assembles from the double integral below [6]. What a MoM code actually computes is this bilinear pairing of basis functions through the dyadic propagator, not a pointwise field evaluation at patch centers unless a separate approximation is introduced. The Galerkin entry for patches Ω_m and Ω_n reads

$$Z_{mn} = -\mathrm{j}\omega\mu_0 \int_{\Omega_m}\int_{\Omega_n} \mathbf{f}_m(\mathbf{r}) \cdot \overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}') \cdot \mathbf{f}_n(\mathbf{r}')\, d^3r\, d^3r', \tag{2.165}$$

for basis functions $\mathbf{f}_m, \mathbf{f}_n$ on patches Ω_m, Ω_n [6]. When $m = n$, the integrand is non-integrable at $\mathbf{r} = \mathbf{r}'$; the Chapter 2 regularized self-term is

$$Z_{mm}^{(\mathrm{reg})} = -\mathrm{j}\omega\mu_0 \int_{\Omega_m}\int_{\Omega_m} \mathbf{f}_m(\mathbf{r}) \cdot \overline{\mathbf{G}}_E^{(\mathrm{reg})}(\mathbf{r},\mathbf{r}') \cdot \mathbf{f}_m(\mathbf{r}')\, d^3r\, d^3r' \;+\; Z_{mm}^{(\mathrm{sing})}, \tag{2.166}$$

where $Z_{mm}^{(\mathrm{reg})}$ is the numerically integrated smooth remainder, $Z_{mm}^{(\mathrm{sing})}$ is the analytic integral of $\overline{\mathbf{G}}_E^{(\mathrm{sing})}$ over the coincident patch—often reduced to closed forms for flat triangles and rectangles—and the sum is the quantity a code must assemble so that neighboring mesh elements do not produce near-singular divergences when observation and source points approach one another [2]. Point matching—evaluating Eq. (2.148) at patch centers with $\overline{\mathbf{G}}_E$ replaced by $\overline{\mathbf{G}}_E^{(\mathrm{reg})}$ only—is *not* equivalent to Eq. (2.166) unless basis overlap and singular subtraction match the same finite-part convention [6]. The distinction is one of source realizability: Galerkin testing defines a bounded bilinear form on $\mathcal{J}_\Omega \times \mathcal{J}_\Omega$; raw point evaluation does not [2]. When $\mathbf{r}$ and $\mathbf{r}'$ lie on distinct patches separated by small distance $d \ll \lambda_0 = 2\pi/k_0$, direct quadrature of Eq. (2.151) is ill-conditioned. Physically, neighboring mesh elements interact through nearly coincident kernel samples: the separation is small enough that reactive R^{-3} and R^{-2} terms vary rapidly across the integration domain, and standard quadrature underestimates or overestimates the coupling unless the singular part is subtracted and integrated analytically [6]. The standard subtraction template is

$$\int_{\Omega_n} \overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}')\cdot\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r' = \int_{\Omega_n} \Big[\overline{\mathbf{G}}_E - \overline{\mathbf{G}}_E^{(\mathrm{near})}(\mathbf{r},\mathbf{r}')\Big]\cdot\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r' + \int_{\Omega_n} \overline{\mathbf{G}}_E^{(\mathrm{near})}(\mathbf{r},\mathbf{r}')\cdot\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r', \tag{2.167}$$

where $\overline{\mathbf{G}}_E^{(\mathrm{near})}$ matches the first few terms of Eq. (2.158) about $\mathbf{r}$ and the integral of $\overline{\mathbf{G}}_E^{(\mathrm{near})}$ is evaluated analytically [6]. Eq. (2.167) does not alter the continuum operator; it stabilizes numerical approximation of $\mathcal{K}_{\mathrm{cpt}}$ in Eq. (2.161). Scalar singularities are weaker than dyadic singularities: the monopole kernel diverges only as R^{-1}, which is barely non-integrable in three dimensions, whereas the dyadic kernel carries an R^{-3} pole that breaks naive integration more severely [14]. This distinction matters computationally because scalar potential formulations can sometimes use simpler self-term corrections, while vector MoM codes must implement full static–dynamic dyadic subtraction [6]. The scalar Green function satisfies, in distributional sense,

$$\int_{B_\varepsilon(\mathbf{r})} G_0(\mathbf{r},\mathbf{r}')\, d^3r' = \frac{\varepsilon}{2} \;+\; \mathcal{O}(\varepsilon^3), \tag{2.168}$$

for small balls $B_\varepsilon(\mathbf{r})$ in $\mathbb{R}^3$, reflecting the R^{-1} integrability borderline in three dimensions [14]. Equation (2.168) explains why monopole self-terms in scalar potential formulations require either finite source extent or explicit regularization; vector dyads inherit stronger R^{-3} divergence captured by Eq. (2.159) [2]. Eqs. (2.164)–(2.166) require patchwise smooth $\mathbf{J}_{\mathrm{imp}}$, Lipschitz Ω, and fixed ω in the passive linear band. They fail for point dipoles modeled as δ-supported $\mathbf{J}_{\mathrm{imp}}$ without accompanying finite-part renormalization, and for non-Lipschitz edges where $\mathcal{H}_{\nabla\times}$ membership of $\mathbf{E}$ requires additional edge singularities not captured by Eq. (2.158) [2]. Lossy media with complex k_0 shift singular expansion coefficients but not the order of the R^{-3} pole; Subsection 2.3.3 does not re-tabulate complex-k subtractors [2]. Regularized convolutions now make $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ well-defined when observation $\mathbf{r}$ approaches source support; retarded time-domain support fixes *when* those spatial kernels may act, completing the causal picture begun in the harmonic dyads with $\mathbf{r}$ as observation and $\mathbf{r}'$ Physically, as source [2].

Finite support compresses accessible channels at fixed ω [7].

Maxwell consistency therefore fixes which admissible traces may carry flux on $\mathcal{F}_{\mathrm{rad}}$ [2].

2.3.4 Retardation and causal propagation

Harmonic Green functions presuppose retarded causality; laboratories measure fields only after source events have had time to propagate. The physical need is to construct time-domain kernels whose support respects outgoing propagation before frequency-domain radiation is read [2, 14]. Maxwell structure requires deciding when a source event may influence the field at a later observation time without violating causality. Harmonic Green functions presuppose that retarded support; this subsection constructs the time-domain kernels that realize outgoing propagation. Retarded time-domain propagation completes the harmonic Green picture: this subsection develops emphretarded time-domain Green operators, their causal support, and the kernel-level upgrade of the resolvent equivalence Eq. (2.123) and harmonic theorem Eq. (2.125) from Subsection 2.2.3 [2]. Outgoing causality and Laplace–Fourier transforms are presupposed through Eqs. (2.118)–(2.121); the display below constructs the Green kernels that realize those statements. Harmonic formulas Eqs. (2.145) and (2.151) are cited; they appear as $s = -\mathrm{j}\omega$ evaluations of causal Laplace transforms under Sommerfeld outgoing selection at infinity [2].

Let $\mathbf{j}_{\mathrm{imp}}(\mathbf{r}, t)$ be causal with $\mathrm{supp}_{(\mathbf{r},t)} \subseteq \Omega_s \times [0, \infty)$, $\Omega_s \Subset \Omega$, and let $\mathcal{B}_\Gamma$, ε, and μ be time-independent. Initial data are homogeneous, $\mathbf{U}_0 = \mathbf{0}$, unless noted for switch-on illustrations. Vacuum regions use $c_0 = 1/\sqrt{\mu_0\varepsilon_0}$ and $k_0 = \omega/c_0$ from Eq. (2.113) [2]. The equivalence class excludes advanced kernels satisfying Eq. (2.128). The causal scalar Green function for the wave operator $(\nabla^2 - \varepsilon_0\mu_0\,\partial_t^2)$ in vacuum satisfies the inhomogeneous wave equation below, with the time argument $t - t'$ measuring elapsed interval between source event and field observation (seconds, SI).

$$\left(\nabla^2 - \varepsilon_0\mu_0\,\partial_t^2\right) G(\mathbf{r}, \mathbf{r}', t - t') = -\delta(\mathbf{r} - \mathbf{r}')\,\delta(t - t'), \tag{2.169}$$

with retarded causality $G = \mathbf{0}$ for $t - t' < 0$: no signal precedes the light cone [2]. Time-harmonic kernels inherit this support; acausal branches appear as standing-wave contamination in open domains [6]. The free-space retarded kernel concentrates its support on the light cone $t - t' = R/c_0$, where $c_0 = 1/\sqrt{\mu_0\varepsilon_0}$ is the vacuum speed of light ($\mathrm{m\,s^{-1}}$) and $R = |\mathbf{r} - \mathbf{r}'|$ is the geometric separation in metres. The free-space outgoing solution is

$$G_0(\mathbf{r}, \mathbf{r}', t - t') = \frac{\delta(t - t' - R/c_0)}{4\pi R}, \qquad R = |\mathbf{r} - \mathbf{r}'|, \tag{2.170}$$

which transports impulse support on the light cone $t - t' = R/c_0$ [2]. Eq. (2.170) is the time-domain counterpart of Eq. (2.145): the locked phasor field $\exp(+\mathrm{j}\omega t)$ selects the $s = -\mathrm{j}\omega$ Laplace transform

of Eq. (2.170) on the outgoing branch [6]. Physically, Eq. (2.170) states that a unit impulse at $(\mathbf{r}', t')$ influences $\mathbf{r}$ only after travel time R/c_0, enforcing finite propagation speed and Eq. (2.102) on potentials generated by scalar convolution [2]. The retarded electric dyadic $\overline{\mathbf{G}}_E(\mathbf{r}, \mathbf{r}', t-t')$ upgrades Eq. (2.148) to the time domain. Time differentiation of the causal kernel enforces that the electric field responds only after the impressed current has acted for a retarded interval; the convolution below is the dyadic analogue of the scalar retarded integral. The causal Maxwell field is

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_{\mathrm{hom}}(\mathbf{r}, t) - \mu_0 \, \partial_t \int_0^t \int_\Omega \overline{\mathbf{G}}_E(\mathbf{r}, \mathbf{r}', t-t') \cdot \mathbf{j}_{\mathrm{imp}}(\mathbf{r}', t') \, d^3 r' \, dt', \tag{2.171}$$

which specializes Eq. (2.101) once the dyadic kernel is identified [2]. Retarded causality requires

$$\overline{\mathbf{G}}_E(\mathbf{r}, \mathbf{r}', t-t') = \mathbf{0} \quad \text{for } t - t' < 0, \tag{2.172}$$

analogous to Eq. (2.119) [2]. For $\mathbf{r} \notin \Omega_s$, Eq. (2.171) produces $\mathbf{E}(\mathbf{r}, t) = \mathbf{0}$ until the earliest retarded arrival time $t_{\mathrm{ret}}(\mathbf{r})$, defined as the infimum over all source events $(\mathbf{r}', t')$ in the support of $\mathbf{j}_{\mathrm{imp}}$ of the sum $t' + R/c_0$:

$$t_{\mathrm{ret}}(\mathbf{r}) = \inf_{(\mathbf{r}', t') \in \operatorname{supp} \mathbf{j}_{\mathrm{imp}}} \{ t' \, + \, R/c_0 \}, \tag{2.173}$$

which is the geometric light-cone support condition underlying Eq. (2.102) [14]. The harmonic dyad Eq. (2.151) is not repeated here; it is the $s = -\mathrm{j}\omega$ symbol of $\overline{\mathbf{G}}_{E,0}$ under Eqs. (2.124) and outgoing radiation condition [6].

Retarded support Eq. (2.172): field at $(\mathbf{r}, t)$ depends only on sources with $t' \leq t - R/c_0$; advanced coupling is excluded.

(2.171). Subsection 2.3.3 treated coincidence in $\mathbf{r}$; the display below completes the causal picture by fixing the temporal ordering enforced by retarded support [2]. Regularized self-terms from Eq. (2.166) apply to harmonic convolutions; time stepping discretizes Eq. (2.171) with the same kernel class [6]. Let $\widetilde{\overline{\mathbf{G}}}_E(\mathbf{r}, \mathbf{r}', s)$ denote the Laplace transform of $\overline{\mathbf{G}}_E(\mathbf{r}, \mathbf{r}', t)$ in $t - t'$ with $\mathrm{Re}\{s\}$ large enough for convergence. The complex variable s (s^{-1}) encodes temporal frequency; evaluation at $s = -\mathrm{j}\omega$ recovers the harmonic dyad used throughout the phasor convention. Causal support Eq. (2.172) implies $\widetilde{\overline{\mathbf{G}}}_E$ is analytic in the right half-plane and, under outgoing radiation condition,

$$\overline{\mathbf{G}}_{E,\omega}(\mathbf{r}, \mathbf{r}') = \widetilde{\overline{\mathbf{G}}}_E(\mathbf{r}, \mathbf{r}', s)\Big|_{s=-\mathrm{j}\omega}, \tag{2.174}$$

which is the kernel form of Eq. (2.123) [2]. Applying the Laplace transform to Eq. (2.171) with causal $\mathbf{j}_{\mathrm{imp}}$ yields

$$\widetilde{\mathbf{E}}(\mathbf{r}, s) = \widetilde{\mathbf{E}}_{\mathrm{hom}}(\mathbf{r}, s) - \mu_0 s \int_\Omega \widetilde{\overline{\mathbf{G}}}_E(\mathbf{r}, \mathbf{r}', s) \cdot \widetilde{\mathbf{j}}(\mathbf{r}', s) \, d^3 r', \tag{2.175}$$

with $\widetilde{\mathbf{j}}$ from Eq. (2.120) [2]. On $s = -\mathrm{j}\omega$ under Eqs. (2.124), Eq. (2.175) reduces to Eq. (2.148) with $\overline{\mathbf{G}}_{E,\omega}$ from Eq. (2.174) and $\mathbf{J}_{\mathrm{imp}} = \widetilde{\mathbf{j}}(\mathbf{r}, s)\Big|_{s=-\mathrm{j}\omega}$ [6]. The Chapter 2 causal propagation theorem at kernel level is

$$\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathcal{S}_t[\mathbf{j}_{\mathrm{imp}}](\mathbf{r}, t)\Big|_{\text{harmonic at } \omega} \quad \Longleftrightarrow \quad \overline{\mathbf{G}}_{E,\omega} = \widetilde{\overline{\mathbf{G}}}_E\Big|_{s=-\mathrm{j}\omega} \tag{2.176}$$

for the same $\mathcal{B}_\Gamma$ and outgoing branch, with $\mathcal{S}_t$ the time-domain map Eq. (2.100) [2]. Eq. (2.176) does not replace Eq. (2.125); it identifies the Green operators whose resolvent relation Eq. (2.123) implements [6]. Advanced kernels and branch failure. Advanced dyads satisfying Eq. (2.128) produce Laplace symbols that cannot equal Eq. (2.174) for any causal $\mathbf{j}_{\mathrm{imp}}$ [2]. Time-harmonic solvers that

implicitly use incoming Green functions therefore fail Eq. (2.176) even when interior Maxwell rows vanish [6]. Exterior problems require outgoing radiation condition consistent with section 2.4; interior cavity problems may require loss or quotienting as in Eq. (2.86) before a causal Laplace contour can be closed [2]. For finite-bandwidth excitation, the harmonic slice at fixed ω is insufficient. Causal fields superpose as

$$\mathbf{E}(\mathbf{r},t)=\int_{-\infty}^{\infty}\widetilde{\mathbf{E}}(\mathbf{r},-\mathrm{j}\omega)\,e^{+\mathrm{j}\omega t}\,\frac{d\omega}{2\pi}, \tag{2.177}$$

with $\widetilde{\mathbf{E}}(\mathbf{r},-\mathrm{j}\omega)$ the Fourier transform induced by Eq. (2.175) on causal data [2]. Eq. (2.177) is the inverse of the single-frequency reduction used in the phasor convention; dispersive $\varepsilon(\omega)$, $\mu(\omega)$ modify $\widetilde{\overline{\mathbf{G}}}_E(\mathbf{r},\mathbf{r}',s)$ and are deferred beyond the locked single-ω harmonic band except where loss enters as $\mathrm{Im}\{k_0\}\neq 0$ [6]. Within the harmonic band, Eq. (2.126) describes the long-time limit after switch-on when outgoing radiation or damping removes transients [2]. Because Π_{rad} commutes with harmonic extraction under the ordering assumed in Subsection 2.2.3,

$$\mathcal{R}_t[\mathbf{j}_{\mathrm{imp}}]=\Pi_{\mathrm{rad}}\circ\mathcal{S}_t[\mathbf{j}_{\mathrm{imp}}],\qquad \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]=\Pi_{\mathrm{rad}}\circ\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}], \tag{2.178}$$

with $\mathcal{R}_\omega$ from Eq. (2.127) [2]. Causality does not promote evanescent content to outgoing radiation: Eq. (2.139) remains the transport criterion after time-domain evolution [2]. Retarded propagation ensures support constraints only; spectral and declared size and distance regimes from Subsection 2.2.3 are additional [6]. Eqs. (2.171)–(2.176) require Eqs. (2.124), outgoing or passive completion on exterior domains, and $\mathbf{j}_{\mathrm{imp}}$ in the causal class of Subsection 2.2.1. They fail for time-varying $\mathcal{B}_\Gamma(t)$, moving media, and nonlinear sources. Coincidence and self-term evaluation of $\overline{\mathbf{G}}_{E,\omega}$ remains governed by Subsection 2.3.3 even when Eq. (2.171) is discretized in time [6]. Retarded support is therefore the temporal counterpart of the spatial singular split in Subsection 2.3.3, causality fixes when propagation may act, while regularization fixes how strongly a source couples to itself at coincidence [2]. Harmonic kernels G and $\overline{\mathbf{G}}_E$ are therefore retarded response functions in space and time; constructive inversion on composite domains asks which resolvent branch those responses realize when interfaces and exterior completions are assembled [2].

Retarded kernels therefore complete the causal picture begun in harmonic theory: radiate on Γ_R is read only after the time-domain support of the source has been qualified. Constructive inversion on composite domains follows next (Subsection 2.3.5).

2.3.5 Inversion and solvability

A compact-range inversion is not complete when a matrix inverts—it is complete when declared support, interface traces, and outgoing completion jointly determine a unique Maxwell field. The physical requirement is constructive solvability on piecewise domains with data from Eq. (1.19). Constructive inversion on piecewise domains closes the laboratory question of resolvability; this subsection develops solvability on single- and multi-region domains with interface data from Eq. (1.19) [2]. The inversion question is the same one compact-range practice faces when a method-of-moments mesh must close against declared support, interface traces, and outgoing completion before coefficients are reported [3]. Weak well-posedness on bounded quotients through Eqs. (2.89) and (2.86) from Subsection 2.1.4 is presupposed; Lax–Milgram and Friedrichs arguments are not repeated. Outgoing uniqueness on exterior domains is finalized in section 2.4.3; here the resolvent branch and its Green realization are stated as hypotheses [2]. Material Helmholtz symbols Eqs. (2.111)–(2.112) and trace spaces Eqs. (2.54)–(2.57) from sections 2.1.1–2.2.2 are fixed input classes. Let $\Omega=\cup_{j=1}^{N_p}\Omega_j$ be a finite union of Lipschitz patches with pairwise interfaces $\Sigma_{ab}=\overline{\Omega_a}\cap\overline{\Omega_b}$, piecewise constants $\varepsilon_j,\mu_j>0$ on Ω_j, and outer boundary $\Gamma=\partial\Omega$. Impressed data satisfy $\mathbf{J}_{\mathrm{imp}}\in\mathcal{J}_\Omega$, charge continuity Eq. (1.8), and

boundary class $\mathcal{B}_\Gamma$ compatible with Eq. (2.56). Regularized convolutions follow Subsection 2.3.3; retarded causality follows Subsection 2.3.4 [2]. Composite domain and patchwise inversion. Inversion separates invertible propagation from compact perturbations and null directions [2]. Ill-posed spherical near-field measurements often fail here: the radiation map is well defined, but the inverse amplifies noise along small singular values [3]. On a composite domain the impressed-source region is partitioned into material patches Ω_j separated by interfaces Σ_{ab}; each patch carries its own wavenumber $k_j = \omega\sqrt{\mu_j\varepsilon_j}$ ($\mathrm{rad\,m^{-1}}$). Decompose the solution map as

$$\Omega = \bigsqcup_{j=1}^{N_p} \Omega_j \;\cup\; \bigsqcup_{a<b} \Sigma_{ab}, \qquad k_j = \omega\sqrt{\mu_j\varepsilon_j}, \tag{2.179}$$

with wavenumbers k_j from Eq. (2.113) on each patch [2]. On Ω_j, the electric field satisfies the patchwise Helmholtz drive of Eq. (2.111) and is represented by the regularized dyadic convolution stated below; the patch kernel $\overline{\mathbf{G}}_E^{(j)}$ incorporates local ε_j, μ_j and the finite-part prescription of Subsection 2.3.3 when observation approaches the source support:

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}_{\mathrm{hom}}(\mathbf{r}) + \mathrm{j}\omega\mu_j \int_{\Omega_j} \overline{\mathbf{G}}_E^{(j)}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r', \qquad \mathbf{r} \in \Omega_j, \tag{2.180}$$

where $\overline{\mathbf{G}}_E^{(j)}$ satisfies Eq. (2.153) with patch coefficients and finite-part evaluation when $\mathbf{r} \in \Omega_j$ approaches $\operatorname{supp}\mathbf{J}_{\mathrm{imp}}$ [6]. In Eq. (2.180), $\mathbf{r}$ is the observation point in patch Ω_j and $\mathbf{r}'$ the source point integrated over the impressed support; $\overline{\mathbf{G}}_E^{(j)}(\mathbf{r},\mathbf{r}')$ is the patchwise outgoing response function [2]. Equation (2.180) is the constructive content of Eq. (2.150) on uniform subdomains; it does not yet enforce interface jumps across Σ_{ab} [2]. Fields in adjacent patches are linked by tangential continuity and normal-flux jumps imported from Eq. (1.19) [2]. The unit normal $\hat{\mathbf{n}}_{ab}$ on interface Σ_{ab} points from patch a into patch b; tangential $\mathbf{E}$ and $\mathbf{H}$ must match across the interface so that the composite field remains a single physical solution. In operator form,

$$\hat{\mathbf{n}}_{ab} \times \mathbf{E}^{(a)}\big|_{\Sigma_{ab}} = \hat{\mathbf{n}}_{ab} \times \mathbf{E}^{(b)}\big|_{\Sigma_{ab}}, \qquad \hat{\mathbf{n}}_{ab} \times \mathbf{H}^{(a)}\big|_{\Sigma_{ab}} = \hat{\mathbf{n}}_{ab} \times \mathbf{H}^{(b)}\big|_{\Sigma_{ab}}, \tag{2.181}$$

with $\mathbf{E}^{(j)}$ from Eq. (2.180) and $\mathbf{H}^{(j)}$ from Eq. (2.152) when valid [6]. Eqs. (2.181) replace free-space Eq. (2.151) by patch Green functions $\overline{\mathbf{G}}_E^{(j)}$ plus interface unknowns on $\mathcal{T}_{\Sigma_{ab}}$ from Eq. (2.54) [2]. The assembled multi-region system is

$$\begin{bmatrix} \mathcal{I}+\mathcal{K}_{11} & \cdots & \mathcal{K}_{1N_p} \\ \vdots & \ddots & \vdots \\ \mathcal{K}_{N_p1} & \cdots & \mathcal{I}+\mathcal{K}_{N_pN_p} \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_{N_p} \end{bmatrix} = \begin{bmatrix} \mathbf{b}_1 \\ \vdots \\ \mathbf{b}_{N_p} \end{bmatrix}, \tag{2.182}$$

where $\mathbf{x}_j$ collects tangential field or equivalent surface currents on patch boundaries, $\mathcal{K}_{jk}$ are dyadic layer potentials generated by $\overline{\mathbf{G}}_E^{(k)}$, and $\mathbf{b}_j$ encodes impressed sources and outer $\mathcal{B}_\Gamma$ data [6]. Eq. (2.182) is the constructive statement of multi-region solvability: invertibility of the block operator is the Green-function analogue of Eq. (2.89) on composite domains [2].

Subsection 2.3.3 develops self-term regularization on each patch; Subsection 2.3.4 records causal time marching when Eq. (2.182) is embedded in $\mathcal{S}_t$ [6]. On a bounded Lipschitz domain with passive $\mathcal{B}_\Gamma$ satisfying Eq. (2.87), coercivity on the quotient $\mathcal{U}^0_\omega/\ker\mathcal{L}_\omega$ from Eq. (2.86) implies existence of a unique weak solution Eq. (2.89) [2]. Green-function inversion is equivalent when the regularized operator

$$\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{u}_\omega \quad\Longleftrightarrow\quad \mathcal{L}_\omega[\mathbf{u}_\omega] = \begin{bmatrix} \mathbf{s}_\omega(\mathbf{J}_{\mathrm{imp}},\rho) \\ \mathbf{g}_\Gamma \end{bmatrix}, \tag{2.183}$$

with $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H})$ obtained from Eqs. (2.180)–(2.166) on simply connected patches and from Eq. (2.182) otherwise [6]. Eq. (2.183) expresses Eq. (2.17) with constructive kernels; stability is controlled by Eq. (2.71) and resolvent bound Eq. (2.76) on the physical branch [2]. On exterior Ω with compact $\operatorname{supp} \mathbf{J}_{\mathrm{imp}}$, existence of $\mathcal{S}_\omega$ requires outgoing Green functions whose far zone matches section 2.4 [2]. Without that selection, the block system Eq. (2.182) admits incoming solutions violating Eq. (2.176) [6]. Uniqueness on the outgoing branch is

$$\mathcal{L}_\omega[\mathbf{u}_\omega] = \mathbf{0} \text{ and outgoing } \mathcal{B}_\Gamma \implies \mathbf{u}_\omega = \mathbf{0}, \tag{2.184}$$

with $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H})$ the total field phasor pair. Eq. (2.184) is the solvability criterion for exterior Green inversion [2]. The split Eq. (2.93) reminds that exterior problems are not Friedrichs self-adjoint on all of $\mathcal{U}_\omega$ [6]. Resonant cavity spectra. Bounded cavities with PEC walls possess discrete spectra $\{\omega_n\}$. At a resonant frequency $\omega = \omega_n$ the Helmholtz operator acquires a non-trivial null space and the Green resolvent ceases to be bounded on the full admissible field space:

$$\ker \mathcal{L}_\omega \neq \{\mathbf{0}\} \quad \Longrightarrow \quad \mathcal{L}_\omega^{-1} \text{ unbounded on } \mathcal{U}_\omega, \tag{2.185}$$

and Eq. (2.183) fails unless quotienting by standing modes as in Eq. (2.86) or introducing loss [2]. Poles of the resolvent appear as

$$\det\left(\mathcal{I} + \mathcal{K}\right) = 0 \quad \Longleftrightarrow \quad \omega = \omega_n, \tag{2.186}$$

for the coupled operator $\mathcal{K}$ in Eq. (2.182) on closed resonant structures [2]. Impressed $\mathbf{J}_{\mathrm{imp}}$ at resonance requires compatibility with $\ker \mathcal{L}_\omega$; otherwise amplitudes diverge even when Green formulas are formally applied [6]. Incoming exterior excitation produces a distinct failure: Eq. (2.184) is violated without additional incident-field subtraction in section 2.5.1 [2]. Numerical equivalence. Finite-element stiffness systems and boundary-element impedance matrices are Galerkin discretizations of Eq. (2.20) and Eq. (2.182), respectively:

$$\mathbf{Z}\mathbf{I} = \mathbf{V} \quad \Longleftrightarrow \quad \mathcal{K}\mathbf{x} = \mathbf{b}, \tag{2.187}$$

when basis functions match trace spaces and self-terms implement Eq. (2.166) [6]. Equation (2.187) is the same inversion problem in matrix form. Ill-conditioning near Eq. (2.186) mirrors practical difficulty at cavity modes and near-spurious resonances in open-region formulations [6]. Eqs. (2.180)–(2.182) require Lipschitz interfaces, passive linear media on each patch, and fixed ω in the harmonic band. They fail for time-varying boundaries, nonlinear constitutive laws, and anisotropic magnetoelectric coupling without retensoring $\overline{\mathbf{G}}_E^{(j)}$ [2]. Point sources without Subsection 2.3.3 regularization produce formal but non-admissible fields; $\mathbf{J}_{\mathrm{imp}}$ must remain in $\mathcal{J}_\Omega$ [6]. With Subsections 2.3.1–2.3.5 complete, section 2.3 has constructed scalar, vector, and dyadic Green kernels, regularized their singular convolutions, tied them to retarded causal propagation, and stated constructive inversion on composite domains compatible with Eq. (2.66). Sommerfeld outgoing selection in section 2.4 fixes the exterior branch assumed in Eq. (2.184); radiation integrals in section 2.5 evaluate far-zone observables from the propagating subspace subject to Eq. (2.139); spectral and Fredholm consequences of Eq. (2.78) continue in section 2.6.4 [2]. Green-function inversion recovers field states from sources under declared completion; whether a desired angular pattern is source-realizable under finite impressed currents is established in Chapter 4 (*Source Realizability and Spectral Truncation*) [17].

2.4 Radiation Conditions and Uniqueness

Green-function inversion in section 2.3 reconstructs fields from sources on compact support. section 2.4 imposes outgoing radiation conditions that separate measurable far-zone content from reactive storage [16]. Sommerfeld's asymptotic selection rule fixed the exterior branch that every open-range measurement implicitly assumes before uniqueness arguments or far-zone integrals are evaluated [16]. A dense near-field map without outgoing filtering can fit non-radiating currents [17].

Maxwell's equations alone do not distinguish fields radiating away from a source from fields arriving from infinity, yet every laboratory measurement assumes the former. Antennas are fed by finite sources, compact ranges suppress external illumination, and receivers attribute detected power to identifiable radiators rather than unexplained incoming waves [16]. section 2.3 supplied Green kernels that map impressed currents to fields; on exterior domains those kernels remain ambiguous until an outgoing branch is fixed at infinity [14]. Retarded causality narrows the admissible class but does not fully replace Sommerfeld selection when harmonic solvers invert $\mathcal{L}_\omega$ at fixed ω [2]. The subsections below state outgoing radiation conditions, embed them as resolvent selection, prove uniqueness on the radiative branch, and record reconstruction limits when that branch is omitted.

Solvability is therefore a joint statement about domains, traces, and outgoing branches—not about matrix condition number alone. Sommerfeld selection and uniqueness of radiative solutions finalize that branch in section 2.4.

2.4.1 Sommerfeld radiation condition

Before Sommerfeld asymptotics are written, the physical ambiguity must be named. Harmonic Helmholtz solutions on open domains superpose $e^{-\mathrm{j}k_0 r}/r$ outgoing and $e^{+\mathrm{j}k_0 r}/r$ incoming spherical waves with equal standing rights in the volume equations alone [16]. A near-field scan or pattern cut on Γ_R cannot distinguish these completions without an explicit outgoing selection: wattmeters and probes then read mixtures of radiation, storage, and energy arriving from the computational or chamber boundary [3]. Sommerfeld radiation condition is the asymptotic statement that restores laboratory identification—radiation launched locally and leaving the domain—before any uniqueness argument or far-zone integral is evaluated [16].

Before any asymptotic formula is written, the range engineer already decides whether the measurement enclosure behaves as an open exterior domain [3]. A compact range places the device under test inside an anechoic chamber whose absorbing linings are designed to let radiation escape instead of reflecting back as multipath [15]. Pattern probes on a large sphere Γ_R and shell wattmeters report tangential field and power only when chamber absorption and computational termination both approximate free-space outgoing behavior [5]. Incomplete absorption produces ripple on azimuth cuts; unstable spherical near-field transforms when spectra are truncated show the same branch failure at the data-processing stage [3]. Until outgoing selection is restored, Γ_R meters read cavity storage and standing-wave content rather than watt-bearing radiation [1]. Sommerfeld radiation condition below is the asymptotic mathematical statement of what the range engineer already assumes when attributing a measured pattern to the device under test rather than to energy arriving from infinity [16]. An open antenna measurement is meaningful only if the field on the range can be identified with radiation from the device under test rather than with energy arriving from infinity. Sommerfeld outgoing selection enforces that identification: disturbances launched locally propagate outward and are not returned from the far boundary of the model [16]. When the condition is omitted, the same near-field scan or pattern cut admits formal completions with incoming or standing-wave content, and inversion may assign equivalent sources that would require excitation mechanisms absent from the physical problem and therefore lie outside the admissible

solution class [17]. Free-space kernels Eqs. (2.145) and (2.151) supply the asymptotic templates; outer-boundary classes from Chapter 1 are presupposed. Uniqueness proofs and reconstruction limits belong to Subsections 2.4.3 and 2.4.4. Let Ω_{ext} denote an exterior region surrounding compact $\operatorname{supp}\mathbf{J}_{\text{imp}} \subset \Omega_s$, with $\omega > 0$ fixed and vacuum constants ε_0, μ_0 in the far zone. Spherical coordinates (r, θ, ϕ) are centered so that $\Omega_s \subset B_R(0)$ for some $R_0 > 0$. Fields are phasors under $\exp(+\mathrm{j}\omega t)$; outgoing means $e^{-\mathrm{j}k_0 r}/r$ decay with $k_0 = \omega\sqrt{\mu_0\varepsilon_0}$ from Eq. (2.113) [2]. Interior interfaces and finite domains use Eq. (1.20); Sommerfeld completion applies on artificial or physical spheres at large r [6]. On Γ_R with $R \gg \lambda_0$, a range engineer expects the measured phase to advance as $k_0 r$ increases—outgoing fronts receding from the enclosure—and the amplitude to fall no faster than r^{-1} in three dimensions [14]. That observation is what Sommerfeld selection encodes at infinity: the scalar template states what every outgoing Helmholtz response must look like at large distance from a localized source. Phase fronts recede with the observer; amplitude decay compatible with energy radiate to infinity is r^{-1} in three dimensions [14]. For scalar potentials and Helmholtz kernels, outgoing radiation requires, with $r = |\mathbf{r}|$ the observation radius from an origin enclosing the source, $k_0 = \omega\sqrt{\mu_0\varepsilon_0}$ the vacuum wavenumber ($\mathrm{rad\,m^{-1}}$), and $A(\theta, \phi, \mathbf{r}')$ the angular amplitude that carries directional information while $\mathbf{r}'$ labels the fixed source point during the limit $r \to \infty$:

$$G(\mathbf{r}, \mathbf{r}') = A(\theta, \phi, \mathbf{r}')\,\frac{e^{-\mathrm{j}k_0 r}}{r} \;+\; \mathcal{O}(r^{-2}), \qquad r = |\mathbf{r}| \to \infty, \tag{2.188}$$

The remainder $\mathcal{O}(r^{-2})$ collects subleading reactive and curvature corrections that need not vanish at finite range [2]. Equation (2.188) is the 2 scalar Sommerfeld condition: it excludes incoming $e^{+\mathrm{j}k_0 r}/r$ and standing $e^{\pm\mathrm{j}k_0 r}/r$ superpositions that degenerate exterior uniqueness [6]. The free-space template Eq. (2.145) satisfies Eq. (2.188) for fixed $\mathbf{r}'$ [2]. Physically, Eq. (2.188) states that monopole-like responses propagate outward with phase fronts $k_0 r = \text{const}$ and amplitude decay r^{-1}, the only radial combination compatible with energy flux to infinity in three dimensions [14]. Vector fields must satisfy a transverse impedance relation at infinity: the electric and magnetic components that carry radiation must stand in the same ratio as a plane wave propagating along $\hat{\mathbf{r}}$, with vacuum impedance η_0 as the proportionality constant [2]. For harmonic Maxwell fields in vacuum far from sources,

$$\mathbf{r} \times \mathbf{E} \;+\; \eta_0\,(\mathbf{r} \times \mathbf{H}) \times \hat{\mathbf{r}} = \mathbf{o}(1), \qquad r \to \infty, \tag{2.189}$$

with $\eta_0 = \sqrt{\mu_0/\varepsilon_0}$ the vacuum wave impedance, $\hat{\mathbf{r}} = \mathbf{r}/r$ the unit radial direction, and $\mathbf{o}(1)$ denoting any vector whose Cartesian components vanish in the limit $r \to \infty$ [2]. Physically, Eq. (2.189) requires that the transverse electric field and the transverse magnetic field on a distant sphere lock phase and magnitude as if energy were leaving the domain along $\hat{\mathbf{r}}$, rather than arriving from infinity [6]. Eq. (2.189) is the 2 vector Sommerfeld radiation condition. It implies transverse radiative fields with asymptotic ratio

$$\mathbf{H}_{\text{rad}} \sim \eta_0^{-1}\,\hat{\mathbf{r}} \times \mathbf{E}_{\text{rad}}, \qquad r \to \infty, \tag{2.190}$$

and matches the outgoing-trace condition Eq. (1.21) from Subsection 1.1.2 when tangential traces are taken on a sphere of large radius [2]. Mathematically, Eq. (2.189) selects the subspace on which the exterior Laplacian/Helmholtz resolvent coincides with retarded Green functions Eq. (2.176); physically, it enforces power flow outward rather than inward at infinity [6]. Far-zone transversality and propagation. Sommerfeld-admissible fields decompose at leading order as

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}_0(\theta, \phi)\,\frac{e^{-\mathrm{j}k_0 r}}{r} \;+\; \mathcal{O}(r^{-2}), \qquad \mathbf{E}_0 \cdot \hat{\mathbf{r}} = 0, \tag{2.191}$$

with $\mathbf{E}_0$ the radiation amplitude pattern and $\mathbf{E}_0 \cdot \hat{\mathbf{r}} = 0$ the transversality constraint [2]. Equation (2.191) connects Sommerfeld completion to Π_{prop} from Subsection 2.2.3, far-zone content lies on the

propagating sheet with real $k_r = k_0$ and outgoing phase [6]. Evanescent and reactive components may still be large at finite r while satisfying Eq. (2.189); regime condition Eq. (2.139) remains separate from Sommerfeld selection [2]. Figure 2.6 shows the schematic: Sommerfeld far zone Eq. :

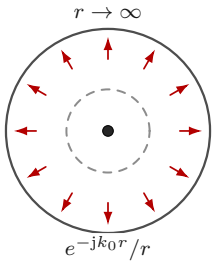


Figure 2.6: Sommerfeld far zone Eq. (2.189): compact sources drive near fields with evanescent content; at large r only outgoing transverse e^{-jk_0r}/r waves satisfy exterior completion.

compact sources drive near fields with evanescent content; at large r only outgoing transv. . . .

Eq. (2.189) is a local impedance statement at infinity: transverse $\mathbf{E}$ and $\mathbf{H}$ must satisfy the plane-wave ratio Eq. (2.190) with η_0 from Eq. (1.12). Mathematically, it selects the eigensheet of the exterior Helmholtz resolvent with outgoing radial phase; physically, it enforces net power flow toward infinity rather than inward from infinity. On a truncation sphere Γ_R with $R = 8\lambda_0$ enclosing a source with $k_0 a = 1.5$, Silver–Müller surrogate Eq. (2.194) typically introduces spurious reflected power below 1% of P_{rad} while reactive near fields at $r = 1.2a$ may differ by $\mathcal{O}(1)$ from the exact Sommerfeld solution [6]. Sommerfeld completion is therefore necessary but not sufficient for transport claims—declared size and distance regime Eq. (2.139) remains mandatory [2]. Figure 2.6 separates Sommerfeld completion from declared size and distance regime. Subsection 2.4.1 fixes the asymptotic *boundary at infinity*; Subsection 2.2.3 and section 1.2.5 govern when far-zone samples may be interpreted as transport [2]. Green-function asymptotics. Dyadic kernels from Subsection 2.3.1 inherit scalar Sommerfeld decay. For $r = |\mathbf{r} - \mathbf{r}'| \to \infty$ at fixed $\mathbf{r}'$,

$$\overline{\mathbf{G}}_{E,0}(\mathbf{r}, \mathbf{r}') = \overline{\mathbf{G}}_{\infty}(\theta, \phi, \mathbf{r}') \, \frac{e^{-\mathrm{j}k_0 r}}{r} \; + \; \mathcal{O}(r^{-2}), \tag{2.192}$$

with $\overline{\mathbf{G}}_{\infty}$ transverse in the $\mathbf{r}$-direction [2]. Eq. (2.192) is the kernel form of Eq. (2.189) under Eq. (2.148): impressed currents that satisfy Eq. (1.22) generate fields whose far zone is built from outgoing dyadic radiation [2]. The leading r^{-1} term in Eq. (2.151) is the explicit vacuum instance of Eq. (2.192) [14]. Subsection 2.4.2 promotes Eq. (2.192) to an operator selector on $\mathcal{S}_\omega$; Subsection 2.4.3 uses it in uniqueness energy integrals [6]. Collecting the asymptotic and impedance conditions above yields the admissible function class on which exterior uniqueness arguments and outgoing radiation are defined. A field pair belongs to this class only if it satisfies the vector Sommerfeld condition and the transverse far-zone template simultaneously; incoming or standing-wave completions are excluded by construction [6]. Define the Sommerfeld outgoing subspace on an exterior shell $\Gamma_R = \{r = R\}$:

$$\mathcal{F}_{\mathrm{SF}} = \{(\mathbf{E}, \mathbf{H}) \in \mathcal{H}_{\nabla\times}(\Omega_{\mathrm{ext}}) : \text{Eqs. (2.189)–(2.191) hold}\}\,. \tag{2.193}$$

In Eq. (2.193), $\mathcal{F}_{\mathrm{SF}} \subset \mathcal{H}_{\nabla\times}(\Omega_{\mathrm{ext}})$ is the Sommerfeld outgoing field class on exterior Ω_{ext}: pairs $(\mathbf{E}, \mathbf{H})$ whose far-zone asymptotics satisfy outgoing phase and transverse radiation templates; incoming and standing-wave completions are excluded by membership in this set. Eq. (2.193) embeds Eq. (1.21) into the 2 function-space setting of Eq. (2.56) [2]. The physical solution operator $\mathcal{S}_\omega$ maps $\mathcal{J}_\Omega$ into $\mathcal{F}_{\mathrm{SF}}$ when $\mathcal{B}_\Gamma$ includes outgoing radiation condition [6]. Incoming solutions lie outside $\mathcal{F}_{\mathrm{SF}}$ even if

they satisfy volume Maxwell rows; that exclusion is the Sommerfeld radiation condition [2]. Finite computations cannot place a boundary at infinity; the Silver–Müller trace below approximates the Sommerfeld impedance relation on a truncation sphere large enough that spurious reflections are small compared with radiated power [6]. On a finite truncation sphere Γ_R, Eq. (2.190) motivates the first-order absorbing condition

$$\hat{\mathbf{n}} \times \mathbf{H} \; - \; \eta_0^{-1}(\hat{\mathbf{n}} \times \mathbf{E}) \times \hat{\mathbf{n}} = \mathbf{0} \quad \text{on } \Gamma_R, \tag{2.194}$$

which approximates Eq. (1.21) at finite R [2]. Eq. (2.194) is not identical to Sommerfeld completion at infinity: spurious reflections from Γ_R contaminate near fields unless R is large compared with source extent and wavelength [6]. section 2.4 treats Eq. (2.194) as a practical surrogate; the continuous target remains Eq. (2.189) [2]. Perfectly matched layers and radiation boundary integrals are discrete refinements of the same outgoing-trace logic [6]. Once Sommerfeld closure is in place, the outgoing radiation map $\mathcal{R}_\omega$ and projector Π_{rad} act on the same outgoing class the range attributes to measured patterns: $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ with full field pair $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ carrying reactive content that Π_{rad} removes before far-zone metering [2]. The outgoing selector Π_{rad} of Eq. (2.2) is idempotent on fields satisfying Eq. (2.193) once uniqueness is established in Subsection 2.4.3 [2]. Prior to that proof, Sommerfeld closure defines the domain on which Π_{rad} acts as projection rather than heuristic windowing [6]. Composition with Eq. (2.27) yields

$$\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}], \qquad \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \in \mathcal{F}_{\mathrm{SF}}, \tag{2.195}$$

with condition for outward angular-momentum flux from section 1.2.5 applied after Eq. (2.195) [2]. Equation (2.195) expresses Eq. (2.127) on the Sommerfeld domain without re-proving causality from Subsection 2.2.3 [6]. Consequence for range practice: pattern meters and gain reports must compose with $\mathcal{R}_\omega$, not with raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$, whenever reactive near fields are non-negligible on the measurement shell [2]. Subsection 2.4.1 therefore states what the compact range already assumes before any far-zone integral is evaluated in section 2.5 [2]. Eqs. (2.188)–(2.193) require linear vacuum far zones, fixed $\omega > 0$, and compact $\operatorname{supp} \mathbf{J}_{\mathrm{imp}}$. They fail for anisotropic or magneto-electric exterior media without modified asymptotics, for lossless closed cavities where Eq. (1.20) uses PEC rather than outgoing radiation condition, and for time-harmonic fields with net incoming power at infinity [2]. Loss shifts Eq. (2.188) to complex k_0 but preserves outgoing *selection* as distinct from incoming [2]. As a worked example, replace Eq. (2.189) by Eq. (2.194) on Γ_R with $R = 5\lambda_0$. Fields inside Ω_s differ from the exact Sommerfeld solution by $\mathcal{O}((k_0 R)^{-1})$ in typical radiative components while reactive near content may differ by order unity [2]. Declaring a transport channel from Γ_R probes without Eq. (2.139) conflates truncation error with angular Physically, transport [2].

2.4.2 Outgoing-wave selection

The laboratory must decide which admissible Maxwell solution corresponds to a source that launches energy into the exterior rather than accepting incoming flux from infinity. Without an explicit outgoing selector, finite meshes and compact ranges cannot distinguish retarded radiation from standing-wave contamination that satisfies the same local equations [2]. Finite computational domains terminate open-region problems artificially. On a physical compact range, anechoic absorbers play the same role as absorbing boundary conditions in equivalent analytical models: they mimic terminations that let radiation escape instead of reflecting into the near field [3]. Silver–Müller and related traces approximate outgoing selection on meshes used for horns, cavities, and reflector feeds [2]. A finite-element model employing inadequate absorbing boundaries permits artificial reflections that contaminate field amplitudes and reconstructed radiation characteristics—the same

standing-wave signatures that appear when chamber absorption is imperfect on a physical range [3]. Absorbing-boundary parameters are therefore a radiation-physics choice about radiated energy, not a mesh-only tuning parameter [2]. The operator $\mathcal{O}_{\mathrm{out}}$ records that choice on $\mathcal{S}_\omega$ before Π_{rad} and $\mathcal{R}_\omega$ attach pattern meters. The outgoing-wave selector $\mathcal{O}_{\mathrm{out}}$ implements Sommerfeld completion Eq. (2.193) as an operator on $\mathcal{F}_\Omega$, on harmonic Green kernels, and on the resolvent branch of $\mathcal{L}_\omega^{-1}$ [2]. The composition relation Eq. (2.137) from Subsection 2.2.3 named $\mathcal{O}_{\mathrm{out}}$ without constructing it; the definitions below supply that operator and its links to Π_{prop}, Π_{rad}, and retarded Green functions Eq. (2.176). Sommerfeld field conditions Eqs. (2.189)–(2.193) are presupposed; uniqueness proofs belong to Subsection 2.4.3. Assumptions. Exterior regions use vacuum far-zone constants with $k_0 > 0$ real in the locked harmonic band, compact $\operatorname{supp} \mathbf{J}_{\mathrm{imp}}$, and time-independent $\mathcal{B}_\Gamma$ including outgoing radiation condition at infinity or on Eq. (2.194) at large finite R [2]. Propagating and evanescent projectors Eqs. (2.133)–(2.134) from Subsection 2.2.3 are fixed on $\widehat{\mathcal{F}}_\Omega$ [6]. Incoming and advanced branches satisfying Eq. (2.128) are excluded. Let $\mathcal{F}_{\mathrm{in}}$ denote the incoming counterpart of $\mathcal{F}_{\mathrm{SF}}$ obtained by reversing outgoing phase in the far zone. The outgoing-wave selector is the idempotent map

$$\mathcal{O}_{\mathrm{out}} : \mathcal{F}_\Omega \to \mathcal{F}_{\mathrm{SF}}, \qquad \mathcal{O}_{\mathrm{out}}[\mathbf{u}_\omega] = \mathbf{u}_{\mathrm{out}}, \tag{2.196}$$

where $\mathbf{u}_{\mathrm{out}}$ is the unique member of $\mathcal{F}_{\mathrm{SF}}$ whose propagating spectral content matches the outgoing k_z branch of Eq. (2.130) and whose incoming content is annihilated [2]. Equation (2.196) is the locked definition for operator realization of Sommerfeld completion: it acts *before* far-zone measurement and *after* volume completion by $\mathcal{L}_\omega$, selecting the physical resolvent branch assumed in Eq. (2.123) [2]. On $\mathcal{F}_{\mathrm{SF}}$,

$$\mathcal{O}_{\mathrm{out}}^2 = \mathcal{O}_{\mathrm{out}}, \qquad \mathcal{O}_{\mathrm{out}}\big|_{\mathcal{F}_{\mathrm{SF}}} = \mathrm{id}, \tag{2.197}$$

while incoming states satisfy $\mathcal{O}_{\mathrm{out}}[\mathbf{u}_{\mathrm{in}}] = \mathbf{0}$ when extended by continuity from the far zone [6]. Physically, $\mathcal{O}_{\mathrm{out}}$ removes incoming spherical and plane-wave components that satisfy Maxwell equations but violate retarded causality [14]. On homogeneous regions, outgoing selection in lateral wavenumber (k_x, k_y) uses the longitudinal root from Eq. (2.130):

$$k_z = \begin{cases} +\sqrt{k_0^2 - k_t^2}, & k_t \le k_0 \text{ (outgoing propagating)}, \\ -\mathrm{j}\,\alpha(k_t), & k_t > k_0 \text{ (decaying evanescent)}, \end{cases} \tag{2.198}$$

with the sign of the propagating root fixed by exterior completion and the locked $\exp(+\mathrm{j}\omega t)$ convention [2]. Equation (2.198) is the spectral action of $\mathcal{O}_{\mathrm{out}}$ on each plane-wave mode; reversing $k_z \to -k_z$ on the propagating sheet produces incoming content annihilated by $\mathcal{O}_{\mathrm{out}}$ [2]. Evanescent modes are unchanged in decay direction but remain outside Π_{rad} through Eq. (2.138) [6]. The propagating projector Π_{prop} from Subsection 2.2.3 retains all outgoing and incoming propagating modes; $\mathcal{O}_{\mathrm{out}}$ acts only after Π_{prop} in the radiation chain Eq. (2.137) [2]. Eq. (2.196) situates outgoing selection between propagating projection and outgoing radiation: $\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}$.

(2.137) with explicit operator order. Regime condition Eq. (2.139) applies after Π_{rad}, not at the $\mathcal{O}_{\mathrm{out}}$ stage Green-function branch selection. Harmonic kernels from section 2.3—with observation coordinate $\mathbf{r}$ and source coordinate $\mathbf{r}'$ in G and $\overline{\mathbf{G}}_E$—admit incoming and outgoing analytic continuations. The outgoing dyadic branch is

$$\overline{\mathbf{G}}_{E,\omega}^{(\mathrm{out})} = \mathcal{O}_{\mathrm{out}}\,\overline{\mathbf{G}}_{E,\omega}, \qquad \overline{\mathbf{G}}_{E,\omega}^{(\mathrm{out})} \text{ satisfies Eq. (2.192)}, \tag{2.199}$$

with $\mathcal{O}_{\mathrm{out}}$ acting on the observation-point dependence [2]. Eq. (2.199) identifies the kernel used in Eq. (2.148) on exterior problems with the retarded symbol of Eq. (2.176) [2]. Incoming kernels $\overline{\mathbf{G}}_{E,\omega}^{(\mathrm{in})}$ are related by $k_0 \to -k_0$ on the propagating Riemann sheet and produce fields outside $\mathcal{F}_{\mathrm{SF}}$

[14]. Subsection 2.4.2 does not re-tabulate Eq. (2.151); it fixes the sheet on which that formula is evaluated [6]. The physical resolvent branch is

$$\mathcal{L}_\omega^{-1}\big|_{\text{phys}} = \mathcal{O}_{\text{out}} \circ \mathcal{L}_\omega^{-1}, \tag{2.200}$$

understood on $\mathcal{D}(\mathcal{L}_\omega)$ with outgoing $\mathcal{B}_\Gamma$ [2]. The solution map on impressed currents becomes

$$\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}] = \mathcal{O}_{\text{out}}\Big(-\mathrm{j}\omega\mu_0 \int_\Omega \overline{\mathbf{G}}_{E,\omega}^{(\text{out})}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\text{imp}}(\mathbf{r}')\, d^3r' \;+\; \mathbf{E}_{\text{hom}}\Big), \tag{2.201}$$

with regularization from Subsection 2.3.3 when $\mathbf{r}$ approaches $\operatorname{supp}\mathbf{J}_{\text{imp}}$ [6]. Equations (2.200) and (2.201) realize Eq. (2.184) once Subsection 2.4.3 proves uniqueness on the selected branch [2]. Causality from Subsection 2.3.4 is consistent with Eq. (2.199) but not sufficient when harmonic solvers bypass time-domain construction [6]. The volume integral in Eq. (2.201) sums impressed current contributions at each source point $\mathbf{r}' \in \Omega$ against the outgoing dyadic evaluated at observation point $\mathbf{r}$; $\mathbf{E}_{\text{hom}}$ carries any exterior illumination not generated by $\mathbf{J}_{\text{imp}}$ and must vanish for pure radiation problems [2]. On propagating content, outgoing selection precedes outgoing radiation in the measurement sequence fixed in section 2.4, $\mathcal{R}_\omega = \Pi_{\text{rad}} \circ \mathcal{S}_\omega$ with $\Pi_{\text{rad}} = \mathcal{O}_{\text{out}} \circ \Pi_{\text{prop}}$ [2]. On propagating content,

$$\Pi_{\text{rad}} = \mathcal{O}_{\text{out}} \circ \Pi_{\text{prop}}, \qquad \mathcal{R}_\omega = \Pi_{\text{rad}} \circ \mathcal{S}_\omega, \tag{2.202}$$

restating Eqs. (2.137) and (2.2) with the locked definition of $\mathcal{O}_{\text{out}}$ [2]. Radiative subspace $\mathcal{F}_{\text{rad}} \subset \mathcal{F}_{\text{SF}}$ consists of fields with nonzero far-zone transverse amplitude Eq. (2.191); $\mathcal{O}_{\text{out}}$ maps into $\mathcal{F}_{\text{SF}}$ but not automatically into $\mathcal{F}_{\text{rad}}$ when evanescent content dominates near sources [2]. Transport reporting therefore requires Eq. (2.139) after Eq. (2.202) [6]. Outgoing selection on a compact range is implemented twice in practice: absorbers suppress multipath reflections that would mimic incoming waves, and harmonic solvers fix the propagating k_z branch of Eq. (2.198) on exterior meshes [3]. When either layer fails, $\mathcal{O}_{\text{out}}$ acts on the wrong field orbit even though volume Maxwell rows still vanish [6]. Incoming–outgoing split. Define $\mathcal{O}_{\text{in}}$ analogously on $\mathcal{F}_{\text{in}}$. On the propagating subspace,

$$\mathrm{id} = \mathcal{O}_{\text{out}} + \mathcal{O}_{\text{in}} \quad \text{on } \Pi_{\text{prop}}\mathcal{F}_\Omega, \qquad \mathcal{O}_{\text{out}}\mathcal{O}_{\text{in}} = 0, \tag{2.203}$$

with evanescent and non-propagating components orthogonal to both when decay direction is fixed by Eq. (2.198) [2]. Eq. (2.203) explains branch degeneracy in unconstrained Helmholtz solvers: both $\mathcal{O}_{\text{out}}[\mathbf{u}]$ and $\mathcal{O}_{\text{in}}[\mathbf{u}]$ satisfy volume equations unless one is selected by boundary or radiation completion [14]. Eqs. (2.196)–(2.201) require linear media, fixed ω, and outgoing $\mathcal{B}_\Gamma$ on exterior problems. They fail on closed PEC cavities where Eq. (1.20) replaces Sommerfeld completion, and on lossless open resonators where discrete spectra require quotienting Eq. (2.86) [2]. Absorbing layers approximate $\mathcal{O}_{\text{out}}$ but introduce spurious modes unless tuned to Eq. (2.189) at large distance [6].

Outgoing selection is therefore the first physical requirement on the radiation map: only after $\mathcal{O}_{\text{out}}$ removes incoming branches does Π_{rad} read a pattern that a laboratory can interpret as launched power. The discussion now turns to uniqueness of that radiative branch under the same boundary data (Subsection 2.4.3).

The lasting point is that outgoing selection is not an optional boundary convention: it is the physical act that turns a Maxwell solution into radiative evidence. Every later coefficient table inherits this choice, because a harmonic label attached before incoming content is removed has not yet become a laboratory statement.

2.4.3 Uniqueness of radiative solutions

Outgoing selection is not a post hoc spatial mask: Sommerfeld radiation and uniqueness on $\mathcal{B}_\Gamma$ decide which Maxwell solutions carry watt-bearing radiation before harmonic charts attach [16, 2]. Reactive near-field energy may be locally large while contributing negligibly to P_{rad} on Γ_R once Π_{rad} is fixed [6]. Without exterior uniqueness, two distinct field pairs could satisfy the same feed data while differing on the measurement sphere—a failure no wattmeter or pattern probe could resolve [16]. The proof below is therefore not a mathematical luxury imposed for elegance; it is the physical condition under which laboratory attribution of radiation to declared sources becomes meaningful. Outgoing selection is not a post hoc spatial mask: Sommerfeld radiation and uniqueness on $\mathcal{B}_\Gamma$ decide which Maxwell solutions carry watt-bearing radiation before harmonic charts attach [16]. On Γ_R, reactive near-field energy may be locally large while contributing negligibly to P_{rad} once $\mathcal{R}_\omega$ and Π_{rad} are fixed [6]. Exterior uniqueness ties measured patterns to declared sources within the radiative class [2]. Solvers that omit outgoing filters fit non-radiating currents that compact ranges cannot distinguish from radiation without watt completion on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ rather than on $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ alone [14]. The proof below imports the difference-field energy method from Subsection 1.1.3 and specializes it to the Sommerfeld outgoing class fixed in Subsection 2.4.1. Subsection 2.4.3 proves that exterior radiative solutions are unique on the outgoing branch, upgrading hypothesis Eq. (2.184) to exterior radiative uniqueness [2]. The difference-field reduction Eqs. (1.29)–(1.30) and energy identity Eq. (1.31) from Subsection 1.1.3 are presupposed; the passive uniqueness argument and coercive estimate Eq. (1.32) are not re-proved. Subsection 2.4.3 specializes the energy method to exterior domains with Eqs. (2.189)–(2.193), outgoing resolvent selection Eqs. (2.200)–(2.201), and radiation refinement Eq. (2.202). Source reconstruction limits belong to Subsection 2.4.4. Let Ω_{ext} denote an exterior domain with compact $\operatorname{supp}\mathbf{J}_{\mathrm{imp}} \subset B_R(0)$, vacuum constants ε_0, μ_0 in the far zone, and fixed $\omega > 0$ in the harmonic band of Subsection 2.2.3. Data are admissible per Eq. (1.23) with outgoing radiation condition Eq. (2.193) at infinity, equivalently Eq. (1.21) on spherical shells [2]. Constitutive response in the source region is passive: $\operatorname{Im}\{\varepsilon\} \geq 0$, $\operatorname{Im}\{\mu\} \geq 0$ or equivalent loss so homogeneous difference energy is dissipative [2]. Incoming and advanced branches are excluded by Eq. (2.196) and Eq. (2.203) [6]. Fields lie in $\mathcal{F}_{\mathrm{SF}}$ as defined in Subsection 2.4.1; outgoing radiation uses $\mathcal{F}_{\mathrm{rad}} \subset \mathcal{F}_{\mathrm{SF}}$ from Eq. (2.191) [2]. Uniqueness is established by showing that two outgoing solutions for the same impressed data cannot differ nontrivially: the proof therefore studies the homogeneous problem satisfied by their difference [2]. Let $(\mathbf{E}_1, \mathbf{H}_1)$ and $(\mathbf{E}_2, \mathbf{H}_2)$ be two harmonic solutions on the outgoing branch for the same impressed data and $\mathcal{B}_\Gamma$. Define $\Delta\mathbf{E} = \mathbf{E}_1 - \mathbf{E}_2$, $\Delta\mathbf{H} = \mathbf{H}_1 - \mathbf{H}_2$, and $\mathbf{u}_\Delta = (\Delta\mathbf{E}, \Delta\mathbf{H})$. Linearity and Eq. (1.29) give the homogeneous exterior system

$$\nabla \times \Delta\mathbf{E} = -\mathrm{j}\omega\mu\,\Delta\mathbf{H}, \quad \nabla \times \Delta\mathbf{H} = \mathrm{j}\omega\varepsilon\,\Delta\mathbf{E}, \quad \mathcal{B}_\Gamma[\Delta\mathbf{E}, \Delta\mathbf{H}] = \mathbf{0}, \tag{2.204}$$

with ε, μ matching the physical model in Ω_s and ε_0, μ_0 outside [2]. Physically, if two outgoing solutions for the same feed existed, their difference would satisfy this homogeneous system while still carrying radiate on Γ_R—a situation the flux argument below excludes [2]. Outgoing radiation condition requires

$$\mathcal{O}_{\mathrm{out}}[\mathbf{u}_\Delta] = \mathbf{u}_\Delta, \qquad \mathbf{u}_\Delta \in \mathcal{F}_{\mathrm{SF}}, \tag{2.205}$$

which is the exterior specialization of the third row of Eq. (1.29) when $\mathcal{B}_\Gamma$ includes Sommerfeld radiation [6]. Equations (2.204)–(2.205) state that uniqueness testing reduces to showing only the trivial pair satisfies the homogeneous outgoing problem [2]. Incoming difference modes would violate Eq. (2.205) even if they satisfied the curl rows [14]. On truncated exterior shells, form the complex Poynting vector for difference fields, $\mathbf{S}_\Delta = \frac{1}{2}\Delta\mathbf{E} \times \Delta\mathbf{H}^*$, as in Eq. (1.31) [2]. For a large sphere

$\Gamma_R = \{r = R\}$ enclosing Ω_s, let $\Omega_R = B_R \setminus \Omega_s$. The divergence theorem on Ω_R yields

$$\int_{\Gamma_R} \hat{\mathbf{r}} \cdot \mathbf{S}_\Delta \, dS = -\mathrm{j}\omega \int_{\Omega_R} \left(u_{\Delta H} - u_{\Delta E} \right) dV \; + \; \mathcal{P}_{\Delta,s}, \tag{2.206}$$

where $u_{\Delta E} = \frac{1}{4}\varepsilon|\Delta\mathbf{E}|^2$, $u_{\Delta H} = \frac{1}{4}\mu|\Delta\mathbf{H}|^2$, and $\mathcal{P}_{\Delta,s}$ collects source-region passive dissipation when $\mathrm{Im}\{\varepsilon\}, \mathrm{Im}\{\mu\} > 0$ [2]. Eq. (2.206) is Eq. (1.31) with $\Delta\mathbf{J}_{\mathrm{imp}} = \mathbf{0}$ and $\Gamma = \Gamma_R$; its real part controls uniqueness through the sign of the boundary flux [2]. On a compact range, the real part of the Γ_R integral is what a wattmeter would read from the difference field if such a superposition were physically present; outgoing selection forces that reading to nonnegative radiation or zero [6]. In lossless vacuum on $\Omega_R \setminus \Omega_s$, the volume term is purely imaginary at steady state, so $\mathrm{Re}\int_{\Gamma_R} \hat{\mathbf{r}} \cdot \mathbf{S}_\Delta \, dS$ is determined entirely by far-zone radiation [6]. Sommerfeld condition Eq. (2.189) implies that on Γ_R with $R \gg a = \max|\mathbf{r}|$ over $\mathrm{supp}\,\mathbf{J}_{\mathrm{imp}}$, the leading transverse contribution to $\Delta\mathbf{E}, \Delta\mathbf{H}$ scales as $e^{-\mathrm{j}k_0 R}/R$ with $k_0 = \omega\sqrt{\varepsilon_0\mu_0}$ [2]. The far-zone transverse radiation law Eq. (2.191) then gives

$$\mathrm{Re}\int_{\Gamma_R} \hat{\mathbf{r}} \cdot \mathbf{S}_\Delta \, dS = P_{\Delta,\mathrm{rad}}(R) = \frac{1}{2\eta_0} \oint_{\Gamma_R} \left|\mathbf{E}_{\Delta,t}\right|^2 dS \; + \; \mathcal{O}(R^{-1}), \tag{2.207}$$

with $\eta_0 = \sqrt{\mu_0/\varepsilon_0}$ and $\mathbf{E}_{\Delta,t}$ the transverse spherical component of $\Delta\mathbf{E}$ [2]. Taking $R \to \infty$ under outgoing radiation condition,

$$\lim_{R\to\infty} P_{\Delta,\mathrm{rad}}(R) = P_{\Delta,\mathrm{rad}} \geq 0, \tag{2.208}$$

where $P_{\Delta,\mathrm{rad}}$ is the net outgoing power carried by the difference field [2]. Equation (2.208) is the exterior boundary term induced by Eq. (1.21): outgoing difference fields radiate nonnegative power to infinity, while incoming difference fields would produce negative flux and are excluded by $\mathcal{O}_{\mathrm{out}}$ [14]. Dyadic asymptotics Eq. (2.192) are presupposed; they are not re-tabulated here [6].

section 1.3, volume terms in Eq. (2.206) are controlled by passivity when loss is present, and vanish in lossless vacuum away from sources [2]. Let $\mathbf{u}_\Delta = (\Delta\mathbf{E}, \Delta\mathbf{H})$ satisfy Eqs. (2.204)–(2.205) on an exterior domain with vacuum far zone. If $P_{\Delta,\mathrm{rad}} = 0$ in Eq. (2.208), then $\mathbf{E}_{\Delta,t} = \mathbf{0}$ on every sphere Γ_R for R large enough, and $\mathbf{u}_\Delta$ has no propagating outgoing content in $\mathcal{F}_{\mathrm{rad}}$ [2]. Eq. (2.207) with $P_{\Delta,\mathrm{rad}} = 0$ forces $\oint_{\Gamma_R} |\mathbf{E}_{\Delta,t}|^2 \, dS = 0$ for all sufficiently large R. Outgoing transverse fields with zero spherical flux cannot carry $1/R$ radiation; the only remaining homogeneous outgoing transverse modes are identically zero on $\mathcal{F}_{\mathrm{rad}}$ [2]. Let Ω_{ext} be an exterior domain with compact impressed sources, vacuum far zone, and outgoing Sommerfeld completion Eqs. (2.193)–(2.196). If $(\mathbf{E}_1, \mathbf{H}_1)$ and $(\mathbf{E}_2, \mathbf{H}_2)$ are harmonic Maxwell solutions on the outgoing branch for the same admissible data, then $\mathbf{E}_1 = \mathbf{E}_2$ and $\mathbf{H}_1 = \mathbf{H}_2$. Equivalently,

$$\mathcal{L}_\omega[\mathbf{u}_\omega] = \mathbf{0} \text{ and } \mathbf{u}_\omega \in \mathcal{F}_{\mathrm{SF}} \implies \mathbf{u}_\omega = \mathbf{0}, \tag{2.209}$$

which upgrades Eq. (2.184) to the Chapter 2 definitive uniqueness statement [2]. Let $\mathbf{u}_\Delta$ be the difference of two outgoing solutions. Then Eqs. (2.204)–(2.205) hold. Apply Eq. (2.206) and take $\mathrm{Re}\{\cdot\}$. In lossless vacuum outside Ω_s, the volume integral is purely imaginary, so $\mathrm{Re}\, P_{\Delta,\mathrm{rad}}(R)$ equals the real boundary flux. Outgoing radiation condition forces $\mathrm{Re}\, P_{\Delta,\mathrm{rad}} \geq 0$; passivity in Ω_s adds nonnegative $\mathrm{Re}\,\mathcal{P}_{\Delta,s}$, so the only way to satisfy Eq. (2.208) with equality throughout is $P_{\Delta,\mathrm{rad}} = 0$ [2]. The auxiliary relation (2.4.3) eliminates outgoing transverse content. Evanescent components remain in $\mathcal{F}_{\mathrm{SF}}$ but are tied to the same source-free outgoing solution operator; with no outgoing radiation and homogeneous forcing, the unique retarded exterior state is trivial by Eq. (2.201) with $\mathbf{J}_{\mathrm{imp}} = \mathbf{0}$ [6]. Hence $\mathbf{u}_\Delta = \mathbf{0}$, proving Eq. (2.209). Exterior radiative uniqueness (Subsection 2.4.3) is the exterior counterpart of the passive uniqueness argument (Subsection 1.1.3): instead of a bounded passive Γ with coercivity Eq. (1.32), the exterior uses Sommerfeld flux Eq. (2.208) as

the coercive boundary channel [2]. The uniqueness index Eq. (1.33) is finite on exterior outgoing problems away from embedded resonances in finite scatterers; it diverges on bounded cavities at Eq. (2.185) [6]. Exterior uniqueness therefore replaces bounded coercivity with an *open* boundary condition that selects decaying outgoing phase [2]. Exterior radiative uniqueness (Subsection 2.4.3) makes the physical resolvent injective on the outgoing subspace:

$$\mathcal{L}_\omega^{-1}\big|_{\mathrm{phys}} : \mathcal{Y}_\omega \to \mathcal{F}_{\mathrm{SF}} \text{ is single-valued}, \qquad \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathcal{L}_\omega^{-1}\big|_{\mathrm{phys}} \begin{bmatrix} \mathbf{s}_\omega(\mathbf{J}_{\mathrm{imp}}, \rho) \\ \mathbf{g}_\Gamma \end{bmatrix}, \tag{2.210}$$

consistent with Eq. (2.183), outgoing Green branch Eq. (2.199), and solution selection Eq. (2.201) [2]. For fixed $\mathbf{J}_{\mathrm{imp}}$ in the admissible class, the exterior field $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ obtained from Eq. (2.201) is unique in $\mathcal{F}_{\mathrm{SF}}$ [2]. Two convolutions differing by a homogeneous outgoing state violate exterior radiative uniqueness (Subsection 2.4.3) [6]. Subsection 2.0.1 declared Π_{rad} idempotent on $\mathcal{F}_{\mathrm{rad}}$; the joint property with outgoing radiation condition can now be stated precisely. On radiative states,

$$\Pi_{\mathrm{rad}}^2 = \Pi_{\mathrm{rad}} \quad \text{on } \mathcal{F}_{\mathrm{rad}}, \tag{2.211}$$

as developed in Subsection 2.0.1 [2]. After exterior radiative uniqueness (Subsection 2.4.3), the measurement sequence Eq. (2.202) defines a single-valued map $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ without branch ambiguity from duplicate outgoing decompositions [6]. Evanescent components remain in $\ker \Pi_{\mathrm{rad}}$ through Eq. (2.138); uniqueness of $\mathcal{S}_\omega$ does not imply uniqueness of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ when near-zone evanescent storage dominates radiation [2]. Outgoing selector idempotency Eq. (2.197) and resolvent selection Eq. (2.200) are consistent with exterior radiative uniqueness (Subsection 2.4.3):

$$\mathcal{O}_{\mathrm{out}} \circ \mathcal{L}_\omega^{-1} = \mathcal{L}_\omega^{-1}\big|_{\mathrm{phys}} \quad \text{is unique on } \mathcal{Y}_\omega, \tag{2.212}$$

because any two selections differing by a homogeneous outgoing mode would violate Eq. (2.209) [2]. Eq. (2.212) closes the excitation-to-flux sequence of Eq. (2.196) on exterior domains: once data are fixed, $\mathcal{O}_{\mathrm{out}}$, $\mathcal{S}_\omega$, and Π_{rad} act on a single field orbit in $\mathcal{F}_{\mathrm{SF}}$ [2]. On bounded domains with passive $\mathcal{B}_\Gamma$, uniqueness follows from Eq. (2.89) and the passive uniqueness argument (Subsection 1.1.3) after quotienting Eq. (2.86) [2]. At cavity eigenfrequencies,

$$\text{bounded PEC cavity: } \ker \mathcal{L}_\omega \neq \{\mathbf{0}\} \quad \text{vs.} \quad \text{exterior outgoing: } \ker \mathcal{L}_\omega|_{\mathcal{F}_{\mathrm{SF}}} = \{\mathbf{0}\}, \tag{2.213}$$

so Eq. (2.209) does not apply to closed resonators where Eq. (2.185) holds [6]. Exterior uniqueness is therefore a property of $\mathcal{F}_{\mathrm{SF}}$ with Sommerfeld completion, not of arbitrary $\mathcal{B}_\Gamma$ from Eq. (1.20) [2]. A finite scatterer embedded in free space may support Mie resonances in the near field without breaking exterior uniqueness: those modes are driven by $\mathbf{J}_{\mathrm{imp}}$ and do not appear as free homogeneous outgoing states [14]. Laboratory errors and non-applicability. Exterior radiative uniqueness (Subsection 2.4.3) requires linear passive media, fixed ω away from pathological embedded spectra, and exact outgoing radiation condition rather than approximate ABC alone [2]. First-order absorbing boundaries that do not enforce Eq. (2.189) at large R can admit spurious incoming content; uniqueness is then a property of the *augmented* truncated problem, not of the physical exterior [6]. Anisotropic, moving, or nonlinear media require modified flux functionals; time-varying $\mathcal{B}_\Gamma(t)$ breaks the harmonic reduction of Subsection 2.2.3 [2]. Uniqueness of $\mathcal{S}_\omega$ does not imply uniqueness of $\mathbf{J}_{\mathrm{imp}}$ in $\mathcal{R}_\omega$— that is Subsection 2.4.4 [6]. declared size and distance regime for transport remains Eq. (2.139), independent of field uniqueness [2]. As a worked example, truncating at finite R with a first-order ABC may return two numerically distinct fields that agree on Γ_R but differ in spurious incoming content reflected between Ω_s and the ABC [6]. Those differences are not counterexamples to exterior radiative uniqueness (Subsection 2.4.3) on Ω_{ext}; they signal that the truncated BVP lacks

Eq. (2.193). Silver–Müller surrogates Physically, eq. (2.194) improve the completion but still require $R \gg \lambda_0$ for uniqueness in practice [2].

Near-field richness therefore does not substitute for far-zone rank once evanescent content is filtered by Π_{prop} [3].

2.4.4 Implications for source reconstruction

Exterior radiative uniqueness (Subsection 2.4.3) fixes uniqueness of $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]$ on the outgoing branch. Subsection 2.4.4 records the inverse-source consequences: field uniqueness does not imply source identifiability. Uniqueness of $(\mathbf{E}, \mathbf{H})$ under outgoing conditions does not make $\mathcal{R}_\omega$ injective—distinct $\mathbf{J}_{\text{imp}}$ can share the same far pattern [17]. Source reconstruction is a conditioned inverse on scan surfaces and range contours, not a direct readout of coefficients [3]. $\mathcal{R}_\omega$ is generally non-injective, and finite observation channels compress rank further [2]. Observation null-space Eq. (2.80), rank composition Eq. (2.84), and non-radiating relation Eq. (2.7) from Subsection 2.1.3 are presupposed; complete structure of $\ker \mathcal{R}_\omega$ belongs to section 2.7.1 [6]. Measurement classification from section 1.5 applies here; angular-momentum observables from Chapter 1 are not re-opened [2]. The forward-inverse split below is the laboratory statement of field uniqueness versus source identifiability.

$$\underbrace{\mathcal{S}_\omega[\mathbf{J}^{(1)}_{\text{imp}}] = \mathcal{S}_\omega[\mathbf{J}^{(2)}_{\text{imp}}]}_{\text{same impressed data}} \implies \mathbf{J}^{(1)}_{\text{imp}} = \mathbf{J}^{(2)}_{\text{imp}} \quad \text{false in general,} \tag{2.214}$$

whereas exterior radiative uniqueness (Subsection 2.4.3) gives

$$\mathcal{S}_\omega[\mathbf{J}^{(1)}_{\text{imp}}] = \mathcal{S}_\omega[\mathbf{J}^{(2)}_{\text{imp}}] \implies (\mathbf{E}_1, \mathbf{H}_1) = (\mathbf{E}_2, \mathbf{H}_2) \quad \text{on } \mathcal{F}_{\text{SF}} \tag{2.215}$$

for the same boundary class and outgoing branch [2]. Equations (2.214)–(2.215) separate *state* uniqueness from *excitation* uniqueness: many admissible currents can produce the same radiative exterior field once reactive degrees of freedom are quotiented [6]. Inverse problems therefore target equivalence classes of $\mathbf{J}_{\text{imp}}$, not individual source distributions [2]. The outgoing radiation map from Subsection 2.0.1 is

$$\mathcal{R}_\omega : \mathcal{J}_\Omega \longrightarrow \mathcal{F}_{\text{rad}}, \qquad \mathcal{R}_\omega[\mathbf{J}_{\text{imp}}] = \Pi_{\text{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\text{imp}}], \tag{2.216}$$

restating Eq. (2.2) after outgoing radiation condition Eqs. (2.202)–(2.212) [2]. The source-side null space recorded in Eq. (2.7) is

$$\ker \mathcal{R}_\omega = \{\mathbf{J}_{\text{imp}} \in \mathcal{J}_\Omega : \Pi_{\text{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\text{imp}}] = \mathbf{0}\}\,, \tag{2.217}$$

generally infinite-dimensional on open domains [6]. section 2.7.1 develops the complete null-space analysis; Subsection 2.4.4 records the reconstruction consequences of Eq. (2.217) without constructing non-radiating bases [2]. Elements of $\ker \mathcal{R}_\omega$ are invisible to far-zone radiative observables yet may carry substantial near-zone reactive energy through $\mathcal{S}_\omega$ [6]. Exact null-space content produces zero radiate on Γ_R by algebraic obstruction; compactness-driven rank decay produces weak tails that fall below tolerance—related mechanisms, not identical failures. For non-radiating currents, energy remains in reactive exchange, stored oscillation, and evanescent overlap near the source region rather than on Γ_R.

Forward chain for reconstruction: $\mathcal{S}_\omega$ is single-valued on $\mathcal{F}_{\text{SF}}$ after exterior radiative uniqueness ; $\mathcal{R}_\omega$ collapses many sources to the same radiative output.

Observation composition and inverse problem. Laboratory readouts compose with $\mathcal{R}_\omega$ before any source inversion is attempted: pattern probes and near-field scanners observe a finite-rank image of impressed currents, not $\mathbf{J}_{\text{imp}}$ itself [3]. From Eq. (2.67) and finite-rank observation Eq. (2.79),

$$\mathcal{M}_\omega = \mathcal{O}_{\text{meas}} \circ \mathcal{O}_{\text{far}} \circ \mathcal{R}_\omega = \mathcal{O}_{\text{meas}} \circ \Lambda_{\text{rad}} : \mathcal{J}_\Omega \longrightarrow \mathbb{C}^N, \tag{2.218}$$

with $\Lambda_{\text{rad}} = \mathcal{O}_{\text{far}} \circ \mathcal{R}_\omega$ [2]. Given measurement vector $\mathbf{y} \in \mathbb{C}^N$, the inverse source problem seeks $\mathbf{J}_{\text{imp}} \in \mathcal{J}_\Omega$ such that

$$\mathcal{M}_\omega[\mathbf{J}_{\text{imp}}] = \mathbf{y}, \qquad \mathcal{D}_J[\mathbf{J}_{\text{imp}}] = 1, \tag{2.219}$$

subject to admissibility Eq. (2.0.1) [6]. Eq. (2.219) is ill-posed whenever $\ker \mathcal{M}_\omega \neq \{\mathbf{0}\}$, which occurs if either $\ker \mathcal{R}_\omega \neq \{\mathbf{0}\}$ or $\ker \mathcal{O}_{\text{meas}}$ is nontrivial on $\operatorname{ran} \mathcal{R}_\omega$ [2]. Rank cap Eq. (2.84) applies directly:

$$\operatorname{rank}(\mathcal{M}_\omega) \leq \min\{N,\, \dim \mathcal{R}(\mathcal{R}_\omega)\}, \tag{2.220}$$

so no measurement conditions with N channels resolves more than N independent radiative degrees of freedom [6]. The composed null space satisfies

$$\ker \mathcal{M}_\omega \supseteq \ker \mathcal{R}_\omega \;\cup\; \mathcal{R}_\omega^{-1}(\ker \mathcal{O}_{\text{meas}}), \tag{2.221}$$

with set inclusion on admissible sources [2]. Identifiable sources modulo null space form the quotient

$$\mathcal{J}_\Omega / \ker \mathcal{M}_\omega \;\cong\; \operatorname{ran} \mathcal{M}_\omega \subset \mathbb{C}^N, \tag{2.222}$$

so reconstruction recovers at best a finite-dimensional image, not a unique $\mathbf{J}_{\text{imp}}(\mathbf{r})$ [6]. Equivalent-current models that expand $\mathbf{J}_{\text{imp}}$ on auxiliary surfaces inherit the same non-uniqueness: distinct surface distributions related by $\ker \mathcal{R}_\omega$ produce identical far-zone patterns [2]. Boundary-element inversion therefore requires regularization or additional constraints beyond Maxwell consistency alone [6].

(2.80) pulled back to source space through Eq. (2.218) [2]. When Eq. (2.219) is ill-posed, stable inversion uses a penalized least-squares or Tikhonov template on admissible sources:

$$\mathbf{J}_{\text{imp}_{\text{reg}}} = \arg \min_{\mathbf{J}_{\text{imp}} \in \mathcal{J}_\Omega} \left\{ \|\mathcal{M}_\omega[\mathbf{J}_{\text{imp}}] - \mathbf{y}\|^2 + \alpha\, \mathcal{P}[\mathbf{J}_{\text{imp}}] \right\}, \tag{2.223}$$

with regularization parameter $\alpha > 0$ and penalty $\mathcal{P}$ enforcing smoothness, minimum norm, or sparsity on an equivalent surface [6]. Equation (2.223) does not remove $\ker \mathcal{M}_\omega$; it selects one representative from the affine coset $\mathcal{M}_\omega^{-1}(\mathbf{y})$ [2]. The choice of $\mathcal{P}$ is physics-adjacent modeling, not a Maxwell identity; different penalties yield different $\mathbf{J}_{\text{imp}_{\text{reg}}}$ with identical radiative predictions when $\delta\mathbf{J}_{\text{imp}} \in \ker \mathcal{R}_\omega$ [6]. Transport and angular reports attach to $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}]$, not to raw $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]$, per Eq. (2.27) [2]. declared size and distance regime Eq. (2.139) criteria whether $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}]$ lies in the propagating sector on which far-zone observables were designed:

$$\mathcal{D}_{J,\text{reg}} = 1 \implies \text{inverse claims target } \mathcal{R}_\omega[\mathbf{J}_{\text{imp}}],\ \text{not reactive near fields}, \tag{2.224}$$

with $\mathcal{D}_{J,\text{reg}}$ from Eq. (2.33) [2]. Attempting to reconstruct $\mathbf{J}_{\text{imp}}$ from near-zone reactive samples without declared size and distance regime conflates storage with radiation, the laboratory error flagged in section 1.2.5 [2]. Outgoing uniqueness from exterior radiative uniqueness (Subsection 2.4.3) stabilizes the *field* side of Eq. (2.219); it does not promote evanescent-dominated states to identifiable radiative sources [6]. Formal adjoint back-projection provides a reconstruction *template* without claiming uniqueness:

$$\widetilde{\mathbf{J}_{\text{imp}}} = \mathcal{M}_\omega^\dagger\, \mathbf{y}, \tag{2.225}$$

where $\mathcal{M}_\omega^\dagger$ is the Hilbert-space adjoint on declared domains [2]. Equation (2.225) is useful for sensitivity analysis and iterative inversion but generally satisfies $\mathcal{M}_\omega[\widetilde{\mathbf{J}_{\mathrm{imp}}}] \neq \mathbf{y}$ when $\ker \mathcal{M}_\omega \neq \{\mathbf{0}\}$ [6]. Consistent reconstruction requires projecting onto $\operatorname{ran} \mathcal{M}_\omega^\dagger$ or applying Eq. (2.223) [2]. Full integral representations for Eq. (2.225) belong to section 2.5.1; they are not repeated here [6]. Equivalent sources and Huygens surfaces. Equivalent-current inversion on a closed surface Γ_{eq} seeks $\mathbf{J}_{\mathrm{eq}}$ on Γ_{eq} such that exterior fields match measured data [2]. Uniqueness of exterior $\mathcal{S}_\omega$ from exterior radiative uniqueness (Subsection 2.4.3) fixes the *field* on $\mathcal{F}_{\mathrm{SF}}$ but not $\mathbf{J}_{\mathrm{eq}}$: any non-radiating tangential distribution on Γ_{eq} can be added without changing far-zone samples [6]. The operator form is

$$\mathbf{J}_{\mathrm{eq},1} - \mathbf{J}_{\mathrm{eq},2} \in \ker \mathcal{R}_\omega\big|_{\Gamma_{\mathrm{eq}}}, \tag{2.226}$$

with restriction to surface currents in the trace class of Subsection 2.1.1 [2]. Numerical equivalence in Eq. (2.187) inherits the same null-space structure at finite matrix rank [6]. Near-field versus far-field data. Near-field scanning samples $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on a surface enclosing Ω_s, not $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ alone [2]. Because $\mathcal{S}_\omega$ is injective on outgoing fields but many sources share the same $\mathcal{R}_\omega$ image, near-field data can *reduce* null-space dimension relative to far-zone-only inversion without eliminating $\ker \mathcal{R}_\omega$ in general [6]. Outgoing radiation condition still required: two incoming/outgoing superpositions with identical near traces on a finite surface need not agree in the far zone unless exterior radiative uniqueness (Subsection 2.4.3) applies to the full exterior problem [2]. Near-field inversion therefore couples Subsection 2.4.3 field uniqueness with Subsection 2.4.4 source non-uniqueness [6]. Eqs. (2.219)–(2.223) require linear Maxwell theory, fixed ω, admissible Eq. (2.0.1), and outgoing forward completion Eq. (2.212) [2]. Nonlinear, time-varying, or dispersive inversion breaks the harmonic reduction of Subsection 2.2.3. Single-frequency inversion cannot resolve broadband excitation without superposition across ω [2]. Loss improves conditioning of Eq. (2.219) by damping resonances but does not make $\mathcal{R}_\omega$ injective on all of $\mathcal{J}_\Omega$ [6]. Complete null-space bases and NR source examples belong to section 2.7 [2]. As a worked example, with N probe channels, Eq. (2.220) limits recoverable modal content to rank N regardless of mesh density [2]. Refining a boundary-element mesh without adding independent measurements does not resolve components in $\ker \mathcal{O}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}}$ [6]. This is the operator reading of Eq. (2.80), distinct from discretization error alone [2]. As a worked example, a source with $\mathcal{D}_{J,\mathrm{reg}} = 0$ may produce strong near-zone $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ while $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \approx \mathbf{0}$ [2]. Inverting near samples without Eq. (2.224) assigns spurious radiative labels to storage-dominated fields [6]. declared size and distance regime Eq. (2.139) must precede angular-transport claims on reconstructed sources [2]. section 2.4 closes with the excitation-to-flux sequence fixed for open regions: Sommerfeld asymptotics and outgoing class Eqs. (2.193)–(2.189) (Subsection 2.4.1), outgoing selector and resolvent branch Eqs. (2.196)–(2.201) (Subsection 2.4.2), exterior radiative uniqueness Eq. (2.209) and resolvent injectivity Eq. (2.210) (Subsection 2.4.3), and inverse reconstruction limits Eqs. (2.219)–(2.217) (Subsection 2.4.4) [2]. Green kernels from section 2.3 are now tied to a single physical branch; $\mathcal{S}_\omega$ is single-valued on $\mathcal{F}_{\mathrm{SF}}$, while $\mathcal{R}_\omega$ and composed measurement readouts remain rank-deficient on $\mathcal{J}_\Omega$ [6]. section 2.5 evaluates radiation integrals, flux meters, and near/intermediate/far partitions on the qualified propagating subspace; section 2.7 completes non-radiating null-space structure [2]. declared size and distance regime Eq. (2.139) and transport criterion Eq. (2.27) remain mandatory before angular-transport claims on operator outputs [2]. Outgoing-wave selection fixes the exterior resolvent branch assumed throughout radiation integrals; realizability limits—which accessible modes finite geometry can excite—are established in 4 (*Source $\mathcal{S}_{\mathrm{real}}$ and Spectral Truncation*).

2.5 Radiation Integrals and Source-to-Field Representations

Uniqueness and radiation conditions in section 2.4 fix the exterior problem. section 2.5 writes the source integrals that range engineers discretize on wire and surface meshes [2]. The same integrals appear in compact-range far-field transforms [3].

section 2.4 fixed the outgoing branch under which exterior solutions are unique. An engineer on a range nonetheless needs evaluated fields and pattern amplitudes at declared observation points—not proof of existence alone [5]. Before any far-zone asymptotic is applied, the field surrounding a compact source occupies three spatial bands: a reactive near zone where $1/r^3$ and $1/r^2$ dyadic terms dominate what probes measure through $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$; an intermediate transition where storage and radiation compete; and a radiative far zone where transverse $e^{-\mathrm{j}k_0 r}/r$ waves support pattern and gain definitions through $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\,\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2]. Distance labels and regime indices are fixed in Subsection 2.5.2 before asymptotic reductions; the integral representations below are the constructive route to $\mathcal{S}_\omega$ in all three bands [2]. A gain trace on Γ_R is a pattern meter reading on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ after outgoing selection; a port $|S_{11}|$ meter on a horn feed reads coupling through $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ and may include reactive storage unrelated to far-zone radiation [6]. A method-of-moments solution on a horn mouth illustrates the distinction: port coupling integrals supply $|S_{11}|$, while a radiation integral on Γ_R supplies far-zone gain; both descend from the same outgoing Green convolution but meter different observables [6]. The excitation-to-flux sequence $\mathbf{J}_{\mathrm{imp}} \mapsto \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \mapsto \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is single-valued on $\mathcal{F}_{\mathrm{SF}}$ at the field level yet rank-deficient at the source level through Eq. (2.217) [2]. The radiation integral Eq. (2.4) is developed here to evaluable outgoing integrals and ties range meters to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\,\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ rather than to the full pair returned by Green convolution Eq. (2.148) [2]. Poynting flux and transport indices from 1 apply here; dyadic kernels and Sommerfeld completion remain in sections 2.3–2.4. Let $\Omega_s = \operatorname{supp}\mathbf{J}_{\mathrm{imp}}$ be compact with characteristic scale a, vacuum wavenumber $k_0 = \omega\sqrt{\varepsilon_0\mu_0}$, and wavelength $\lambda_0 = 2\pi/k_0$. section 2.5 organizes evaluation in four stages:

$$\mathbf{J}_{\mathrm{imp}} \xrightarrow{\mathcal{S}_\omega} (\mathbf{E},\mathbf{H}) \in \mathcal{F}_{\mathrm{SF}} \xrightarrow{\Pi_{\mathrm{rad}}} \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \xrightarrow{\mathcal{F}_{\mathrm{flux}}} \text{power, pattern, norm reports,} \tag{2.227}$$

where $\mathcal{F}_{\mathrm{flux}}$ denotes the flux and operator-norm functionals developed below by reference to 1 flux identities [2]. Eq. (2.227) is the section-level restatement of Eq. (2.10): stage (iii) radiation must precede stage (iv) metering [6]. Applying $\mathcal{F}_{\mathrm{flux}}$ to $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ without Π_{rad} imports reactive circulation into quantities designed for outgoing transverse radiation (section 1.2.5) [2]. The outgoing integral realization of $\mathcal{R}_\omega$ for $\mathbf{r} \notin \Omega_s$ is

$$\mathbf{E}_{\mathrm{rad}}(\mathbf{r}) = -\mathrm{j}\omega\mu_0 \int_{\Omega_s} \overline{\mathbf{G}}^{(\mathrm{out})}_{E,\omega}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r', \qquad \mathbf{H}_{\mathrm{rad}} = -\frac{1}{\mathrm{j}\omega\mu_0}\,\nabla \times \mathbf{E}_{\mathrm{rad}}, \tag{2.228}$$

In Eq. (2.228), $\mathbf{r}$ is the observation point, $\mathbf{r}'$ ranges over the source volume Ω_s, $\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')$ is the impressed electric current density at the source point, μ_0 is the vacuum permeability, ω is the angular frequency, and $\overline{\mathbf{G}}^{(\mathrm{out})}_{E,\omega}$ is the outgoing electric dyadic from Eq. (2.199); the magnetic field $\mathbf{H}_{\mathrm{rad}}$ follows from $\mathbf{E}_{\mathrm{rad}}$ by the source-free Maxwell curl relation in the observation variable $\mathbf{r}$ [2]. Equation (2.228) upgrades Eq. (2.4) after sections 2.3–2.4; the integral subsection below supplies magnetic couplings, equivalent surfaces, and routes that reduce to the same $\mathcal{R}_\omega$ image [6]. Near coincidence $\mathbf{r} \to \Omega_s$ requires regularization from Subsection 2.3.3; evaluation uses Eq. (2.228) only where the observation point lies outside Ω_s or on declared equivalent surfaces with trace matching [2]. declared size and distance regime Eq. (2.139) must be satisfied before flux meters attach to any shell integral [2]. As a worked example, for $\mathbf{J}_{\mathrm{imp}}$ supported in a ball of radius $a \ll \lambda_0$, $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ includes dominant reactive $1/r^2$, $1/r^3$ terms on spheres with $r \sim a$, while $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ carries leading

$1/r$ transverse radiation with $k_0 r$ phase for $r \gg a$ [14]. Integrating Poynting flux on S_R with $R \sim a$ conflates storage with radiation; only when $k_0 R \gg 1$ and $R \gg a$ does shell power approach the R-independent value predicted by far-zone reduction [6]. As a worked example, near-field probes sample $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on a surface enclosing Ω_s [2]. Transforming those samples to a far-zone pattern implements Eq. (2.228) only when outgoing radiation condition and equivalent-source uniqueness on the exterior side are satisfied—that is, when exterior radiative uniqueness (Subsection 2.4.3) holds [6]. Distinct impressed currents in $\ker \mathcal{R}_\omega$ produce identical patterns while differing in reactive near fields; near scans alone do not resolve that null space without Physically, additional constraints (Subsection 2.4.4) [2].

The reference case is therefore radiate on Γ_R: reactive storage cannot substitute for propagating rank [5].

2.5.1 Source-to-field integral representations

Retarded source integrals are what method-of-moments codes discretize on antennas [6]. Far-zone extraction is a limit on the same integral, valid only after outgoing selection and Fraunhofer distance criterion [14, 3]. The representation begins from retarded potentials driven by $\mathbf{J}_{\mathrm{imp}}$ on compact Ω_s, not from a plane-wave expansion [1]. The quantity being organized is the outgoing field pair $(\mathbf{E}, \mathbf{H})$; angular labels attach only after Π_{rad} [16]. An alternative route through static potentials would omit radiation damping and misstate reactive storage [14]. On a horn range, this step is what a MoM mesh implements before any pattern plot [6]. Before any volume quadrature is written, the range engineer already knows which observable the measurement will report [6]. A calibrated pattern probe on Γ_R integrates tangential $\mathbf{E}$ or $\mathbf{H}$ against a known polarization and yields a voltage proportional to a directional amplitude—a discrete sample of the far-zone pattern carried by $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, not of the full pair $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2]. A shell wattmeter closes the same budget through $\mathcal{P}_{\mathrm{rad}}$ on Γ_R after Π_{rad} [2]. Near-field scanners on a hull enclosing Ω_s record $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ and require a transform before pattern meters apply [3]. The volume integrals below are the constructive route to those meter readings: they assemble $(\mathbf{E}, \mathbf{H})$ from impressed $\mathbf{J}_{\mathrm{imp}}$, after which radiation selection maps the result to the pattern quantities the range records [2]. Radiation integrals assemble far-zone fields as superpositions of current moments weighted by outgoing phase [2]. Four laboratory settings motivate the representations that follow. On an antenna range, probes on a large sphere Γ_R sample the field produced by a feed in the compact range; the measured voltage is a discretization of a surface integral over Γ_R, not a direct reading of impressed $\mathbf{J}_{\mathrm{imp}}$ [6]. Horn illumination fixes $\mathbf{J}_{\mathrm{imp}}$ on the mouth through a waveguide mode or aperture integral; the pattern on the range wall appears only after those currents convolve against the outgoing Green function [15]. Aperture radiation replaces the volume support by equivalent currents on Γ_{eq} when the interior is inaccessible [2]. Near-field scanners reconstruct far-zone quantities from dense samples on a hull enclosing Ω_s—a transform that succeeds only when outgoing uniqueness and radiation selection are respected [3]. The representation begins from retarded potentials driven by $\mathbf{J}_{\mathrm{imp}}$ on compact Ω_s, organizes the outgoing pair $(\mathbf{E}, \mathbf{H})$ before angular labels attach, and coincides with what method-of-moments codes discretize before far-zone extraction [1]. Static-potential routes omit radiation damping and misstate reactive storage; they are not used here [14]. Outgoing kernels, regularization, and branch selection are presupposed from sections 2.3–2.4; distance-regime asymptotics and flux meters follow in the subsections below. Assumptions. Fix $\omega > 0$, compact $\operatorname{supp} \mathbf{J}_{\mathrm{imp}} = \Omega_s \Subset \Omega$, admissible data Eq. (1.23), charge continuity Eq. (1.8), and outgoing radiation condition Eqs. (2.193)–(2.196) on exterior problems [2]. Observation points $\mathbf{r}$ lie outside Ω_s or on equivalent surfaces with trace matching declared below; coincidence limits use Subsection 2.3.3 [6]. Kernels $\overline{\mathbf{G}}^{(\mathrm{out})}_{E,\omega}$, $\overline{\mathbf{G}}^{(\mathrm{out})}_{H,\omega}$, and

scalar $G_\omega^{(\mathrm{out})}$ denote outgoing branches fixed by Eq. (2.199) [2]. The electric volume integral is the constructive form of the dyadic convolution route: each source element $\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\,d^3r'$ contributes a dipole-like response that propagates outward with retarded phase, and the total field is their coherent sum [2]. On a MoM mesh, patch currents discretize the same sum; on Γ_R, pattern probes sample the far-zone limit after $\mathcal{R}_\omega$ has removed reactive storage that the volume integral still carries at finite r [6]. In the symbols below, $\mathbf{r}$ is the observation coordinate, $\mathbf{r}'$ the source coordinate integrated over Ω_s, $\mathbf{E}_{\mathrm{hom}}$ any exterior illumination not generated by $\mathbf{J}_{\mathrm{imp}}$, and the prefactor $-\mathrm{j}\omega\mu_0$ ties the integral to the locked phasor convention $\exp(+\mathrm{j}\omega t)$ [2]. The primary volume representation is

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0 \int_{\Omega_s} \overline{\mathbf{G}}_{E,\omega}^{(\mathrm{out})}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\,d^3r' \;+\; \mathbf{E}_{\mathrm{hom}}(\mathbf{r}), \tag{2.229}$$

$$\mathbf{H}(\mathbf{r}) = \frac{1}{\mathrm{j}\omega\mu_0}\,\nabla \times \mathbf{E}(\mathbf{r}), \tag{2.230}$$

Eqs. (2.229)–(2.230) package the dyadic route as an evaluable radiation map: the electric field is a volume convolution against the *outgoing* dyad, while the magnetic field is a curl derived from that electric field in source-free observation regions [2]. For a square loop of side $2a$ with $k_0 a = 1.2$ observed at $\mathbf{r}$ with $k_0|\mathbf{r}| = 6$, the integral typically requires 16–64 wavelength-scale subdivisions of Ω_s for 2% amplitude accuracy when raw Eq. (2.151) is used patchwise; self-term corrections from Subsection 2.3.3 dominate error when $|\mathbf{r}| \to a$ [6]. The homogeneous term $\mathbf{E}_{\mathrm{hom}}$ carries exterior illumination and must vanish for pure radiation problems [2]. Equations (2.229)–(2.230) specialize Eqs. (2.148)–(2.152) with outgoing dyads for $\mathbf{r} \notin \Omega_s$ on simply connected observation regions; they coincide with Eq. (2.228) when $\mathbf{E}_{\mathrm{hom}} = \mathbf{0}$ [2]. The prefactor $-\mathrm{j}\omega\mu_0$ matches Eq. (2.115) and the locked phasor axis [2]. What propagates to a distant probe is the coherent sum of outgoing dyadic responses $\overline{\mathbf{G}}_{E,\omega}^{(\mathrm{out})}(\mathbf{r},\mathbf{r}')$ from each source element; what remains reactive at finite r is the non-transverse and inverse-power content retained in $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ until Π_{rad} and declared size and distance regime isolate radiation [2]. A near-field scanner records the full pair $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$; a pattern meter on Γ_R with $k_0 R \gg 1$ approximates $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [3]. When magnetic impressed sources $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{mag})}$ are primary, duality maps $\varepsilon_0\mathbf{E} \leftrightarrow \mu_0\mathbf{H}$ and $\mathbf{J}_{\mathrm{imp}} \leftrightarrow \mathbf{J}_{\mathrm{imp}}^{(\mathrm{mag})}$ yields

$$\mathbf{H}(\mathbf{r}) = \mathrm{j}\omega\varepsilon_0 \int_{\Omega_s} \overline{\mathbf{G}}_{H,\omega}^{(\mathrm{out})}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}^{(\mathrm{mag})}(\mathbf{r}')\,d^3r' \;+\; \mathbf{H}_{\mathrm{hom}}(\mathbf{r}), \tag{2.231}$$

$$\mathbf{E}(\mathbf{r}) = -\frac{1}{\mathrm{j}\omega\varepsilon_0}\,\nabla \times \mathbf{H}(\mathbf{r}), \tag{2.232}$$

with $\overline{\mathbf{G}}_{H,\omega}^{(\mathrm{out})}$ the magnetic dyad defined duality-wise from Subsection 2.3.1 without re-tabulating its closed form [2]. Equations (2.231)–(2.232) complete the dual route to Eqs. (2.229)–(2.230) for aperture magnetic currents and dual-equivalent models [2]. When both electric and magnetic impressed sources are present, superposition assembles the full Maxwell pair from the dual volume routes without re-opening gauge or static-potential paths [2]. Superposition gives the coupled volume representation

$$\begin{aligned}
\mathbf{E}(\mathbf{r}) &= -\mathrm{j}\omega\mu_0 \int_{\Omega_s} \overline{\mathbf{G}}_{E,\omega}^{(\mathrm{out})}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\,d^3r' - \frac{1}{\mathrm{j}\omega\varepsilon_0}\,\nabla \times \mathbf{H}^{(\mathrm{mag})}(\mathbf{r}) + \mathbf{E}_{\mathrm{hom}}(\mathbf{r}), \\
\mathbf{H}(\mathbf{r}) &= \mathrm{j}\omega\varepsilon_0 \int_{\Omega_s} \overline{\mathbf{G}}_{H,\omega}^{(\mathrm{out})}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}^{(\mathrm{mag})}(\mathbf{r}')\,d^3r' + \frac{1}{\mathrm{j}\omega\mu_0}\,\nabla \times \mathbf{E}^{(\mathrm{elec})}(\mathbf{r}) + \mathbf{H}_{\mathrm{hom}}(\mathbf{r}),
\end{aligned} \tag{2.233}$$

where $\mathbf{E}^{(\mathrm{elec})}, \mathbf{H}^{(\mathrm{mag})}$ denote the single-source contributions from Eqs. (2.229) and (2.231) [2]. Eq. (2.233) is the first-order Maxwell integral system on outgoing kernels; it reduces to either

branch when one impressed source vanishes [2]. Linearity Eq. (2.5) follows from Eq. (2.233) on fixed $\mathcal{B}_\Gamma$ [6]. The combined integral returns the full admissible Maxwell pair $\mathcal{S}_\omega[\mathfrak{D}]$: electric and magnetic impressed sources superpose at observation $\mathbf{r}$ with each contribution evaluated at source $\mathbf{r}'$ through the outgoing dyads G and $\overline{\mathbf{G}}_E$, $\overline{\mathbf{G}}_H$. Pattern, gain, and shell wattmeters do not read this sum directly; they act on $\mathcal{R}_\omega[\mathfrak{D}] = \Pi_{\mathrm{rad}}\,\mathcal{S}_\omega[\mathfrak{D}]$ after outgoing selection removes reactive and non-propagating fractions that the volume integral still carries at finite range [2]. Figure 2.7 maps

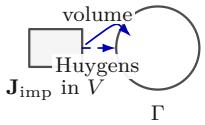


Figure 2.7: Integral routes: electric volume, magnetic volume, and Huygens surface representations converge to the same $\mathcal{S}_\omega$ image on the outgoing branch when trace matching holds.

the schematic structure: Integral routes: electric volume, magnetic volume, and Huygens surface representations converge to the same imag. . . .

When $\nabla \cdot \mathbf{J}_{\mathrm{imp}} = -\mathrm{j}\omega\rho$ with nonzero ρ, the complete electric field includes a scalar-potential gradient driven by ρ:

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0 \int_{\Omega_s} \overline{\mathbf{G}}^{(\mathrm{out})}_{E,\omega}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r' - \nabla \int_{\Omega_s} G^{(\mathrm{out})}_{\omega}(\mathbf{r},\mathbf{r}')\, \frac{\rho(\mathbf{r}')}{\varepsilon_0}\, d^3r' \;+\; \mathbf{E}_{\mathrm{hom}}(\mathbf{r}), \tag{2.234}$$

with $G^{(\mathrm{out})}_\omega$ the outgoing scalar kernel from Eq. (2.144) [2]. Equation (2.234) enforces Gauss coupling without leaving $\mathcal{J}_\Omega$; when Eq. (1.8) holds and ρ is determined by $\mathbf{J}_{\mathrm{imp}}$, the gradient term is fixed by the same impressed data [2]. The charge-augmented integral still evaluates $\mathcal{S}_\omega[\mathfrak{D}]$: the scalar kernel $G^{(\mathrm{out})}_\omega(\mathbf{r},\mathbf{r}')$ propagates charge density from $\mathbf{r}'$ to scalar potential at $\mathbf{r}$, while the dyadic term propagates $\mathbf{J}_{\mathrm{imp}}$; shell meters on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ remain insensitive to how reactive near storage partitions between the two contributions at finite r [6]. Lorenz-gauge vector-potential reduction Eq. (2.146) collapses to Eq. (2.234) on uniform exterior domains; gauge freedom is not re-opened [6]. When the interior of a source region is inaccessible, equivalent currents on a enclosing surface reproduce the exterior field without resolving volume $\mathbf{J}_{\mathrm{imp}}$; the surface integrals below are the operator form of that laboratory substitution [2]. Let Γ_{eq} be a closed surface enclosing Ω_s, with outward normal $\hat{\mathbf{n}}$ and equivalent surface densities $\mathbf{J}_{s,\mathrm{eq}}$, $\mathbf{K}_{s,\mathrm{eq}}$ on Γ_{eq} in the trace class of Subsection 2.1.1. The exterior Huygens representation is

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0 \int_{\Gamma_{\mathrm{eq}}} \overline{\mathbf{G}}^{(\mathrm{out})}_{E,\omega}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{s,\mathrm{eq}}(\mathbf{r}')\, dS' + \int_{\Gamma_{\mathrm{eq}}} \nabla' G^{(\mathrm{out})}_{\omega}(\mathbf{r},\mathbf{r}') \times \mathbf{K}_{s,\mathrm{eq}}(\mathbf{r}')\, dS', \tag{2.235}$$

$$\mathbf{H}(\mathbf{r}) = \mathrm{j}\omega\varepsilon_0 \int_{\Gamma_{\mathrm{eq}}} \overline{\mathbf{G}}^{(\mathrm{out})}_{H,\omega}(\mathbf{r},\mathbf{r}') \cdot \mathbf{K}_{s,\mathrm{eq}}(\mathbf{r}')\, dS' - \int_{\Gamma_{\mathrm{eq}}} \nabla' G^{(\mathrm{out})}_{\omega}(\mathbf{r},\mathbf{r}') \times \mathbf{J}_{s,\mathrm{eq}}(\mathbf{r}')\, dS', \tag{2.236}$$

for $\mathbf{r}$ outside the closed surface bounded by Γ_{eq} [2]. Equations (2.235)–(2.236) are the surface form of the same outgoing propagation operator; ∇' acts on the source coordinate $\mathbf{r}'$ [2]. Tangential field matching

$$\hat{\mathbf{n}} \times \mathbf{E}\big|_{\Gamma^+_{\mathrm{eq}}} = \hat{\mathbf{n}} \times \mathbf{E}\big|_{\Gamma^-_{\mathrm{eq}}}, \qquad \hat{\mathbf{n}} \times \mathbf{H}\big|_{\Gamma^+_{\mathrm{eq}}} = \hat{\mathbf{n}} \times \mathbf{H}\big|_{\Gamma^-_{\mathrm{eq}}} \tag{2.237}$$

fixes $\mathbf{J}_{s,\mathrm{eq}} = \hat{\mathbf{n}} \times (\mathbf{H}^- - \mathbf{H}^+)$, $\mathbf{K}_{s,\mathrm{eq}} = \hat{\mathbf{n}} \times (\mathbf{E}^- - \mathbf{E}^+)$ in the stratified sense of Eq. (1.19) [2]. Equivalent-source non-uniqueness Eq. (2.226) applies: distinct $(\mathbf{J}_{s,\mathrm{eq}}, \mathbf{K}_{s,\mathrm{eq}})$ pairs related by $\ker \mathcal{R}_\omega$ yield identical far-zone radiation [6]. The Huygens surface integrals reproduce exterior fields from equivalent currents on Γ_{eq}: observation $\mathbf{r}$ lies outside the closed surface, source $\mathbf{r}'$ ranges over

Γ_{eq}, and the same outgoing kernels $G_\omega^{(\mathrm{out})}$ and $\overline{\mathbf{G}}_{E,\omega}^{(\mathrm{out})}$ propagate tangential data outward. What is measured on a distant range is still $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ after Π_{rad}; equivalent currents that differ only by $\ker \mathcal{R}_\omega$ leave that measurement unchanged while altering reactive near fields through $\mathcal{S}_\omega$ [2]. Let $\mathfrak{D} = (\mathbf{J}_{\mathrm{imp}}, \mathbf{J}_{\mathrm{imp}}^{(\mathrm{mag})}, \rho, \mathcal{B}_\Gamma)$ be admissible with outgoing radiation condition on exterior problems. Then the physical solution $\mathcal{S}_\omega[\mathfrak{D}] = (\mathbf{E}, \mathbf{H})$ is given by Eqs. (2.233)–(2.234) with outgoing kernels, or by Eqs. (2.235)–(2.237) when equivalent surface data replace volume sources [2]. The representation agrees with Eq. (2.201) and Green–BVP equivalence Eq. (2.183) on the outgoing branch fixed by exterior radiative uniqueness (Subsection 2.4.3). Subsection 2.3.1 defined dyadic kernels satisfying Eq. (2.147); Subsection 2.3.3 justified convolution on $\mathcal{J}_\Omega$; section 2.4 selected the outgoing branch. Substituting $\overline{\mathbf{G}}_{E,\omega}^{(\mathrm{out})}$ into Eq. (2.148) yields Eq. (2.229). Magnetic and coupled forms follow from duality and superposition [2]. Huygens formulas follow by replacing volume sources with surface distributions that reproduce the same tangential traces [6]. Uniqueness on $\mathcal{F}_{\mathrm{SF}}$ when exterior radiative uniqueness (Subsection 2.4.3) holds identifies the integral representation with $\mathcal{S}_\omega$ [2]. The volume and surface integrals above return the full admissible pair $\mathcal{S}_\omega[\mathfrak{D}]$; pattern meters and gain reports act only after outgoing selection through $\mathcal{R}_\omega$ and Π_{rad} [2]. radiation selection follows evaluation:

$$\mathcal{R}_\omega[\mathfrak{D}] = \Pi_{\mathrm{rad}} \Big(\mathbf{E}[\mathfrak{D}],\, \mathbf{H}[\mathfrak{D}] \Big), \tag{2.238}$$

with Π_{rad} from Eq. (2.202) [2]. Pattern cuts, gain reports, and shell power meters act on Eq. (2.238), not on the raw volume integral alone [6]. On the outgoing branch,

$$\mathcal{S}_\omega[\mathfrak{D}] = \mathcal{O}_{\mathrm{out}}\, \mathcal{L}_\omega^{-1} \begin{bmatrix} \mathbf{s}_\omega(\mathfrak{D}) \\ \mathbf{g}_\Gamma \end{bmatrix} = \mathcal{O}_{\mathrm{out}} \Big(\mathcal{I}_{\mathrm{vol}}[\mathfrak{D}] \Big), \tag{2.239}$$

with $\mathcal{I}_{\mathrm{vol}}[\mathfrak{D}]$ the combined volume integrals Eqs. (2.233)–(2.234), connecting integral and operator routes through Eqs. (2.123)–(2.200) [2]. Eq. (2.239) is the constructive content of Eq. (2.183) for evaluation purposes: a method-of-moments matrix implements the resolvent route while patch quadrature implements the integral route, and both must reference the same outgoing branch [6]. When $\Omega = \cup_j \Omega_j$ with interface traces from Subsection 2.3.5, Eqs. (2.233)–(2.234) hold patchwise with local k_j and outgoing exterior completion on the infinite exterior region [2]. Impressed $\mathbf{J}_{\mathrm{imp}}$ may vanish in passive exterior regions; equivalent currents on material boundaries enter through Eqs. (2.235)–(2.237) with Γ_{eq} on scatterer surfaces [6]. Interior cavity problems replace outgoing G with standing-mode Green functions and require quotienting Eq. (2.86); those representations are excluded from the outgoing integral family developed here [2]. Eqs. (2.229)–(2.234) require linear media, fixed ω, admissible $\mathfrak{D}$, outgoing radiation condition on exterior problems, and observation points outside singular coincidence unless regularization from Subsection 2.3.3 is applied [2]. Incoming kernels produce different integrals violating Eq. (2.176) [14]. Anisotropic, moving, or nonlinear media require modified kernels beyond the locked patchwise-scalar class [2]. Near-zone reactive dominance does not invalidate the integral; it signals that $\mathcal{R}_\omega[\mathfrak{D}]$ may differ sharply from $\mathcal{S}_\omega[\mathfrak{D}]$ until Eq. (2.139) is satisfied [6]. One operational distinction closes the display below: volume integrals construct the full Maxwell pair $\mathcal{S}_\omega[\mathfrak{D}]$, whereas pattern meters and watt integrators on Γ_R read only after outgoing selection has isolated $\mathcal{R}_\omega[\mathfrak{D}]$. A method-of-moments matrix must not be read as a radiation report; Maxwell theory shows that the matrix implements whichever Green branch and boundary completion were declared, and reactive near-zone content may dominate local probes without contributing to outgoing radiation [2]. A measurement on a closed scan surface therefore records the complete field state unless a transform and declared size and distance regime explicitly project onto the radiative sector. Synthesis, diagnosis, and inversion measurements must name the same outgoing branch before any two routes—resolvent inversion and direct quadrature—are

compared [6]. Subsection 2.5.2 next fixes distance labels that tell the engineer when far-zone asymptotics may replace the full integral without misreading reactive storage as radiated power.

2.5.2 Near, intermediate, and far regimes

Multipole expansions order far-zone fields by angular complexity: the $\ell = 1$ dipole term dominates electrically small feeds, whereas reflector measurements require many ℓ before sidelobe floors stabilize [2]. The far-field approximation replaces distance-dependent amplitudes by $e^{-\mathrm{j}k_0 r}/r$ and angular factors; range calibrations must sit in the Fraunhofer zone for that step to be valid [14]. A pattern meter on Γ_R with $k_0 R \ll 1$ still reads $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$; only when $k_0 R \gg 1$ and $R \gg a$ does the same probe approximate $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ through the far-zone reduction developed below [6].

The same source produces all three regions simultaneously; the distinction arises from how different terms of the radiation integral scale with distance [2]. Near the feed, reactive $1/r^3$ and $1/r^2$ terms dominate shell integrals; at intermediate distances those terms compete with the leading $e^{-\mathrm{j}k_0 r}/r$ radiation; only on sufficiently large spheres does the transverse $1/r$ term alone support pattern and gain definitions [6]. Let $\Omega_s = \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$ be compact with characteristic scale a. For observation radius $r = |\mathbf{r}|$ from a reference point in Ω_s, define

$$\begin{aligned} \text{near: } & k_0 r \lesssim 1, \\ \text{intermediate: } & 1 \lesssim k_0 r \lesssim (k_0 a)^{-1}, \\ \text{far: } & k_0 r \gg k_0 a, \quad r \gg a, \end{aligned} \tag{2.240}$$

In Eq. (2.240), $r = |\mathbf{r}|$ is the observation radius (m) from a reference point in $\Omega_s = \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$; a is the source characteristic scale (m); $k_0 = \omega\sqrt{\varepsilon_0 \mu_0}$ is the vacuum wavenumber (rad m^{-1}); and the inequalities label reactive near, transition, and radiative far bands before the constants $C_{\mathrm{n}}, C_{\mathrm{i}}, C_{\mathrm{f}}$ are fixed in Eq. (2.241). with implicit constants fixed below [2]. Eqs. (2.240) are scale labels, not sharp cutoffs: reactive and radiative terms coexist in the near zone, and intermediate zones require explicit asymptotic bookkeeping before source integrals reduce to transverse $1/r$ radiation [6]. Pattern and gain definitions used in antenna theory attach only after the far-zone reduction in Eq. (2.191) [6]. Field-magnitude scalings, far-zone reduction of Eqs. (2.229)–(2.233), Fresnel–Fraunhofer transition scales, and power-flux convergence on large spheres are developed below [2]. Sommerfeld asymptotics Eqs. (2.192) and (2.191) are presupposed; regime indices Eqs. (2.30)–(2.31) are invoked by reference [2]. Let $\Omega_s = \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$ be compact with characteristic radius $a = \max_{\mathbf{r},\mathbf{r}' \in \Omega_s} |\mathbf{r} - \mathbf{r}'|/2$, vacuum wavenumber $k_0 = \omega\sqrt{\varepsilon_0 \mu_0}$, wavelength $\lambda_0 = 2\pi/k_0$, and electrical size $k_0 a$ [2]. Observation points $\mathbf{r}$ satisfy $|\mathbf{r}| = r \notin \Omega_s$; reference origin is chosen in Ω_s. Outgoing kernels from the source integrals above and dyadic structure from Eq. (2.151) are presupposed without re-tabulation [2]. Constants $C_1, C_2, \ldots$ below depend on k_0, a and source geometry but not on r in the asymptotic limits stated [6]. The locked regime partition is

$$\begin{aligned} &\textit{near (reactive) zone: } k_0 r \le C_{\mathrm{n}}, \\ &\textit{intermediate (transition) zone: } C_{\mathrm{n}} < k_0 r \le C_{\mathrm{i}}(k_0 a), \\ &\textit{far (radiative) zone: } k_0 r > C_{\mathrm{i}}(k_0 a), \quad r > C_{\mathrm{f}} a^2/\lambda_0, \end{aligned} \tag{2.241}$$

with declared positive constants $C_{\mathrm{n}} = \mathcal{O}(1)$, $C_{\mathrm{i}} = \mathcal{O}(1)$ when $k_0 a = \mathcal{O}(1)$, and Fraunhofer scale $r_{\mathrm{F}} = 2a^2/\lambda_0$ entering $C_{\mathrm{f}} = \mathcal{O}(1)$ [2]. Equations (2.241) upgrade Eq. (2.240): they are diagnostic bands, not disjoint sharp cutoffs [2]. For $k_0 a \ll 1$ (electrically small sources), C_{i} may be taken near unity and the intermediate zone is narrow; for extended apertures with $k_0 a \gg 1$, the Fresnel–Fraunhofer condition $r \gtrsim r_{\mathrm{F}}$ controls pattern formation [6]. Near-zone field scalings. Evaluating

Eq. (2.229) with Eq. (2.151) for $k_0 r \le C_{\mathrm{n}}$ yields

$$|\mathbf{E}(\mathbf{r})| = \mathcal{O}\big(\mu_0\omega\|\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{J}}\, k_0^{-2} r^{-3}\big) \;+\; \mathcal{O}\big(\mu_0\omega\|\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{J}}\, k_0^{-1} r^{-2}\big), \tag{2.242}$$

with $\|\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{J}}$ a source amplitude scale on Ω_s [2]. Eq. (2.242) reflects the R^{-3} and R^{-2} terms in Eq. (2.151); the leading r^{-3} contribution is reactive (stored) in the sense of section 1.2.5 [2]. Magnetic field follows from Eq. (2.230) with the same near-zone polynomial-in-$k_0 r$ ordering [6]. On shells with Eq. (2.30) $\chi(r) \gg 1$, Poynting flux through Eq. (2.6) is not a credible radiation meter [2]. When $C_{\mathrm{n}} < k_0 r \le C_{\mathrm{i}}(k_0 a)$, reactive and radiative terms in Eq. (2.151) are comparable:

$$|\mathbf{E}_{\mathrm{reactive}}(\mathbf{r})| = \mathcal{O}\big(|\mathbf{E}_{\mathrm{rad}}(\mathbf{r})|\, k_0 r\big), \tag{2.243}$$

with $\mathbf{E}_{\mathrm{rad}}$ the leading transverse $e^{-\mathrm{j}k_0 r}/r$ component and $\mathbf{E}_{\mathrm{reactive}}$ the remainder [2]. Equation (2.243) diagnoses regime mixing: storage and radiation coexist in the full field pair until radiation is isolated by $\mathcal{R}_\omega$ [2]. For extended sources, the Fresnel parameter

$$F = \frac{\pi a^2}{\lambda_0 r} = \frac{k_0 a^2}{2r} \tag{2.244}$$

separates Fresnel ($F \gtrsim 1$) from Fraunhofer ($F \ll 1$) pattern regions on observation shells [14]. The parameter F marks the transition between those pattern regimes; complete aperture diffraction theory is not re-developed [2].

section 2.4 [2]. The far-zone reduction of integrals proceeds as follows: at distances large compared with source extent and wavelength, a pattern probe on Γ_R samples the transverse $e^{-\mathrm{j}k_0 r}/r$ wave whose angular shape is fixed by a Fourier transform of $\mathbf{J}_{\mathrm{imp}}$ over Ω_s [14]. The volume integral at finite r still evaluates $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$: reactive terms propagate mathematically but do not radiation watt-bearing power; only after the far-zone limit does the meter reading coincide with $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ through Eqs. (2.245)–(2.246) [2]. When $k_0 r \gg 1$ and $r \gg r_{\mathrm{F}}$ for extended sources, outgoing dyadic asymptotics Eq. (2.192) reduce Eq. (2.229) to

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}_0(\theta,\phi)\,\frac{e^{-\mathrm{j}k_0 r}}{r} \;+\; \mathcal{O}(r^{-2}), \qquad \mathbf{E}_0 \cdot \hat{\mathbf{r}} = 0, \tag{2.245}$$

In watt units, the display fixes how much of $\|\mathcal{S}_\omega[\mathfrak{D}]\|_{\mathcal{E}}^2$ passes through Π_{rad} on Γ_R [2].

$$\mathbf{E}_0(\theta,\phi) = -\mathrm{j}\omega\mu_0\,\frac{1}{4\pi}\int_{\Omega_s} \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, e^{-\mathrm{j}k_0 \hat{\mathbf{r}}\cdot\mathbf{r}'}\, d^3 r', \tag{2.246}$$

with $\hat{\mathbf{r}} = \mathbf{r}/r$ and the transversality constraint enforced by $\overline{\mathbf{G}}_\infty$ in Eq. (2.192) [2]. Equation (2.245) is the far-zone limit of Eq. (2.191); the amplitude integral Eq. (2.246) is the vector radiation integral used in antenna far-field post-analysis [6]. The amplitude integral is what a Fraunhofer-range pattern probe measures after Π_{rad}: it Fourier-transforms $\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')$ over the source volume while the reactive inverse-power terms that $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ carries at finite r no longer enter at leading order [6]. Magnetic field in the far zone satisfies Eq. (2.190) with the same $\mathbf{E}_0$ [2]. Let $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ be supported in $B_a(0)$ with outgoing radiation condition. For $r > r_{\min}(a,\lambda_0)$ as in Eq. (2.241), the field from Eq. (2.229) satisfies Eqs. (2.245)–(2.246) with error $\mathcal{O}\big((k_0 a)^2/(k_0 r)\,|\mathbf{E}_0|\,r^{-1}\big)$ for extended sources and $\mathcal{O}\big((k_0 r)^{-1}\,|\mathbf{E}_0|\,r^{-1}\big)$ for $k_0 a \ll 1$ [2]. Substitute Eq. (2.192) into Eq. (2.229) and expand $|\mathbf{r}-\mathbf{r}'|^{-1}$ for $|\mathbf{r}'| \le a \ll r$. The leading $e^{-\mathrm{j}k_0 r}/r$ term factorizes into $\mathbf{E}_0(\theta,\phi)$ times phase $e^{-\mathrm{j}k_0 \hat{\mathbf{r}}\cdot\mathbf{r}'}$ over the source; transversality follows from the dyadic structure Eq. (2.151) at leading order [2]. Higher orders in a/r and $1/(k_0 r)$ give the stated error bounds [14]. On spheres $S_r = \{|\mathbf{r}| = r\}$, define shell power $P(R) = \mathrm{Re}\oint_{S_R} \mathbf{S}\cdot\hat{\mathbf{r}}\, dS$ as in Eq. (2.6) [2]. In the far zone,

$$|P(r) - P_\infty| = \mathcal{O}(r^{-1}), \qquad P_\infty = \frac{1}{2\eta_0}\oint_{S_r} |\mathbf{E}_0(\theta,\phi)|^2\, d\Omega, \tag{2.247}$$

with $\mathbf{E}_0$ from Eq. (2.246) and η_0 from Eq. (2.207) [2]. Equation (2.247) is the operator form of Eq. (2.31): $P(r)$ stabilizes once reactive r^{-2}, r^{-3} contributions no longer dominate the shell integral [2]. Evaluating Eq. (2.6) on $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ before $r \gtrsim r_c$ from Eq. (2.32) confuses storage with radiation [6]. Distance regimes induce an observation-shell map on operator outputs:

$$\begin{aligned} r \in \text{near} &\implies \text{meter } \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}], \\ r \in \text{far} &\implies \text{meter } \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ via Eqs. (2.245)–(2.246)}, \end{aligned} \tag{2.248}$$

with intermediate shells requiring explicit $\chi(r)$ from Eq. (2.30) before transport claims [2]. When $k_0 a \ll 1$, Eq. (2.246) reduces to the electric-dipole pattern proportional to $\widehat{\boldsymbol{\theta}} \times (\widehat{\boldsymbol{\theta}} \times \mathbf{p}_e)$ with $\mathbf{p}_J = \int \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3 r'$ and $\mathbf{p}_e = -\mathrm{j}\mathbf{p}_J/\omega$ [14]. The near zone extends to $r \sim \lambda_0$; far-zone reduction applies for $k_0 r \gg 1$ [6]. When $k_0 a \gg 1$, multiple lobes appear in $\mathbf{E}_0(\theta, \phi)$ and Fraunhofer distance $r_{\mathrm{F}} = 2a^2/\lambda_0$ dominates the far-zone threshold [2]. Angular-spectrum action in section 2.9 filters $\mathbf{E}_0(\theta, \phi)$ without altering the integral reduction above [2]. Eqs. (2.241)–(2.247) require linear vacuum or piecewise-constant media, fixed ω, outgoing radiation condition, and observation outside Ω_s [2]. They fail in cutoff waveguides where propagating modes do not exist (Eq. (2.140)), in cavity standing-wave fields, and in lossless open resonances without outgoing selection [2]. Anisotropic beams and surface waves require modified asymptotics beyond the locked free-space dyad template [6]. The far-field reduction asserts field asymptotics on the outgoing branch only; it does not assert uniqueness of sources [2]. The regime partition is not a sharp geometric cutoff but a diagnostic on which terms of the same outgoing integral dominate what an instrument measures. Near and intermediate zones may appear to contradict far-zone pattern theory because substantial field energy is visible on small spheres; on the bench, that energy is largely reactive circulation diagnosed through $\mathcal{S}_\omega$, while pattern definitions attach only after $k_0 r \gg 1$ and Fresnel–Fraunhofer qualifications are met [6]. Shell watt readings must not be identified with radiated power until Eq. (2.139) is satisfied: what changes across bands is the relative weight of storage and radiation in the *same* Maxwell solution, not the governing equations themselves [5]. Subsection 2.5.3 connects these spatial bands to flux, momentum, and operator norms imported from Chapter 1, always downstream of the integral evaluation chain established here.

2.5.3 Energy, momentum, angular-momentum flux, power, and operator norms

With outgoing integrals and distance labels in place, the display below evaluates what an observer can meter on large shells. The radiation operator does not merely transport fields: it transports measurable physical quantities—energy, momentum, and angular momentum—which ultimately determine what an observer can detect [5]. On a calibrated compact range, the wattmeter on Γ_R is the primary pattern-adjacent meter: it integrates $\hat{\mathbf{r}} \cdot \mathbf{S}$ on the radiative branch $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ after Π_{rad}, not on the full $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ pair that near-field probes sample at $r \sim a$ [2]. Power budgets on Γ_R must close before torque or angular-flux claims are promoted [12]. Complex Poynting flux from Eq. (1.34) applies to any admissible field pair, but power budgets for transport claims use the radiative branch:

$$P_{\mathrm{rad}}(R) = \mathrm{Re} \oint_{\Gamma_R} \hat{\mathbf{r}} \cdot \mathbf{S}[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]]\, dS, \tag{2.249}$$

with $\mathbf{S}[\cdot]$ the Poynting vector functional imported from Subsection 1.1.4 and $\Gamma_R = \{r = R\}$ large enough for far-zone reduction when reporting radiation [2]. Eq. (2.249) specializes Eq. (2.6) to the outgoing selector sequence; it is not equivalent to integrating $\mathbf{S}[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]]$ unless reactive content has been shown negligible or orthogonal to the declared observable [6]. Poynting balance Eq. (1.34), Maxwell stress Eq. (1.37), angular balance Eqs. (1.41)–(1.42), and transport indices Eqs. (1.64)–(1.67) are presupposed from Chapter 1 [2]. Radiative–non-radiative splitting is taken up in the

closing subsection below. Let $(\mathbf{E},\mathbf{H}) = \mathcal{S}_\omega[\mathfrak{D}]$ and $(\mathbf{E}_{\mathrm{rad}},\mathbf{H}_{\mathrm{rad}}) = \mathcal{R}_\omega[\mathfrak{D}]$ be as in the evaluation chain Eq. (2.227) [2]. Flux meters on spherical shells $S_r = \{|\mathbf{r}| = r\}$ assume r lies in the far zone of Eq. (2.241) unless noted otherwise, with $\mathcal{D}_{J,\mathrm{reg}} = 1$ from Eq. (2.33) [2]. Complex Poynting vector $\mathbf{S} = \frac{1}{2}\mathbf{E}\times\mathbf{H}^*$ and stress tensor $\mathbf{T}$ follow Eqs. (1.34) and (1.37) [2]. Angular-flux tensor $\mathbf{M} = \mathbf{r}\times\mathbf{T}$ uses origin $\mathbf{r}_0$ declared in Subsection 1.1.5 [12]. Flux functionals on the radiative branch. Define the radiative Poynting functional

$$\mathbf{S}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}] = \frac{1}{2}\,\mathbf{E}_{\mathrm{rad}}\times\mathbf{H}_{\mathrm{rad}}^*, \qquad (\mathbf{E}_{\mathrm{rad}},\mathbf{H}_{\mathrm{rad}}) = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}], \tag{2.250}$$

with $\mathbf{E}_{\mathrm{rad}},\mathbf{H}_{\mathrm{rad}}$ the field components selected by Eq. (2.202) [2]. Equation (2.250) specializes Eq. (1.34) to the operator output on which radiation claims are allowed per Eq. (2.27) [6]. The reactive counterpart $\mathbf{S}_{\mathrm{re}} = \frac{1}{2}(\mathbf{E}-\mathbf{E}_{\mathrm{rad}})\times(\mathbf{H}-\mathbf{H}_{\mathrm{rad}})^*$ is not a transport meter unless separately qualified [2]. Stress and angular-flux functionals on the radiative branch are

$$\mathbf{T}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}] = \mathbf{T}(\mathbf{E}_{\mathrm{rad}},\mathbf{H}_{\mathrm{rad}}), \qquad \mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}] = \mathbf{r}\times\mathbf{T}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}], \tag{2.251}$$

with $\mathbf{T}(\cdot,\cdot)$ the Maxwell stress map Eq. (1.37) [2]. Equations (2.250)–(2.251) specialize 1 flux objects; they introduce no new conservation law [2]. Shell power on the radiative branch is the bounded linear functional

$$\mathcal{P}_{\mathrm{rad}}(r;\mathbf{J}_{\mathrm{imp}}) = \mathrm{Re}\oint_{S_r}\hat{\mathbf{r}}\cdot\mathbf{S}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}]\,dS, \tag{2.252}$$

which coincides with Eq. (2.249) and Eq. (2.6) when $(\mathbf{E},\mathbf{H})$ in those equations is replaced by $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2]. In the far zone, Eqs. (2.245)–(2.246) give

$$\mathcal{P}_{\mathrm{rad}}(r;\mathbf{J}_{\mathrm{imp}}) = \frac{1}{2\eta_0}\oint_{S_r}\left|\mathbf{E}_0(\theta,\phi)\right|^2 d\Omega \;+\; \mathcal{O}(r^{-1}), \tag{2.253}$$

with $\eta_0 = \sqrt{\mu_0/\varepsilon_0}$ and $\mathbf{E}_0$ from Eq. (2.246) [2]. Equation (2.253) is the operator form of Eq. (2.247) and Eq. (2.31) [2]. On a calibrated compact range, $\mathcal{P}_{\mathrm{rad}}$ is the wattmeter reading on Γ_R after radiation selection—the quantity an antenna engineer reports as radiated power [6]. Evaluating $\mathcal{P}_{\mathrm{rad}}$ on $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ before $r \gtrsim r_c$ from Eq. (2.32) imports reactive storage into the power budget [6]. Linear momentum flux meter. Net momentum flux through S_r is

$$\mathbf{F}_{\mathrm{rad}}(r;\mathbf{J}_{\mathrm{imp}}) = \mathrm{Re}\oint_{S_r}\mathbf{T}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}]\cdot\hat{\mathbf{r}}\,dS, \tag{2.254}$$

the time-averaged force functional inherited from Eq. (1.38) [2]. In the far zone with Eq. (2.190),

$$\mathbf{F}_{\mathrm{rad}}(r;\mathbf{J}_{\mathrm{imp}}) = \frac{1}{c}\,\hat{\mathbf{r}}\,\mathcal{P}_{\mathrm{rad}}(r;\mathbf{J}_{\mathrm{imp}}) \;+\; \mathcal{O}(r^{-1}), \tag{2.255}$$

linking momentum radiate to radiated power on the propagating branch [2]. A scattering analyst interpreting radiation pressure on a target uses $\mathbf{F}_{\mathrm{rad}}$ as the time-averaged force carried by the outgoing wave [2]. Eq. (2.255) expresses Eq. (1.39) in the far-zone limit for unit $\hat{\mathbf{q}} = \hat{\mathbf{r}}$ [2]. Momentum meters attach to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, not to reactive near-zone circulation in $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]\setminus\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [6]. Angular-momentum flux meter. Axis torque about unit direction $\hat{\mathbf{u}}$ is the functional

$$\tau_{\hat{\mathbf{u}}}(r;\mathbf{J}_{\mathrm{imp}}) = -\hat{\mathbf{u}}\cdot\mathrm{Re}\oint_{S_r}\mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}]\cdot\hat{\mathbf{r}}\,dS, \tag{2.256}$$

specializing Eq. (1.44) at harmonic steady state to the radiative branch [12]. Net angular-momentum flux through S_r is

$$\mathbf{J}_{\mathrm{rad}}(r;\mathbf{J}_{\mathrm{imp}}) = \mathrm{Re}\oint_{S_r}\mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}]\cdot\hat{\mathbf{r}}\,dS, \tag{2.257}$$

with $\tau_{\hat{\mathbf{u}}} = -\hat{\mathbf{u}} \cdot \mathbf{J}_{\mathrm{rad}}$ [2]. Orbital–spin splits Eq. (1.43) are not re-derived; any angular report must close through Eq. (1.42) on the declared control volume [12]. Directional concentration uses Eq. (1.45) evaluated on $\mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}]$ [2].

Subsection 1.2.4 fixed transport $\mathcal{D}_J = 1$ through conditions $\mathcal{C}_1$–$\mathcal{C}_3$ on field states. On operator outputs,

$$\begin{aligned} \mathcal{C}_1[\mathbf{J}_{\mathrm{imp}}] &\iff \text{Eq. (1.62) holds on } \mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}], \\ \mathcal{C}_2[\mathbf{J}_{\mathrm{imp}}] &\iff \text{Eq. (1.42) holds for } (\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}}), \\ \mathcal{C}_3[\mathbf{J}_{\mathrm{imp}}] &\iff \text{Eq. (1.63) holds with } \Pi = \Pi_{\mathrm{rad}} \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}], \end{aligned} \tag{2.258}$$

with indices Λ_J, $\mathcal{K}_J$, and $\mathcal{R}_\tau$ from Eqs. (1.64)–(1.66) evaluated on the same radiative fields [2]. Operational acceptance $\mathcal{D}_J = 1$ in Eq. (1.67) therefore means transport interpretation is attached to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ after $\mathcal{D}_{J,\mathrm{reg}} = 1$, not to raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2]. The condition for outward angular-momentum flux of section 1.2.4 applies unchanged once $(\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}})$ replaces the full pair on the measurement shell [2]. Radiation operator norms. Field energy on the radiative branch uses the inherited bilinear form Eq. (2.19) restricted to $\mathcal{F}_{\mathrm{rad}}$:

$$\|\mathbf{u}_{\mathrm{rad}}\|_{\mathcal{E}}^2 = \mathfrak{a}_\omega\big(\mathbf{u}_{\mathrm{rad}}, \mathbf{u}_{\mathrm{rad}}\big), \qquad \mathbf{u}_{\mathrm{rad}} = (\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}}), \tag{2.259}$$

with $\mathfrak{a}_\omega$ from Subsection 2.0.2 [2]. Shell L^2 power density norms for pattern reporting are

$$\|\mathbf{E}_0\|_{L^2(S_r)}^2 = \oint_{S_r} \big|\mathbf{E}_0(\theta,\phi)\big|^2 \, d\Omega, \qquad \|\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{J}} = \Big(\int_{\Omega_s} |\mathbf{J}_{\mathrm{imp}}|^2 \, dV \Big)^{1/2}, \tag{2.260}$$

linking source amplitude to far pattern through Eq. (2.246) [6]. The radiation operator norm on admissible sources is

$$\|\mathcal{R}_\omega\|_{\mathrm{op}} = \sup_{\|\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{J}}=1} \|\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\|_{\mathcal{E}}, \tag{2.261}$$

bounded on fixed outgoing domains when resolvent bound Eq. (2.76) holds [2]. Observation maps compose on the right, yielding

$$\Lambda_{\mathrm{meas}} = \mathcal{O}_{\mathrm{meas}} \circ \mathcal{O}_{\mathrm{far}} \circ \mathcal{R}_\omega, \qquad \|\Lambda_{\mathrm{meas}}\|_{\mathrm{op}} \le \|\mathcal{O}_{\mathrm{meas}}\|_{\mathrm{op}} \, \|\mathcal{R}_\omega\|_{\mathrm{op}}, \tag{2.262}$$

consistent with Eq. (2.218) and rank cap Eq. (2.220) [2]. For a range engineer, $\|\mathcal{R}_\omega\|_{\mathrm{op}}$ bounds how strongly a unit-amplitude feed can excite the far pattern; the composition Λ_{meas} in Eq. (2.262) is the formal image of gain calibration followed by pattern sampling on S_r [6]. Let $\mathbf{J}_{\mathrm{imp}}$ be admissible with $\mathcal{D}_{J,\mathrm{reg}} = 1$ and let r lie in the far zone of Eq. (2.241). Then $\mathcal{P}_{\mathrm{rad}}(r; \mathbf{J}_{\mathrm{imp}})$, $\mathbf{F}_{\mathrm{rad}}(r; \mathbf{J}_{\mathrm{imp}})$, and $\mathbf{J}_{\mathrm{rad}}(r; \mathbf{J}_{\mathrm{imp}})$ satisfy Eqs. (2.253)–(2.255) with errors $\mathcal{O}(r^{-1})$, and $\mathcal{P}_{\mathrm{rad}}(r; \mathbf{J}_{\mathrm{imp}}) \to P_\infty > 0$ whenever Eq. (2.31) holds on the radiative branch [2]. Substitute Eqs. (2.245)–(2.246) into Eqs. (2.250)–(2.254) and integrate on S_r. Sommerfeld ratio Eq. (2.190) gives Eq. (2.255); power integral reduces to Eq. (2.253) [2]. Convergence follows from Eq. (2.247) [2]. Eqs. (2.250)–(2.261) require harmonic steady state, linear passive media, outgoing radiation condition, and regime-appropriate shells [2]. They fail on reactive-dominated shells with $\chi(r) \gg 1$ from Eq. (2.30), in cutoff guides, and when $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is used without $\mathcal{R}_\omega$ projection [2]. Material stress debates (Minkowski versus Abraham) are not re-opened; vacuum-side shells use Eq. (1.37) [2]. Gauge-dependent near-zone angular densities are excluded from $\mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}]$ unless closure through Eq. (1.42) is verified [12]. As a worked example, compute $\Lambda_J(r)$ from Eq. (1.64) using $\mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}]$ on S_r with $k_0 r = 10$ and compare to $k_0 r = 0.5$ [2]. Only the outer shell with $\mathcal{D}_{J,\mathrm{reg}} = 1$ and $\mathcal{C}_1$–$\mathcal{C}_3$ satisfied admits transport interpretation per Eq. (1.67) [2]. Flux meters are therefore downstream observables: they presuppose the integral evaluation chain, the far-zone reduction when applicable, and outgoing selection through $\mathcal{R}_\omega$ before any watt, momentum, or angular-momentum budget is reported on Γ_R [6]. Flux and norm functionals are not

independent postulates added to Maxwell theory; they are laboratory readouts that presuppose the same outgoing selection already enforced on pattern meters. Power measured on a reactive shell must be distinguished from power reported after Π_{rad}: the former samples $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$, the latter the transverse outgoing sector carried by $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2]. Momentum and angular-momentum budgets add not new dynamics but conserved-flux accounting on the qualified radiative image once regime restrictions from 1 are satisfied [12]. Subsection 2.5.4 decomposes the same solution map into radiative and non-radiative components.

2.5.4 Radiative and non-radiative components

A compact-range engineer can measure strong electric and magnetic fields within one wavelength of a horn mouth while the far-zone gain remains modest: reactive energy stored near the feed is large, yet little of it crosses a distant measurement sphere as radiation [5]. Conversely, currents in $\ker \mathcal{R}_\omega$ can sustain elaborate near-field structure while producing no far-field signature—the classical non-radiating source [6]. Inverse synthesis and scattering measurements inherit this split: near-field scans and angle-resolved data constrain only the radiative image $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, leaving non-radiative content invisible until additional excitation or realizability constraints are imposed; Chapter 4 develops that $\mathcal{S}_{\mathrm{real}}$ question spectrally [2].

The component partition below imports reactive–radiative logic from section 1.2.5 and spectral projectors Eqs. (2.133)–(2.138) without re-proving projector algebra [2]. Complete non-radiating null-space classification belongs to section 2.7. Let $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H}) = \mathcal{S}_\omega[\mathfrak{D}]$ and $\mathbf{u}_{\mathrm{rad}} = \mathcal{R}_\omega[\mathfrak{D}]$ be as in Eq. (2.227) [2]. Spectral projectors Π_{prop}, Π_{ev}, Π_{non} act on $\widehat{\mathcal{F}}_\Omega$ as in Subsection 2.2.3; outgoing selector $\mathcal{O}_{\mathrm{out}}$ and radiation map satisfy Eqs. (2.202)–(2.137) [2]. declared size and distance regime uses $\mathcal{D}_{J,\mathrm{reg}}$ from Eq. (2.33) and criterion Eq. (2.139) [2]. On admissible $\mathfrak{D}$,

$$\mathcal{S}_\omega[\mathfrak{D}] = \Pi_{\mathrm{prop}}\mathcal{S}_\omega[\mathfrak{D}] + \Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathfrak{D}] + \Pi_{\mathrm{non}}\mathcal{S}_\omega[\mathfrak{D}], \tag{2.263}$$

restating Eq. (2.136) on the integral outputs constructed above [2]. Define the *non-propagating* component

$$\mathbf{u}_{\mathrm{np}} = \Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathfrak{D}] + \Pi_{\mathrm{non}}\mathcal{S}_\omega[\mathfrak{D}], \tag{2.264}$$

and the *propagating* component $\mathbf{u}_{\mathrm{prop}} = \Pi_{\mathrm{prop}}\mathcal{S}_\omega[\mathfrak{D}]$ [2]. Only $\mathbf{u}_{\mathrm{prop}}$ is eligible for outgoing radiation selection; evanescent and non-propagating content satisfies Eq. (2.138) [6]. Equation (2.263) applies the 1 partition to integral outputs: reactive dominance on near shells corresponds to large $\|\mathbf{u}_{\mathrm{np}}\|$ relative to $\|\mathbf{u}_{\mathrm{prop}}\|$ in the sense of Eq. (2.142) [2]. The radiative component is

$$\mathbf{u}_{\mathrm{rad}} = \mathcal{R}_\omega[\mathfrak{D}] = \Pi_{\mathrm{rad}}\,\mathcal{S}_\omega[\mathfrak{D}] = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}\,\mathcal{S}_\omega[\mathfrak{D}], \tag{2.265}$$

using Eqs. (2.202) and (2.137) [2]. The complementary *non-radiative* component is

$$\mathbf{u}_{\mathrm{nr}} = \mathcal{S}_\omega[\mathfrak{D}] - \mathbf{u}_{\mathrm{rad}} = (\mathrm{id} - \Pi_{\mathrm{rad}})\mathcal{S}_\omega[\mathfrak{D}]. \tag{2.266}$$

Eqs. (2.265)–(2.266) partition every admissible field pair into radiation and non-radiation parts without altering Maxwell dynamics: $\mathbf{u}_\omega = \mathbf{u}_{\mathrm{rad}} + \mathbf{u}_{\mathrm{nr}}$ [2]. Non-radiative content includes evanescent, non-propagating, and incoming-propagating fractions annihilated by Π_{rad}; large near-field magnitude therefore does not establish radiation [2]. Sources in $\ker \mathcal{R}_\omega$ produce $\mathbf{u}_{\mathrm{rad}} = \mathbf{0}$ while $\mathbf{u}_{\mathrm{nr}} \neq \mathbf{0}$ (Eq. (2.217)) [2].

Radiative/non-radiative split: $\mathbf{u}_{\mathrm{rad}}$ is $\Pi_{\mathrm{rad}}\mathcal{S}_\omega$; $\mathbf{u}_{\mathrm{nr}}$ collects all non-radiating content including Eq. (2.264).

Figure 2.5 from Subsection 2.2.3 on integral outputs [2]. Using energy norm Eq. (2.47), define

$$\mathcal{E}_{\mathrm{rad}}[\mathfrak{D}] = \|\mathbf{u}_{\mathrm{rad}}\|_{\mathcal{E}}^2, \qquad \mathcal{E}_{\mathrm{nr}}[\mathfrak{D}] = \|\mathbf{u}_{\mathrm{nr}}\|_{\mathcal{E}}^2, \tag{2.267}$$

with $\|\cdot\|_{\mathcal{E}}$ from Eq. (2.259) on each component [2]. Shell power splits analogously:

$$\mathcal{P}_{\mathrm{tot}}(r) = \mathcal{P}_{\mathrm{rad}}(r;\mathbf{J}_{\mathrm{imp}}) + \mathcal{P}_{\mathrm{nr}}(r;\mathbf{J}_{\mathrm{imp}}), \tag{2.268}$$

where $\mathcal{P}_{\mathrm{rad}}$ is Eq. (2.252) and

$$\mathcal{P}_{\mathrm{nr}}(r;\mathbf{J}_{\mathrm{imp}}) = \mathrm{Re}\oint_{S_r} \hat{\mathbf{r}}\cdot\mathbf{S}_{\mathrm{nr}}[\mathbf{J}_{\mathrm{imp}}]\,dS, \qquad \mathbf{S}_{\mathrm{nr}} = \tfrac{1}{2}\,\mathbf{E}_{\mathrm{nr}}\times\mathbf{H}_{\mathrm{nr}}^{*}, \tag{2.269}$$

with $(\mathbf{E}_{\mathrm{nr}},\mathbf{H}_{\mathrm{nr}}) = \mathbf{u}_{\mathrm{nr}}$ [2]. Eq. (2.268) is not a new conservation law; it decomposes Eq. (2.6) when both components are evaluated on the same shell [2]. In the far zone, $\mathcal{P}_{\mathrm{nr}}(r)\to 0$ and $\mathcal{P}_{\mathrm{rad}}(r)\to P_\infty$ under Eq. (2.31) [2]. Near-zone shells with $\chi(r)\gg 1$ from Eq. (2.30) typically have $|\mathcal{P}_{\mathrm{nr}}|\gg|\mathcal{P}_{\mathrm{rad}}|$ [2]. declared size and distance regime from section 1.2.5 becomes

$$\mathcal{D}_{J,\mathrm{reg}} = 1 \implies \text{transport uses } \mathbf{u}_{\mathrm{rad}},\ \mathcal{P}_{\mathrm{rad}},\ \mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}], \tag{2.270}$$

and forbids interpreting $\mathbf{u}_{\mathrm{nr}}$, $\mathcal{P}_{\mathrm{nr}}$, or $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathfrak{D}]$ as radiative channels [2]. Eq. (2.270) is the component form of Eq. (2.139) after Eqs. (2.265)–(2.266). Laboratory errors from section 1.2.5—near-zone reactive scans, cutoff-guide modes, evanescent dominance—appear as large $\mathcal{E}_{\mathrm{nr}}$ or $\|\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathfrak{D}]\|$ with small $\mathcal{P}_{\mathrm{rad}}$ [2]. Link to radiation null space. Non-radiative components encode source-side invisible content:

$$\mathbf{J}_{\mathrm{imp}}\in\ker\mathcal{R}_\omega \implies \mathbf{u}_{\mathrm{rad}} = \mathbf{0}, \quad \mathbf{u}_{\mathrm{nr}} = \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \neq \mathbf{0} \text{ in general}, \tag{2.271}$$

as recorded in Eq. (2.7) [6]. Conversely, $\mathbf{u}_{\mathrm{nr}} = \mathbf{0}$ does not imply trivial $\mathcal{S}_\omega[\mathfrak{D}]$: a pure outgoing propagating state may have $\mathbf{u}_{\mathrm{rad}} = \mathbf{u}_\omega$ [2]. section 2.7.1 classifies $\ker\mathcal{R}_\omega$; here only the field-side split Eq. (2.266) is identified as the operator image of non-radiating content [6]. Let $\mathcal{D}_{J,\mathrm{reg}} = 1$ and let r lie in the far zone of Eq. (2.241). Then

$$\frac{\mathcal{E}_{\mathrm{nr}}[\mathfrak{D}]}{\mathcal{E}_{\mathrm{rad}}[\mathfrak{D}]+\epsilon}\to 0, \qquad \frac{|\mathcal{P}_{\mathrm{nr}}(r)|}{\mathcal{P}_{\mathrm{rad}}(r)+\epsilon}\to 0, \tag{2.272}$$

as $r\to\infty$ for fixed $\mathfrak{D}$, with $\epsilon>0$ regularizing zero radiation [2]. Equivalently, $\mathbf{u}_\omega\to\mathbf{u}_{\mathrm{rad}}$ in far-zone observables metered by Eqs. (2.245)–(2.252). Eq. (2.138) annihilates evanescent and non-propagating contributions in Π_{rad}. Far-zone asymptotics and flux stabilization on large shells give $\mathcal{P}_{\mathrm{rad}}(r)\to P_\infty>0$ while reactive terms decay relatively [2]. Regime condition Eq. (2.31) removes reactive dominance in the limit. Below guide cutoff Eq. (2.140), modal content lies in $\Pi_{\mathrm{non}}\mathcal{S}_\omega[\mathfrak{D}]$ or $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathfrak{D}]$ [2]. Applying Eq. (2.246) or transport indices to such modes violates Eq. (2.270) as in the TE_{10} illustration of section 1.1.2 [6]. Spectral projectors and regime logic must therefore be applied together on the same integral outputs [2]. Eqs. (2.263)–(2.272) require linear media, fixed ω, outgoing radiation condition, and spectral completion sufficient for Eqs. (2.133)–(2.134) [2]. Sharp boundaries between subspaces may blur when loss, anisotropy, or coupled interfaces mix propagating and evanescent sheets [2]. Component splits are diagnostic on shells, not substitutes for exterior radiative uniqueness (Subsection 2.4.3) on the full field [6]. Outgoing source-to-field integrals map impressed data to evaluable fields; distance-regime labels fix where asymptotic reductions apply; flux and operator-norm meters act on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ after declared size and distance regime; and the radiative–non-radiative split Eqs. (2.265)–(2.270) separates radiate from storage on the same integral outputs [2]. section 2.6 develops spectral decomposition of radiation operators on the representations

constructed here [6]. The radiative–non-radiative split is what separates a wattmeter that has been qualified on Γ_R from a near-field probe that still sees storage and evanescent overlap in $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [6]. Both components coexist in one Maxwell solution until projectors and outgoing selection name which part radiation meters may integrate [5]. Evanescent sheets must not be dismissed as mere mathematical artifacts; however, Maxwell theory shows they carry reactive energy measurable on enclosing scans while remaining null under far-zone angular accessibility maps developed in section 2.9 [2]. section 2.6 now organizes the radiative image itself into eigenmodes and recoverable rank on the integral representations constructed in sections 2.3–2.5.

2.6 Spectral Theory of Radiation Operators

The development has followed impressed current through Maxwell completion, outgoing selection, and evaluable radiation integrals. Spectral theory treats outgoing radiation not as a single scalar amplitude but as a countable nesting of modes on compact geometries, each with its own coupling to sources and its own observability limit on finite instruments [6]. Pattern shape must not be identified with a single dominant mode; Maxwell propagation on bounded supports shows instead that many directions can be degenerate under rotation while remaining distinguishable under excitation, and that Fredholm structure controls which target patterns are reachable at all [20]. A laboratory measurement on a circular aperture measures a finite slice of that nesting—never the full operator spectrum—so rank and degeneracy statements below are as much measurement discipline as abstract mathematics [15].

Radiation integrals in section 2.5 map sources to outgoing fields. section 2.6 analyzes that map spectrally: which directions carry power and how degeneracy splits under broken symmetry [6]. Sidelobe floors on finite apertures are the visible face of spectral decay [3]. Every antenna engineer learns, long before operator theory appears in the curriculum, that a finite structure cannot radiate an unlimited set of independent far-zone patterns. A dish whose rim spans only a few wavelengths resolves only a modest number of principal beam directions on a compact range; a spherical near-field scanner with finite probe count tells the same story after transformation to far-zone coefficients [6]. Collin and Harrington treat that compression as structural: it is imposed by Maxwell propagation from compact sources through outgoing selection to a finite observation shell, not by a particular receiver model [15]. Stratton and Harrington organized outgoing fields by angular structure on large spheres; Chu sharpened the consequence that electrical size bounds directional radiate at fixed frequency [2]. Subsection 2.6.1 introduces the laboratory reading—singular values σ_n of $\mathcal{O}_{\mathrm{meas}} \circ \mathcal{R}_\omega$ and the recoverable rank they define—before any eigenpair formula is written; later subsections refer back rather than restate that principle. Why spectra arise in radiation physics is therefore not a mathematical accident. Maxwell's equations are linear in impressed sources on the outgoing branch fixed in section 2.4; superposition of two feeds superposes far-zone amplitudes. On a large sphere S_r, orthogonality of angular patterns under an appropriate energy inner product later organizes the radiation map $\mathcal{R}_\omega$ into radiation modes; each carries an *radiation eigenvalue* λ_n of the radiation operator—ranking qualified power on Γ_R and distinct from any correlation-operator spectrum unless explicitly stated [2]. Some feeds couple strongly to dominant modes; others produce elaborate reactive near fields through $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ yet register negligibly on a far-zone wattmeter after Π_{rad} [5]. Observable far-zone content remains primary; formal harmonic indices organize radiation but are not themselves physical modes—that distinction is developed once in Subsection 2.6.1 and thereafter [2]. The radiation map $\mathcal{R}_\omega$ from section 2.5 inherits this ordering. Spectral analysis names dominant and weak radiative directions, explains why finite dishes and finite scan grids lose information, and states when pattern synthesis and source inversion are well posed [2]. 3 will

supply explicit harmonic bases and tabulated matrix elements; the analysis below asks what the radiation map itself permits at fixed ω before any particular angular basis is chosen [2]. Measured patterns appear low-rank in practice because finite apertures, finite scan grids, and finite electrical size impose rank limits; some synthesis targets remain unrealizable regardless of numerical effort; rotation of an antenna reorients coefficients without changing radiated power when frames are consistently handled [6]. Spectral analysis begins from a fixed fact: measured patterns never carry unlimited independent information. Finite apertures, finite scan grids, and finite electrical size impose rank limits long before any harmonic chart exhausts its index range [3]. On a compact range the dominant beams are the few principal cuts that survive above the noise floor; weak modes decay in singular-value cliffs and become experimentally invisible even when formal expansions can still list them [2]. Measurement rank and aperture geometry fix how many independent far-zone directions an instrument can resolve; the spectral decomposition of $\mathcal{R}_\omega$ then explains why additional modes decay below declared thresholds [3]. Finite observation on S_r therefore precedes modal interpretation, not the reverse. Finite aperture observation on S_r fixes how many independent far-zone directions an observation operator can resolve—that bound is *finite radiative complexity*. Compactness of the composed map $\mathcal{K} = \mathcal{O}_{\text{meas}} \circ \mathcal{R}_\omega$ then explains why singular values σ_n, ranking the radiation efficiency of each independent source organization through the composed observation map, decay and define recoverable rank [3]. Only after that ordering does the section construct radiation eigenvalues λ_n of $\mathcal{R}_\omega$, decompose radiation modes $\mathbf{u}_n$ from their exciting source modes $\mathbf{J}_{\text{imp}}^{(n)}$, and complete symmetry, Fredholm, and geometric filters on what finite hardware can recover [2]. Let $\Omega_s = \operatorname{supp} \mathbf{J}_{\text{imp}}$ be compact, $k_0 = \omega\sqrt{\varepsilon_0\mu_0}$, and outgoing field pair $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H}) \in \mathcal{F}_{\text{SF}}$ from Eq. (2.56). The four stages below record the same program in compact notation; Subsection 2.6.1 develops σ_n decay and finite recoverable rank once, and later subsections cite it by reference:

$$\mathcal{R}_\omega \xrightarrow{\text{eigenmodes}} \{\lambda_n, \mathbf{u}_n\} \xrightarrow{\text{symmetry}} \text{degeneracy classes} \xrightarrow{\text{Fredholm}} \text{solvability / index} \xrightarrow{\text{geometry}} \text{radiative spectr} \tag{2.273}$$

Read λ_n as a coupling-efficiency scalar, $\mathbf{u}_n$ as the radiation pattern on $\mathcal{F}_{\text{rad}}$, and the exciting source mode $\mathbf{J}_{\text{imp}}^{(n)}$ as the coefficient-side preimage in $\mathcal{J}_\Omega$. from radiation-eigenmode ordering through rotational degeneracy, Fredholm solvability, and geometric filters on recoverable content [2]. The primary map remains

$$\mathcal{R}_\omega : \mathcal{J}_\Omega \longrightarrow \mathcal{F}_{\text{rad}}, \qquad \mathcal{R}_\omega[\mathbf{J}_{\text{imp}}] = \Pi_{\text{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\text{imp}}], \tag{2.274}$$

from Eq. (2.2), with Π_{rad} selecting outgoing radiative solutions per Eq. (2.137). In plain language: $\mathcal{R}_\omega$ is the radiation map from impressed currents to far-zone radiation; $\mathcal{J}_\Omega$ is the admissible source region where $\mathbf{J}_{\text{imp}}$ may be placed; $\mathcal{F}_{\text{rad}}$ is the space of outgoing radiative field pairs observable on Γ_R after outgoing selection; and Π_{rad} is the filter that removes reactive and non-propagating content before pattern measurement [2]. Equation (2.274) represents the entire electromagnetic experiment beginning with impressed current inside the source region and ending with the radiative content observable on a far-zone measurement surface [6]. Subsection 2.6.1 fixes σ_n decay and finite recoverable rank on $\mathcal{K}$; the subsections below construct radiation eigenpairs, split degeneracy under symmetry, diagnose Fredholm solvability, and apply geometric filters. A sidelobe floor on a measured pattern is often the visible face of Physically, the aperture-limited stage [3].

Physically realizable radiation is therefore bounded by geometry and frequency before optimization adjusts weights [7].

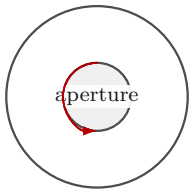


Figure 2.8: Finite circular aperture on the observation sphere: compact $\mathcal{K} = \mathcal{O}_{\text{meas}} \circ \mathcal{R}_\omega$; singular-value decay marks harmonics below the declared radiation tolerance.

2.6.1 Compact circular aperture and singular-value decay

Before any eigenvalue formula is written, it helps to watch the phenomenon on a range. On a dish compact range, place a feed above a circular scan disk on plane $z = 0$ and sample tangential field on S_r; on a spherical near-field scanner, place probes on a hull at $R \approx 2\lambda_0$ and transform to far-zone coefficients [3]. In either measurement, count how many independent beam directions remain above the receiver noise floor. The answer depends on disk or scan radius a_{ap} and source electrical size, not on how many spherical harmonics the harmonic chart can list [7]. Electrical size and scan extent fix a recoverable rank; singular values σ_n of $\mathcal{O}_{\text{meas}} \circ \mathcal{R}_\omega$ measure the radiation efficiency of each independent source organization through the aperture and observation composition, and fall off once the modal index exceeds what the geometry can resolve [2]. Synthesis harmonic caps therefore overstate physical rank when σ_n has already fallen below the noise floor—harmonic indices are labels, not independent radiative channels. Compact decay of $\sigma_n(\mathcal{K})$ and the finite recoverable rank it defines are stated here once; later subsections invoke Subsection 2.6.1 by reference rather than repeating the measurement narrative [6].

Consider a circular scan aperture of radius a_{ap} on plane $z = 0$ sampling the far-zone tangential field from a compact source with $k_0 a = 1.5$. The measurement composition $\mathcal{K} = \mathcal{O}_{\text{meas}} \circ \mathcal{R}_\omega$ is compact on the source and field spaces declared in section 2.1.3; its singular values σ_n rank radiation efficiency of source organizations and decay, defining a practical recoverable rank and finite effective dimension before any harmonic label is attached [2]. Figure 2.8 displays the singular-value cascade for a finite circular observation aperture: compact support compresses the composed map $\mathcal{K} = \mathcal{O}_{\text{meas}} \circ \mathcal{R}_\omega$; modes whose singular values fall below the declared tolerance carry negligible far-zone flux [2, 7].

Compactness from Subsection 2.1.3 yields $\sigma_n(\mathcal{K}) \to 0$; for disk apertures the decay is approximately exponential in modal index when $k_0 a_{\text{ap}}$ is moderate [2]. In practice the engineer reads off a knee in the singular-value curve: retain modes above the knee for synthesis or inversion; discard the tail and accept the residual sidelobe floor as a geometry-limited artifact rather than a tuning defect [3]. What practical criterion tells the engineer where to truncate the spectrum? Retaining every mode is impossible: instrument noise, finite aperture extent, and finite scan count cap recoverable rank, and singular-value decay controls which source organizations remain physically realizable through the composed observation map [2]. Truncation is therefore inevitable; the physical question is where the knee lies. Truncating at N modes with

$$\frac{\|\mathcal{K} - \mathcal{K}_N\|}{\|\mathcal{K}\|} \leq \frac{\sigma_{N+1}}{\sigma_1}, \tag{2.275}$$

where $\mathcal{K} = \mathcal{O}_{\text{meas}} \circ \mathcal{R}_\omega$ is the full observation composition from source to field samples on the declared shell, $\mathcal{K}_N$ is its rank-N truncated approximation, σ_1 is the largest singular value (dominant radiative channel), and σ_{N+1} is the first discarded singular value [6]. The ratio σ_{N+1}/σ_1 is the relative strength of the strongest discarded radiation channel compared with the dominant channel—a direct laboratory readout of how much pattern error truncation will introduce [3]. Eq. (2.83) bounds

pattern error on S_r. At $k_0 a_{\mathrm{ap}} = 3$, retaining $N = 9$ singular modes typically captures 95% of the radiated power resolvable through the disk; harmonics beyond $N \approx 12$ lie below a $-20\,\mathrm{dB}$ threshold relative to σ_1 [6]. Figure 2.8 foreshadows Chapter 4: realizability and spectral truncation ask which angular components survive compression by $\mathcal{K}$, not which indices a formal expansion can list [17]. Laboratory reports therefore inherit a finite rank from geometry before source inversion is attempted; improving SNR without enlarging a_{ap} cannot recover modes annihilated by σ_n below measurement noise [3]. Local notation (σ_n, recoverable rank). Compose the measurement operator $\mathcal{K} = \mathcal{O}_{\mathrm{meas}} \circ \mathcal{R}_\omega$ on declared source and field spaces. Its singular values σ_n—ordered $\sigma_1 \geq \sigma_2 \geq \cdots \to 0$—are the radiation efficiencies of independent source organizations as seen through the composed aperture and observation map [2]. The *recoverable rank* is the largest N for which σ_N remains above the instrument noise floor; modes beyond that cliff are mathematically listable in a harmonic chart yet physically indistinguishable from clutter on Γ_R [7]. Compact σ_n decay and finite dimension on $\mathcal{K}$ are fixed in the display below; elsewhere invoke Subsection 2.6.1 by reference only [6]. The spectral relations developed below restate three laboratory facts in compact form before the eigenbasis is constructed. A *radiation mode* is a normalized far-zone pattern direction $\mathbf{u}_n$; its *source mode* $\mathbf{J}_{\mathrm{imp}}^{(n)}$ is an admissible current that excites it. The *radiation eigenvalue* λ_n of the radiation operator $\mathcal{R}_\omega$—not a correlation-operator eigenvalue unless later stated—ranks how efficiently that radiation mode reaches qualified flux on Γ_R [2]. each radiating mode carries radiation efficiency $|\lambda_n|$ measured by coupled power in that direction. Second, not every desired far-zone target can be synthesized: obstructions appear as orthogonality conditions on outgoing solutions. Third, when geometry possesses rotational symmetry, rotating the hardware reorients modal coefficients without changing eigenvalue ordering within degenerate blocks. The introductory relation equations record these facts; the subsections that follow interpret them in measurement language [2]. On a compact range, a *radiation mode* appears as a principal far-zone pattern direction —a normalized cut on S_r that carries outgoing radiation power—while its *source mode* is the admissible current distribution that excites it [6]. Dominant beams correspond to modes with large $|\lambda_n|$ because those organizations couple most efficiently to the wattmeter on Γ_R [2]. The eigenpair relation below states the experimentally familiar observation that certain current organizations radiate much more effectively than others [15]. On a family of radiating modes, the eigenvalue template reads

$$\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(n)}] = \lambda_n \, \mathbf{u}_n, \qquad \mathbf{u}_n = P_n \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(n)}], \tag{2.276}$$

with λ_n measuring outgoing radiation efficiency of mode n from finite impressed current $\mathbf{J}_{\mathrm{imp}}^{(n)}$ [2]. In synthesis and inversion, $\mathbf{g}$ denotes the desired far-zone radiated field one wishes to realize; solvability fails when $\mathbf{g}$ carries a component along cokernel directions—an *orthogonality obstruction* that no admissible source can remove [2]. Experimentally, such failure appears as a synthesis target that refuses to converge, a measured residual orthogonal to every reachable pattern, or a pattern null that persists despite port tuning [6]. Solvability on outgoing radiative solutions satisfies

$$\mathcal{R}_\omega + K \text{ solvable on } \mathcal{J}_\Omega \iff \mathbf{g} \perp \ker(\mathcal{R}_\omega + K)^\dagger, \tag{2.277}$$

with K compact on the graph domain of impressed sources [2]. When a symmetry group G commutes with $\mathcal{R}_\omega$, the equation below expresses a simple laboratory fact: rotating the hardware and then measuring the pattern gives the same result as measuring first and rotating the resulting pattern [6].

$$\mathcal{R}_\omega \, \mathcal{D}(R)[\mathbf{J}_{\mathrm{imp}}] = \mathcal{D}(R) \, \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}], \qquad R \in G, \tag{2.278}$$

with $\mathcal{D}(R)$ the field representation induced from Eq. (1.70) on $\mathcal{F}_{\mathrm{rad}}$ [2]. Modal projectors compose with radiation selection through Π_{rad} acting as the radiation filter that passes only radiative content,

P_n as mode-selection lenses that isolate individual beam families from a measured pattern, and Π_{ev}, Π_{non} as projectors onto evanescent and non-propagating sectors that qualified far-zone meters cannot resolve [2].

$$\Pi_{\mathrm{rad}} = \sum_{n \in \mathcal{N}_{\mathrm{rad}}} P_n, \qquad P_n \Pi_{\mathrm{ev}} = P_n \Pi_{\mathrm{non}} = \mathbf{0}, \tag{2.279}$$

using Eqs. (2.133)–(2.138) by reference [2]. Spectral decompositions *organize* radiation; they do not *create* angular structure—see Subsection 2.6.1 for the harmonics-versus-modes distinction stated once in measurement terms [2].

Catalogue membership does not imply laboratory access on finite supports [6].

2.6.2 Eigenfunctions and spectral decomposition

Discrete modal sums on exterior domains organize radiated power into separable angular sectors only after outgoing Sommerfeld condition and flux gauge are declared [2, 14]. Continuous-spectrum bookkeeping runs in parallel with discrete ℓ blocks; mixing the two without measure conventions produces apparent mode creation on finite meshes [3]. Source integrals from section 2.5 return amplitudes on Γ_R. The question an experienced range engineer asks next is practical: how many of those amplitudes remain independent above the noise floor for a given electrical size and scan extent? [2]. The eigenpair construction below answers that question in normalized form; recoverable rank and σ_n cliffs are already fixed in Subsection 2.6.1. The radiation map $\mathcal{R}_\omega$ admits a spectral decomposition once outgoing selection and declared size and distance regime have fixed the admissible radiative class; the eigenpairs below name those principal radiative directions and their source couplings. What an eigenmode means physically. A radiation mode is a principal far-zone pattern direction $\mathbf{u}_n$; its source mode $\mathbf{J}_{\mathrm{imp}}^{(n)}$ is the admissible current distribution that excites it. The radiation eigenvalue λ_n of $\mathcal{R}_\omega$ measures radiation efficiency $|\lambda_n|$ of that radiation mode on Γ_R [2]. Modes with large $|\lambda_n|$ are the dominant radiative channels on Γ_R: they are the beams an array designer weights first and the cuts a diagnostician examines before worrying about sidelobe structure. Modes with small $|\lambda_n|$ appear only after long integration, narrowband filtering, or exceptional dynamic range; on a typical compact range they never rise above the clutter floor. Modes with $\lambda_n = 0$ may still produce strong reactive near fields—the balanced-loop illustration from section 2.5 is the archetype—yet contribute nothing to the far-zone wattmeter [17]. Eigenvalue ordering is therefore a statement about measurable radiation, not about how many terms a numerical chart can index. On a dish range, dominant eigenmodes appear as the principal cuts in a principal-component decomposition of sampled $\mathbf{E}_t$ on S_r: the first few components carry most resolvable power [3]. Rank truncation in synthesis follows the recoverable rank fixed in Subsection 2.6.1; the decomposition below passes through $\mathcal{R}_\omega$ on physically realizable source classes, with coupled power on Γ_R as the physical quantity [2]. Antenna-range observations. Harrington's synthesis practice already exploited the ordering: one designs or diagnoses feeds by how they excite a modest number of dominant modes, not by attempting independent control of every harmonic index a harmonic listing can enumerate [6]. On a compact range, rotating a horn reorients the principal cuts while leaving total radiated power unchanged when probe and source frames rotate together—a fact developed formally in the rotation discussion below, but visible in raw pattern data long before covariance formulas appear [15]. When a measured pattern refuses to match a synthesis target despite ample numerical effort, the fault is often spectral rather than computational: the target lies in a weakly coupled or forbidden sector, not in the subspace the aperture and source geometry can reach [2]. The eigenpair definition that follows encodes that experience in normalized form. Propagation projectors Eqs. (2.133)–(2.138), compactness templates from section 2.1.3, and far-zone amplitudes Eqs. (2.245)–(2.246) are presupposed [2]. Explicit vector spherical harmonic bases belong to Chapter 3; here modal

coordinates are fixed at the operator level compatible with Eq. (2.6.2) [6]. Assumptions. Fix $\omega > 0$, physically realizable impressed data $\mathfrak{D}$ on $\mathcal{J}_\Omega$ Eq. (2.48), outgoing radiation condition on exterior problems Eqs. (2.193)–(2.196), and radiation map Eq. (2.274) [2]. Field pairs lie in $\mathcal{F}_{\mathrm{SF}} \subset \mathcal{F}_\Omega$ from Eq. (2.56); outgoing radiative images lie in $\mathcal{F}_{\mathrm{rad}} \subset \operatorname{ran} \mathcal{R}_\omega$ with energy norm Eq. (2.47) [2]. Spectral completion $\widehat{\mathcal{F}}_\Omega$ supporting Eqs. (2.133)–(2.134) is assumed sufficient for the expansions below [6]. Radiation eigenpairs. Eigenmodes are introduced because measured patterns on S_r are rarely arbitrary: principal-component analysis of sampled $\mathbf{E}_t$ reveals a small number of dominant radiative directions, and the eigenbasis below names those directions in normalized form [3]. What experiment reveals is rank ordering—a few strong beams, a long tail of weak modes below the noise floor, and radiation-invisible reactive organizations that never register on a far-zone wattmeter [17]. The laboratory observations above condense into a normalized triple: radiation eigenvalue λ_n of $\mathcal{R}_\omega$, radiation mode $\mathbf{u}_n$, and source mode $\mathbf{J}_{\mathrm{imp}}^{(n)}$ linked through $\mathcal{R}_\omega$ [2]. Physically, $|\lambda_n|$ ranks how much qualified power each normalized radiation mode carries to Γ_R; mathematically, the triple satisfies the locked eigenpair relation below. A *radiation eigenpair* is a triple $(\lambda_n, \mathbf{u}_n, \mathbf{J}_{\mathrm{imp}}^{(n)})$ with $\lambda_n \in \mathbb{C}$, $\mathbf{u}_n = (\mathbf{E}_n, \mathbf{H}_n) \in \mathcal{F}_{\mathrm{rad}}$, and $\mathbf{J}_{\mathrm{imp}}^{(n)} \in \mathcal{J}_\Omega$ such that

$$\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(n)}] = \lambda_n \, \mathbf{u}_n, \qquad \|\mathbf{u}_n\|_{\mathcal{U}} = 1, \tag{2.280}$$

where λ_n is the radiation eigenvalue (radiation-efficiency score of mode n), $\mathbf{u}_n$ is the normalized radiation mode (principal far-zone pattern direction), $\mathbf{J}_{\mathrm{imp}}^{(n)}$ is the source mode (admissible current that excites $\mathbf{u}_n$), and $\|\mathbf{u}_n\|_{\mathcal{U}} = 1$ fixes unit radiative energy on $\mathcal{F}_{\mathrm{rad}}$ [2]. λ_n is a radiation-efficiency score assigned to the nth radiative channel [6]. Here $\|\cdot\|_{\mathcal{U}}$ is the field energy norm Eq. (2.47) on $\mathcal{F}_{\mathrm{rad}}$ [2]. Eq. (2.280) develops Eq. (2.276) to a normalized locked definition: λ_n measures radiation gain of mode n relative to unit radiative energy, and $\ker \mathcal{R}_\omega$ corresponds to $\lambda_n = 0$ with $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(n)}] \neq \mathbf{0}$ through Eq. (2.217) [2]. Eigenpairs are defined on the outgoing branch only; incoming propagating sheets excluded by Eq. (2.202) [6]. Interpretation: modes with $|\lambda_n| \gg 1$ dominate compact-range cuts; modes with $\lambda_n = 0$ are radiation-invisible yet may carry reactive near fields through $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(n)}]$ [17]. Modal projectors on field space. Projector $P_n[\mathbf{u}]$ extracts a single beam family from a measured pattern: it isolates the component of field $\mathbf{u}$ aligned with radiation mode $\mathbf{u}_n$ and discards the orthogonal complement [2]. Modal filtering matters because synthesis, diagnosis, and interference analysis proceed mode by mode rather than on raw harmonic indices [6]. For each index n in a family of radiating modes $\mathcal{N}$, define a rank-one (or finite-dimensional) projector $P_n : \mathcal{F}_\Omega \to \mathcal{F}_{\mathrm{rad}}$ by

$$P_n[\mathbf{u}] = \langle \mathbf{u}, \mathbf{u}_n \rangle_{\mathcal{U}} \, \mathbf{u}_n, \qquad P_n \circ \Pi_{\mathrm{ev}} = P_n \circ \Pi_{\mathrm{non}} = 0, \tag{2.281}$$

using dual pairing inherited from Eq. (2.25) on qualified Maxwell states and annihilation of non-propagating content through Eq. (2.138) [2]. Equation (2.281) specializes resolution Eq. (2.279) once an orthonormal family of radiating modes $\{\mathbf{u}_n\}_{n \in \mathcal{N}_{\mathrm{rad}}}$ is chosen. Projectors satisfy $P_n P_m = \delta_{nm} P_n$ when modes are orthogonal in $\mathcal{U}$ [2]. Once eigenmodes are fixed, any physically realizable excitation decomposes into modal coefficients. The quantity a compact-range measurement ultimately recovers is the radiation-side pattern carried by $\mathbf{u}_{\mathrm{rad},n} = P_n \mathcal{R}_\omega[\mathfrak{D}]$; the source-side amplitude $c_n[\mathfrak{D}]$ measures how strongly the full solution $\mathcal{S}_\omega[\mathfrak{D}]$ aligns with mode n before radiation [2]. For general physically realizable $\mathfrak{D}$, define source coefficients

$$c_n[\mathfrak{D}] = \langle \mathcal{S}_\omega[\mathfrak{D}], \mathbf{u}_n \rangle_{\mathcal{U}}, \qquad \mathbf{u}_{\mathrm{rad},n} = P_n \, \mathcal{R}_\omega[\mathfrak{D}] = \lambda_n \, c_n[\mathfrak{D}] \, \mathbf{u}_n, \tag{2.282}$$

when the expansion converges in $\mathcal{U}$ [2]. Physically, c_n is the excitation strength of mode n on the radiative branch before outgoing selection is applied; $\lambda_n c_n$ is the coefficient on the normalized far-zone pattern carried by $\mathbf{u}_n$ [14]. Linearity Eq. (2.5) implies $c_n[a_1\mathfrak{D}_1 + a_2\mathfrak{D}_2] = a_1 c_n[\mathfrak{D}_1] + a_2 c_n[\mathfrak{D}_2]$

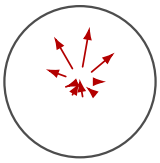


Figure 2.9: Spectral: $\mathcal{S}_\omega$ supplies coefficients c_n; $\mathcal{R}_\omega$ maps them to radiation amplitudes $\lambda_n c_n$ on projectors P_n Eq. (2.281).

for physically realizable superpositions after splitting each term through $\mathcal{S}_\omega$ [2]. Figure 2.9 displays the modal spectral decomposition: $\mathcal{S}_\omega$ supplies source coefficients c_n; $\mathcal{R}_\omega$ maps them to radiation amplitudes $\lambda_n c_n$ on the projectors P_n of Eq. (2.281) [2].

How can a complicated radiation pattern be reconstructed from simpler building blocks? The modal expansion below decomposes any radiative field into orthonormal radiation modes, each carrying a fraction of the radiation energy budget [2]. When the radiation eigenbasis is complete on the radiation subspace, every radiative field admits a countable expansion in normalized modes; the Parseval identity below states that radiative energy equals the sum of squared modal coefficients [2]. When $\{\mathbf{u}_n\}_{n\in\mathcal{N}_{\rm rad}}$ is a complete orthonormal family in $\mathcal{F}_{\rm rad}$ for the problem class, every radiative field $\mathbf{v} \in \operatorname{ran}\mathcal{R}_\omega$ admits

$$\mathbf{v} = \sum_{n\in\mathcal{N}_{\rm rad}} \langle \mathbf{v}, \mathbf{u}_n\rangle_{\mathcal{U}}\, \mathbf{u}_n, \qquad \|\mathbf{v}\|_{\mathcal{U}}^2 = \sum_{n\in\mathcal{N}_{\rm rad}} \left|\langle \mathbf{v}, \mathbf{u}_n\rangle_{\mathcal{U}}\right|^2, \tag{2.283}$$

with convergence in $\mathcal{U}$ [2]. Applying Eq. (2.283) to $\mathbf{v} = \mathcal{R}_\omega[\mathfrak{D}]$ gives

$$\mathcal{R}_\omega[\mathfrak{D}] = \sum_{n\in\mathcal{N}_{\rm rad}} \lambda_n\, c_n[\mathfrak{D}]\, \mathbf{u}_n, \tag{2.284}$$

the radiation eigenbasis restatement of compression operator Eq. (2.82) with explicit eigenvalues and normalized modes [2]. The Parseval identity in Eq. (2.283) states the modal energy budget: total radiative energy equals the sum of squared modal coefficients. Beam decomposition and recoverable complexity are therefore fixed by how many terms carry appreciable weight before truncation [3]. Truncation to $|\mathcal{N}| \le N$ yields $\mathcal{R}_\omega^{(N)}$ from Eq. (2.82); error Eq. (2.83) applies when the tail is compact on source space [6]. Let outgoing radiation condition and domain class be fixed as above. Suppose $\{\mathbf{u}_n\}_{n\in\mathcal{N}_{\rm rad}}$ is a complete orthonormal radiation eigenbasis in $\mathcal{F}_{\rm rad}$ with Eq. (2.280) and projectors Eq. (2.281). Then only modes with $\lambda_n \neq 0$ contribute to observable radiation: modes with $\lambda_n = 0$ lie in $\ker\mathcal{R}_\omega$ and are invisible to far-zone meters even when they carry reactive near fields [17].

$$\mathcal{R}_\omega = \sum_{n\in\mathcal{N}_{\rm rad}} \lambda_n\, P_n \circ \mathcal{S}_\omega, \qquad \Pi_{\rm rad} = \sum_{n\in\mathcal{N}_{\rm rad}} P_n, \tag{2.285}$$

on physically realizable $\mathfrak{D}$, with convergence in $\mathcal{U}$ for fields in $\operatorname{ran}\mathcal{R}_\omega$ [2]. The range satisfies

$$\operatorname{ran}\mathcal{R}_\omega = \overline{\operatorname{span}\{\mathbf{u}_n :\ n \in \mathcal{N}_{\rm rad},\ \lambda_n \neq 0\}}. \tag{2.286}$$

Eq. (2.284) follows by applying Eq. (2.283) to $\mathcal{R}_\omega[\mathfrak{D}]$ and using Eqs. (2.282)–(2.280). Linearity of $\mathcal{S}_\omega$ and $\mathcal{R}_\omega$ extends the sum from finite superpositions to all physically realizable impressed data by completion in $\mathcal{U}$ on the outgoing radiative subspace [2]. Eq. (2.285) is obtained by recognizing each term as $P_n\mathcal{S}_\omega$ acting with gain λ_n. Eq. (2.286) follows because eigenvectors with $\lambda_n = 0$ contribute no outgoing radiation; non-radiating sources lie in $\ker\mathcal{R}_\omega$ Eq. (2.217) [6]. Resolution Eq. (2.279)

matches $\Pi_{\mathrm{rad}} = \sum_n P_n$ on $\operatorname{ran} \mathcal{R}_\omega$ [2]. Resonant cavities, microwave filters, and waveguide resonators store energy in interior modes that need not radiate to a radiating port: stored versus radiated energy is the physical distinction the cavity specialization encodes [2]. On a bounded domain with passive $\mathcal{B}_\Gamma$ and interior resonances, Eq. (2.6.2) yields a discrete accumulating set $\{\lambda_n\}_{n=1}^{\infty}$ when $\mathcal{R}_\omega$ is restricted to cavity outgoing-propagating subspaces compatible with Eq. (2.59) [2]. The cavity specialization reads

$$\mathcal{R}_\omega^{(\mathrm{cav})} = \sum_{n=1}^{\infty} \lambda_n \, P_n^{(\mathrm{cav})} \circ \mathcal{S}_\omega, \qquad \lambda_n \to 0 \text{ or } \lambda_n \to \infty \text{ according to mode class,} \tag{2.287}$$

with $P_n^{(\mathrm{cav})}$ projectors onto normalized cavity radiation modes [2]. Eq. (2.287) is the constructive content of Eq. (2.82) in closed volumes; storage resonances with $\lambda_n = 0$ on $\mathcal{R}_\omega$ remain in $\mathbf{u}_{\mathrm{nr}}$ Eq. (2.266) [6]. Using Eq. (2.287) on open exterior problems misstates continuous angular content Eq. (2.6.2) [2]. In free space the discrete modal sum becomes continuous: every outgoing direction on the sphere S^2 carries spectral weight, and polarization index $\sigma \in \{1, 2\}$ labels the two transverse polarizations perpendicular to propagation direction $\hat{\mathbf{k}}$ [2]. Physically, $\hat{\mathbf{k}}$ is the unit propagation direction observed on a far-zone scanner; σ selects which transverse polarization component is sampled; $\mathbf{e}_\sigma$ is the corresponding unit polarization vector with $\mathbf{e}_\sigma \cdot \hat{\mathbf{k}} = 0$; and $d\Omega(\hat{\mathbf{k}})$ is the solid-angle measure that weights how much radiated power each direction receives [3]. For compact sources in free space with outgoing radiation condition, angular eigenmodes are labeled by propagation direction $\hat{\mathbf{k}} \in S^2$ and polarization index $\sigma \in \{1, 2\}$ transverse to $\hat{\mathbf{k}}$ [2]. A normalized plane-wave radiation mode is

$$\mathbf{u}_{\hat{\mathbf{k}},\sigma}(\mathbf{r}) = \big(\mathbf{E}_{\hat{\mathbf{k}},\sigma}(\mathbf{r}),\, \mathbf{H}_{\hat{\mathbf{k}},\sigma}(\mathbf{r})\big), \qquad \mathbf{E}_{\hat{\mathbf{k}},\sigma} \sim \mathbf{e}_\sigma \, e^{-\mathrm{j}k_0 \hat{\mathbf{k}}\cdot\mathbf{r}}/r, \tag{2.288}$$

with $\mathbf{e}_\sigma \cdot \hat{\mathbf{k}} = 0$ and outgoing phase [2]. The continuous spectral measure $d\Omega(\hat{\mathbf{k}})$ replaces discrete summation in Eq. (2.283):

$$\mathcal{R}_\omega[\mathfrak{D}](\mathbf{r}) = \int_{S^2} \sum_{\sigma=1}^{2} a_\sigma(\hat{\mathbf{k}}) \, \mathbf{u}_{\hat{\mathbf{k}},\sigma}(\mathbf{r}) \, d\Omega(\hat{\mathbf{k}}), \tag{2.289}$$

with amplitudes a_σ determined by $\mathcal{S}_\omega[\mathfrak{D}]$ through Eq. (2.282) in the rigged limit [2]. Rigged Hilbert space completion and measure normalization are developed in 3 Subsection 3.1.3 records the continuous angular target Eq. (2.289) compatible with far-zone reduction [6]. Each radiation mode $\mathbf{u}_n$ carries a far-zone fingerprint $\mathbf{f}_n(\theta, \phi)$—the angular pattern shape that mode n contributes on S^2— while $\mathbf{E}_0(\theta, \phi)$ is the measurable far-zone electric pattern a compact range records after probe correction and transformation [3]. Let $\mathcal{D}_{J,\mathrm{reg}} = 1$ and let $\mathbf{u}_n$ be radiation eigenmodes with far amplitudes $\mathbf{f}_n(\theta, \phi)$ defined through Eq. (2.246). Then the far pattern of $\mathcal{R}_\omega[\mathfrak{D}]$ satisfies

$$\mathbf{E}_0(\theta, \phi) = \sum_{n \in \mathcal{N}_{\mathrm{rad}}} \lambda_n \, c_n[\mathfrak{D}] \, \mathbf{f}_n(\theta, \phi), \tag{2.290}$$

with convergence in mean-square on S^2 when the modal sum converges in $\mathcal{U}$ [2]. Equation (2.290) is the upgrade of Eq. (2.6.2); explicit $\mathbf{f}_n$ in vector harmonic form belong to 3 [2]. Apply the far-field approximation of source integrals (Subsection 2.5.2) to each modal component $\lambda_n c_n \, \mathbf{u}_n$ and superpose using linearity Eq. (2.5) on outgoing radiative solutions. Coefficients match Eq. (2.282) because far-zone reduction depends linearly on $\mathcal{R}_\omega[\mathfrak{D}]$ through Eqs. (2.245)–(2.246) [2]. declared size and distance regime Eq. (2.270) must hold before interpreting Eq. (2.290) as transport data [6]. Source reconstruction is harder than forward radiation because inversion requires adjoint modes

$\widetilde{\mathbf{u}}_n$ that test whether a desired pattern $\mathbf{g}$ lies in the reachable range of $\mathcal{R}_\omega$ [2]. When $\mathcal{R}_\omega$ is not self-adjoint on $\mathcal{J}_\Omega$, define adjoint modes $\widetilde{\mathbf{u}}_n$ through the source pairing Eq. (2.55) such that

$$\langle \widetilde{\mathbf{u}}_m,\, \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(n)}] \rangle_{\mathcal{U}} = \lambda_n\, \delta_{mn}, \tag{2.291}$$

for normalized radiation eigenpairs [2]. Equation (2.291) is the field-side restatement of solvability relation Eq. (2.277): cokernel directions annihilated by Eq. (2.291) obstruct inversion $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{g}$ [2]. Complete adjoint-mode catalogs on exterior domains belong to Subsection 2.6.4 [6]. Modes with $\lambda_n = 0$ satisfy $P_n \mathcal{R}_\omega = \mathbf{0}$ but may have $P_n \mathcal{S}_\omega \neq \mathbf{0}$ when reactive content projects onto non-radiation subspaces Eq. (2.264) [2]. Evanescent and non-propagating sheets remain annihilated: spectral labels cannot convert evanescent storage into qualified far-zone radiation—declared size and distance regime and outgoing selection remain mandatory before transport claims [2].

$$P_n \circ \Pi_{\mathrm{ev}} = P_n \circ \Pi_{\mathrm{non}} = \mathbf{0} \quad \text{for all } n \in \mathcal{N}_{\mathrm{rad}}, \tag{2.292}$$

by Eq. (2.138) and modal definition Eq. (2.281) [2]. Spectral labels therefore do not override declared size and distance regime Eq. (2.139) [6]. The eigenbasis expansions and far-pattern coefficients in Eq. (2.290) require linear media at fixed ω, outgoing radiation condition, and completeness of $\{\mathbf{u}_n\}$ appropriate to the problem class [2]. Loss shifts λ_n into the complex plane without promoting evanescent storage to outgoing radiation via Eq. (2.292) [2]. Recoverable coefficients in Eq. (2.290) are further capped by finite observation rank Eq. (2.220) as established in Subsection 2.6.1 [6]; consult that subsection for σ_n cliffs rather than restating finite-rank principle here. As a worked example, consider a rectangular cavity fed by an electrically small probe at $f = 10\,\mathrm{GHz}$. The physical configuration is familiar from filter and resonator laboratories: a closed metal box with a coaxial penetration, interior standing waves visible in $|S_{11}|$ and in probe-coupled power at ports [2]. Interior modes produce a dense discrete spectrum in $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$—many field patterns exist inside the volume—but only propagating outgoing sectors contribute to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on a measurement aperture or radiating port [2]. Truncating Eq. (2.287) at $N = 20$ modes with error bounded by Eq. (2.83) yields a stable interior power budget on a port wattmeter: the expected observation is a clean $|S_{11}|$ trace with identifiable resonances and predictable coupling to the feed. The lesson sharpens by contrast. Apply the same truncated cavity sum to an exterior half-wave dipole in free space and sidelobe structure misaligns: the exterior problem requires the continuous angular measure Eq. (2.289), not a cavity eigenbasis [14]. On a compact range the dipole presents a smooth far-zone pattern cut; in the cavity it presents port physics entirely governed by enclosure modes. Conflating the two domain classes is a common numerical error in aperture codes and in student projects that reuse modal truncations without checking outgoing selection [6]. The significance is operational: spectral truncation is domain-specific, outgoing selection must precede modal counting, and a dense interior spectrum is not the same object as a recoverable far-zone pattern family [2]. As a worked example, a balanced loop pair with vanishing electric dipole moment is the cleanest laboratory reminder that near-field strength does not imply far-zone radiation [17]. The physical setup places two loops with opposing dipole moments so that radiation from electric dipole terms cancels on Γ_R while reactive $\mathbf{H}$ and $\mathbf{E}$ remain large within one wavelength of the structure. On a spherical near-field scan at $R = 0.4\lambda_0$, probes record strong tangential fields; on Γ_R at $R = 10\lambda_0$ the wattmeter reads negligible radiation. Coefficients c_n on radiating modes vanish ($\lambda_n = 0$), yet $\Pi_{\mathrm{ev}} \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}]$ may be substantial [2]. The observable behavior confuses teams who report patterns from $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ without Π_{rad}: the near field looks radiative locally while the far-zone signature is null. The interpretation is spectral, not a failure of probes. Membership in $\ker \mathcal{R}_\omega$ is invisible to far-zone pattern meters yet measurable in the reactive sector. For inverse synthesis, the consequence is equally concrete: two distinct feeds can match the same far pattern while differing in reactive

storage—a realizability question developed in 4 [2]. The example should be remembered as the archetype of radiation efficiency $\lambda_n = 0$ with non-trivial Maxwell storage. Subsection 2.6.2 fixed radiation eigenpairs, modal projectors, and the spectral expansion of $\mathcal{R}_\omega$ on $\mathcal{F}_{\mathrm{rad}}$. Recoverable rank and σ_n cliffs remain anchored in Subsection 2.6.1; the eigenbasis organizes radiative directions that finite apertures can in principle resolve once regime condition holds [2]. Rotational degeneracy and Fredholm Physically, solvability develop next on the same operator.

2.6.3 Rotational symmetry and spectral degeneracy

Rotational degeneracy is visible on every compact range before covariance formulas appear: rotating a horn reorients principal cuts while total radiated power stays fixed when probe and source frames rotate together [6]. The formal commutation relations below encode that laboratory symmetry on the radiation eigenbasis from Subsection 2.6.2.

Rotational symmetry on isotropic ranges organizes degenerate radiative directions into irreducible blocks; the SO(3) action below is the operator formalization of that laboratory fact. Let $R \in \mathrm{SO}(3)$, field rotation $\mathcal{D}(R)$ on qualified Maxwell pairs $\mathbf{u} = (\mathbf{E}, \mathbf{H}) \in \mathcal{F}_{\mathrm{SF}}$, and radiation map Eq. (2.274) on outgoing data [2]. Isotropic vacuum or statistically isotropic environments are assumed unless stated otherwise: constitutive tensors commute with R in the sense of section 1.3 [2]. Impressed sources transform as polar vectors under active rotation, $(\mathcal{A}_R \mathbf{J}_{\mathrm{imp}})(\mathbf{r}) = R\,\mathbf{J}_{\mathrm{imp}}(R^{-1}\mathbf{r})$, consistent with Eq. (1.70) [2]. declared size and distance regime Eq. (2.139) and condition for outward angular-momentum flux Eq. (2.27) remain mandatory before interpreting rotated reports as radiation data [6]. Rotation in the laboratory. Rotating a horn on a compact range is the everyday experiment that motivates covariance relations. Mount the horn on a rotary stage, record $\mathbf{E}_t$ on S_r at several azimuths, and superpose the cuts: total radiated power remains unchanged when probe and source frames rotate together, but the principal beam walks across the sphere as expected for a steerable feed [6]. Rotating a dish feed scans a different arc on the reflector surface; modal coefficients mix according to the same rules that rotate a measured $\mathbf{E}_t$ distribution on the rim [15]. Spinning a corrugated aperture distribution about its axis redistributes directional content within fixed ℓ blocks without creating new radiation unless the source itself changes [2]. Laboratory reports must distinguish active rotation of hardware from passive relabeling of a fixed field in a rotated coordinate chart; confusing the two conventions splits degeneracies artificially in processed $a_n(\hat{\mathbf{k}})$ data and can make a symmetric structure appear artificially asymmetric in synthesis charting [6]. The covariance identity below is the formal statement of what range technicians already enforce by co-rotating probes and references. Field rotation means physically reorienting the $\mathbf{E}$ and $\mathbf{H}$ vectors with the hardware: what changes is the direction of the pattern on S_r; what remains unchanged, when frames are handled consistently, is total radiated power and the eigenvalue ordering within degenerate blocks [6]. Define the Maxwell field representation

$$\mathcal{D}(R)\,\mathbf{u} = (\mathcal{A}_R\mathbf{E},\, \mathcal{A}_R\mathbf{H}), \qquad \mathbf{u} = (\mathbf{E}, \mathbf{H}) \in \mathcal{F}_\Omega, \tag{2.293}$$

using active rotation Eq. (1.70) on each component [2]. Eq. (2.293) specializes section 1.3 to phasor Maxwell pairs: passive relabeling Eq. (1.72) produces the same physics with rotated components related by Eq. (1.73) [2]. Group composition Eq. (1.76) yields $\mathcal{D}(R_1R_2) = \mathcal{D}(R_1)\mathcal{D}(R_2)$ on qualified field pairs [2]. Covariance of the radiation map. This equation is perhaps the most important symmetry statement in antenna engineering: active rotation of the feed in a co-rotated measurement frame leaves physical radiation content unchanged while reorienting modal coefficients through the representation $\mathcal{D}(R)$ [6]. When sources and observation frames are co-rotated, the radiation chain must commute with rotation up to the field representation $\mathcal{D}(R)$: rotating the feed and evaluating

on a co-rotated far-zone frame must yield the same physics as evaluating first and then rotating the pattern [2]. That requirement is visible on every compact range where probe calibration tracks mechanical azimuth. Formally,

$$\mathcal{R}_\omega\big[\mathcal{A}_R \mathbf{J}_{\mathrm{imp}}\big] = \mathcal{D}(R)\,\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}], \tag{2.294}$$

on physically realizable $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ with outgoing radiation condition and isotropic constitutive response [2]. Equation (2.294) develops Eq. Eq. (2.278) and aligns with covariance test Eq. (2.43) for $\mathbf{a} = \mathbf{0}$ [2]. Residual $\mathcal{R}_\mathcal{F}$ from Eq. (2.36) diagnoses broken commutation in discretizations [6]. Rotating $\mathbf{J}_{\mathrm{imp}}$ while evaluating flux on an unrotated $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ violates origin and covariance criteria from section 1.76 even when Eq. (2.294) holds for power alone [2]. Figure 2.10 states rotational

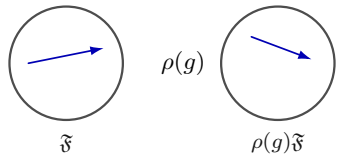


Figure 2.10: Rotational covariance Eq. (2.294): active source rotation commutes with $\mathcal{R}_\omega$ up to field representation $\mathcal{D}(R)$ Eq. (2.293).

covariance Eq. (2.294): active rotation of impressed sources commutes with $\mathcal{R}_\omega$ up to the field representation $\mathcal{D}(R)$ of Eq. (2.293) [2].

(2.294) in the same layout as Figure 2.0.3: geometry and components must be rotated consistently before modal coefficients are compared [2]. Under rotation, modal coefficients mix through the matrix $D^{(\mathrm{rad})}_{nm}(R)$: an engineer sees principal cuts walk across the sphere while total power on Γ_R stays fixed when probe and source frames rotate together [6]. Let $\{\mathbf{u}_n\}_{n\in\mathcal{N}_{\mathrm{rad}}}$ be the orthonormal radiation basis developed above and $c_n[\mathfrak{D}]$ source amplitudes Eq. (2.282). Under active rotation of impressed data,

$$c_n[\mathcal{A}_R\mathfrak{D}] = \sum_{m\in\mathcal{N}_{\mathrm{rad}}} D^{(\mathrm{rad})}_{nm}(R)\, c_m[\mathfrak{D}], \tag{2.295}$$

whenever Eq. (2.294) holds and the family of radiating modes is closed under $\mathcal{D}(R)$,

$$\mathcal{D}(R)\,\mathbf{u}_m = \sum_{n\in\mathcal{N}_{\mathrm{rad}}} D^{(\mathrm{rad})}_{nm}(R)\,\mathbf{u}_n. \tag{2.296}$$

Eqs. (2.295)–(2.296) are the spectral restatement of vector covariance Eq. (1.72) on $\mathcal{F}_{\mathrm{rad}}$ [2]. Matrix elements $D^{(\mathrm{rad})}_{nm}(R)$ are not tabulated here; Chapter 3 (*Angular Momentum Operators and SO(3)*) supplies them on vector spherical harmonic bases [6]. Eigenvalues λ_n in Eq. (2.280) are unchanged within a degenerate block because $\mathcal{R}_\omega$ commutes with $\mathcal{D}(R)$ on that block [2]. Repeated eigenvalues occur because rotational symmetry makes distinct angular orientations physically equivalent within an irreducible block: full SO(3) symmetry produces $(2\ell{+}1)$-fold degeneracy, and symmetry breaking—anisotropic materials, finite facets, or broken ground-plane parity—splits those blocks into distinct λ_n [2]. Suppose Eq. (2.294) holds for all $R \in \mathrm{SO}(3)$ in a compact subgroup $G \subseteq \mathrm{SO}(3)$ leaving the problem geometry invariant. Then $\mathcal{R}_\omega$ and $\mathcal{D}(R)$ generate a commuting family on $\mathcal{F}_{\mathrm{rad}}$: for each $R \in G$,

$$\mathcal{D}(R)\,\mathcal{R}_\omega = \mathcal{R}_\omega\,\mathcal{D}(R) \quad \text{on physically realizable sources,} \tag{2.297}$$

and eigenspaces of $\mathcal{R}_\omega\mathcal{S}_\omega$ decompose into irreducible G-representations. If an irreducible block carries angular index ℓ, its degeneracy multiplicity satisfies

$$\dim \mathcal{H}_\ell = 2\ell + 1 \quad \text{for full SO(3) completion on } S^2, \tag{2.298}$$

with $\mathcal{H}_\ell$ the ℓ-indexed radiative subspace [2]. Equation (2.298) explains repeated eigenvalues in Eq. (2.283) without duplicating mode physics. Eq. (2.297) follows by applying Eq. (2.294) after Eq. (2.285) and using linearity of $\mathcal{S}_\omega$ [2]. Standard spectral theory for commuting unitary or orthogonal representations implies simultaneous block diagonalization of $\mathcal{R}_\omega$ and $\mathcal{D}(R)$ on invariant subspaces [2]. For full rotational symmetry on angular patterns, irreducible representations of SO(3) on $L^2(S^2)$ have dimension $2\ell+1$; Eq. (2.298) records that count on $\mathcal{F}_{\mathrm{rad}}$ once outgoing transversality and Eq. (2.138) are imposed [14]. Detailed construction of $\mathcal{H}_\ell$ is developed in 3 [6]. Total radiated power P_{rad} is the scalar quantity a wattmeter or sphere integral reports on Γ_R; it cannot change under rigid rotation when frames are co-rotated [5]. Modal coefficients c_n redistribute directional content, while radiation eigenvalues λ_n set how efficiently each mode reaches the meter [6]. Rigid rotation cannot change total radiated power when meters act on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ with co-rotated frames. Using Eq. (2.252) and Eq. (2.34),

$$P_{\mathrm{rad}}[\mathcal{A}_R\mathfrak{D}] = P_{\mathrm{rad}}[\mathfrak{D}], \qquad P_{\mathrm{rad}}[\mathfrak{D}] = \sum_{n\in\mathcal{N}_{\mathrm{rad}}} |\lambda_n|^2\,|c_n[\mathfrak{D}]|^2 \quad \text{when modes are orthogonal and normalized,} \tag{2.299}$$

for lossless outgoing radiation [2]. Equation (2.299) is the modal restatement of scalar invariance Eq. (2.35) on integrated power: rotated coefficients redistribute directional content but preserve the scalar budget when $\mathcal{D}_{J,\mathrm{reg}} = 1$ [2]. Net angular-flux magnitude obeys Eq. (1.41) on the same qualified subspace Active rotation reorients sources and patterns in space; passive rotation re-expresses fixed physics in a rotated chart Eqs. (1.70)–(1.72) [2]. Modal coefficients transform as Eq. (2.295) under active action, while passive relabeling conjugates the same $D^{(\mathrm{rad})}_{nm}(R)$ through Eq. (1.73) without altering Eq. (2.299) [2]. Confusing the two conventions produces spurious degeneracy splitting in measured $a_n(\hat{\mathbf{k}})$ from Eq. (2.290) even when rotational symmetry of the spectrum applies [2]. Laboratory rotation of an antenna rotates measured coefficients through Eq. (2.295); radiated power on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ remains invariant in the far zone [6]. Origin translation $\mathbf{a}$ shifts torque and angular-flux reports through section 1.76 while leaving far-zone radiative power invariant when $\mathcal{D}_{J,\mathrm{reg}} = 1$ [2]. At the spectral level, origin change mixes modal coefficients within fixed ℓ blocks through addition of lower-ℓ content from shifted multipole expansions; it does not create new $\lambda_n \neq 0$ radiation unless sources change [14]. The rotation analysis above therefore separates *rotational* degeneracy from *translational* bookkeeping in section 1.76; intrinsic angular observables use Eq. (2.37) by reference [2]. Let $G_0 \subset \mathrm{SO}(3)$ be the exact symmetry of a source–boundary configuration and let $\mathcal{H}_\ell$ be an irreducible block from rotational degeneracy of the spectrum. Any symmetry-breaking perturbation (reduced G_0, anisotropic material, or finite facetization) splits Eq. (2.298) into smaller blocks with distinct λ_n within the same ℓ sector [2]. The splitting order is controlled by representation coupling rules imported from section 1.3 on tensor perturbations; explicit Clebsch–Gordan content belongs to 3 Subsection 3.2.5 [2]. When G_0 is proper, $\mathcal{D}(R)$ block-diagonalizes with fixed λ_n on each irreducible piece. Perturbations that fail to commute with all $R \in G_0$ remove simultaneous eigenstructure; standard degenerate perturbation theory applies to the restricted radiation operator on $\mathcal{H}_\ell$ [2]. Rotational degeneracy of the spectrum requires isotropic or symmetry-matched media, outgoing radiation condition, and radiating-mode families closed under $\mathcal{D}(R)$ [2]. Anisotropic crystals reduce G and split Eq. (2.298) when symmetry is broken [2]. Observation rank Eq. (2.220) resolves only a finite subset of each degenerate block even when continuous symmetry is exact [6]. declared size and distance regime Eq. (2.270) still governs whether rotated near fields may be Physically, reported as transport [2].

Uniqueness of the radiative branch therefore fixes which admissible solution corresponds to launched power rather than standing-wave contamination [2].

2.6.4 Fredholm structure and pattern solvability

Pattern synthesis and source inversion are the inverse problems every antenna laboratory eventually confronts. One specifies a desired far-zone amplitude distribution and asks which feed currents will produce it; or one measures near fields on a scanner and asks which sources could have generated them [6]. When these inversions fail, the failure is rarely mysterious in hindsight: non-radiating components lie in $\ker \mathcal{R}_\omega$—the set of admissible impressed currents annihilated by radiation—and leave no far-zone signature, so no finite measurement set can resolve them without additional excitation or constraints [17]. Recoverable rank from Subsection 2.6.1 caps the truncated modal system even when a continuous target is mathematically consistent [3]. Fredholm theory names kernel, cokernel, and index after the eigenbasis and rotational structure above are in place [2].

Assumptions. Fix $\omega > 0$, bounded or exterior problem class with outgoing radiation condition on open problems, physically realizable $\mathcal{J}_\Omega$ Eq. (2.48), radiation map Eq. (2.274), and the radiation eigenbasis developed above [2]. Source pairing uses dual space Eq. (2.55); field pairing uses Eq. (2.25) [2]. Compact perturbations are taken in operator norm on $\mathcal{J}_\Omega$ for sources with compact support Ω_s [6]. declared size and distance regime Eq. (2.139) governs whether recovered $\mathbf{g}$ may be reported as transport data [2]. The *radiation kernel* $\ker \mathcal{R}_\omega$ collects impressed currents invisible to qualified far-zone meters. The *cokernel* $\operatorname{coker} \mathcal{R}_\omega = \mathcal{F}_{\mathrm{rad}}/\operatorname{ran} \mathcal{R}_\omega$ labels far-zone pattern targets that no admissible source can realize—equivalently, directions orthogonal to every radiative image under the adjoint pairing [2]. The *Fredholm index* $\operatorname{ind} \mathcal{R}_\omega = \dim \ker \mathcal{R}_\omega - \dim \operatorname{coker} \mathcal{R}_\omega$ counts the net imbalance between invisible source degrees and unreachable pattern obstructions; it is stable under the compact perturbations recorded below [6]. In the laboratory, kernel membership explains why two distinct feeds can match the same far pattern; cokernel membership explains why a synthesis target refuses to converge no matter how many ports are tuned [2]. Source inversion is difficult because radiation smooths fine source detail: the far-zone map retains only radiative content and discards non-radiating degrees in $\ker \mathcal{R}_\omega$ [17]. The radiation map inherits the Green-function split of section 2.3, a leading outgoing propagation part $\mathcal{R}_{0,\omega}$ plus a compact smoothing part K_{rad} at finite separation on bounded source classes [2]. Physically, $\mathcal{R}_{0,\omega}$ is the dominant far-zone propagation law and K_{rad} collects the compact corrections that smooth fine source structure into resolvable pattern content [6]. That split is the operator-side restatement of the range observation that far-zone data smooth fine source detail. Formally,

$$\mathcal{R}_\omega = \mathcal{R}_{0,\omega} + K_{\mathrm{rad}}, \qquad K_{\mathrm{rad}} \text{ compact on } \mathcal{J}_\Omega, \tag{2.300}$$

where $\mathcal{R}_{0,\omega}$ captures leading outgoing singular propagation and K_{rad} collects compact smoothing at finite separation on bounded source classes [2]. Eq. (2.300) specializes Eq. (2.85) to the radiation image Eq. (2.238) after sections 2.3–2.5 [2]. On exterior domains with compact Ω_s, K_{rad} is compact while $\mathcal{R}_{0,\omega}$ carries the non-compact far-zone propagation; on bounded cavities, both parts reduce to discrete modal books Eq. (2.287) [6]. The Fredholm index counts the net imbalance between invisible source degrees ($\ker \mathcal{R}_\omega$) and unreachable pattern targets ($\operatorname{coker} \mathcal{R}_\omega$). On the range, kernel membership means different feeds may produce identical far patterns; cokernel membership means a synthesis target lies outside the reachable subspace no matter how many ports are tuned [2]. For closed operators with finite-dimensional kernel and cokernel, define the Fredholm index

$$\operatorname{ind} \mathcal{R}_\omega = \dim \ker \mathcal{R}_\omega - \dim \operatorname{coker} \mathcal{R}_\omega, \tag{2.301}$$

with $\operatorname{coker} \mathcal{R}_\omega = \mathcal{F}_{\mathrm{rad}}/\operatorname{ran} \mathcal{R}_\omega$ on the declared quotient by outgoing radiation condition [2]. Equation (2.301) counts independent non-radiating source degrees Eq. (2.217) minus obstructions to reaching a target far pattern $\mathbf{g} \in \mathcal{F}_{\mathrm{rad}}$ [2]. Index is stable under compact perturbations in Eq. (2.300) [6].

Observation rank cap Eq. (2.220) is independent of $\operatorname{ind}\mathcal{R}_\omega$ but composes with it in inversion budgets [2].

(2.84) [2]. The Fredholm alternative states that a target far-zone field is synthesizable only if it carries no component along cokernel directions invisible to the adjoint radiation map [2]. Consequence: a measured pattern residual that is orthogonal to every reachable image is not a calibration fault—it signals membership in $\operatorname{coker}\mathcal{R}_\omega$ or in a geometrically forbidden sector from Subsection 2.6.5 [6]. The forward radiation equation $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{g}$ is solvable iff the target $\mathbf{g} \in \mathcal{F}_{\mathrm{rad}}$ is orthogonal to the cokernel of the adjoint map on declared pairings:

$$\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{g} \text{ solvable} \iff \langle \mathbf{g},\, \mathbf{w}\rangle_{\mathcal{U}} = 0 \quad \forall\ \mathbf{w} \in \ker \mathcal{R}_\omega^\dagger, \tag{2.302}$$

with $\mathcal{R}_\omega^\dagger$ the adjoint on $(\mathcal{J}_\Omega, \mathcal{F}_{\mathrm{rad}})$ induced by Eqs. (2.55) and (2.25) [2]. Eq. (2.302) develops Eq. (2.277) and aligns with biorthogonality Eq. (2.291) on modal adjoints [2]. When $\mathbf{g}$ arises from far-pattern data, solvability is further capped by Eq. (2.220) even if Eq. (2.302) holds in the continuous model [6]. Let $\mathcal{R}_\omega$ satisfy Eq. (2.300) on $\mathcal{J}_\Omega$ with $\mathcal{R}_{0,\omega}$ Fredholm of index zero on the outgoing quotient and K_{rad} compact. Then $\mathcal{R}_\omega$ is Fredholm,

$$\operatorname{ind}\mathcal{R}_\omega = \operatorname{ind}\mathcal{R}_{0,\omega}, \tag{2.303}$$

and Eq. (2.302) holds. For finite-rank observation $\mathcal{O}_{\mathrm{meas}} : \mathcal{F}_{\mathrm{rad}} \to \mathbb{C}^N$,

$$\mathcal{O}_{\mathrm{meas}}\mathcal{R}_\omega \text{ is Fredholm on } \mathcal{J}_\Omega \iff \mathcal{R}_\omega \text{ Fredholm and } \operatorname{rank}(\mathcal{O}_{\mathrm{meas}}) = N, \tag{2.304}$$

with rank bound Eq. (2.84) Compact perturbation stability of Fredholm property and index follows from standard operator theory applied to Eq. (2.300); Subsection 2.1.3 recorded the embedding inputs without computing index here [2]. Solvability criterion Eq. (2.302) is the Fredholm alternative for $\mathcal{R}_\omega$. Composition with finite-rank $\mathcal{O}_{\mathrm{meas}}$ preserves Fredholm class and adjusts index by at most N on the range side, yielding Eq. (2.304) and Eq. (2.84) [2]. The modal fredholm reduction proceeds as follows: on the orthonormal radiation eigenbasis developed above,

$$\mathcal{R}_\omega = \sum_{n \in \mathcal{N}_{\mathrm{rad}}} \lambda_n\, P_n \circ \mathcal{S}_\omega, \qquad \lambda_n \neq 0 \iff \text{mode } n \text{ radiating}, \tag{2.305}$$

so solvability of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{g}$ reduces to matching coefficients $c_n[\mathfrak{D}]$ Eq. (2.282) on modes with $\lambda_n \neq 0$ [2]. Modes with $\lambda_n = 0$ lie in $\ker\mathcal{R}_\omega$ at the source level Eq. (2.217); modes with $\mathbf{g}$ unsupported on $\operatorname{ran}\mathcal{R}_\omega$ generate cokernel obstructions in Eq. (2.302) [2]. Truncation to $\mathcal{R}_\omega^{(N)}$ from Eq. (2.82) yields approximate solvability with error Eq. (2.83) controlled by $\sigma_{N+1}(K_{\mathrm{rad}})$ [6]. Let $\mathbf{g} \in \operatorname{ran}\mathcal{R}_\omega$ and expand $\mathbf{g} = \sum_n g_n\, \mathbf{u}_n$ with $\{\mathbf{u}_n\}$ from Subsection 2.6.2. Then Eq. (2.302) holds if and only if $g_n = 0$ whenever $\lambda_n = 0$. A particular solution is determined by

$$c_n = \frac{g_n}{\lambda_n} \quad \text{for } \lambda_n \neq 0, \tag{2.306}$$

with arbitrary coefficients on $\ker\mathcal{R}_\omega$ not fixed by far-zone data [2]. Eq. (2.306) is the spectral form of Eq. (2.219) on the radiative range; Tikhonov and regularized inversions from Subsection 2.4.4 damp ill-conditioned λ_n without altering Fredholm index [2]. Substitute Eq. (2.284) into $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{g}$ and use orthogonality of $\{\mathbf{u}_n\}$ in $\mathcal{U}$ [2]. Modes with $\lambda_n = 0$ do not appear in the range; modes with $\lambda_n \neq 0$ invert mode-wise yielding Eq. (2.306). Kernel directions remain free because $\mathcal{R}_\omega$ is not injective on $\mathcal{J}_\Omega$ Eq. (2.217) [6]. Cokernel elements pair with unreachable far-zone content: if $\mathbf{w} \in \ker\mathcal{R}_\omega^\dagger$, then $\langle \mathbf{g}, \mathbf{w}\rangle_{\mathcal{U}} = 0$ is required for solvability Eq. (2.302) [2]. On exterior domains, continuous angular spectra Eq. (2.289) imply infinite-dimensional cokernel constraints in principle; finite apertures

replace them with rank-N projections through Eq. (2.304) [6]. Far-pattern coefficients from the eigenbasis express $\mathbf{g}$ through $\mathbf{f}_n$; solvability requires consistency of measured coefficients with Eq. (2.306) on radiating modes only [2]. On the outgoing branch, solvability of $\mathcal{L}_\omega[\mathbf{u}_\omega] = \mathbf{s}_\omega$ and solvability of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{g}$ are linked through Eq. (2.239): forward field construction is single-valued on $\mathcal{F}_{\mathrm{SF}}$ while source inversion remains rank-deficient [2]. (2.86) is not identical to $\operatorname{ind}\mathcal{R}_\omega$; outgoing selection removes non-radiative degrees before pattern inversion [2]. The Fredholm analysis above treats $\mathcal{R}_\omega$ as the primary object for far-zone inverse problems [6]. Fredholm structure on $\mathcal{R}_\omega$ requires linear media, fixed ω, outgoing radiation condition, and compactness of K_{rad} on the declared source class [2]. Resonant cavities violate invertibility of $\mathcal{L}_0$ unless quotiented Eq. (2.86); radiation index must be computed on the outgoing subspace after Eq. (2.202) [2]. Loss regularizes small λ_n but does not remove $\ker\mathcal{R}_\omega$ Eq. (2.217) [6]. declared size and distance regime Eq. (2.270) remains mandatory before transport claims on recovered $\mathbf{g}$ [2]. As a worked example, a spherical near-field scanner with $N = 50$ probe channels is a concrete inversion budget. The physical configuration places fifty probes on a sphere of radius $R \approx 2\lambda_0$ around a compact horn feed; each channel records tangential $\mathbf{E}$ or $\mathbf{H}$ depending on probe design [3]. After transformation to far-zone coefficients, the composition $\mathcal{O}_{\mathrm{meas}}\mathcal{R}_\omega$ has rank at most 50 Eq. (2.84) [2]. The observable behavior is instructive: the first few reconstructed modes match direct far-field cuts taken on the compact range, while coefficients beyond the fiftieth singular direction track noise rather than new beam structure. Even when Eq. (2.302) holds for a continuous target $\mathbf{g}$, discrete recovery satisfies only a truncated modal system; tail error is bounded by Eq. (2.83) with the first neglected singular value of the measurement composition cited from Subsection 2.6.1 [6]. The lesson for laboratory planning is direct: adding probes without enlarging scan radius or improving SNR cannot recover modes the geometry has already annihilated. Scanner rank is a hard cap that composes with, but does not replace, Fredholm obstructions from $\ker\mathcal{R}_\omega$ [2]. As a worked example, suppose a synthesis target $\mathbf{g} \neq \mathbf{0}$ specifies a far-zone pattern that is orthogonal to every radiating mode reachable from the declared source class—for example, a forbidden parity combination above a ground plane when only even sectors are physically accessible [15]. No amount of feed optimization will satisfy forward radiation: the obstruction appears as inconsistent far-zone coefficients in Eq. (2.302), not as slow convergence of a numerical solver [6]. By contrast, when $\mathbf{g} = \mathbf{0}$, the equation is always solvable with $\mathbf{J}_{\mathrm{imp}} \in \ker\mathcal{R}_\omega$ [2]; canonical non-radiating families in section 2.7.1 (when their additional outgoing-cancellation conditions are verified) show that such sources may be strongly measurable in the near zone yet invisible on Γ_R. Mode-wise invertibility on the radiative range detects obstructions through g_n on $\lambda_n = 0$ sectors [2]. Geometric sector constraints on recoverable spectra are developed next; complete non-radiating classification belongs to section 2.7.1 [2]. Subsection 2.6.4 closed the Fredholm account:

compact perturbation, kernel and cokernel pairings, solvability criterion, and modal reduction on the radiation eigenbasis. Kernel membership explains invisible source degrees; cokernel membership explains unreachable pattern targets; recoverable rank from Subsection 2.6.1 caps what Physically, finite instruments can resolve even when continuous solvability holds [2].

2.6.5 Geometric constraints on radiative spectra

Geometry can forbid a modal sector before instrumentation or inversion enter the story. On a reflector compact range, a slot above a ground plane may exhibit a persistent pattern null in a parity-forbidden sector: the null survives probe upgrades and SNR improvements because the symmetry class, not receiver noise, annihilates that angular content [15]. An electrically small feed above the same plane supports only low-order multipole radiation; formal harmonic expansions may list high-ℓ indices that the support condition suppresses long before $\mathcal{O}_{\mathrm{meas}}$ enters [14]. The geometric operator below encodes that selection on modal amplitudes after eigenmodes and Fredholm structure

are known [2].

Finite electrical size suppresses high-order radiating modes long before the measurement operator enters; the geometric filter below encodes that support constraint on modal amplitudes. Let $\Omega_s = \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$, outer boundary Γ, piecewise material data (ε, μ), outgoing radiation condition, and radiation eigenbasis $\{\mathbf{u}_n, \mathbf{f}_n\}_{n \in \mathcal{N}_{\mathrm{rad}}}$ from the developments above [2]. Modal coefficients a_n in Eq. (2.290) are identified with $\lambda_n c_n[\mathfrak{D}]$ on radiating modes [2]. Geometric symmetry group $G_{\mathrm{geo}} \subseteq \mathrm{SO}(3) \times \{+\}$ acts on sources, boundaries, and constitutive tensors as in section 1.3 by reference [2]. declared size and distance regime Eq. (2.139) remains mandatory before transport claims on any surviving modal sector [6]. When symmetry and boundary class are fixed, modal amplitudes do not vary independently: some linear combinations are forced to zero by image parity or by support even when the radiation map would otherwise allow them [2]. The matrix Γ_{nm} in Eq. (2.307) encodes how a symmetry operation on the physical configuration maps one modal sector into another; a forbidden sector is one whose unit amplitude vector lies in the kernel of $\mathcal{C}_{\mathrm{geo}}$ [2]. Define a bounded operator $\mathcal{C}_{\mathrm{geo}}$ on modal amplitudes by

$$\left(\mathcal{C}_{\mathrm{geo}}\mathbf{a}\right)_n = \sum_{m \in \mathcal{N}_{\mathrm{rad}}} \Gamma_{nm}\, a_m, \qquad \mathbf{a} = \{a_m\}_{m \in \mathcal{N}_{\mathrm{rad}}}, \tag{2.307}$$

where Γ_{nm} encodes how geometry and boundary class map modal sectors under G_{geo} [2]. Equation (2.307) is the constructive account of realization of Eq. (2.307): when $\mathcal{C}_{\mathrm{geo}}\mathbf{e}_n = \mathbf{0}$ for a unit vector $\mathbf{e}_n$, mode n is *geometrically forbidden* even if $\lambda_n \neq 0$ on the radiation map. Geometric symmetry can annihilate modal sectors before instrumentation enters: a forbidden mode has zero coupling to physically realizable sources even when formal eigenvalues are non-zero [2]. Broken symmetry restores coupling—a practical lever in reflector and array design [15]. Eq. (2.302) holds for other modes [2]. Rotation commutation Eq. (2.294) acts on $\mathbf{f}_n$; $\mathcal{C}_{\mathrm{geo}}$ acts on physically accessible *sectors* after symmetry reduction [6]. Compact support $\Omega_s \subset \mathbb{R}^3$ implies entire-function structure in spatial wavenumber: for each outgoing mode $\mathbf{f}_n(\theta, \phi)$,

$$a_n(\mathbf{J}_{\mathrm{imp}}) = 0 \text{ unless } \mathbf{f}_n \text{ has non-negligible overlap with } \mathcal{F}_\omega[\mathcal{J}_{\Omega_s}], \tag{2.308}$$

with $\mathcal{J}_{\Omega_s}$ the restriction of admissible sources to Ω_s [2]. Eq. (2.308) is a support-induced geometric filter distinct from Fredholm cokernel constraints Eq. (2.302): electrically small Ω_s suppresses high-ℓ sectors in Eq. (2.290) through $(k_0 a)^{2\ell}$ scaling established in Chapter 3 (*Angular Momentum Operators and SO(3)*) [14]. Rank cap Eq. (2.220) limits measurable a_n separately from support annihilation [6]. Perfect conductors, symmetry planes, and equivalent image configurations impose linear relations among modal amplitudes:

$$\sum_{m \in \mathcal{N}_{\mathrm{rad}}} M_{nm}(\Gamma, \mathcal{B}_\Gamma)\, a_m = 0 \quad \Longrightarrow \quad a_n = 0 \text{ for forbidden parity sectors,} \tag{2.309}$$

with M_{nm} fixed by tangential boundary class Eq. (1.20) and image rules on outgoing Green branches Eq. (2.199) [2]. Equation (2.309) specializes degeneracy lifting under broken symmetry (Subsection 2.6.3): a ground plane replaces full SO(3) by a reflection group and annihilates odd or even combinations per declared image parity [2]. Sommerfeld outgoing radiation condition from section 2.4 is presupposed; incoming image branches are excluded from $\mathcal{F}_{\mathrm{rad}}$ [6]. When G_{geo} contains mirror reflection σ, modal sectors split into parity classes $\Pi_\pm$ with

$$a_n = 0 \text{ for all } n \in \mathcal{N}_{\mathrm{rad}} \text{ with } \Pi_\pm \mathbf{f}_n = \mp \mathbf{f}_n \text{ and forbidden parity,} \tag{2.310}$$

as in the ground-plane illustration of Subsection 2.6.3 [2]. Eq. (2.310) explains pattern nulls that are not explained by $\ker \mathcal{O}_{\mathrm{meas}}$ or $\ker \mathcal{R}_\omega$ alone [2]. Slot and patch symmetries reduce Eq. (2.310) to discrete sector labels on $\mathcal{N}_{\mathrm{rad}}$ without re-tabulating dyads from Subsection 2.3.1 [6].

Geometric, Fredholm, and observation constraints multiply:

$$\mathcal{N}_{\text{recoverable}} \subseteq \mathcal{N}_{\text{acc}} \cap \{n : \text{ resolvable by } \mathcal{O}_{\text{meas}}\} \cap \{n : \lambda_n \neq 0\}, \tag{2.311}$$

with the last two factors from Subsection 2.6.4 and Eq. (2.220) [2]. Equation (2.311) prevents attributing a pattern null solely to finite aperture when Eq. (2.307) already forbids the sector [2]. Inverse source Eq. (2.219) on $\mathcal{N}_{\text{recoverable}}$ remains rank-deficient through $\ker \mathcal{R}_\omega$ Eq. (2.217) even when geometry is exact [6]. Geometric constraints on radiative spectra assume linear media, fixed ω, and symmetry of $(\Omega_s, \Gamma, \varepsilon, \mu)$ under declared G_{geo} [2]. Weak symmetry breaking perturbs Eq. (2.9.3) through lifted degeneracy rather than strict zero [2]. Loss and anisotropy modify Γ_{nm} and may reopen partially forbidden sectors at finite amplitude [2]; declared size and distance regime Eq. (2.270) is unchanged, for geometric accessibility does not promote reactive near fields to radiation [6]. Eigenmode ordering, rotational degeneracy, Fredholm solvability, and geometric filters now compose the spectral chain of Eq. (2.273) on the integral representations from section 2.5 [2]. The thread is laboratory-first: finite aperture and σ_n decay fix recoverable rank once in Subsection 2.6.1; radiation eigenvalues λ_n of $\mathcal{R}_\omega$ then organize dominant directions; rotation, Fredholm structure, and geometry explain what inversion can and cannot recover. section 2.7 completes non-radiating null spaces at the source level; 3 supplies explicit harmonic bases; 4 addresses which eigenstructures finite sources can realize after truncation [6].

2.7 Non-Radiating Sources and Null Spaces

Spectral theory in section 2.6 ranked radiative directions and recoverable rank. section 2.7 addresses the complementary laboratory question: *How can current exist without observable radiation?* [17]

Laboratory signatures recur in three configurations before any kernel is displayed. A balanced wire loop in vacuum, driven at fixed ω with vanishing net electric dipole moment, stores reactive $\mathbf{E}$ and $\mathbf{H}$ within λ_0 of the conductor while a qualified wattmeter on Γ_R at $R \gg \lambda_0$ records zero outgoing transverse flux once Π_{rad} selection is enforced [2]. A counter-directed dipole pair separated by $d \ll \lambda_0$ can annihilate leading $1/r$ transverse fields on S^2 while reactive overlap remains in the gap [14]. A waveguide section below cutoff carries standing reactive fields at ports yet contributes no propagating sheet to exterior flux meters [2]. In each case Maxwell permits substantial local structure while $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}] = \mathbf{0}$: rich near-zone data, null far-zone radiation—not a solver artifact [14]. The *radiation kernel* collects every admissible impressed current whose outgoing radiative image vanishes: currents that reorganize near-zone energy without qualified far-zone flux [17]. Near-field scanners sample $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]$ on enclosing shells; far-zone flux meters and angular integrators read only after Π_{rad} has removed non-radiating content [3]. A source invisible to the latter may remain vivid to the former—the central diagnostic that separates reactive storage from outgoing radiation. Measurements that report singular values without kernel awareness merge synthesis with invisible modes [14]. By the *radiation null space* $\ker \mathcal{R}_\omega$ we mean precisely this kernel: admissible impressed currents whose radiative image vanishes under the outgoing map $\mathcal{R}_\omega = \Pi_{\text{rad}} \circ \mathcal{S}_\omega$—physically realizable sources that store energy locally yet export no qualified far-zone power [17]. Membership in $\ker \mathcal{R}_\omega$ is an radiation statement; it does not require $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}] = \mathbf{0}$. The balanced loop above illustrates electric-dipole cancellation per Eq. (2.329); $\ker \mathcal{R}_\omega$ membership requires verified complete outgoing cancellation under the declared family while $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}^{(\text{loop})}]$ remains measurable on near probes [17]. Non-radiating sources are the central electromagnetic proof that existence does not imply observability. A current distribution may generate substantial near-zone $\mathbf{E}$ and $\mathbf{H}$ while radiating zero far-zone power: $\mathcal{R}_\omega$ is not injective, and $\ker \mathcal{R}_\omega$ is generally infinite-dimensional on open

domains. Different impressed currents may therefore produce identical radiative images, angular spectra, and reciprocity readouts while differing in reactive storage and evanescent overlap. Hidden electromagnetic structure is not a modelling defect but a consequence of Maxwell propagation from compact sources through outgoing selection to finite instruments [2]. The absence of radiated signatures does not imply the absence of electromagnetic structure; it identifies source configurations whose field content remains inaccessible to far-zone observation. section 2.6 closed the spectral chain Eq. (2.273): radiation eigenmodes, rotational degeneracy, Fredholm solvability, and geometric filters on admissible modal sectors [2]. That program organized $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ in modal coordinates and explained which far-zone patterns are reachable, degenerate, or symmetry-forbidden. section 2.7 addresses the complementary *source-side* question named above: which impressed currents produce no radiative radiate at all, how those currents relate to evanescent and non-propagating field content, and what null-space structure implies for inverse problems and angular spectra [2]. Eq. (2.217) from Subsection 2.4.4 named $\ker \mathcal{R}_\omega$; the subsections below develop constructive catalog, canonical non-radiating configurations, and reconstructability limits without re-opening Green construction, Sommerfeld completion, or radiation integrals from sections 2.3–2.5 [2]. Reciprocity constraints on coupled modes belong to section 2.8; plane-wave angular-spectrum action to section 2.9; explicit harmonic bases to 3 [6]. Nullity, $\mathcal{E}_{\mathrm{acc}}^{(N)}$, and realizability. Three deficits are often merged in inverse problems and structured-beam reporting [17]. *Nullity* is source-side membership in $\ker \mathcal{R}_\omega$: far data cannot detect additions satisfying Eq. (2.349) [20]. *probe rank* is field-side: which directions in the angular spectrum survive geometric, transport, and declared size and distance regimes even when $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \neq \mathbf{0}$ [6]. $\mathcal{S}_{\mathrm{real}}$—whether a target pattern is $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ for some admissible $\mathbf{J}_{\mathrm{imp}}$—is not decided by rank-$N$ recovery alone; 4 develops synthesizable proofs while section 2.7 supplies the forward null-space catalog that constrains every claim [21]. A source may store reactive energy nearby yet radiation no far-zone power—the radiation map annihilates whole families of impressed currents. Field-level uniqueness after outgoing radiation condition (exterior radiative uniqueness (Subsection 2.4.3)) does not imply source identifiability: many admissible $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ can share the same $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ through Eq. (2.217) [2]. Spectral mode $\lambda_n = 0$ in Eq. (2.276) marks a radiative eigen-direction with vanishing radiation, yet $\ker \mathcal{R}_\omega$ is generally infinite-dimensional on open domains and is not exhausted by finitely many modal labels [6]. Using cavity modal books or truncated harmonic sums to *approximate* $\ker \mathcal{R}_\omega$ misstates identifiability even when outgoing source-to-field integrals from Subsection 2.5.1 are evaluated exactly [14]. section 2.7 separates operator null spaces, field-side non-radiative components Eqs. (2.265)–(2.266), and observation null spaces Eq. (2.80), then forwards reciprocity and angular-spectrum filters to sections 2.8–2.9 [2]. Let $\Omega_s = \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$ be compact, $\omega > 0$ fixed, and outgoing field pairs lie in $\mathcal{F}_{\mathrm{SF}}$ from Eq. (2.56). section 2.7 organizes analysis in four stages:

$$\ker \mathcal{R}_\omega \xrightarrow{\text{operator structure}} \text{null-space bases} \xrightarrow{\text{canonical sources}} \text{NR current families} \xrightarrow{\text{evanescent overlap}} \text{non-reconstruct} \tag{2.312}$$

where the first arrow is Subsection 2.7.1, the second is Subsection 2.7.1, the third is Subsection 2.7.2, and the fourth is Subsection 2.7.4 [2]. Eq. (2.312) expresses Eq. (2.273) at the source level: spectral $\lambda_n = 0$ modes from Eq. (2.279) lie in $\mathbf{u}_{\mathrm{nr}}$, but complete $\ker \mathcal{R}_\omega$ requires impressed-data coordinates on $\mathcal{J}_\Omega$, not only far-zone modal shapes [2]. Eq. (2.312) is the source-side restatement of the spectral chain Eq. (2.273): operator nullity precedes constructive NR catalogs, which precede evanescent non-reconstructability, which precede angular-spectrum consequences. Mathematically, each arrow preserves linearity of $\mathcal{R}_\omega$; physically, each stage removes a distinct laboratory error from inverse-source charting. On a cube of side $2a$ with $k_0 a = 2$, evanescent overlap with $\Pi_{\mathrm{ev}} \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ can exceed 25% of the near-field energy on S_R with $R = a$ while contributing zero to $a(\widehat{\mathbf{k}})$; mistaking that storage for NR nullity violates Eq. (2.335) [2]. 1 flux laws enter only after Π_{rad} [6]. Composition law

Eq. (2.311) still applies: geometric annihilation Eq. (2.307) and observation rank Eq. (2.220) are independent of, and composable with, source null-space membership [6]. The laboratory examples and kernel reading above fix the measurement ordering before any operator display. Near-field probes sample $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$; flux meters read only after Π_{rad} [2]. With that ordering in place, keep three spaces in view throughout section 2.7, impressed currents $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ on compact support Ω_s, full Maxwell fields $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \in \mathcal{F}_\Omega$, and the radiative image on which far-zone instruments read. The primary object remains the radiation map that projects the latter from the former:

$$\mathcal{R}_\omega : \mathcal{J}_\Omega \longrightarrow \mathcal{F}_{\mathrm{rad}}, \qquad \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}], \tag{2.313}$$

from Eqs. (2.2)–(2.202) [2]. The displayed map is the mathematical encoding of the wattmeter reading: only $\Pi_{\mathrm{rad}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ enters $\mathcal{F}_{\mathrm{rad}}$, so near-zone energy in $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \setminus \mathcal{F}_{\mathrm{rad}}$ cannot be inferred from far data alone [2]. Secondary targets include the observation map $\mathcal{M}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}}$ from Eq. (2.218), the non-radiative field projector $\mathrm{id} - \Pi_{\mathrm{rad}}$ on $\mathcal{F}_\Omega$, and evanescent/non-propagating selectors Eqs. (2.133)–(2.264) from Subsection 2.2.3 [2]. Inverse source Eq. (2.219) is ill-posed whenever $\ker \mathcal{M}_\omega \neq \{\mathbf{0}\}$, which occurs if $\ker \mathcal{R}_\omega \neq \{\mathbf{0}\}$ or $\ker \mathcal{O}_{\mathrm{meas}}$ is nontrivial on $\operatorname{ran} \mathcal{R}_\omega$ [6]. reciprocity-based solvability Eq. (2.302) governs reachability of a *target* radiative field; section 2.7 classifies sources that map to $\mathbf{0} \in \mathcal{F}_{\mathrm{rad}}$ regardless of solvability for nonzero targets [2]. Radiation null space. Subsection 2.7.1 develops Eq. (2.217) to structural statements on $\ker \mathcal{R}_\omega$ and $\ker \mathcal{R}_\omega^\dagger$. Locally, $\ker \mathcal{R}_\omega$ collects impressed currents annihilated by radiation; $\ker \mathcal{R}_\omega^\dagger$ collects far-side test directions orthogonal to every radiative image; and $\operatorname{coker} \mathcal{R}_\omega = \mathcal{F}_{\mathrm{rad}}/\operatorname{ran} \mathcal{R}_\omega$ labels pattern targets that no admissible source can realize [2]. The reference pattern below records admissibility and typical infinite dimensionality on open domains:

$$\ker \mathcal{R}_\omega = \{\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega : \Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0},\ \mathcal{D}_J[\mathbf{J}_{\mathrm{imp}}] = 1\}, \qquad \dim \ker \mathcal{R}_\omega = \infty \text{ typ. on open domains}, \tag{2.314}$$

with $\mathcal{D}_J = 1$ Eq. (2.0.1) enforced throughout [2]. Eq. (2.314) does not re-define Eq. (2.217); it records the 2 primary target for constructive characterization of nullity, dimensionality, and pairing with cokernel directions obstructing inversion Eq. (2.277) [2]. Modal sectors with $\lambda_n = 0$ in Eq. (2.279) supply a finite-dimensional *slice* of $\ker \mathcal{R}_\omega$ only when eigenmodes are declared on outgoing subspaces [6]. Non-radiating current configurations. Subsection 2.7.1 develops canonical impressed currents that realize Eq. (2.7):

$$\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega, \qquad \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}] \neq \mathbf{0}, \qquad \|\mathbf{u}_{\mathrm{nr}}\| \text{ may be large near } \Omega_s, \tag{2.315}$$

with $\mathbf{u}_{\mathrm{nr}}$ from Eq. (2.266) [2]. Eq. (2.315) specializes Eq. (2.271) to constructive families—loop modes, counter-propagating pairs, and equivalent-surface cancellations compatible with Eq. (1.8)—without tabulating dyadic kernels from Subsection 2.3.1 [2]. Such currents are invisible to far-zone flux meters Eq. (2.252) yet may dominate near-zone energy budgets Eq. (2.268) [6]. Evanescent overlap and non-reconstructability. Subsection 2.7.2 links $\ker \mathcal{R}_\omega$ to evanescent subspaces and finite observation rank. Evanescent content is field-side reactive storage with $\Pi_{\mathrm{rad}}\mathbf{u}_{\mathrm{ev}} = \mathbf{0}$; it must not be conflated with source-side $\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$, although both deficits can coexist on the same excitation [2]. The relation records the measurement inclusion and the evanescent non-radiation law:

$$\ker \mathcal{O}_{\mathrm{meas}} \circ \mathcal{R}_\omega \supseteq \ker \mathcal{R}_\omega, \qquad \Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ never radiates through } \Pi_{\mathrm{rad}}, \tag{2.316}$$

using Eqs. (2.138) and (2.220) [2]. Eq. (2.316) separates *radiation nullity* (no outgoing transverse radiation) from *measurement nullity* (unobservable even when radiation is nonzero) [2]. Tikhonov stabilization Eq. (2.223) does not shrink $\ker \mathcal{R}_\omega$; regularization only damps ill-conditioned directions

on ran $\mathcal{R}_\omega$ [6]. Accessible modal sector Eq. (2.9.3) from Subsection 2.6.5 further restricts recoverable content independently of evanescent storage [2]. Angular-spectrum consequences. Subsection 2.7.4 records how null-space structure propagates to continuous angular spectra Eq. (2.289):

$$a(\widehat{\mathbf{k}}) = 0 \text{ on suppressed sheets } \mathcal{K}_{\text{sup}} \implies \mathbf{J}_{\text{imp}} \text{ may still carry reactive content in } \Pi_{\text{ev}}\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}], \tag{2.317}$$

with $a(\widehat{\mathbf{k}})$ the far angular amplitude and $\mathcal{K}_{\text{sup}}$ the complement of accessible sector Eq. (2.9.3) [2]. Equation (2.317) links section 2.7 to section 2.9, plane-wave filters and retained/suppressed components in Subsections 2.9.1–2.9.3 presuppose the source-side null-space catalog developed here [2]. Angular probe rank on the far shell (Subsection 2.6.5) is not re-proved here; Subsection 2.7.4 states angular-spectrum *consequences* of Eq. (2.217) [6].

$$\mathbf{J}_{\text{imp}} \in \ker \mathcal{R}_\omega \implies a_n(\mathbf{J}_{\text{imp}}) = 0 \text{ for all radiating modes with } \lambda_n \neq 0, \tag{2.318}$$

while geometric constraints Eq. (2.9.3) may force additional $a_n = 0$ for $\mathbf{J}_{\text{imp}} \notin \ker \mathcal{R}_\omega$ [2]. Eq. (2.318) follows from Eq. (2.290) and linearity Eq. (2.5); it is cited in section 2.7 [2]. Inverse problems must budget $\ker \mathcal{R}_\omega$, $\ker \mathcal{O}_{\text{meas}}$, and forbidden sectors from geometric constraints on radiative spectra (Subsection 2.6.5) separately [6]. Several constructions deliberately remain outside section 2.7. Reciprocity identities and modal orthogonality belong to section 2.8; plane-wave angular-spectrum operators and near-to-far transforms to section 2.9; vector spherical harmonic completeness and explicit $\mathbf{f}_n$ to 3; range membership of unreachable modes to 4 [2]. section 2.7 Subsection 2.6.2 develops spectral decomposition; Fredholm structure of radiation operators (Subsection 2.6.4), or geometric constraints on radiative spectra (Subsection 2.6.5); it classifies invisible sources on the maps those results presuppose [2]. Worked non-radiating examples appear in Subsection 2.10.2 by reference to Subsection 2.7.1 [6]. Regime logic from section 1.2.5 is unchanged. Null-space membership does not override condition for outward angular-momentum flux: a current in $\ker \mathcal{R}_\omega$ has $\mathcal{P}_{\text{rad}} = 0$ on far shells, while reactive near fields from $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]$ may still violate naive flux interpretations if Eq. (2.139) is ignored [2]. Far-zone radiative dominance (Subsection 2.5.4) applies only to radiating sources; it does not trivialize $\ker \mathcal{R}_\omega$ [2]. Compression error Eq. (2.83) bounds modal truncation on ran $\mathcal{R}_\omega$ independently of null-space dimension [6]. As a worked example, consider a thin rectangular wire loop of perimeter $2\pi b$ with $b \ll \lambda_0$, embedded in free space and driven at angular frequency ω so that the impressed current completes one closed circulation with equal and opposite arms along the loop's long sides. The physical setup is a balanced dipole pair folded into a single closed contour: magnetic flux linkage is negligible at ω, yet the loop stores reactive energy in the near zone. A closed admissible loop carrying oppositely directed arms can cancel far-zone transverse radiation through Eq. (2.228) while maintaining nonzero $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]$ near the loop [2]. Subsection 2.7.1 classifies such families; here the point is that Eq. (2.7) is realizable with $\mathcal{D}_J[\mathbf{J}_{\text{imp}}] = 1$, not merely a formal kernel direction [6]. Pattern-matching algorithms that fit only ran $\mathcal{R}_\omega$ assign spurious modal labels to invisible loop content [2]. As a worked example, take a compact spherical source of radius a with $k_0 a = 1.5$ and truncate the exterior modal book at $N = 12$ radiation eigenpairs. A single $\lambda_n = 0$ eigenmode from Eq. (2.280) contributes to $\ker \mathcal{R}_\omega$ when realized by admissible $\mathbf{J}_{\text{imp}}^{(n)}$ [2]. On the range, the truncated index $n \leq N$ labels only a finite slice of invisible degrees; the full kernel on open $\mathcal{J}_\Omega$ is generally much larger. On exterior problems the modal index set is not finite; truncating to the first N modes underestimates $\dim \ker \mathcal{R}_\omega$ and overstates identifiability of inverse Eq. (2.219) [6]. Subsection 2.7.1 states the correct operator bookkeeping [2].

Watt closure on the radiative shell therefore precedes any angular label comparison in measurement reports [2].

2.7.1 Null spaces of radiation operators

On a declared compact-range or anechoic measurement, technicians record frequency, outline, and probe geometry before any harmonic fit is trusted [3]. Non-radiating currents store reactive energy without far-zone radiation [2]. Near-field richness must not be reported as far-zone diversity [3]. Maxwell's retarded radiation to Γ_R remains the governing constraint [1]. section 2.7 opened with laboratory signatures—balanced loops, counter-directed pairs, below-cutoff guides—and named the radiation kernel in plain language. Subsection 2.7.1 now supplies the operator account: structural properties of $\ker \mathcal{R}_\omega$, adjoint null spaces, cokernel pairings, and modal null sectors, developing Eq. (2.217) from Subsection 2.4.4 [2]. The loop and scanner patterns cited in the section introduction are not re-derived here; they motivate the locked definition below.

Assumptions are fixed as follows. Fix $\omega > 0$, admissible source space $\mathcal{J}_\Omega$ Eq. (2.48), charge continuity and boundary compatibility Eq. (2.0.1), outgoing radiation condition on exterior problems Eqs. (2.193)–(2.202), and radiation map $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ from Eqs. (2.2)–(2.202) [2]. Dual pairings on sources and fields are Eqs. (2.55) and (2.25) [2]. declared size and distance regime Eq. (2.139) governs transport reports on any recovered radiative component [6]. Radiation null space (locked definition). Physically, $\ker \mathcal{R}_\omega$ is the set of impressed currents that produce no qualified far-zone pattern: the wattmeter reading is null even when near probes are not [2]. Formally, the *radiation null space* is the source-side kernel

$$\ker \mathcal{R}_\omega = \{\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega : \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0} \in \mathcal{F}_{\mathrm{rad}},\ \mathcal{D}_J[\mathbf{J}_{\mathrm{imp}}] = 1\}\,, \tag{2.319}$$

equivalently $\Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0}$ on the outgoing branch [2]. Eq. (2.319) is the Subsection 2.7.1 development of Eq. Eq. (2.217): $\mathcal{D}_J = 1$ is explicit, and membership is defined on $\mathcal{J}_\Omega$ rather than on formal test directions [2]. Elements of $\ker \mathcal{R}_\omega$ produce no qualified far-zone outgoing radiation through Eq. (2.228) yet may satisfy $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \neq \mathbf{0}$ Eq. (2.271) [6]. Consequence for inversion: any far-zone datum fixes $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ only up to additions in $\ker \mathcal{R}_\omega$. Linearity Eq. (2.5) makes $\ker \mathcal{R}_\omega$ a linear subspace of $\mathcal{J}_\Omega$ closed under admissible superposition [2]. Adjoint null space and cokernel. Targets in $\operatorname{coker} \mathcal{R}_\omega$ are far-zone patterns orthogonal to every reachable image; they obstruct solvability even when no near-field anomaly is visible [2]. Let $\mathcal{R}_\omega^\dagger : \mathcal{F}_{\mathrm{rad}} \to \mathcal{J}_\Omega$ denote the adjoint induced by Eqs. (2.55) and Eq. (2.25) defines the adjoint pairing. Define

$$\ker \mathcal{R}_\omega^\dagger = \{\mathbf{w} \in \mathcal{F}_{\mathrm{rad}} : \langle \mathbf{w},\, \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\rangle_{\mathcal{U}} = 0 \ \ \forall\ \mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega\}\,, \tag{2.320}$$

and the cokernel quotient

$$\operatorname{coker} \mathcal{R}_\omega = \mathcal{F}_{\mathrm{rad}} / \operatorname{ran} \mathcal{R}_\omega, \tag{2.321}$$

with $\operatorname{ran} \mathcal{R}_\omega \subset \mathcal{F}_{\mathrm{rad}}$ the radiative image [2]. Eqs. (2.320)–(2.321) pair with $\ker \mathcal{R}_\omega$ through reciprocity-based solvability Eq. (2.302) imported from Subsection 2.6.4, targets $\mathbf{g} \in \mathcal{F}_{\mathrm{rad}}$ orthogonal to $\ker \mathcal{R}_\omega^\dagger$ are reachable, while obstructions lie in cokernel directions [2]. Biorthogonality Eq. (2.291) expresses the same pairing on modal coordinates without re-deriving adjoint modes here [6]. When $\mathcal{R}_\omega$ is Fredholm in the sense of Fredholm structure of radiation operators (Subsection 2.6.4),

$$\dim \ker \mathcal{R}_\omega < \infty \iff \dim \ker \mathcal{R}_\omega^\dagger < \infty, \tag{2.322}$$

and index Eq. (2.301) counts the imbalance [2]. Eq. (2.322) does not re-prove Fredholm structure of radiation operators (Subsection 2.6.4); it records the Chapter 2 null-space partition used in inversion Eq. (2.219) [2]. On exterior domains with compact Ω_s, $\dim \ker \mathcal{R}_\omega$ is typically infinite even when $\mathcal{R}_\omega$ is Fredholm on truncated subspaces Eq. (2.82) [6]. On the orthonormal basis of spectral decomposition of the radiation operator (Subsection 2.6.2), define the *modal null index set*

$$\mathcal{N}_{\mathrm{null}} = \{n \in \mathcal{N}_{\mathrm{rad}} : \lambda_n = 0\}\,, \tag{2.323}$$

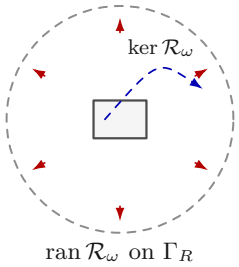


Figure 2.11: Invisible source modes in $\ker \mathcal{R}_\omega$ produce no outgoing flux on Γ_R; the radiative range $\operatorname{ran} \mathcal{R}_\omega$ is paired with adjoint kernel $\ker \mathcal{R}_\omega^\dagger$ by solvability Eq. (2.302).

with λ_n from Eq. (2.280) [2]. Eq. (2.323) is the spectral slice of $\ker \mathcal{R}_\omega$ on declared eigenmodes:

$$\sum_{n \in \mathcal{N}_{\text{null}}} c_n[\mathfrak{D}]\, \mathbf{J}_{\text{imp}}^{(n)} \in \ker \mathcal{R}_\omega \quad \text{when admissible superpositions exist,} \tag{2.324}$$

using source coefficients Eq. (2.282) and eigenpairs Eq. (2.280) [2]. Equation (2.324) does not exhaust $\ker \mathcal{R}_\omega$ on exterior problems because $\mathcal{N}_{\text{rad}}$ need not span all invisible source degrees [6]. modal invertibility on the radiative range (Subsection 2.6.4) already states that coefficients on $\ker \mathcal{R}_\omega$ remain free in Eq. (2.306); Subsection 2.7.1 identifies the modal-labeled portion [2]. Figure interpretation. Read $\mathcal{J}_\Omega$ as the impressed-source Hilbert space and $\mathcal{F}_{\text{rad}}$ as the outgoing radiative image. The forward arrow $\mathcal{R}_\omega = \Pi_{\text{rad}} \circ \mathcal{S}_\omega$ maps admissible impressed currents to far-zone radiation; dashed $\ker \mathcal{R}_\omega$ feeds back into $\mathcal{J}_\Omega$ as invisible source degrees that annihilate far-zone patterns Eq. (2.344). The lower adjoint kernel $\ker \mathcal{R}_\omega^\dagger$ pairs with cokernel directions through reciprocity-based solvability Eq. (2.302): a nonzero cokernel obstruction means no impressed current can realize the target far field on declared support [2]. Figure 2.11 separates kernel and cokernel data before observation composition Eq. (2.304). Let $\mathcal{R}_\omega$ satisfy Eq. (2.319) on admissible $\mathcal{J}_\Omega$ with outgoing radiation condition. Then:

(i) $\ker \mathcal{R}_\omega$ is a linear subspace of $\mathcal{J}_\Omega$ and $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}] = \mathbf{0}$ iff $\mathbf{J}_{\text{imp}} \in \ker \mathcal{R}_\omega$.

(ii) Solvability of $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}] = \mathbf{g}$ is equivalent to Eq. (2.302) on declared pairings.

(iii) On the spectral basis of Subsection 2.6.2, $\mathcal{N}_{\text{null}}$ Eq. (2.323) labels modes with $P_n \mathcal{R}_\omega = \mathbf{0}$; modal superpositions Eq. (2.324) lie in $\ker \mathcal{R}_\omega$ when admissible.

(iv) Domain class fixes dimensionality:

$$\begin{gathered} \text{bounded cavity + discrete book: } \dim \ker \mathcal{R}_\omega \text{ finite on truncated } \mathcal{R}_\omega^{(N)}; \\ \text{exterior + compact } \Omega_s : \ \dim \ker \mathcal{R}_\omega = \infty \text{ typ. on full } \mathcal{J}_\Omega. \end{gathered} \tag{2.325}$$

. Part (i) follows from linearity Eq. (2.5) and definition Eq. (2.319) [2]. Part (ii) is Fredholm structure of radiation operators (Subsection 2.6.4) and Eq. (2.302), cited without re-proof [6]. Part (iii) uses Eq. (2.305) and $\lambda_n = 0$ in Eq. (2.280) [2]. Part (iv) contrasts discrete cavity spectra Eq. (2.287) with continuous exterior angular content Eq. (2.289) [14]. If $\mathbf{J}_{\text{imp}} \in \ker \mathcal{R}_\omega$, then radiative and non-radiative components satisfy $\mathbf{u}_{\text{rad}} = \mathbf{0}$ and $\mathbf{u}_{\text{nr}} = \mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]$ from Eqs. (2.265)–(2.266); near-zone energy in $\mathbf{u}_{\text{nr}}$ may be large while outgoing radiation vanishes [2]. Composition with spectral and geometric filters. Null-space membership implies modal annihilation on radiating sectors:

$$\mathbf{J}_{\text{imp}} \in \ker \mathcal{R}_\omega \implies a_n(\mathbf{J}_{\text{imp}}) = 0 \text{ for all } n \text{ with } \lambda_n \neq 0, \tag{2.326}$$

with far coefficients a_n from Eq. (2.290) [2]. Eq. (2.326) expresses Eq. (2.318) from section 2.7 without re-deriving the modal expansion [2]. Geometric constraints on radiative spectra (Subsection 2.6.5) may force additional $a_n = 0$ for sources outside $\ker \mathcal{R}_\omega$ [6]. Composition law Eq. (2.311) budgets identifiability deficits separately [2]. Fredholm index Eq. (2.301) is unchanged under compact perturbations Eq. (2.300) and under Tikhonov regularization Eq. (2.223) on the range side [2]. Regularization stabilizes Eq. (2.306) for small $|\lambda_n|$ but leaves $\ker \mathcal{R}_\omega$ unchanged: invisible source components cannot be recovered from far-zone data at any regularization level [2]. Compression truncation Eq. (2.83) approximates $\operatorname{ran} \mathcal{R}_\omega$, not $\ker \mathcal{R}_\omega$ [6]. Null-space structure of radiation operators (Subsection 2.7.1) assumes linear media at fixed ω, outgoing radiation condition, and admissible $\mathcal{J}_\Omega$ Eq. (2.0.1) [2]. Loss modifies λ_n and may merge modal null sectors without promoting evanescent content to outgoing radiation via Eq. (2.138) [2]. Nonlinear or time-varying sources require the causal chain of Subsection 2.2.3; harmonic null spaces here are fixed-frequency slices [6]. declared size and distance regime Eq. (2.270) remains mandatory before interpreting near-zone reactive dominance as null-space membership [2]. Subsection 2.7.1 fixed operator null-space structure through Eq. (2.319), adjoint and cokernel pairings, and the modal null sector Eqs. (2.323)–(2.324) [2]. Constructive non-radiating current families that realize the kernel in the laboratory develop next.

The memorable laboratory requirement is therefore Π_{rad}: incoming branches must be removed before flux is integrated [2].

2.7.2 Non-radiating current configurations

The section introduction displayed balanced loops, counter-directed pairs, and below-cutoff guides as laboratory signatures of $\ker \mathcal{R}_\omega$. Subsection 2.7.1 gives the definitive constructive account: admissible $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega$ with $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}] \neq \mathbf{0}$, developing Eq. (2.315) [2]. Radiation integrals, Huygens surface forms Eqs. (2.235)–(2.237), and equivalent-source ambiguity Eq. (2.226) from Subsections 2.5.1 and 2.4.4 are presupposed; dyadic kernel tabulations from Subsection 2.3.1 are not repeated [2]. Evanescent overlap and measurement nullity belong to Subsection 2.7.2; worked example examples appear in Subsection 2.10.2 by reference [6]. Assumptions. Fix $\omega > 0$, compact support Ω_s, admissible impressed data Eq. (2.0.1), charge continuity Eq. (1.8), outgoing radiation condition Eqs. (2.202)–(2.212), and null space Eq. (2.319) [2]. All configurations below are required to satisfy $\mathcal{D}_J[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}] = 1$ and $\nabla \cdot \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} = -\mathrm{j}\omega\rho^{(\mathrm{NR})}$ with $\rho^{(\mathrm{NR})}$ compatible with boundary class Eq. (1.20) [2]. declared size and distance regime Eq. (2.270) governs whether near-zone reactive content may be reported as transport data [6]. Non-radiating current (locked definition). A *non-radiating current configuration* is an admissible impressed current

$$\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega, \qquad \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}] \neq \mathbf{0} \text{ in general}, \tag{2.327}$$

with radiative component $\mathbf{u}_{\mathrm{rad}} = \mathbf{0}$ and non-radiative component $\mathbf{u}_{\mathrm{nr}} = \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}]$ from Eqs. (2.265)–(2.266) [2]. Eq. (2.327) is the Subsection 2.7.1 development of Eq. (2.315) and expresses Eq. (2.7) with explicit $\mathcal{D}_J = 1$: far-zone radiation vanishes through Eq. (2.228), while near-zone fields may remain substantial kernel image on field space (Subsection 2.7.1) [2]. Linearity Eq. (2.5) implies admissible superpositions of non-radiating configurations remain non-radiating when charge continuity is preserved [2]. Maxwell-closed constraints. Non-radiating configurations are not arbitrary curl fields. Impressed data must satisfy

$$\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in L^2(\Omega_s)^3, \quad \nabla \cdot \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} = -\mathrm{j}\omega\rho^{(\mathrm{NR})}, \quad \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \text{ BC-compatible on } \Gamma, \tag{2.328}$$

as in Eq. (2.0.1) [2]. Equation (2.7.2) excludes discontinuous $\mathbf{J}_{\rm imp}^{(\rm NR)}$ that violate Eq. (1.8) before Maxwell field specification is attempted [2]. Surface-equivalent realizations on $\Gamma_{\rm eq}$ inherit trace class constraints from Subsection 2.1.1 [6]. Membership in $\ker\mathcal{R}_\omega$ is an *radiation* statement; Eq. (2.328) is a *source* statement—both are required [2]. Let $\mathcal{C}\subset\Omega_s$ be a closed admissible circulation path enclosing negligible magnetic flux linkage at ω, with oppositely directed arms $\mathcal{C}_+$ and $\mathcal{C}_-$ carrying equal magnitude currents. Physically, picture a wire loop in the xy-plane with counter-directed segments along $\pm\hat{\mathbf{x}}$: each arm radiates a leading $1/r$ transverse field, but the net electric dipole moment vanishes when the circulation closes. When the opposing-arm circulation closes with equal magnitude currents,

$$\int_{\mathcal{C}_+}\mathbf{J}_{\rm loop}\cdot d\boldsymbol{\ell}+\int_{\mathcal{C}_-}\mathbf{J}_{\rm loop}\cdot d\boldsymbol{\ell}=0\implies\mathbf{p}=\frac{1}{\mathrm{j}\omega}\int_{\Omega_s}\mathbf{J}_{\rm imp}^{(\rm loop)}\,d^3r=\mathbf{0},\tag{2.329}$$

for the associated volume current $\mathbf{J}_{\rm imp}^{(\rm loop)}$ obtained by compact support regularization on $\mathcal{C}$ [2]. Equation (2.329) is the electric-dipole cancellation law at the source level: vanishing net electric dipole moment removes the leading electric-dipole term in Eq. (2.246) but does *not* by itself establish $\mathcal{R}_\omega[\mathbf{J}_{\rm imp}^{(\rm loop)}]=\mathbf{0}$—magnetic-dipole and higher-multipole outgoing radiation may remain [14]. For the canonical opposing-arm filament pair declared here—equal currents on $\mathcal{C}_\pm$ with negligible magnetic-flux linkage—Eq. (2.329) establishes electric-dipole cancellation only [14]. Magnetic-dipole and higher-multipole outgoing terms may remain unless additional geometry and phase constraints are met, as in the counter-propagating pair Eq. (2.330) with $\mathbf{E}_{0,+}+\mathbf{E}_{0,-}=\mathbf{0}$ on S^2 [17]. Membership $\mathbf{J}_{\rm imp}^{(\rm loop)}\in\ker\mathcal{R}_\omega$ therefore requires explicit verification of complete outgoing cancellation—not inference from $\mathbf{p}=\mathbf{0}$ alone [2]. Near-zone reactive fields from $\mathcal{S}_\omega[\mathbf{J}_{\rm imp}^{(\rm loop)}]$ need not vanish; Eq. (2.270) still applies [2]. Two compact sources $\mathbf{J}_{\rm imp}^{(+)}$, $\mathbf{J}_{\rm imp}^{(-)}$ with supports $\Omega_\pm$ separated by distance $d\ll\lambda_0$ can form a non-radiating pair when their outgoing far patterns interfere destructively. A concrete arrangement is a pair of equal-strength electric dipoles along $\hat{\mathbf{z}}$, separated by $d=\lambda_0/20$, driven in opposite phase so the leading $1/r$ terms cancel on S^2 while reactive overlap remains in the gap:

$$\mathbf{J}_{\rm imp}^{(\rm NR,cp)}=\mathbf{J}_{\rm imp}^{(+)}+\mathbf{J}_{\rm imp}^{(-)},\qquad\mathbf{E}_{0,+}(\theta,\phi)+\mathbf{E}_{0,-}(\theta,\phi)=\mathbf{0}\text{ on }S^2,\tag{2.330}$$

with $\mathbf{E}_{0,\pm}$ far amplitudes from Eq. (2.246) for each component [2]. Equation (2.330) realizes Eq. (2.327) when phase and polarization of $\mathbf{J}_{\rm imp}^{(\pm)}$ are tuned so $\Pi_{\rm rad}\mathcal{S}_\omega[\mathbf{J}_{\rm imp}^{(\pm)}]$ cancel on the outgoing branch [2]. The pair may store substantial reactive energy in the overlap region through Eq. (2.268) while $\mathcal{P}_{\rm rad}=0$ on far shells [6]. On a closed equivalent surface $\Gamma_{\rm eq}$ enclosing radiating sources, tangential non-radiating distributions satisfy Eq. (2.226):

$$\mathbf{J}_{s,\rm eq}^{(\rm NR)}\in\ker\mathcal{R}_\omega\big|_{\Gamma_{\rm eq}},\qquad\mathbf{J}_{s,\rm eq}^{(\rm NR)}\cdot\hat{\mathbf{n}}=0\text{ on }\Gamma_{\rm eq},\tag{2.331}$$

with $\hat{\mathbf{n}}$ the outward normal [2]. Equation (2.331) is the Subsection 2.7.1 constructive realization of Huygens equivalence ambiguity: exterior fields on $\mathcal{F}_{\rm SF}$ are fixed by exterior radiative uniqueness (Subsection 2.4.3), but equivalent surface currents are unique only up to additions in Eq. (2.331) [2]. Integrals Eqs. (2.235)–(2.237) realize the same $\mathcal{S}_\omega$ image for equivalent and non-radiating tangential perturbations [6]. When radiation eigenpairs with $\lambda_n=0$ are declared in Eq. (2.280),

$$\mathbf{J}_{\rm imp}^{(\rm NR,mod)}=\sum_{n\in\mathcal{N}_{\rm null}}\alpha_n\,\mathbf{J}_{\rm imp}^{(n)},\qquad\mathcal{R}_\omega[\mathbf{J}_{\rm imp}^{(\rm NR,mod)}]=\mathbf{0},\tag{2.332}$$

for coefficients $\alpha_n\in\mathbb{C}$ such that the superposition satisfies Eq. (2.7.2) [2]. Equation (2.332) specializes Eq. (2.324) from Subsection 2.7.1 to constructive sources: each $\mathbf{J}_{\rm imp}^{(n)}$ with $\lambda_n=0$ is

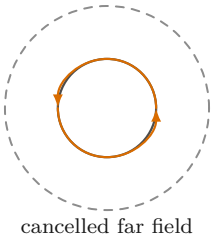


Figure 2.12: Canonical non-radiating families: opposing-arm loop cancellation Eq. (2.329) (electric-dipole cancellation under the stated geometry; $\ker \mathcal{R}_\omega$ membership requires additional outgoing cancellation conditions), counter-propagating pair Eq. (2.330), surface tangential null Eq. (2.331), and modal eigenrealization Eq. (2.332), each mapping to $\ker \mathcal{R}_\omega$ only under its declared additional conditions.

a modal non-radiating generator when admissible [2]. On exterior domains $\mathcal{N}_{\text{null}}$ need not span all of $\ker \mathcal{R}_\omega$ Eq. (2.325) [6]. The four upper nodes are constructive *families*, not interchangeable templates. Loop cancellation annihilates the leading $1/r$ transverse moment; counter-propagating pairs cancel on S^2 through phase interference; surface tangential nulls exploit Huygens equivalence ambiguity; modal $\lambda_n = 0$ generators realize the spectral slice Eq. (2.324). Each family can realize $\ker \mathcal{R}_\omega$ only under its declared additional conditions; the opposing-arm loop supplies electric-dipole cancellation first, yet each leaves distinct reactive near fields $\mathcal{S}_\omega[\mathbf{J}^{(\text{NR})}_{\text{imp}}] \neq \mathbf{0}$—the schematic warns that near-zone scans cannot distinguish families once far patterns vanish [2]. Figure 2.12 orders constructive families before evanescent and measurement extensions in Subsection 2.7.2. Equivalent-current solvers on Γ_{eq} seek $\mathbf{J}_{\text{eq}}$ matching near or far data [2]. Canonical non-radiating configurations (Subsection 2.7.1) and Eq. (2.331) show why such inversions are ill-posed even when forward fields are unique: any solution may be shifted by tangential $\mathbf{J}^{(\text{NR})}_{s,\text{eq}}$ without changing $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}]$ [2]. Tikhonov regularization Eq. (2.223) selects a particular representative but does not eliminate the equivalence class [6]. Subsection 2.7.2 separates this source nullity from finite-rank observation nullity Eq. (2.337) [2]. Subsection 2.7.1 assumes linear Maxwell theory at fixed ω, outgoing radiation condition, and admissible sources Eq. (2.7.2) [2]. Lossy media modify loop and pair conditions through complex $k(\mathbf{r})$ without promoting evanescent storage to radiation Eq. (2.138) [2]. Canonical families generate *explicit* members of $\ker \mathcal{R}_\omega$; they do not exhaust $\dim \ker \mathcal{R}_\omega = \infty$ on exterior domains Eq. (2.325) [6]. declared size and distance regime Eq. (2.139) remains mandatory: large near-zone $|\mathbf{E}|$ from $\mathcal{S}_\omega[\mathbf{J}^{(\text{NR})}_{\text{imp}}]$ is not radiation without Π_{rad} condition [2].

Physical interpretation therefore closes when $\mathbf{E} \times \mathbf{H}$ flux on Γ_R matches the outgoing pair used above [5].

2.7.3 Evanescent overlap and non-reconstructability

Near-field scans resolve evanescent sheets whose lateral wavenumber exceeds k_0: they store reactive energy but never pass Π_{rad}. Before coefficients are interpreted, the issue is how much of that overlap is invisible to finite-rank reconstruction maps, and when residual nullity is field-side rather than source-side [2, 6]. Subsections 2.7.1–2.7.1 fixed $\ker \mathcal{R}_\omega$ and canonical non-radiating impressed currents Eq. (2.327). Evanescent sheets are a distinct field-side deficit: reactive storage that never passes Π_{rad} even when the source radiates. Subsection 2.7.2 develops the account of evanescent subspaces, their overlap with non-radiating source content, and reconstructability limits under finite observation rank—developing Eq. (2.316) from section 2.7 [2]. Propagation projectors Eqs. (2.133)–(2.138) from Subsection 2.2.3, non-propagating split Eq. (2.264) from Subsection 2.5.4, measurement null-space extension measurement null-space extension (Subsection 2.7.1) from Subsection 2.7.1, and rank cap Eq. (2.220) from Subsection 2.4.4 are presupposed; projector algebra and inverse-

source templates are not re-proved [2]. Angular-spectrum consequences belong to Subsection 2.7.4; plane-wave filters to section 2.9 [6]. Assumptions. Fix $\omega > 0$, admissible $\mathcal{J}_\Omega$ Eq. (2.0.1), outgoing radiation condition Eqs. (2.202)–(2.212), field pairs in $\mathcal{F}_{\mathrm{SF}} \subset \mathcal{F}_\Omega$, and spectral completion supporting Eqs. (2.133)–(2.134) [2]. Observation map $\mathcal{M}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}}$ from Eq. (2.218) is finite rank with $\operatorname{rank}(\mathcal{O}_{\mathrm{meas}}) = N$ Eq. (2.79) [2]. declared size and distance regime Eq. (2.139) governs transport reports on any recovered radiative component [6]. On the spectral completion $\widehat{\mathcal{F}}_\Omega$, the *evanescent subspace* is

$$\mathcal{F}_{\mathrm{ev}} = \left\{ \mathbf{u} \in \widehat{\mathcal{F}}_\Omega : \Pi_{\mathrm{ev}} \mathbf{u} = \mathbf{u} \right\}, \tag{2.333}$$

with Π_{ev} from Eq. (2.133) [2]. Eq. (2.333) is the Subsection 2.7.2 field-side label for modes with imaginary longitudinal wavenumber Eq. (2.130) and decay rate Eq. (2.131) [2]. The companion *non-propagating* subspace combines evanescent and below-cutoff content through Eq. (2.264) [6]. Evanescent sheets carry reactive structure bound to apertures; they decay exponentially with scan height and satisfy Eq. (2.138) without re-derivation here [2]. For admissible $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$, the evanescent projector Π_{ev} selects field content whose lateral wavenumber exceeds the vacuum propagation circle $k_t > k_0$: such modes decay exponentially with distance from the source and carry reactive storage rather than outgoing radiation. Define the evanescent field content

$$\mathbf{u}_{\mathrm{ev}} = \Pi_{\mathrm{ev}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}], \qquad \|\mathbf{u}_{\mathrm{ev}}\|_{\mathcal{E}} = \|\Pi_{\mathrm{ev}} \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]\|_{\mathcal{E}}, \tag{2.334}$$

with energy norm Eq. (2.47) [2]. Eq. (2.334) measures reactive storage excited by $\mathbf{J}_{\mathrm{imp}}$ that never passes through Π_{rad} because

$$\Pi_{\mathrm{rad}}\, \mathbf{u}_{\mathrm{ev}} = \mathbf{0}, \qquad \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ depends only on } \Pi_{\mathrm{prop}} \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ after outgoing selection}, \tag{2.335}$$

by Eqs. (2.138) and (2.137) [2]. Eq. (2.335) upgrades the second line of Eq. (2.316): evanescent overlap is field-side non-radiation, distinct from source-side $\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$ Eq. (2.319) [6]. A source may satisfy $\|\mathbf{u}_{\mathrm{ev}}\| \gg \|\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\|$ while still radiating [2]. Eq. (2.316) separates radiation annihilation from finite-rank readout loss. Source nullity $\ker \mathcal{R}_\omega$ is strict radiation failure; measurement nullity $\ker \mathcal{M}_\omega$ additionally loses directions that radiate but the aperture cannot resolve [2]:

$$\underbrace{\ker \mathcal{R}_\omega}_{\text{no radiation}} \subseteq \underbrace{\ker \mathcal{M}_\omega}_{\text{no finite readout}}, \qquad \ker \mathcal{M}_\omega \supsetneq \ker \mathcal{R}_\omega \text{ when } \ker \mathcal{O}_{\mathrm{meas}}\big|_{\operatorname{ran} \mathcal{R}_\omega} \neq \{\mathbf{0}\}, \tag{2.336}$$

with strict inclusion typical for finite apertures Eq. (2.80)

$$\ker \mathcal{M}_\omega = \ker \mathcal{R}_\omega \;\cup\; \mathcal{R}_\omega^{-1}\big(\ker \mathcal{O}_{\mathrm{meas}}\big|_{\operatorname{ran} \mathcal{R}_\omega}\big), \tag{2.337}$$

[2]. Equation (2.336) expresses measurement null-space extension (Subsection 2.7.1) and Eq. (2.337) without re-proof: evanescent storage contributes to the first factor only when it annihilates radiation; finite rank contributes to the second factor even when radiation is nonzero [2]. Tikhonov reconstruction Eq. (2.223) stabilizes ill-conditioned inversions on $\operatorname{ran} \mathcal{M}_\omega^\dagger$ but does not shrink $\ker \mathcal{M}_\omega$ [6]. Define the *non-reconstructable set* at observation rank N:

$$\mathcal{K}_{\mathrm{nrec}}(N) = \{\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega : \mathcal{M}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0}\} = \ker \mathcal{M}_\omega, \tag{2.338}$$

equivalently the kernel of Eq. (2.219) for generic $\mathbf{y}$ [2]. Equation (2.338) is the Subsection 2.7.2 constructive account of reconstructability label: sources in $\mathcal{K}_{\mathrm{nrec}}(N)$ are invisible to the declared measurement bank regardless of near-zone $|\mathbf{E}|$ [2]. Rank bound

$$\dim \operatorname{ran} \mathcal{M}_\omega \leq N < \infty \implies \dim \mathcal{K}_{\mathrm{nrec}}(N) = \infty \text{ on typical open domains}, \tag{2.339}$$

follows from Eq. (2.220) and infinite-dimensional $\mathcal{J}_\Omega$ [2]. Geometric sector constraints Eq. (2.9.3) and composition law Eq. (2.311) further restrict recoverable coordinates independently of Eq. (2.338) [2].

The measurement sequence orders radiation selection, evanescent overlap, and finite-rank observation. Π_{ev} content may dominate $|\mathbf{E}|$ on inner shells yet satisfies Eq. (2.335) on qualified flux meters. $\mathcal{O}_{\mathrm{meas}}$ with rank N collapses recoverable modes into $\mathcal{K}_{\mathrm{nrec}}(N)$: radiation can exist while the aperture cannot resolve it. Dashed branches mark observation deficits distinct from source nullity $\ker \mathcal{R}_\omega$ [2]. Let $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ be admissible with evanescent content Eq. (2.334) and observation map $\mathcal{M}_\omega$ of rank N. Then:

(i) $\Pi_{\mathrm{rad}}\mathbf{u}_{\mathrm{ev}} = \mathbf{0}$ and $\mathbf{u}_{\mathrm{ev}}$ carries no qualified far-zone export Eq. (2.335).

(ii) $\ker \mathcal{R}_\omega \subseteq \ker \mathcal{M}_\omega = \mathcal{K}_{\mathrm{nrec}}(N)$.

(iii) Near-field data on a finite enclosing surface reduce but do not in general eliminate $\mathcal{K}_{\mathrm{nrec}}(N)$: if $\mathcal{M}_{\mathrm{near}}$ samples $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on Γ_{enc}, then

$$\ker \mathcal{M}_{\mathrm{near}} \subsetneq \ker \mathcal{M}_\omega \text{ when } \Gamma_{\mathrm{enc}} \text{ resolves more modes than far-zone aperture,} \tag{2.340}$$

while $\ker \mathcal{R}_\omega$ is unchanged [2].

(iv) Reconstructability budget:

$$\mathcal{N}_{\mathrm{recoverable}} \subseteq \mathcal{N}_{\mathrm{acc}} \cap \{n : \text{ resolvable by } \mathcal{M}_\omega\} \cap \{n : \ \lambda_n \neq 0\}, \tag{2.341}$$

with $\mathcal{N}_{\mathrm{acc}}$ from Eq. (2.9.3) and modal indices from Subsection 2.6.2 [2].

Part (i) is Eq. (2.138) [2]. Part (ii) follows from measurement null-space extension (Subsection 2.7.1) and definition Eq. (2.338) [6]. Part (iii) uses near- versus far-field sampling logic from Subsection 2.4.4, $\mathcal{S}_\omega$ is single-valued on outgoing fields while $\mathcal{R}_\omega$ remains non-injective [2]. Part (iv) composes Eq. (2.311) with rank cap Eq. (2.220) [2]. For all admissible $\mathbf{J}_{\mathrm{imp}}$,

$$\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\,\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\,\Pi_{\mathrm{prop}}\,\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}], \tag{2.342}$$

and $\Pi_{\mathrm{rad}}\Pi_{\mathrm{ev}} = \mathbf{0}$ on $\mathcal{F}_\Omega$ [2]. Eq. (2.342) implies $\mathbf{u}_{\mathrm{ev}}$ from Eq. (2.334) does not alter $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$; adding evanescent excitation to an otherwise radiating source changes $\mathbf{u}_{\mathrm{nr}}$ and local power Eq. (2.268) without changing far patterns when radiation is unchanged [2]. declared size and distance regime Eq. (2.270) still forbids reporting $\mathbf{u}_{\mathrm{ev}}$ as transport [6]. Substitute Eqs. (2.138) and (2.137) into $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ [2]. Near-zone energy follows from Eq. (2.264) [6]. Let Γ_{enc} be a closed surface enclosing Ω_s and let $\mathcal{M}_{\mathrm{near}}$ sample tangential $\mathbf{E}$, $\mathbf{H}$ on Γ_{enc}. Then $\ker \mathcal{M}_{\mathrm{near}} \subseteq \ker \mathcal{M}_\omega$ when far-zone observations are a subset of near-field functionals, but

$$\ker \mathcal{R}_\omega \subseteq \ker \mathcal{M}_{\mathrm{near}} \subseteq \ker \mathcal{M}_\omega \tag{2.343}$$

with strict inclusions possible at each step [2]. Equation (2.343) expresses Subsection 2.4.4 near- versus far-field data without re-opening uniqueness exterior radiative uniqueness (Subsection 2.4.3) [2]. Outgoing radiation condition remains mandatory: identical near traces on a finite surface do not imply identical far zones without exterior selection [6]. If $\mathcal{M}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0}$, then far functionals vanish; near samples may still vanish on Γ_{enc} for non-radiating additions in $\ker \mathcal{R}_\omega$ [2]. Additional near resolution shrinks the observation null space on $\operatorname{ran} \mathcal{R}_\omega$ but leaves $\ker \mathcal{R}_\omega$ unchanged Eq. (2.319) [6]. When Eq. (2.140) fails, modal content lies in $\Pi_{\mathrm{non}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ or $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2]. Such storage satisfies Eq. (2.335) and may dominate Eq. (2.268) on sub-wavelength shells while $\mathcal{P}_{\mathrm{rad}} = 0$ [2].

Applying far-zone pattern formulas to below-cutoff modes violates Eq. (2.139) as in the TE_{10} illustration referenced from section 1.1.2 [6]. Subsection 2.7.2 records operator consequences; guide nomenclature is not re-tabulated [2]. Non-radiating configurations Eq. (2.327) may carry large $\|\mathbf{u}_{\mathrm{ev}}\|$ when $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}]$ is evanescent-dominated [2]. Conversely, large $\|\mathbf{u}_{\mathrm{ev}}\|$ does not imply $\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$: a radiating source may store substantial evanescent energy near Ω_s while radiating through Π_{prop} [2]. Subsections 2.7.1 and 2.7.2 address orthogonal deficits—source nullity versus reactive storage versus finite-rank readout [6]. Subsection 2.7.2 assumes linear Maxwell theory at fixed ω, outgoing radiation condition, and declared finite-rank $\mathcal{O}_{\mathrm{meas}}$ [2]. Loss modifies evanescent decay Eq. (2.131) without promoting Π_{ev} content to Π_{rad} radiation [2]. Dispersive or broadband superposition requires Subsection 2.2.3; harmonic evanescent labels here are fixed-frequency slices [6]. Subsection 2.7.4 translates non-reconstructability to continuous angular spectra Eq. (2.289) [2].

Evanescent overlap therefore never enlarges far-zone rank: reactive scan content lies in $\mathcal{F}_{\mathrm{ev}}$ and is filtered before $\mathbf{c}$ is reported. Angular-spectrum consequences of those null spaces are developed next (Subsection 2.7.4).

2.7.4 Angular-spectrum consequences of null spaces

Null-space structure on $\ker \mathcal{R}_\omega$ becomes a directional deficit only after it is translated to the continuous angular measure laboratories report. The physical question is which far-zone directions remain absent when source-side non-radiating content and evanescent sheets are both present [2, 6]. Far-zone angular amplitudes $a(\widehat{\mathbf{k}})$ are what compact ranges and synthesis codes ultimately report. Subsections 2.7.1–2.7.2 fixed $\ker \mathcal{R}_\omega$, canonical non-radiating impressed currents, and evanescent non-reconstructability; Subsection 2.7.4 records how those source-side deficits propagate to the continuous angular measure Eq. (2.289) without re-opening plane-wave operators from section 2.9 [2].

Assumptions. Fix $\omega > 0$, outgoing radiation condition, admissible $\mathcal{J}_\Omega$, and far amplitude $\mathbf{E}_0(\theta, \phi)$ from Eq. (2.246) when $\mathcal{D}_{J,\mathrm{reg}} = 1$ [2]. Continuous angular coefficients $a_\sigma(\widehat{\mathbf{k}})$ are defined through the rigged limit of Eq. (2.289) compatible with modal amplitudes Eq. (2.282) [2]. Finite-rank observation $\mathcal{M}_\omega$ and non-reconstructable set $\mathcal{K}_{\mathrm{nrec}}(N)$ from Eq. (2.338) are presupposed [6]. declared size and distance regime Eq. (2.139) governs transport claims on any retained angular content [2]. Let $a(\widehat{\mathbf{k}})$ denote the scalar angular spectrum obtained from Eq. (2.289) on the outgoing transverse sheet [2]. For $\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$,

$$a(\widehat{\mathbf{k}}) = 0 \quad \text{for all } \widehat{\mathbf{k}} \in S^2 \text{ on the radiating sheet,} \tag{2.344}$$

equivalently $\mathbf{E}_0(\theta, \phi) = \mathbf{0}$ in Eq. (2.290) [2]. Eq. (2.344) is the Subsection 2.7.4 development of the first clause of Eq. Eq. (2.317): null-space membership annihilates the entire far-zone angular measure, not merely selected harmonics [6]. Near-zone reactive content in $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ may remain nonzero without contributing to $a(\widehat{\mathbf{k}})$ Eq. (2.335) [2]. Suppressed and accessible angular sheets. Define the *suppressed angular sheet*

$$\mathcal{K}_{\mathrm{sup}} = S^2 \setminus \mathcal{K}_{\mathrm{acc}}, \qquad \mathcal{K}_{\mathrm{acc}} \text{ from Eq. (2.9.3)}, \tag{2.345}$$

as directions annihilated by geometric, support, or parity constraints before observation [2]. Eq. (2.345) expresses Eq. (2.317): $a(\widehat{\mathbf{k}}) = 0$ on $\mathcal{K}_{\mathrm{sup}}$ does not imply $\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$, because geometric forbidding Eq. (2.307) and radiation nullity Eq. (2.344) are distinct mechanisms [2]. Accessible sheet $\mathcal{K}_{\mathrm{acc}}$ intersects recoverable modal sector Eq. (2.341) [6].

$$\mathcal{R}_\omega\left[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}\right] = \mathcal{R}_\omega\left[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}\right], \qquad \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega, \tag{2.346}$$

Maxwell admissibility fixes which field components may carry flux through Γ_R before harmonic labels are read [2].

$$\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega \implies a_n\Big(\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}\Big) = 0, \quad \lambda_n = 0 \text{ on radiating modes,} \tag{2.347}$$

Watt on Γ_R changes only through the outgoing field pair already constrained by Maxwell structure on the selected branch [2].

$$\Big\|\mathcal{S}_\omega\Big[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}\Big]\Big\|_{\mathcal{E}} \neq \Big\|\mathcal{S}_\omega\Big[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}\Big]\Big\|_{\mathcal{E}} \text{ while } \mathcal{R}_\omega\Big[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}\Big] = \mathcal{R}_\omega\Big[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}\Big], \tag{2.348}$$

For radiating $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}$ and non-radiating $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega$,

$$a(\widehat{\mathbf{k}}; \mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}) = a(\widehat{\mathbf{k}}; \mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}), \tag{2.349}$$

on the outgoing transverse sheet, restating superposition law Eq. (2.346) in angular coordinates [2]. Eq. (2.349) is the continuous form of modal invisibility Eq. (2.347) and explains why inverse angular-spectrum solvers cannot detect $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}$ from far data alone [2]. Canonical families canonical non-radiating configurations (Subsection 2.7.1) therefore induce *invisible* angular perturbations at fixed $a(\widehat{\mathbf{k}})$ [6]. Evanescent decoupling from angular spectra. Evanescent field content does not enter $a(\widehat{\mathbf{k}})$:

$$a(\widehat{\mathbf{k}}) \text{ depends only on } \Pi_{\mathrm{prop}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ after } \Pi_{\mathrm{rad}}, \qquad \Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \not\mapsto a(\widehat{\mathbf{k}}), \tag{2.350}$$

by Eqs. (2.335) and (2.342) [2]. Eq. (2.350) specializes the second line of Eq. (2.317): reactive storage may be large in $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ while $a(\widehat{\mathbf{k}})$ reports only outgoing radiation [2]. declared size and distance regime Eq. (2.270) forbids interpreting $\mathbf{u}_{\mathrm{ev}}$ as angular-transport data [6]. Observation rank on angular spectra. Finite aperture collapses recoverable angular content:

$$\{\, a(\widehat{\mathbf{k}}) \text{ recoverable from } \mathcal{M}_\omega \,\} \subseteq \{\, a(\widehat{\mathbf{k}}) \text{ supported on } \mathcal{K}_{\mathrm{acc}} \cap \operatorname{ran} \mathcal{R}_\omega \,\}, \tag{2.351}$$

with strict inclusion when $\ker \mathcal{O}_{\mathrm{meas}}\big|_{\operatorname{ran}\mathcal{R}_\omega} \neq \{\mathbf{0}\}$ Eq. (2.80) [2]. Equation (2.351) links angular spectra to $\mathcal{K}_{\mathrm{nrec}}(N)$: sources invisible to $\mathcal{M}_\omega$ need not have $a(\widehat{\mathbf{k}}) = \mathbf{0}$ pointwise [2]. Rank cap Eq. (2.220) limits resolved harmonics independently of Eq. (2.344) [6].

Figure interpretation. Source excitation $\mathcal{J}_\Omega$ maps through $\mathcal{R}_\omega$ to angular amplitude $a(\widehat{\mathbf{k}})$; dashed $\ker \mathcal{R}_\omega$ forces $a = 0$ on the entire radiating sheet Eq. (2.344). The suppressed sheet $\mathcal{K}_{\mathrm{sup}}$ removes directions by geometry before $\mathcal{E}_{\mathrm{acc}}^{(N)}$ is tested: $a = 0$ on $\mathcal{K}_{\mathrm{sup}}$ does not place $\mathbf{J}_{\mathrm{imp}}$ in $\ker \mathcal{R}_\omega$ [2]. Only the intersection with $\mathcal{K}_{\mathrm{acc}}$ carries laboratory-recoverable angular content [17]. Let $a(\widehat{\mathbf{k}})$ be the outgoing angular spectrum from Eq. (2.289) and let $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ be admissible. Then:

(i) $\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega \Rightarrow a(\widehat{\mathbf{k}}) = 0$ on the radiating sheet Eq. (2.344).

(ii) Superposition with $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega$ leaves $a(\widehat{\mathbf{k}})$ unchanged Eq. (2.349).

(iii) Evanescent content satisfies Eq. (2.350) and does not alter $a(\widehat{\mathbf{k}})$ when radiation is unchanged.

(iv) Full angular budget:

$$\operatorname{support}\big(a(\widehat{\mathbf{k}})\big) \subseteq \mathcal{K}_{\mathrm{acc}} \cap \{\widehat{\mathbf{k}} : \text{ resolvable by } \mathcal{M}_\omega\} \cap \{\widehat{\mathbf{k}} : \ \lambda_n \neq 0 \text{ modal labels}\}, \tag{2.352}$$

composing geometric, observation, and Fredholm constraints from Eqs. (2.311)–(2.341) [2].

Part (i) follows from $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0}$ and far-zone reduction the far-field approximation of source integrals (Subsection 2.5.2) [2]. Part (ii) uses canonical non-radiating configurations (Subsection 2.7.1) and linearity of angular extraction [6]. Part (iii) is evanescent content never radiates (Subsection 2.7.2) in angular coordinates [2]. Part (iv) composes Eq. (2.9.3), angular probe rank on the far shell (Subsection 2.6.5), and Eq. (2.351) [2]. Modal coefficients in Eq. (2.290) and continuous $a(\widehat{\mathbf{k}})$ satisfy

$$\lambda_n \, c_n[\mathfrak{D}] = 0 \text{ for all radiating } n \iff a(\widehat{\mathbf{k}}) = 0 \text{ on } S^2 \tag{2.353}$$

when the modal basis is complete on $\operatorname{ran}\mathcal{R}_\omega$ in the rigged sense [2]. Eq. (2.353) connects Eq. (2.326) to Eq. (2.344) without re-proving completeness in Chapter 3 (*Angular Momentum Operators and SO(3)*) [2]. Truncated modal books misstate Eq. (2.353) on exterior domains [14]. far-pattern coefficients from spectral modes (Subsection 2.6.2) and Eq. (2.289) identify discrete and continuous representations on outgoing transverse modes [2]. Null modal coefficients imply vanishing $\mathbf{E}_0$; conversely $\mathbf{E}_0 = \mathbf{0}$ forces all radiating modal weights to zero [6]. Given target angular spectrum $a_{\mathrm{tgt}}(\widehat{\mathbf{k}})$, the source inversion problem seeks $\mathbf{J}_{\mathrm{imp}}$ with $a(\widehat{\mathbf{k}}; \mathbf{J}_{\mathrm{imp}}) = a_{\mathrm{tgt}}(\widehat{\mathbf{k}})$. Any solution is unique only up to additions in $\ker\mathcal{R}_\omega$:

$$\mathbf{J}^{(1)}_{\mathrm{imp}} - \mathbf{J}^{(2)}_{\mathrm{imp}} \in \ker\mathcal{R}_\omega \quad \text{whenever} \quad a(\widehat{\mathbf{k}}; \mathbf{J}^{(1)}_{\mathrm{imp}}) = a(\widehat{\mathbf{k}}; \mathbf{J}^{(2)}_{\mathrm{imp}}), \tag{2.354}$$

and up to $\mathcal{K}_{\mathrm{nrec}}(N)$ at finite observation rank [2]. Eq. (2.354) is the angular restatement of Eq. (2.219) ill-posedness; Tikhonov regularization Eq. (2.223) selects a representative without eliminating the quotient [2]. If $a(\widehat{\mathbf{k}}; \mathbf{J}^{(1)}_{\mathrm{imp}}) = a(\widehat{\mathbf{k}}; \mathbf{J}^{(2)}_{\mathrm{imp}})$, then $\mathcal{R}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}} - \mathbf{J}^{(2)}_{\mathrm{imp}}] = \mathbf{0}$ on $\mathcal{F}_{\mathrm{rad}}$ [2]. Finite $\mathcal{M}_\omega$ adds measurement null directions measurement null-space extension (Subsection 2.7.1) [6]. The nullity–rank-N recovery–realizability triad introduced in section 2.7 applies unchanged in angular coordinates: far data cannot detect additions in $\ker\mathcal{R}_\omega$ Eq. (2.349); accessible directions survive geometric and regime filters even when radiation is nonzero; $\mathcal{S}_{\mathrm{real}}$ of a target $a_{\mathrm{target}}(\widehat{\mathbf{k}})$ is decided in 4 [17]. Evanescent overlap Eq. (2.334) changes near fields without altering a when radiation is fixed [2]. Forward link to angular-spectrum action. section 2.9 develops plane-wave operators that act on $a(\widehat{\mathbf{k}})$; Subsection 2.9.3 refines far-field angular reports on $\mathcal{K}_{\mathrm{acc}}$ [2]. Subsection 2.7.4 fixes the *source-side* constraints those operators inherit: Eqs. (2.344)–(2.352) must be composed with declared size and distance regime Eq. (2.139) before transport claims [2]. Reciprocity identities on coupled angular modes belong to section 2.8 [6]. Null-space consequences for angular spectra (Subsection 2.7.4) assumes linear Maxwell theory at fixed ω, outgoing radiation condition, and rigged-sense convergence of Eq. (2.289) [2]. Loss shifts phase of $a(\widehat{\mathbf{k}})$ without converting evanescent storage to radiation [2]. Vector harmonic explicit forms of $\mathbf{f}_n(\theta, \phi)$ belong to 3; Subsection 2.7.4 records operator consequences only [6]. As a worked example, place the canonical opposing-arm loop of Subsection 2.7.1 with radius $b = \lambda_0/8$ in vacuum and excite it with a uniform phasor current at ω. The canonical opposing-arm loop realization $\mathbf{J}^{(\mathrm{NR})}_{\mathrm{imp}}$ from Subsection 2.7.1 annihilates the leading electric-dipole term per Eq. (2.329); a wattmeter on a sphere at $R = 10\lambda_0$ registers zero outgoing radiation only when the complete outgoing field cancels under the declared family (e.g. Eq. (2.330))—while $|\mathbf{E}|$ may exceed the free-space impedance value by an order of magnitude at $r = b$ [2]. The near-field probe therefore sees rich reactive structure that the angular spectrum cannot detect [6]. Angular-spectrum probes of $\mathcal{S}_\omega[\mathbf{J}^{(\mathrm{NR})}_{\mathrm{imp}}]$ without Π_{rad} misstate null-space membership [6]. As a worked example, when $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \neq \mathbf{0}$ but the aperture resolves only N harmonics, Eq. (2.351) leaves unresolved $\widehat{\mathbf{k}}$ in the measurement null space even when $a(\widehat{\mathbf{k}})$ is nonzero [2]. This is distinct from Eq. (2.349) [2]. section 2.7 closes with the null-space sequence begun in its introduction: operator structure and $\ker\mathcal{R}_\omega$ (Subsection 2.7.1), canonical non-radiating current families (Subsection 2.7.1), evanescent subspaces and non-reconstructability (Subsection 2.7.2), and

angular-spectrum consequences (Subsection 2.7.4) with accessible and suppressed sheets Eqs. (2.9.3)–(2.345) (Subsection 2.7.4) [2]. Spectral and geometric filters from section 2.6 now meet source-side invisibility: far-zone reports use only $a(\widehat{\mathbf{k}})$ on the composed budget Eq. (2.352), while $\ker \mathcal{R}_\omega$, $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$, and $\mathcal{K}_{\mathrm{nrec}}(N)$ explain deficits that no angular label alone resolves [2]. section 2.8 adds reciprocity on the same modal coordinates; section 2.9 and Subsection 2.9.3 develop plane-wave action and far-field angular rank-N recovery on $\mathcal{K}_{\mathrm{acc}}$; 3 supplies explicit harmonic bases [6]. Transport and angular-transport claims must compose with Eq. (2.312), Eq. (2.352), and declared size and distance regime Eq. (2.139), not with raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ or unqualified near-zone reactive content [2]. Source null spaces explain what far-zone observation cannot detect; synthesizable—whether accessible angular content is generable by finite $\mathbf{J}_{\mathrm{imp}}$—is decided in 4 alongside source equivalence and effective rank.

2.8 Reciprocity and Radiative Coupling

Null-space structure explains invisible near-field rearrangements; reciprocity explains why passive hardware nevertheless reports the same complex coupling when transmit and receive roles are exchanged [2]. On a calibrated vector network analyzer the practitioner excites port A, reads the complex response at port B, then exchanges cables, standards, and probe normalization so that port B drives and port A receives; on reciprocal antennas and reciprocal media the normalized scattering entries S_{21} and S_{12} must agree once outgoing Sommerfeld condition and impressed-current normalization are held fixed [3]. Two-port impedance matrices $\mathbf{Z}$ and scattering matrices $\mathbf{S}$ inherit the same interchange symmetry whenever terminations and reference planes are declared consistently [15]. That bench fact precedes every volume identity below: reciprocity is what the network analyzer already presupposes when a passive feed is tuned by comparing forward and reverse coupling [2].

section 2.7 closed the null-space sequence Eq. (2.312): operator kernels, canonical non-radiating currents, evanescent non-reconstructability, and angular-spectrum consequences (Subsection 2.7.4) [2]. That program classified invisible and partially observable source content on $\mathcal{R}_\omega$ and $\mathcal{M}_\omega$. section 2.8 addresses exchange symmetry: when transmit and receive maps coincide, how radiative modes couple under bilinear pairings, and which orthogonality relations constrain coupled spectra on $\mathcal{F}_{\mathrm{rad}}$ [2]. Reciprocity ties transmission and reception on passive hardware before angular spectra are filtered; broken reciprocity after probe correction usually traces to inconsistent normalization or missing outgoing tails [3]. Preview swap Eq. (2.154) from Subsection 2.3.1 named kernel interchange; section 2.8 upgrades reciprocity to operator form on $\mathcal{S}_\omega$, $\mathcal{R}_\omega$, and modal coordinates from sections 2.6–2.7 [2]. Plane-wave angular-spectrum action belongs to section 2.9; explicit harmonic orthogonality and completeness belong to 3 Subsection 3.4.3; continuous-spectrum normalization belongs to Subsection 3.5.2 by reference to Subsection 2.8.2 [6]. Spectral decomposition in Subsection 2.6.2 and biorthogonality Eq. (2.291) fixed modal coordinates on $\mathcal{F}_{\mathrm{rad}}$; null-space structure in Subsection 2.7.1 showed that many sources share the same radiative image [2]. Reciprocity does not remove $\ker \mathcal{R}_\omega$, but it constrains how coupling coefficients between declared modes behave when impressed sources and test probes are swapped [2]. Treating antenna transmit and receive as unrelated maps violates Eq. (2.154) even when outgoing source-to-field integrals from Subsection 2.5.1 are numerically accurate [6]. section 2.8 separates Lorentz reciprocity in operator form, modal orthogonality and coupling, and radiation–reception interchange without re-opening Green construction from section 2.3 [2]. Fix angular frequency $\omega > 0$, impressed sources in $\mathcal{J}_\Omega$, and outgoing field pairs in $\mathcal{F}_{\mathrm{SF}}$ Eq. (2.56). The section develops reciprocity in three stages that mirror the measurement

sequence excite–couple–swap:

$$\text{Lorentz reciprocity} \xrightarrow{\text{operator form}} \mathcal{S}_\omega, \mathcal{R}_\omega \xrightarrow{\text{modal}} \text{orthogonality / coupling} \xrightarrow{\text{systems}} \text{radiation \& reception}, \tag{2.355}$$

where the first arrow is Subsection 2.8.1, the second is Subsection 2.8.2, and the third is Subsection 2.8.3 [2]. Equation (2.355) expresses Eq. (2.273) and Eq. (2.312) at the bilinear level: spectral modes from section 2.6 supply coordinates; null-space quotients from section 2.7 explain non-uniqueness; reciprocity constrains *symmetric* parts of coupling on $\operatorname{ran}\mathcal{R}_\omega$ [2]. Angular budget Eq. (2.352) and declared size and distance regime Eq. (2.139) remain mandatory before transport claims on coupled modes [6]. Denote by $\mathcal{M}_A$ and $\mathcal{M}_B$ the finite-rank port maps that compose radiation selection with probe functionals on $\mathcal{F}_{\mathrm{rad}}$; the port-swap experiment asserts $\mathcal{M}_B[\mathbf{J}^{(\mathrm{tx})}_{\mathrm{imp}}] = \mathcal{M}_A[\mathbf{J}^{(\mathrm{rx})}_{\mathrm{imp}}]$ on reciprocal excitations before any volume contraction is invoked [6]. Maxwell adjoint pairing Eq. (2.25), source dual pairing Eq. (2.55), and radiation map Eq. (2.274) supply the continuous objects that finite port matrices sample; dyadic reciprocity swap Eq. (2.154) and modal biorthogonality Eq. (2.291) specialize the same symmetry on outgoing branches fixed by Eqs. (2.202)–(2.212) [2]. Subsection 2.8.1 extends the bench fact to sesquilinear form on $\mathcal{S}_\omega$ and $\mathcal{R}_\omega$; the identity below states the operator identity the VNA experiment already presupposes. Lorentz reciprocity. Subsection 2.8.1 states the operator law: for admissible source pairs $(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}})$ and induced fields $(\mathbf{u}_1, \mathbf{u}_2)$,

$$\langle \mathbf{u}_1,\, \mathcal{S}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}]\rangle_{\mathcal{U}} = \langle \mathbf{u}_2,\, \mathcal{S}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}]\rangle_{\mathcal{U}} \tag{2.356}$$

on passive linear media without magneto-electric coupling, with pairing induced from Eq. (2.25) on outgoing admissible states [2]. Eq. (2.356) packages the port-swap symmetry as a sesquilinear identity on admissible Maxwell pairs: field one tested against source two equals field two tested against source one when media and boundaries belong to the reciprocal class [2]. It develops Eq. Eq. (2.154) from kernel swap to full source–field reciprocity on $\mathcal{S}_\omega$; restrictions on loss, anisotropy, and boundary class are stated in Subsection 2.8.1 [2]. Reciprocity on $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ follows by projection when both sides satisfy declared size and distance regime Eq. (2.270) [6]. On radiation eigenmodes Eq. (2.280), Subsection 2.8.2 develops orthogonality and coupling coefficients:

$$\langle \mathbf{u}_m,\, \mathbf{u}_n\rangle_{\mathcal{U}} = \delta_{mn}, \qquad C_{mn} = \langle \widetilde{\mathbf{u}}_m,\, \mathcal{R}_\omega[\mathbf{J}^{(n)}_{\mathrm{imp}}]\rangle_{\mathcal{U}}, \tag{2.357}$$

with $\widetilde{\mathbf{u}}_m$ adjoint modes tied to Eq. (2.291) [2]. Equation (2.357) is the section 2.8 target for Subsection 2.8.2 orthogonality and coupling on $\mathcal{F}_{\mathrm{rad}}$; explicit vector harmonic normalization belongs to 3 [2]. Modes with $\lambda_n = 0$ lie in null sectors Eq. (2.323); reciprocity constraints apply on radiating modes with $\lambda_n \neq 0$ [6]. Radiation and reception. Subsection 2.8.3 records transmit–receive interchange for coupled systems:

$$\mathcal{M}_{\mathrm{rx}}[\mathbf{J}^{(\mathrm{tx})}_{\mathrm{imp}}] = \mathcal{M}_{\mathrm{tx}}[\mathbf{J}^{(\mathrm{rx})}_{\mathrm{imp}}] \quad \text{under reciprocity-qualified pairings,} \tag{2.358}$$

with $\mathcal{M}_{\mathrm{tx}}$, $\mathcal{M}_{\mathrm{rx}}$ finite-rank compositions of $\mathcal{R}_\omega$ and $\mathcal{O}_{\mathrm{meas}}$ Eq. (2.218) [2]. Eq. (2.358) does not eliminate $\ker\mathcal{R}_\omega$ or $\mathcal{K}_{\mathrm{nrec}}(N)$; it constrains *symmetric* coupling on observable coordinates after projection onto $\operatorname{ran}\mathcal{R}_\omega$ [2]. Geometric forbidden sectors Eq. (2.9.3) and angular budget Eq. (2.352) compose independently [6]. Radiative coupling bilinear form. Coupling between two radiating subsystems is tracked by

$$\mathfrak{C}_\omega(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}}) = \langle \mathcal{R}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}],\, \mathcal{R}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}]\rangle_{\mathcal{U}} \quad \text{on } \mathcal{F}_{\mathrm{rad}}, \tag{2.359}$$

with reciprocity implying $\mathfrak{C}_\omega(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}}) = \mathfrak{C}_\omega(\mathbf{J}^{(2)}_{\mathrm{imp}}, \mathbf{J}^{(1)}_{\mathrm{imp}})$ under Eq. (2.356) and passive media [2]. Eq. (2.368) measures radiation overlap, not raw near-zone reactive overlap $\langle \mathcal{S}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}],\, \mathcal{S}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}]\rangle$,

which may include Π_{ev} content annihilated by Eq. (2.138) [2]. declared size and distance regime Eq. (2.139) qualifies when $\mathfrak{C}_\omega$ may be reported as transport data [6]. The next three subsections treat Lorentz reciprocity, modal orthogonality and coupling, and transmit–receive exchange.

Angular-spectrum nulls therefore propagate source-side rank limits to $a(\widehat{\mathbf{k}})$: a rich near-field chart cannot restore directions already removed by $\ker \mathcal{R}_\omega$. Plane-wave filtering on finite supports completes that picture in section 2.9.

2.8.1 Lorentz reciprocity in operator form

On a declared compact-range or anechoic measurement, technicians record frequency, outline, and probe geometry before any harmonic fit is trusted [3]. Reciprocity links transmission and reception on the same passive structure [2]. Two-port calibrations depend on it [14]. Maxwell's retarded radiation to Γ_R remains the governing constraint [1]. Reciprocity is the laboratory fact that passive linear antennas exchange the same complex transfer when transmit and receive roles swap on a calibrated vector network analyzer [2]. The adjoint radiation map extends that symmetry for continuous sources and traces, enabling source reconstruction and equivalence checks without re-solving unrelated models [17]. Reflector feeds, horn coupling, and patch arrays are routinely tuned by comparing S_{21} and S_{12}; persistent asymmetry after calibration usually signals a broken outgoing radiation condition or incorrect impressed-current normalization rather than a genuine non-reciprocal medium [3]. Reciprocity cannot be set aside merely because "only the radiative image matters": pattern, gain, and scattering matrices on passive hardware are reciprocity-linked observables tied to the same Maxwell operator [2]. Lorentz reciprocity is therefore a consequence of passive linear Maxwell theory, not an optional post-check applied after the forward solve [2]. Two-port calibrations and bistatic radar-cross-section matrices inherit the same exchange symmetry whenever media and boundaries satisfy the reciprocal class stated below [14]. The volume identity that follows states the same swap observed on the VNA once impressed sources and test fields are paired through the Maxwell adjoint. section 2.8 established bilinear exchange symmetry through Eqs. (2.355)–(2.356). Subsection 2.8.1 develops Lorentz reciprocity on $\mathcal{S}_\omega$ and $\mathcal{R}_\omega$, including the full proof of dyadic swap Eq. (2.154) deferred from Subsection 2.3.1 [2]. Eq. (2.25), source dual pairing Eq. (2.55), outgoing radiation condition Eqs. (2.202)–(2.212), and outgoing source-to-field integrals of Subsection 2.5.1 are presupposed; Green construction and Sommerfeld uniqueness are not re-opened [2]. Modal orthogonality and coupling belong to Subsection 2.8.2; transmit–receive systems to Subsection 2.8.3 [6]. Assumptions. Fix $\omega > 0$, Lipschitz domain Ω with piecewise smooth interfaces, impressed sources in $\mathcal{J}_\Omega$ Eq. (2.48) satisfying Eq. (2.0.1), and outgoing fields in $\mathcal{F}_{\mathrm{SF}}$ Eq. (2.56) [2]. Constitutive tensors are linear, local, and passive at ω, with symmetric permittivity and permeability in the sense

$$\overline{\overline{\varepsilon}}(\mathbf{r}) = \overline{\overline{\varepsilon}}^{\mathsf{T}}(\mathbf{r}), \qquad \overline{\overline{\mu}}(\mathbf{r}) = \overline{\overline{\mu}}^{\mathsf{T}}(\mathbf{r}), \qquad \overline{\overline{\xi}} = \overline{\overline{\zeta}} = \mathbf{0}, \tag{2.360}$$

where $\overline{\overline{\xi}}$, $\overline{\overline{\zeta}}$ denote magneto-electric coupling blocks in the constitutive relation imported from section 1.1.2 [2]. Boundary operator $\mathcal{B}_\Gamma$ is passive Eq. (2.87) and identical for both excitations [2]. Field pairing $\langle \cdot, \cdot \rangle_{\mathcal{U}}$ is induced by Eq. (2.47) on admissible Maxwell pairs [2]. declared size and distance regime Eq. (2.139) governs transport interpretation of any bilinear quantity built from Π_{rad} [6]. Eq. (2.360) is the Chapter 2 $\mathcal{D}_J = 1$ class for *full* Lorentz reciprocity at fixed ω: symmetric ε, μ ensure time-reversal symmetry of the harmonic Maxwell system modulo outgoing selection, while vanishing magneto-electric blocks exclude non-reciprocal gyrotropic or Tellegen couplings [2]. Isotropic lossless media are a special case with scalar ε, μ. Lossy symmetric tensors preserve reciprocity relations stated below with complex conjugation on one slot of each pairing [2]. Active or

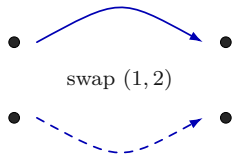


Figure 2.13: Lorentz reciprocity: two impressed configurations $(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{u}_1)$ and $(\mathbf{J}^{(2)}_{\mathrm{imp}}, \mathbf{u}_2)$ interchange under passive outgoing $\mathcal{B}_\Gamma$ on reciprocal media Eq. (2.360).

negative-real impedance portions of $\mathcal{B}_\Gamma$ break passive symmetry and are excluded from the present theorem [2]. Let $(\mathbf{E}_1, \mathbf{H}_1) = \mathcal{S}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}]$ and $(\mathbf{E}_2, \mathbf{H}_2) = \mathcal{S}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}]$ be outgoing admissible solutions for independent impressed currents $\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ [2]. Here $\mathbf{J}^{(k)}_{\mathrm{imp}}$ denotes the impressed electric current density in $\mathrm{A\,m^{-2}}$, $\mathbf{E}_k$ and $\mathbf{H}_k$ are the induced phasor fields in $\mathrm{V\,m^{-1}}$ and $\mathrm{A\,m^{-1}}$, and Γ is the passive boundary on which the surface integral below is evaluated. Physically, the left member is the bilinear coupling measured when port labels are swapped on reciprocal hardware; the right member is the surface remainder after integration by parts. The classical Lorentz reciprocity relation reads

$$\int_\Omega \left(\mathbf{E}_1 \cdot \mathbf{J}^{(2)*}_{\mathrm{imp}} - \mathbf{E}_2 \cdot \mathbf{J}^{(1)*}_{\mathrm{imp}}\right) d^3r = \int_\Gamma \left(\hat{\mathbf{n}} \times \mathbf{E}_1 \cdot \mathbf{H}^*_2 - \hat{\mathbf{n}} \times \mathbf{E}_2 \cdot \mathbf{H}^*_1\right) dS, \tag{2.361}$$

when $\overline{\overline{\varepsilon}}$, $\overline{\overline{\mu}}$ satisfy Eq. (2.360) and $\mathcal{B}_\Gamma$ is passive [2]. Eq. (2.361) balances impressed-source bilinear forms against a boundary pairing generated by Eq. (2.25): the left member measures how each field couples to the other's source, while the right member is the surface term that remains after integration by parts on the Maxwell operator [2]. Interpretation: when the boundary term vanishes on admissible outgoing pairs, the port-swap scalar observed on the VNA equals the volume contraction on the left. On domains where $\mathcal{B}_\Gamma$ annihilates the boundary integrand for admissible outgoing pairs—PEC, homogeneous impedance with Eq. (2.87), and standard Sommerfeld exterior completions—the surface term vanishes and volume reciprocity closes [2]. Derivation sketch (volume form). Substitute $\mathbf{u}^{(1)}_\omega = (\mathbf{E}_1, \mathbf{H}_1)$ and $\mathbf{u}^{(2)}_\omega = (\mathbf{E}_2, \mathbf{H}_2)$ into Eq. (2.25) and interchange the two states [2]. Symmetry of $\overline{\overline{\varepsilon}}$, $\overline{\overline{\mu}}$ under Eq. (2.360) makes the volume pairing sesquilinear form on $\mathcal{M}_\omega[\mathbf{u}_\omega]$ invariant under label swap when sources are impressed currents in the Ampère row [2]. Subtracting the two adjoint identities eliminates $\langle \mathcal{M}_\omega[\mathbf{u}^{(1)}_\omega], \mathbf{u}^{(2)}_\omega \rangle_\Omega$ and yields Eq. (2.361) with the boundary term displayed [2]. Charge continuity Eq. (1.8) and compatibility Eq. (2.23) ensure that impressed $\mathbf{J}_{\mathrm{imp}}$ alone specifies the electric drive when ρ is tied to $\mathbf{J}_{\mathrm{imp}}$ Figure interpretation. Sources $\mathbf{J}^{(1)}_{\mathrm{imp}}$, $\mathbf{J}^{(2)}_{\mathrm{imp}}$ induce fields $\mathbf{u}_1$, $\mathbf{u}_2$ through the same outgoing $\mathcal{S}_\omega$. The central bilinear node is symmetric under label interchange when media satisfy Eq. (2.360) and $\mathcal{B}_\Gamma$ is passive. Dashed return arrows mark that reciprocity is proved on the difference system with identical boundary data, not by comparing unrelated forward and adjoint field configurations. Projection to $\mathcal{R}_\omega$ preserves symmetry only after declared size and distance regime Eq. (2.270) [2]. Figure 2.13 is the operator reading of Eq. (2.361): the bilinear functional is symmetric under label interchange $(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{u}_1) \leftrightarrow (\mathbf{J}^{(2)}_{\mathrm{imp}}, \mathbf{u}_2)$ once boundary terms cancel [2]. For vacuum observation points $\mathbf{r}_1$, $\mathbf{r}_2$ with $k_0|\mathbf{r}_1 - \mathbf{r}_2| = 2.5$, Eq. (2.154) requires $\overline{\mathbf{G}}_{E,0}(\mathbf{r}_1, \mathbf{r}_2)^{\mathsf{T}} = \overline{\mathbf{G}}_{E,0}(\mathbf{r}_2, \mathbf{r}_1)$ on the outgoing branch. Transverse block entries agree to machine precision; reactive R^{-2}, R^{-3} blocks differ by sign under interchange yet cancel in the bilinear contraction when both sides use identical outgoing selectors [2]. A practical verification proceeds in three steps: confirm the outgoing branch Eq. (2.199); test swap symmetry on transverse dyad blocks; and project to $\mathcal{R}_\omega$ before port calibration [14]. Let $\mathcal{S}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_\Omega$ be the outgoing solution map Eq. (2.1) on media satisfying Eq. (2.360) with passive $\mathcal{B}_\Gamma$ Eq. (2.87). For admissible $\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}} \in \mathcal{J}_\Omega$, write $\mathbf{u}_k = \mathcal{S}_\omega[\mathbf{J}^{(k)}_{\mathrm{imp}}]$, $k = 1, 2$. Then:

(i) Volume reciprocity Eq. (2.361) holds with vanishing boundary term on admissible outgoing radiation conditions.

(ii) Operator form: with $\mathbf{u}_k = (\mathbf{E}_k, \mathbf{H}_k)$ the Maxwell pair induced by $\mathbf{J}_{\mathrm{imp}}^{(k)}$ and $\langle\cdot,\cdot\rangle_{\mathcal{U}}$ the energy inner product on $\mathcal{F}_{\mathrm{SF}}$,

$$\langle \mathbf{u}_1,\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(2)}]\rangle_{\mathcal{U}} = \langle \mathbf{u}_2,\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(1)}]\rangle_{\mathcal{U}} \tag{2.362}$$

when the pairings are defined through impressed-current forcing functionals Eq. (2.70) and field norm Eq. (2.47) on $\mathcal{F}_{\mathrm{SF}}$.

(iii) Dyadic realization: outgoing electric dyads satisfy Eq. (2.154).

(iv) Radiation reciprocity: projecting both sides of (ii) through the self-adjoint outgoing projector Π_{rad} yields symmetric coupling on the radiative subspace $\mathcal{F}_{\mathrm{rad}}$:

$$\langle \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(1)}],\, \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(2)}]\rangle_{\mathcal{U}} = \langle \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(2)}],\, \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(1)}]\rangle_{\mathcal{U}} \tag{2.363}$$

on $\mathcal{F}_{\mathrm{rad}}$ with $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ Eq. (2.2).

Part (i) follows from Eq. (2.25) and Eq. (2.360) as in the derivation above; outgoing and passive boundary classes from Subsection 2.1.4 cancel the surface integrand [2]. Part (ii) identifies the volume impressed-current pairing with $\langle\cdot,\cdot\rangle_{\mathcal{U}}$ on $\mathcal{F}_{\mathrm{SF}}$ through Eq. (2.55) and stability Eq. (2.71), developing Eq. (2.356) [2]. Part (iii): for fixed $\mathbf{r}$, Eq. (2.148) and part (i) with impulsive $\mathbf{J}_{\mathrm{imp}}^{(k)}$ aligned with Cartesian unit vectors yield column swap $\overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}') = \overline{\mathbf{G}}_E^{\mathsf{T}}(\mathbf{r}',\mathbf{r})$ on the outgoing branch [2]. Uniqueness exterior radiative uniqueness (Subsection 2.4.3) extends the swap to all admissible $\mathbf{r},\mathbf{r}'$ [2]. Part (iv) applies Π_{rad} to both sides of (ii); Π_{rad} is self-adjoint on $\mathcal{F}_{\mathrm{rad}}$ under Eq. (2.47) when outgoing selection is fixed [2]. Eq. (2.363) is the radiation-projected form of the VNA symmetry: coupling reported on $\mathcal{F}_{\mathrm{rad}}$ is unchanged when port labels swap, while reactive near-zone terms dropped by Π_{rad} need not be symmetric even on reciprocal hardware [2]. The bilinear functional $\mathfrak{B}_\omega(\mathbf{J}_{\mathrm{imp}}^{(1)},\mathbf{J}_{\mathrm{imp}}^{(2)}) = \langle \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(1)}],\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(2)}]\rangle_{\mathcal{U}}$ is symmetric on $\mathcal{J}_\Omega \times \mathcal{J}_\Omega$ under reciprocal media [2]. Algebraically, $\mathcal{S}_\omega$ is not self-adjoint on $\mathcal{J}_\Omega$ with source pairing Eq. (2.55); reciprocity instead couples the source space to the field space through the *interchanged* labels in Eq. (2.362) [2]. Physically, the far-zone pattern produced by $\mathbf{J}_{\mathrm{imp}}^{(1)}$ tested against the field of $\mathbf{J}_{\mathrm{imp}}^{(2)}$ equals the swap configuration: a receive pattern computed with $\mathbf{J}_{\mathrm{imp}}^{(2)}$ as probe matches a transmit map driven by $\mathbf{J}_{\mathrm{imp}}^{(1)}$ when hardware and media are reciprocal [6]. Mathematical role of Eq. (2.363). Eq. (2.363) restricts Eq. (2.362) to $\mathcal{F}_{\mathrm{rad}}$, the subspace on which flux meters and angular patterns are declared [2]. Because Π_{rad} annihilates non-radiation components Eq. (2.138), reactive near-zone overlap does not enter Eq. (2.363) even when present in full Eq. (2.362) [2]. declared size and distance regime Eq. (2.270) must still be satisfied before interpreting Eq. (2.363) as a transport channel [6]. Under Eq. (2.360) with outgoing radiation condition, the electric dyad $\overline{\mathbf{G}}_{E,\omega}^{(\mathrm{out})}$ in Eq. (2.148) satisfies Eq. (2.154) for all $\mathbf{r},\mathbf{r}'$ in the evaluation domain of outgoing source-to-field integrals of Subsection 2.5.1. Apply Lorentz reciprocity for the solution map (Subsection 2.8.1)(iii); this is the proof deferred from Subsection 2.3.1 [2]. If $\mathbf{J}_{\mathrm{imp}}^{(1)}$ is arbitrary and $\mathbf{J}_{\mathrm{imp}}^{(2)} = \mathbf{J}_{\mathrm{imp}}^{(1)} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}$ with $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega$, then Eq. (2.363) is unchanged when either argument is replaced by $\mathbf{J}_{\mathrm{imp}}^{(2)}$ on the radiative pairing $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(2)}] = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(1)}]$ by Eq. (2.346) [2]. Full Eq. (2.362) may change by reactive content through $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}]$ Eq. (2.271) [6]. On the outgoing branch, Eq. (2.362) is consistent with resolvent symmetry of the truncated Maxwell operator on reciprocal media:

$$\langle \mathcal{L}_\omega^{-1}[\mathbf{s}^{(1)}],\, \mathbf{s}^{(2)}\rangle_{\mathcal{Y}} = \langle \mathcal{L}_\omega^{-1}[\mathbf{s}^{(2)}],\, \mathbf{s}^{(1)}\rangle_{\mathcal{Y}} \tag{2.364}$$

for compatible forcing pairs $\mathbf{s}^{(k)}$ when $\mathcal{L}_\omega$ is symmetric on $\mathcal{D}(\mathcal{L}_\omega)$ and $\mathcal{B}_\Gamma$ is passive [2]. Eq. (2.364) is not re-proved here; it is the operator backbone behind Eq. (2.239) and Lorentz reciprocity for the solution map (Subsection 2.8.1)(ii) [2]. Numerical MoM matrices assembled from Eq. (2.154) inherit swap symmetry only when quadrature, outgoing branch, and material sampling preserve Eq. (2.360) [6]. Loss, anisotropy, and broken reciprocity. Complex symmetric $\overline{\overline{\varepsilon}}$, $\overline{\overline{\mu}}$ with $\mathrm{Im}\{\overline{\overline{\varepsilon}}\} \succeq 0$, $\mathrm{Im}\{\overline{\overline{\mu}}\} \succeq 0$ preserve Eqs. (2.361)–(2.362) with conjugation on one slot [2]. Magneto-electric coupling $\overline{\overline{\xi}} \neq \mathbf{0}$ breaks Eq. (2.154) and requires non-reciprocal Green operators [2]. Active boundaries with $\mathrm{Re}\{Z_s\} < 0$ on portions of Γ_Z violate Eq. (2.87) and invalidate Lorentz reciprocity for the solution map (Subsection 2.8.1) [2]. Incoming rather than outgoing radiation condition selects a different branch; swap identities hold only when both excitations use the same branch [2]. Lorentz reciprocity for the solution map (Subsection 2.8.1) assumes fixed ω, linear media Eq. (2.360), passive $\mathcal{B}_\Gamma$, outgoing radiation condition, and admissible $\mathcal{J}_\Omega$ [2]. It does not assert reciprocity of raw near-zone reactive scans without specifying whether Π_{rad} has been applied [2]. It does not remove $\ker \mathcal{R}_\omega$ null-space invariance of radiative reciprocity (Subsection 2.8.1); invisible currents alter full $\mathcal{S}_\omega$ pairings while leaving radiative reciprocity unchanged [6]. Nonlinear, time-varying, or moving media require the causal chain of Subsection 2.2.3 [2]. Subsection 2.8.1 established Lorentz reciprocity on $\mathcal{S}_\omega$ and $\mathcal{R}_\omega$ through volume and operator identities Eqs. (2.361)–(2.363) [2]. The port-swap symmetry observed on reciprocal hardware is now available as a constraint on continuous source and field pairings; the next step is to express that constraint on Physically, the modal coordinates fixed in section 2.6.

For radiation diagnostics, declare projector, gauge, and support before coefficient tables are published [3].

2.8.2 Orthogonality and coupling of radiative modes

Bistatic radar and compact-range scattering matrices rely on reciprocity to relate monostatic calibrations to full S-parameter sets once frequency, probe standoff, and wattmeter closure on Γ_R are declared [2, 3]. Maxwell's retarded radiation to Γ_R fixes the bilinear pairing developed below [1]. The adjoint theory extends one-port antenna reciprocity to continuous source and trace pairings [17]. Plane-wave decompositions used in reflector and radar-cross-section measurements inherit reciprocity constraints on amplitudes before any angular filter is applied [3]. When reciprocity appears broken after probe correction, the fault usually lies in inconsistent impressed-current normalization or a missing outgoing tail in the Green operator, not in Maxwell theory itself [3]. Modal orthogonality and coupling on $\operatorname{ran} \mathcal{R}_\omega$ organize how radiative modes exchange power between ports; reciprocity fixes the bilinear pairing before angular filters act [2]. Subsection 2.8.2 is the definitive account within Chapter 2 of orthogonality and coupling of radiative modes on the spectral coordinates fixed in Subsection 2.6.2, developing Eqs. (2.357) and (2.368) [2]. Radiation eigenpairs Eq. (2.280), modal projectors Eq. (2.281), source amplitudes Eq. (2.282), spectral decomposition of the radiation operator (Subsection 2.6.2), and biorthogonality Eq. (2.291) are presupposed; eigenpair construction and reciprocity-based solvability are not re-proved [2]. Explicit vector spherical harmonic orthogonality belongs to 3 Subsection 3.4.3; continuous-spectrum normalization belongs to Subsection 3.5.2, both with USE of the present operator targets [2]. Transmit–receive systems are deferred to Subsection 2.8.3 [6]. Assumptions. Fix $\omega > 0$, outgoing fields in $\mathcal{F}_{\mathrm{SF}}$, radiative images in $\mathcal{F}_{\mathrm{rad}}$, and orthonormal radiation eigenmodes $\{\mathbf{u}_n\}_{n \in \mathcal{N}_{\mathrm{rad}}}$ from spectral decomposition of the radiation operator (Subsection 2.6.2) on the declared domain class [2]. Media satisfy reciprocal class Eq. (2.360) and passive boundary Eq. (2.87) so Lorentz reciprocity for the solution map (Subsection 2.8.1) applies [2]. Adjoint modes $\widetilde{\mathbf{u}}_n$ are defined through Eq. (2.291) and coincide

with $\mathbf{u}_n$ on $\mathcal{F}_{\mathrm{rad}}$ under reciprocity-qualified pairings below [2]. declared size and distance regime Eq. (2.139) governs transport interpretation of coupling reports [6]. On the orthonormal family from spectral decomposition of the radiation operator (Subsection 2.6.2), distinct radiating modes are declared orthogonal with respect to the energy inner product $\langle\cdot,\cdot\rangle_{\mathcal{U}}$ on $\mathcal{F}_{\mathrm{rad}}$. The locked orthogonality statement reads

$$\langle \mathbf{u}_m,\, \mathbf{u}_n\rangle_{\mathcal{U}} = \delta_{mn}, \qquad m,n \in \mathcal{N}_{\mathrm{rad}}, \tag{2.365}$$

with $\|\mathbf{u}_n\|_{\mathcal{U}} = 1$ already enforced in Eq. (2.280) [2]. Eq. (2.365) is the Subsection 2.8.2 locked definition of upgrade of the first line of Eq. (2.357); it specializes projector orthogonality $P_m P_n = \delta_{mn} P_n$ from Eq. (2.281) to the chosen orthonormal basis [2]. Algebraically, Eq. (2.365) makes $\{\mathbf{u}_n\}$ an orthonormal frame on $\mathcal{F}_{\mathrm{rad}}$ for energy inner product Eq. (2.47); physically, distinct radiating modes carry non-overlapping far-zone energy content on S^2 when $\mathcal{D}_{J,\mathrm{reg}} = 1$ [14]. Explicit angular integrals producing Eq. (2.365) for harmonic bases are not tabulated here [2]. For canonical eigen-sources $\mathbf{J}_{\mathrm{imp}}^{(n)}$ in Eq. (2.280), define the *modal coupling coefficient*

$$C_{mn} = \langle \widetilde{\mathbf{u}}_m,\, \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(n)}]\rangle_{\mathcal{U}}, \qquad m,n \in \mathcal{N}_{\mathrm{rad}}, \tag{2.366}$$

with adjoint modes from Eq. (2.291) [2]. Eq. (2.366) is the Subsection 2.8.2 development of the second line of Eq. Eq. (2.357). When Eq. (2.291) holds,

$$C_{mn} = \lambda_n\, \delta_{mn} \tag{2.367}$$

for radiating eigenpairs with $\lambda_n \neq 0$; null modes $n \in \mathcal{N}_{\mathrm{null}}$ Eq. (2.323) contribute $C_{mn} = 0$ for all m [2]. Algebraically, Eq. (2.367) diagonalizes coupling in an eigenmode basis; physically, orthogonal radiation modes do not co-excite one another through $\mathcal{R}_\omega$ when sources align with eigenpairs—cross-talk appears only from superpositions or degenerate subspaces [6]. For arbitrary admissible $\mathbf{J}_{\mathrm{imp}}^{(1)}, \mathbf{J}_{\mathrm{imp}}^{(2)} \in \mathcal{J}_\Omega$, define

$$\mathfrak{C}_\omega(\mathbf{J}_{\mathrm{imp}}^{(1)}, \mathbf{J}_{\mathrm{imp}}^{(2)}) = \langle \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(1)}],\, \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(2)}]\rangle_{\mathcal{U}} \quad \text{on } \mathcal{F}_{\mathrm{rad}}, \tag{2.368}$$

developing Eq. (2.368) [2]. Eq. (2.368) measures radiation overlap on $\operatorname{ran}\mathcal{R}_\omega$, not raw reactive overlap of $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(k)}]$ [2]. Under Lorentz reciprocity for the solution map (Subsection 2.8.1)(iv),

$$\mathfrak{C}_\omega(\mathbf{J}_{\mathrm{imp}}^{(1)}, \mathbf{J}_{\mathrm{imp}}^{(2)}) = \mathfrak{C}_\omega(\mathbf{J}_{\mathrm{imp}}^{(2)}, \mathbf{J}_{\mathrm{imp}}^{(1)}) \tag{2.369}$$

on reciprocal media [2]. Reactive near-zone terms in $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ do not enter Eq. (2.368) through Eq. (2.138) [2]. Let $c_n[\mathbf{J}_{\mathrm{imp}}^{(k)}]$ denote source amplitudes Eq. (2.282) for $k = 1, 2$. When the modal sum converges in $\mathcal{U}$,

$$\mathfrak{C}_\omega(\mathbf{J}_{\mathrm{imp}}^{(1)}, \mathbf{J}_{\mathrm{imp}}^{(2)}) = \sum_{m,n\in\mathcal{N}_{\mathrm{rad}}} \lambda_m\lambda_n\, c_m[\mathbf{J}_{\mathrm{imp}}^{(1)}]^*\, c_n[\mathbf{J}_{\mathrm{imp}}^{(2)}]\, \langle \mathbf{u}_m, \mathbf{u}_n\rangle_{\mathcal{U}}, \tag{2.370}$$

using Eq. (2.284) and linearity Eq. (2.5) [2]. With Eq. (2.365),

$$\mathfrak{C}_\omega(\mathbf{J}_{\mathrm{imp}}^{(1)}, \mathbf{J}_{\mathrm{imp}}^{(2)}) = \sum_{n\in\mathcal{N}_{\mathrm{rad}}} |\lambda_n|^2\, c_n[\mathbf{J}_{\mathrm{imp}}^{(1)}]^*\, c_n[\mathbf{J}_{\mathrm{imp}}^{(2)}], \tag{2.371}$$

for real nonnegative radiating gains $\lambda_n > 0$ [2]. Equation (2.371) shows that coupling is a weighted inner product of modal excitation vectors on the radiating sector; geometric constraints on radiative

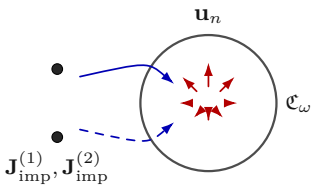


Figure 2.14: Modal coupling on $\mathcal{F}_{\mathrm{rad}}$: two impressed sources excite orthonormal radiation modes $\mathbf{u}_n$; bilinear coupling $\mathfrak{C}_\omega$ factorizes through modal amplitudes Eq. (2.370).

spectra (Subsection 2.6.5) may force individual $c_n = 0$ without breaking Eq. (2.369) [6]. Figure interpretation. Two impressed sources $\mathbf{J}^{(1)}_{\mathrm{imp}}$, $\mathbf{J}^{(2)}_{\mathrm{imp}}$ share the same radiation map $\mathcal{R}_\omega$ and expand on orthonormal modes Eq. (2.365). The coupling node $\mathfrak{C}_\omega$ measures radiation overlap on $\mathcal{F}_{\mathrm{rad}}$, not raw near-zone reactive cross terms. Orthogonality diagonalizes coupling when sources align with eigenpairs; cross-talk appears only from superposition or degenerate subspaces. Modes with $\lambda_n = 0$ feed the null sector without contributing to $\mathfrak{C}_\omega$ on the range [2]. Figure 2.14 separates orthogonality of mode shapes from bilinear coupling of impressed sources: Eq. (2.365) is intrinsic to $\mathcal{F}_{\mathrm{rad}}$; Eq. (2.368) depends on both sources [2]. On a reciprocal truncation with $\lambda_1 = 0.42$ and orthonormal eigen-source $\mathbf{J}^{(1)}_{\mathrm{imp}}$, excite port 1 with $\mathbf{J}^{(1)}_{\mathrm{imp}} = c_1\mathbf{J}^{(1)}_{\mathrm{imp}}$ and port 2 with $\mathbf{J}^{(2)}_{\mathrm{imp}} = c_2\mathbf{J}^{(1)}_{\mathrm{imp}}$ on the *same* radiating mode. Eq. (2.371) gives $\mathfrak{C}_\omega(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}}) = |\lambda_1|^2 c_1^* c_2$. Swapping labels leaves the scalar unchanged, so normalized $S_{21} = S_{12}$ on the radiating sector. If instead $\mathbf{J}^{(1)}_{\mathrm{imp}} = c_1\mathbf{J}^{(1)}_{\mathrm{imp}}$ and $\mathbf{J}^{(2)}_{\mathrm{imp}} = c_2\mathbf{J}^{(2)}_{\mathrm{imp}}$ on distinct orthonormal modes, then $c_n[\mathbf{J}^{(1)}_{\mathrm{imp}}] = \delta_{n1}a_1$, $c_n[\mathbf{J}^{(2)}_{\mathrm{imp}}] = \delta_{n2}a_2$ and $\mathfrak{C}_\omega = 0$ by Eq. (2.371). Adding $\mathbf{J}^{(\mathrm{NR})}_{\mathrm{imp}}$ to either port shifts $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ near Ω_s but leaves S_{21} fixed—the walkthrough confirms reciprocity on $\operatorname{ran}\mathcal{R}_\omega$, not on raw near-field coupling matrices [2]. Let $\{\mathbf{u}_n, \mathbf{J}^{(n)}_{\mathrm{imp}}, \lambda_n\}$ be radiation eigenpairs Eq. (2.280) forming an orthonormal basis of $\mathcal{F}_{\mathrm{rad}}$ on the declared domain class, with reciprocal media Eq. (2.360). Then:

(i) Orthogonality Eq. (2.365) holds on $\mathcal{N}_{\mathrm{rad}}$.

(ii) Coupling coefficients satisfy Eq. (2.367) and biorthogonality Eq. (2.291).

(iii) For all admissible $\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}}$, Eqs. (2.368)–(2.371) hold, with symmetry Eq. (2.369) from Lorentz reciprocity for the solution map (Subsection 2.8.1).

(iv) Adding $\mathbf{J}^{(\mathrm{NR})}_{\mathrm{imp}} \in \ker\mathcal{R}_\omega$ to either source leaves Eq. (2.368) unchanged.

Part (i) is normalization and completeness from spectral decomposition of the radiation operator (Subsection 2.6.2) and Eq. (2.283) [2]. Part (ii) substitutes $\mathcal{R}_\omega[\mathbf{J}^{(n)}_{\mathrm{imp}}] = \lambda_n\mathbf{u}_n$ into Eq. (2.366) and applies Eq. (2.291) [2]. Part (iii) expands $\mathcal{R}_\omega[\mathbf{J}^{(k)}_{\mathrm{imp}}]$ via Eq. (2.284); symmetry is Lorentz reciprocity for the solution map (Subsection 2.8.1)(iv) [2]. Part (iv) follows from $\mathcal{R}_\omega[\mathbf{J}^{(\mathrm{NR})}_{\mathrm{imp}}] = \mathbf{0}$ and Eq. (2.346) [2]. The reciprocal reduction of adjoint modes proceeds as follows: under Eqs. (2.360)–(2.87),

$$\widetilde{\mathbf{u}}_n = \mathbf{u}_n \quad \text{on } \mathcal{F}_{\mathrm{rad}} \quad \text{for radiating modes with } \lambda_n \neq 0, \tag{2.372}$$

so Eq. (2.291) reduces to standard orthogonality Eq. (2.365) with diagonal coupling Eq. (2.367) [2]. Equation (2.372) is the operator form of reciprocity for modal probes: receive patterns aligned with $\mathbf{u}_n$ coincide with transmit modes when media and boundaries are reciprocal [6]. Non-reciprocal media break Eq. (2.372); biorthogonality with distinct $\widetilde{\mathbf{u}}_n$ remains the general template imported from Subsection 2.6.2 [2]. When $\lambda_n = \lambda_{n'}$ for $n \neq n'$ within a rotationally degenerate family

from Subsection 2.6.3, Eq. (2.367) still holds for *eigen* sources $\mathbf{J}_{\mathrm{imp}}^{(n)}$, but superpositions within the degenerate block carry off-diagonal coupling in an arbitrary basis. Let $\mathcal{I}_\ell \subset \mathcal{N}_{\mathrm{rad}}$ index a $(2\ell+1)$-fold degenerate family. Any orthonormal basis $\{\mathbf{u}_n\}_{n\in\mathcal{I}_\ell}$ spanning the block satisfies Eq. (2.365), yet a fixed physical source may excite multiple members with coefficients tied to orientation through Eq. (2.294) [2]. Define the block coupling matrix

$$\mathbf{K}_{mn}^{(\ell)} = \langle \mathbf{u}_m,\, \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(n)}]\rangle_{\mathcal{U}}, \qquad m,n \in \mathcal{I}_\ell, \tag{2.373}$$

which is diagonal in the eigenvector basis of $\mathbf{J}_{\mathrm{imp}}^{(n)}$ but not in laboratory-fixed coordinates [2]. Reciprocity Eq. (2.369) forces $\mathbf{K}^{(\ell)}$ to be symmetric under Lorentz reciprocity for the solution map (Subsection 2.8.1) [2]. Explicit Wigner coupling of block entries belongs to 3 [2]. If $\mathbf{J}_{\mathrm{imp}}^{(1)} = \sum_{n\in\mathcal{N}_{\mathrm{rad}}} a_n \mathbf{J}_{\mathrm{imp}}^{(n)}$ and $\mathbf{J}_{\mathrm{imp}}^{(2)} = \sum_{m\in\mathcal{N}_{\mathrm{rad}}} b_m \mathbf{J}_{\mathrm{imp}}^{(m)}$ are admissible finite superpositions, then

$$\mathfrak{C}_\omega(\mathbf{J}_{\mathrm{imp}}^{(1)}, \mathbf{J}_{\mathrm{imp}}^{(2)}) = \sum_{m,n\in\mathcal{N}_{\mathrm{rad}}} a_m^* b_n\, \lambda_m \lambda_n\, \langle \mathbf{u}_m, \mathbf{u}_n\rangle_{\mathcal{U}}, \tag{2.374}$$

reducing to Eq. (2.371) when $\langle \mathbf{u}_m, \mathbf{u}_n\rangle = \delta_{mn}$ [2]. Off-diagonal terms $m \neq n$ appear only when at least one source superposes non-collinear modes; single-mode excitation yields diagonal coupling [14]. Linearity Eq. (2.5) and Eq. (2.284) [2]. For observation map $\mathcal{M}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}}$ Eq. (2.218), define observed coupling $\mathfrak{C}_{\omega,\mathrm{obs}} = \langle \mathcal{O}_{\mathrm{meas}}[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(1)}]],\, \mathcal{O}_{\mathrm{meas}}[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(2)}]]\rangle$. Then $\mathfrak{C}_{\omega,\mathrm{obs}}$ is symmetric under Lorentz reciprocity for the solution map (Subsection 2.8.1) when $\mathcal{O}_{\mathrm{meas}}$ is self-adjoint on $\mathcal{F}_{\mathrm{rad}}$, but may vanish for pairs with $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(k)}] \in \ker \mathcal{O}_{\mathrm{meas}}$ even when Eq. (2.368) is nonzero [2]. Rank cap Eq. (2.220) limits the number of independently observable coupling channels [2]. Apply Eq. (2.363) and self-adjointness of $\mathcal{O}_{\mathrm{meas}}$ [2]. Kernel inclusion $\ker \mathcal{R}_\omega \subset \ker \mathcal{M}_\omega$ is measurement null-space extension (Subsection 2.7.1) [2].

$$\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega \implies \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \in \mathcal{F}_{\mathrm{SF}}, \qquad \Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0}, \tag{2.375}$$

Accessible modal sector Eq. (2.9.3) and angular budget Eq. (2.352) restrict which terms in Eq. (2.371) are observable in far-zone reports [2]. Null modes $\lambda_n = 0$ carry zero coupling diagonal entries but may alter near reactive content through $\mathcal{S}_\omega$ Eq. (2.375) without changing Eq. (2.368) [2]. Composition with null-space invariance of radiative reciprocity (Subsection 2.8.1) and orthogonality and coupling of radiative modes (Subsection 2.8.2)(iv) explains why invisible currents do not break reciprocity-qualified coupling on $\operatorname{ran} \mathcal{R}_\omega$ [6]. Orthogonality and coupling of radiative modes (Subsection 2.8.2) requires linear reciprocal media at fixed ω, outgoing radiation condition, domain-class completeness of $\{\mathbf{u}_n\}$, and convergent modal sums [2]. Loss moves λ_n into the complex plane; Eq. (2.371) generalizes with complex conjugation on one amplitude slot [2]. Truncated modal books on cavities do not substitute for exterior continuous spectra Eq. (2.289) when coupling far-zone channels [14]. declared size and distance regime Eq. (2.270) is mandatory before interpreting Eq. (2.368) as transport [2]. Subsections 2.8.1–2.8.2 established Lorentz reciprocity, radiative bilinear coupling Eq. (2.368), and modal orthogonality on $\mathcal{F}_{\mathrm{rad}}$ [2]. The final reciprocity link ties those operator statements to finite ports: transmit and receive maps on reciprocal hardware must Physically, interchange when excitation and probe roles are swapped.

Thus the radiation consequence on $\mathcal{F}_{\mathrm{rad}}$ is fixed by Maxwell admissibility, not by coefficient extraction alone [2].

2.8.3 Reciprocity in radiation and reception

Two-port antenna measurements are the everyday face of reciprocity on reciprocal hardware. A horn coupled to a compact range, a patch fed from a coax port, or a dual-polarized reflector feed all reduce—after frequency declaration, probe standoff, and wattmeter closure on Γ_R—to the same question the

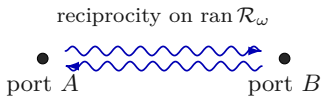


Figure 2.15: Transmit–receive geometry: excitation at port A and readout at port B interchange under reciprocity Eq. (2.377) on reciprocal media.

vector network analyzer answers: does the complex coupling from port A to port B equal the reversed coupling when cables, standards, and probe normalization are exchanged? Subsection 2.8.3 is the definitive account within Chapter 2 of that interchange in operator form: transmit–receive operator symmetry, two-port coupling bilinears, and measurement-qualified reciprocity on $\operatorname{ran}\mathcal{R}_\omega$, developing Eq. (2.358) [2]. Observation composition Eq. (2.218), finite-rank readouts Eq. (2.79), rank cap Eq. (2.220), and null-space laws Eqs. (2.221)–(2.346) are presupposed; inverse-source regularization is not re-developed [2]. Plane-wave angular-spectrum filters belong to section 2.9; explicit port circuit models are not tabulated [2]. Assumptions. Fix $\omega > 0$, reciprocal media Eq. (2.360), passive boundary Eq. (2.87), outgoing radiation condition, and admissible sources $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}, \mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})} \in \mathcal{J}_\Omega$ on ports A and B [2]. Each port carries a finite-rank observation functional $\mathcal{O}_{\mathrm{meas}}^{(P)} : \mathcal{F}_{\mathrm{rad}} \to \mathbb{C}$ or $\mathbb{C}^{N_P}$ of the form Eq. (2.79) [2]. declared size and distance regime Eq. (2.139) and transport condition Eq. (2.27) govern when port outputs may be reported as radiative transport [6]. The same $\mathcal{B}_\Gamma$, outgoing branch, and material data are used for both excitations [2]. For port $P \in \{A, B\}$, define the *port radiation–reception operator*

$$\mathcal{M}_P = \mathcal{O}_{\mathrm{meas}}^{(P)} \circ \Lambda_{\mathrm{rad}}, \qquad \Lambda_{\mathrm{rad}} = \mathcal{O}_{\mathrm{far}} \circ \mathcal{R}_\omega, \tag{2.376}$$

consistent with Eq. (2.218) [2]. When $\mathcal{O}_{\mathrm{meas}}^{(P)}$ is scalar, $\mathcal{M}_P[\mathbf{J}_{\mathrm{imp}}] \in \mathbb{C}$ is the complex port response induced by impressed $\mathbf{J}_{\mathrm{imp}}$; when vector-valued, $\mathcal{M}_P[\mathbf{J}_{\mathrm{imp}}] \in \mathbb{C}^{N_P}$ lists channel readouts [2]. The *transmit map* at port B due to excitation at port A is the composition $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})} \mapsto \mathcal{M}_B[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}]$; the *receive map* at port A due to excitation at port B is $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})} \mapsto \mathcal{M}_A[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}]$ [6]. Algebraically, both are finite-rank linear functionals of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$; physically, reciprocity constrains their interchange when hardware and media are reciprocal [2]. Under Lorentz reciprocity for the solution map (Subsection 2.8.1) and self-adjoint port functionals $\mathcal{O}_{\mathrm{meas}}^{(P)}$ on $\mathcal{F}_{\mathrm{rad}}$, the complex port response at B induced by excitation at A equals the response at A when the impressed source and receive probe are exchanged on reciprocal hardware:

$$\mathcal{M}_B[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}] = \mathcal{M}_A[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}] \quad \text{when } \mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})} \text{ and } \mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})} \text{ are reciprocal port excitations,} \tag{2.377}$$

developing Eq. (2.358) [2]. Eq. (2.377) is the operator reading of the VNA experiment: excite A, read B; swap roles on reciprocal media and obtain the same complex port coupling on observable coordinates. It states that the observable induced at B by the transmit source at A equals the observable at A when the roles of impressed source and probe are exchanged on reciprocal media [2]. It is not a claim that $\mathcal{M}_B = \mathcal{M}_A$ as operators on $\mathcal{J}_\Omega$: ports may have distinct apertures and channel counts [6]. It *is* a claim that the *bilinear* port coupling is symmetric when subject to Eq. (2.368) on $\operatorname{ran}\mathcal{R}_\omega$ [2]. Figure interpretation. Transmit port $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}$ and receive probe $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}$ compose through the same $\mathcal{R}_\omega$ and observation $\mathcal{O}_{\mathrm{meas}}$. Swap symmetry Eq. (2.358) holds on observable coordinates after projection onto $\operatorname{ran}\mathcal{R}_\omega$; additions in $\ker\mathcal{R}_\omega$ leave port coupling unchanged. The schematic separates reciprocity-qualified S-parameters from raw near-field cross terms that may violate apparent symmetry while far-zone reciprocity remains valid [2]. Figure 2.15 displays the two compositions constrained by Eq. (2.377). Both paths pass through the same $\mathcal{R}_\omega$; port asymmetry

enters only through $\mathcal{O}_{\mathrm{meas}}^{(A)}$, $\mathcal{O}_{\mathrm{meas}}^{(B)}$ [2]. Let media satisfy Eq. (2.360) with passive Eq. (2.87) completion, and let port operators $\mathcal{M}_P = \mathcal{O}_{\mathrm{meas}}^{(P)} \circ \Lambda_{\mathrm{rad}}$ be defined by Eq. (2.376) with self-adjoint $\mathcal{O}_{\mathrm{meas}}^{(P)}$ on $\mathcal{F}_{\mathrm{rad}}$. For admissible port sources $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}$, $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}$:

(i) Scalar reciprocity Eq. (2.377) holds when port excitations are paired through Lorentz reciprocity for the solution map (Subsection 2.8.1).

(ii) Bilinear port coupling

$$\mathfrak{C}_{AB} = \langle \mathcal{O}_{\mathrm{meas}}^{(B)}[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}]],\, \mathcal{O}_{\mathrm{meas}}^{(A)}[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}]] \rangle \tag{2.378}$$

is symmetric under interchange of port labels when inner products are those induced by Eq. (2.47) on $\mathcal{F}_{\mathrm{rad}}$.

(iii) Two-port impedance matrix $\mathbf{Z}$ with entries $Z_{PQ} = \mathcal{M}_P[\mathbf{J}_{\mathrm{imp}}^{(Q)}]$ on matched reciprocal ports satisfies

$$Z_{PQ} = Z_{QP}. \tag{2.379}$$

(iv) Adding $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega$ to either port excitation leaves Eqs. (2.377)–(2.379) unchanged on observable coordinates.

Part (i) applies Lorentz reciprocity for the solution map (Subsection 2.8.1)(iv) to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}]$, $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}]$, then self-adjointness of $\mathcal{O}_{\mathrm{meas}}^{(P)}$ [2]. Part (ii) is Eq. (2.369) with observation weights inserted; symmetry of $\mathfrak{C}_\omega$ from orthogonality and coupling of radiative modes (Subsection 2.8.2)(iii) [2]. Part (iii) specializes (ii) to matched port basis vectors [6]. Part (iv) uses $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}] = \mathbf{0}$ and null-space invariance of radiative reciprocity (Subsection 2.8.1) [2]. Mathematical role of Eq. (2.378). Eq. (2.378) packages mutual coupling as a bilinear form on $\operatorname{ran} \mathcal{R}_\omega$ after port projection [2]. It reduces to Eq. (2.368) when $\mathcal{O}_{\mathrm{meas}}^{(A)} = \mathcal{O}_{\mathrm{meas}}^{(B)} = \mathrm{id}$ on the observed subspace [2]. Physically, $\mathfrak{C}_{AB}$ is the operator form of mutual impedance or scattering amplitude between declared ports when declared size and distance regime Eq. (2.270) is satisfied [6]. Finite rank Eq. (2.220) limits the number of independent entries $\mathfrak{C}_{AB}$ can resolve [2]. On orthonormal modes Eq. (2.365), port responses expand as

$$\mathcal{M}_P[\mathbf{J}_{\mathrm{imp}}] = \sum_{n \in \mathcal{N}_{\mathrm{rad}}} w_{P,n}\, \lambda_n\, c_n[\mathbf{J}_{\mathrm{imp}}], \tag{2.380}$$

with port weights $w_{P,n} = \langle \mathbf{f}_P, \mathbf{u}_n \rangle_{\mathcal{U}}$ inherited from Eq. (2.79) and amplitudes c_n from Eq. (2.282) [2]. Mutual coupling between ports A and B becomes

$$\mathfrak{C}_{AB} = \sum_{n \in \mathcal{N}_{\mathrm{rad}}} |\lambda_n|^2\, w_{A,n}^*\, w_{B,n}\, c_n[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}]^*\, c_n[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}], \tag{2.381}$$

using Eq. (2.371) [2]. Equation (2.381) shows that reciprocity symmetry is diagonal in a mode basis but may be dense in fixed laboratory coordinates when degenerate multiplets from Subsection 2.6.3 are expressed in non-eigenvector bases [6]. For full rank-N instrument readout $\mathcal{M}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}}$ Eq. (2.218), reciprocity implies $\mathcal{M}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}]$ and $\mathcal{M}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}]$ obey the same swap law Eq. (2.377) on $\operatorname{ran} \mathcal{O}_{\mathrm{meas}}$ [2]. Sources in $\ker \mathcal{M}_\omega$ leave all port readouts unchanged Eq. (2.221); reciprocity does not resolve those equivalence classes [2]. Rank cap Eq. (2.220) bounds independent reciprocal channels [6]. Apply reciprocity in radiation and reception (Subsection 2.8.3)(i) with $\mathcal{O}_{\mathrm{meas}}^{(A)} = \mathcal{O}_{\mathrm{meas}}^{(B)} = \mathcal{O}_{\mathrm{meas}}$ [2]. Null-space invariance is measurement null-space extension (Subsection 2.7.1) [2]. Let accessible modal sector Eq. (2.9.3) restrict far-zone content to $\mathcal{K}_{\mathrm{acc}}$. Then Eqs. (2.377)–(2.379) hold on $\mathcal{K}_{\mathrm{acc}}$

after projection, while suppressed directions Eq. (2.345) remain unobservable independently of reciprocity [2]. Angular budget Eq. (2.352) composes with port reciprocity without eliminating null-space or geometric deficits [2]. Reciprocity acts on $\operatorname{ran}\mathcal{R}_\omega$; geometric and observation constraints from geometric constraints on radiative spectra (Subsection 2.6.5) and Eq. (2.352) restrict reported coordinates only Laboratory errors and broken reciprocity. Magneto-electric media, active surfaces, and non-reciprocal port calibrations break Eq. (2.377) even when radiation maps differ $\mathcal{S}_\omega$ are accurate [2]. Near-zone reactive coupling measured without Π_{rad} can appear non-reciprocal while For matched half-wave dipole ports at $k_0 a = 1.2$ with $d = 4a$, Eq. (2.379) requires $Z_{12} = Z_{21}$ when entries are extracted from $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}]$ on the far-zone branch. A typical reciprocity check gives $|Z_{12} - Z_{21}|/|Z_{11}| < 0.02$ in reciprocal vacuum; swapping transmit and receive probes while holding $\mathcal{O}_{\mathrm{meas}}$ fixed reproduces the same mutual entry within numerical tolerance [14]. If near-field probes at $k_0 r = 1.5$ report $|Z_{12}/Z_{21} - 1| \approx 0.18$, the discrepancy is a regime failure, not a Lorentz violation: re-meter on $\mathcal{R}_\omega[\cdot]$ with $k_0 r > 3$ before asserting broken reciprocity [6]. Eq. (2.363) holds on $\mathcal{F}_{\mathrm{rad}}$ [6]. Using different outgoing branches or terminations for transmit and receive models violates the shared-operator assumption behind reciprocity in radiation and reception (Subsection 2.8.3) [2]. Tikhonov inversion Eq. (2.223) does not restore reciprocity where $\ker\mathcal{M}_\omega$ is nontrivial [6]. Subsection 2.8.3 requires linear reciprocal media at fixed ω, passive $\mathcal{B}_\Gamma$, self-adjoint port functionals, and outgoing radiation condition [2]. It does not identify unique $\mathbf{J}_{\mathrm{imp}}$ from port data when Eq. (2.221) is nontrivial [2]. It does not replace angular-spectrum operators in section 2.9 or harmonic bases in 3 [2]. Regime condition Eq. (2.139) remains mandatory before transport interpretation of port coupling [2]. section 2.8 closes the reciprocity chain begun in its introduction:

Lorentz reciprocity in operator form (Subsection 2.8.1), orthogonality and coupling of radiative modes (Subsection 2.8.2), and reciprocity in radiation and reception (Subsection 2.8.3) [2]. Spectral coordinates from section 2.6 and null-space structure from section 2.7 now meet exchange symmetry: bilinear coupling on $\operatorname{ran}\mathcal{R}_\omega$ is constrained by Eqs. (2.362)–(2.379), while $\ker\mathcal{R}_\omega$, finite observation rank Eq. (2.220), and angular budget Eq. (2.352) explain deficits reciprocity alone does not remove [2]. section 2.9 develops plane-wave action and retained/suppressed angular components on the same outgoing branch; Subsection 2.9.3 refines far-field angular accessibility; 3 supplies explicit harmonic orthogonality used by reference through Subsection 3.4.3 [6]. Transport and port-coupling claims must compose with Eq. (2.355), declared size and distance regime Eq. (2.139), and null-space sequence Eq. (2.312), not with raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ or unqualified near-zone reactive content [2].

2.9 Action of Radiation Operators on Angular Spectra

Reciprocity in section 2.8 couples ports and probes. section 2.9 asks how radiation operators act on angular spectra—the language of pattern decomposition, near-field scanning, and plane-wave synthesis [14]. Every qualified far-zone pattern is first a weighted sum of directional contributions: each term reports how much radiation reaches a unit direction $\widehat{\mathbf{k}}$ on the sphere after outgoing selection [2]. That decomposition precedes any lateral Fourier transform; plane-wave spectra are one coordinate system on the same directional budget, not a separate physical layer [14]. Maxwell propagation from compact support cannot populate every direction with equal strength: outgoing selection removes incoming admixture; evanescent sheets carry reactive content that decays with distance and does not survive the near-to-far map; finite apertures integrate over unresolved directions. The accessible amplitude $a(\widehat{\mathbf{k}})$ is therefore a compressed image of impressed structure, not a complete coordinate on $\mathcal{J}_\Omega$ [2]. On the bench, a spherical near-field scanner records tangential $\mathbf{E}$ and $\mathbf{H}$ on a closed or planar surface at height h above the device under test; the angular-spectrum chart synthesizes outgoing plane-wave trials, back-propagates to the scan plane, and

accepts residuals only after scan pitch, bandlimit, and probe correction are stated [3]. The sequence scan–synthesize–bandlimit–report is what the theory encodes once radiation has been selected [2].

section 2.8 closed the reciprocity chain Eq. (2.355): Lorentz reciprocity (Subsection 2.8.1), modal orthogonality and coupling (Subsection 2.8.2), and radiation–reception interchange (Subsection 2.8.3) [2]. That program fixed exchange symmetry on $\operatorname{ran}\mathcal{R}_\omega$ and on port coordinates after projection. section 2.9 develops filter action in plane-wave coordinates: how $\mathcal{R}_\omega$, Π_{rad}, and observation maps retain or suppress continuous angular content on S^2, and how near-zone fields map to far-zone amplitudes $a(\widehat{\mathbf{k}})$ [2]. Preview expansion Eq. (2.141) from Subsection 2.2.3 named the lateral spectral template; here that reference pattern is developed to locked statements compatible with outgoing criterion Eq. (2.198) and far amplitude Eq. (2.246) [2]. Explicit basis change to vector harmonics is developed in 3 Subsection 3.8.1; harmonic normalization belongs to Subsection 3.5.2 with citation of Subsection 2.8.2 [6]. Spectral decomposition in Subsection 2.6.2 and continuous representation Eq. (2.289) fixed modal and plane-wave coordinates on $\mathcal{F}_{\mathrm{rad}}$; null-space consequences in Subsection 2.7.4 and angular budget Eq. (2.352) classified which directions $a(\widehat{\mathbf{k}})$ may carry radiation [2]. Reciprocity from section 2.8 constrains bilinear coupling on observable coordinates but does not replace angular filters: a reciprocal antenna may still have suppressed sheets Eq. (2.345) and finite observation rank Eq. (2.351) [2]. Treating $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ near sources as if it were already $a(\widehat{\mathbf{k}})$ repeats the regime error class of section 1.2.5 [2]. section 2.9 separates plane-wave angular spectra, retained and suppressed components, near-to-far transformation, and far-field angular $\mathcal{E}^{(N)}_{\mathrm{acc}}$ without re-opening Green construction from section 2.3 [2]. Let $\omega > 0$ be fixed, $\Omega_s = \operatorname{supp}\mathbf{J}_{\mathrm{imp}}$ be compact, and outgoing pairs lie in $\mathcal{F}_{\mathrm{SF}}$ Eq. (2.56). Section 2.9 organizes analysis in four stages:

$$\text{plane-wave spectrum} \xrightarrow{\mathcal{R}_\omega,\Pi_{\mathrm{rad}}} \text{retained / suppressed} \xrightarrow{\text{near-to-far}} a(\widehat{\mathbf{k}}) \xrightarrow{\text{probe rank}} \mathcal{K}_{\mathrm{acc}}, \tag{2.382}$$

where the first arrow is Subsection 2.9.1, the second is Subsection 2.9.2, the third is Subsection 2.9.2, and the fourth is Subsection 2.9.3 [2]. Eq. (2.382) expresses Eq. (2.312) and Eq. (2.355) at the angular level: null-space and reciprocity constraints from sections 2.7–2.8 bound which $a(\widehat{\mathbf{k}})$ are physically meaningful; section 2.9 develops the operators that enforce those bounds on plane-wave coordinates [2]. Transport criteria from section 1.2.4 and angular observables from section 1.2.2 enter Subsection 2.9.3 by reference, not by re-derivation [2]. Modal representation, not separate physics (Sections 2.6 and 2.9). Spectral and angular decompositions organize information about $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$; they do not define angular structure [2]. Observable angular structure remains primary: modes, plane-wave sheets, and harmonic labels are coordinates on radiation, not substitutes for Maxwell flux or qualified transport [1]. A nonzero modal coefficient or angular amplitude reports surviving source-generated content after propagation and outgoing selection; renaming that content does not create new physical degrees of freedom [6]. The primary objects are radiation map Eq. (2.313), outgoing plane-wave criterion Eq. (2.198), and continuous angular amplitude $a(\widehat{\mathbf{k}})$ from Eq. (2.289) [2]. Secondary targets include propagating projectors Eqs. (2.133)–(2.138), observation composition Eq. (2.218), accessible sector Eq. (2.9.3), and composition law Eq. (2.352) [2]. section 2.9 develops filters on the same outgoing branch fixed by Eqs. (2.202)–(2.212); incoming sheets remain excluded from $\mathcal{F}_{\mathrm{rad}}$ [6]. Plane-wave angular spectrum and radiation operators. Subsection 2.9.1 develops plane-wave angular spectra and the action of $\mathcal{R}_\omega \circ \Pi_{\mathrm{prop}}$ on spectral coefficients:

$$\widetilde{\mathbf{E}}_{\mathrm{rad}}(k_x, k_y) = \mathcal{F}_{\mathrm{rad}}\big[\Pi_{\mathrm{rad}}\,\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](k_x, k_y), \qquad k_z = k_z(k_t) \text{ from Eq. (2.198)}, \tag{2.383}$$

with $\mathcal{F}_{\mathrm{rad}}$ the lateral Fourier transform on outgoing admissible fields [2]. Eq. (2.392) develops Eq. (2.141) by inserting radiation selection; evanescent sheets satisfy $\widetilde{\mathbf{E}}_{\mathrm{rad}} = 0$ under Eq. (2.138) [2]. Rigged-space normalization of plane-wave modes belongs to 3 Subsection 3.1.3 [6]. Subsection 2.9.2

partitions angular content into retained, suppressed, and observation-filtered sheets:

$$a(\widehat{\mathbf{k}}) = a_{\mathrm{ret}}(\widehat{\mathbf{k}})\,\chi_{\mathcal{K}_{\mathrm{acc}}}(\widehat{\mathbf{k}}) \;+\; a_{\mathrm{sup}}(\widehat{\mathbf{k}})\,\chi_{\mathcal{K}_{\mathrm{sup}}}(\widehat{\mathbf{k}}), \qquad a_{\mathrm{filt}} = \mathcal{O}_{\mathrm{meas}}[a], \tag{2.384}$$

using accessible sector Eq. (2.9.3) and suppressed sheet Eq. (2.345) from Subsection 2.7.4 [2]. Eq. (2.384) is the section 2.9 target for Subsection 2.9.2 partition; geometric constraints on radiative spectra (Subsection 2.6.5) and observation rank Eq. (2.351) compose independently with Eq. (2.352) [2]. Null modes annihilate $a(\widehat{\mathbf{k}})$ through Eq. (2.344) without altering near reactive content [6]. Near-to-far transformation. Subsection 2.9.2 records the operator map from near-zone admissible fields to far angular amplitudes:

$$a(\widehat{\mathbf{k}}) = \lim_{r\to\infty} \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](r,\widehat{\mathbf{k}}), \tag{2.385}$$

with $\mathcal{H}_{\mathrm{ff}}$ the far-zone extractor tied to Eq. (2.246) and the far-field approximation of source integrals (Subsection 2.5.2) [2]. Eq. (2.385) does not promote evanescent storage to radiation: $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is annihilated before Eq. (2.385) when $\mathcal{R}_\omega$ is unchanged [2]. declared size and distance regime Eq. (2.270) qualifies when the limit defines transport data [6]. Subsection 2.9.3 develops probe rank on S^2 importing section 1.2.2 angular observables and section 1.2.4 transport criteria by reference [2]:

$$\widehat{\mathbf{k}} \in \mathcal{K}_{\mathrm{acc}} \iff a(\widehat{\mathbf{k}}) \text{ observable after } \mathcal{M}_\omega \text{ and subject to } \mathcal{D}_{J,\mathrm{reg}} = 1, \tag{2.386}$$

upgrading angular $\mathcal{E}^{(N)}_{\mathrm{acc}}$ on the far shell (Subsection 2.6.5) and composing with Eq. (2.352) [2]. Equation (2.386) is the section 2.9 completion target; explicit plane-wave filters and measurement condition is developed in Subsection 2.9.3 [6]. Reciprocity in radiation and reception (Subsection 2.8.3) constrains coupling on $\mathcal{K}_{\mathrm{acc}}$ but does not enlarge it [2].

$$\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega \implies a(\widehat{\mathbf{k}}) = 0 \text{ on radiation}, \qquad \mathfrak{C}_\omega \text{ symmetric on } \mathcal{K}_{\mathrm{acc}}, \tag{2.387}$$

importing Eq. (2.344) and reciprocity in radiation and reception (Subsection 2.8.3) [2]. Modal coordinates Eq. (2.284) and plane-wave coordinates Eq. (2.289) describe the same $\operatorname{ran}\mathcal{R}_\omega$ on exterior domains; section 2.9 fixes plane-wave-side filters, while 3 Subsection 3.8.1 supplies basis change law [2]. Finite-rank observation Eq. (2.220) may remove angular content even when $a(\widehat{\mathbf{k}}) \neq 0$ on $\mathcal{F}_{\mathrm{rad}}$ [2]. Several constructions deliberately remain outside section 2.9. Vector spherical harmonic completeness and normalization are developed in 3; section 2.9 records plane-wave radiation action without tabulating Y_ℓ^m [2]. Non-radiating catalogs remain in section 2.7; reciprocity constraints remain in section 2.8 [2]. section 2.9 uses Fredholm structure, geometric constraints, and null-space structure from sections 2.6–2.7 [6]. Regime logic from section 1.2.5 is unchanged: near-zone reactive dominance does not create far-zone angular transport without $\mathcal{D}_{J,\mathrm{reg}} = 1$ and outgoing condition [2]. Angular-transport claims must compose with Eq. (2.382), angular budget Eq. (2.352), and declared size and distance regime Eq. (2.139) [6].

Coefficient reports must declare projector and gauge before angular labels are compared [3].

Radiation on Γ_R therefore remains the laboratory requirement: watt must close through outgoing flux before angular labels are compared [2].

2.9.1 Plane-wave angular spectrum and radiation operators

Plane-wave angular spectra connect near fields to far patterns on scanners [14]. Evanescent content must be filtered before far-zone claims [3]. Maxwell's retarded radiation to Γ_R remains the governing constraint [1]. Before lateral wavenumbers enter, fix what a pattern measurement actually resolves. Qualified far-zone radiation is a directional budget: each direction $\widehat{\mathbf{k}}$ on S^2 carries an amplitude

$a(\widehat{\mathbf{k}})$ reporting surviving power flux after radiation selection [2]. Propagating directions ($k_t \le k_0$) contribute to that budget and can reach the far zone; evanescent directions ($k_t > k_0$) store reactive energy that decays exponentially with distance from the source and is annihilated on radiation [6]. A spherical near-field scanner or planar phased-array diagnostic begins on a measurement surface, not in (k_x, k_y) space [3]. Probes record tangential $\mathbf{E}$ and $\mathbf{H}$ on a sheet at height h; the angular-spectrum chart synthesizes outgoing plane-wave trials, back-propagates to the scan plane, and accepts the residual only after scan height, probe pitch, and bandlimit are declared [17]. Aperture measurements favor plane spectra; compact ranges favor spherical amplitudes, but both express the same directional decomposition in different coordinates [6]. Denser scan grids sharpen interpolation; they do not invent propagating modes absent from $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [17]. *Observation versus source coordinates.* Probes observable field quantities in laboratory Cartesian coordinates (x, y, z) on the scan plane $z = z_0$. Lateral wavenumbers (k_x, k_y) index plane-wave trials in the (x, y) plane and are *observation-side* spectral labels attached to tangential traces on $z = z_0$, not source coordinates on compact $\Omega_s = \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$ [2]. Transverse magnitude $k_t = \sqrt{k_x^2 + k_y^2}$ partitions the spectrum: sheets with $k_t \le k_0$ carry propagating radiation, while $k_t > k_0$ carry evanescent reactive content whose decay length scales with scan height [6]. Impressed current $\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')$ enters the reported spectrum only through $\mathcal{R}_\omega$ after radiation selection; treating raw scan data as if it were already $a(\widehat{\mathbf{k}})$ amplifies high-k_t noise into false superdirective claims [17]. section 2.9 developed angular-spectrum action through Eqs. (2.382)–(2.392). Subsection 2.9.1 develops plane-wave angular spectra and the action of $\mathcal{R}_\omega$, Π_{rad}, and outgoing selection on lateral spectral coefficients [2]. Dispersion split Eqs. (2.129)–(2.130), spectral projectors Eqs. (2.133)–(2.138), outgoing selector Eq. (2.196), plane-wave criterion Eq. (2.198), radiation map Eq. (2.313), and lateral template Eq. (2.141) are presupposed; Subsection 2.2.3 and section 2.4.3 proofs are not repeated [2]. Continuous angular representation Eq. (2.289) and far amplitude Eq. (2.246) are cited as consistency targets; rigged normalization belongs to Chapter 3 (*Angular Momentum Operators and SO(3)*) Subsection 3.1.3 [6]. Retained/suppressed partitions belong to Subsection 2.9.2. Assumptions. Fix $\omega > 0$, homogeneous exterior region with symbol k_0 from Eq. (2.113), compact $\Omega_s = \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$, admissible $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ Eq. (2.0.1), and outgoing fields $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H}) \in \mathcal{F}_{\mathrm{SF}}$ Eq. (2.56) [2]. Lateral wavenumbers (k_x, k_y) use $k_t = \sqrt{k_x^2 + k_y^2}$ and outgoing longitudinal root Eq. (2.198) [2]. Transverse polarization satisfies Eq. (2.132) on each spectral sheet [2]. declared size and distance regime Eq. (2.139) governs transport interpretation of spectral outputs [6]. On a plane $z = z_0$ in an outgoing admissible region, the SNF back-propagation step seeks a superposition of plane-wave trials whose tangential field matches the measured trace on $z = z_0$. Each trial is labeled by (k_x, k_y); the synthesis charting propagates it along z with outgoing longitudinal wavenumber $k_z(k_t)$ from Eq. (2.198) [2]. The formal decomposition packages that matching problem as a lateral Fourier transform on the observation plane. Define the *lateral angular spectrum* of $\mathbf{E}$ by

$$\widetilde{\mathbf{E}}(k_x, k_y; z_0) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathbf{E}(x, y, z_0)\, e^{\mathrm{j}(k_x x + k_y y)}\, dx\, dy, \tag{2.388}$$

with inverse

$$\mathbf{E}(x, y, z_0) = \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \widetilde{\mathbf{E}}(k_x, k_y; z_0)\, e^{-\mathrm{j}(k_x x + k_y y)}\, dk_x\, dk_y, \tag{2.389}$$

under standard Fourier conventions [2]. Helmholtz propagation along z multiplies each spectral component by

$$\widetilde{\mathbf{E}}(k_x, k_y; z) = \widetilde{\mathbf{E}}(k_x, k_y; z_0)\, e^{-\mathrm{j}k_z(k_t)(z - z_0)}, \qquad k_z \text{ from Eq. (2.198)}, \tag{2.390}$$

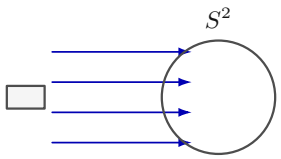


Figure 2.16: Plane-wave angular spectrum: compact source radiates outgoing plane waves observed on S^2 after $\mathcal{R}_\omega$ and lateral transform Eq. (2.388).

for homogeneous media [2]. Eqs. (2.388)–(2.390) upgrade Eq. (2.141) to explicit forward/inverse transforms with outgoing k_z selection [2]. Mathematically, $\widetilde{\mathbf{E}}(k_x, k_y; z_0)$ is the lateral Fourier transform of the transverse field sampled on the scan plane; each (k_x, k_y) sheet is an independent plane-wave channel whose longitudinal behavior is fixed by Eq. (2.130) [2]. Physically, propagating sheets ($k_t \le k_0$) carry energy that can reach the far zone after Eq. (2.385), whereas evanescent sheets ($k_t > k_0$) decay exponentially with distance from z_0 and remain bound to the reactive shell unless a separate source excites them [6]. The inverse transform Eq. (2.389) reconstructs the tangential field on $z = z_0$ from its spectral coefficients; probe interpolation error and finite aperture enter here, before any claim about far-zone pattern [3]. Propagation Eq. (2.390) shifts each sheet along z by the phase factor $e^{-\mathrm{j}k_z(k_t)(z-z_0)}$, which is the operator-side statement of SNF back-propagation between scan planes at different heights [14]. Consequence for measurement: reported angular-spectrum charts are only as trustworthy as the declared scan height and bandlimit on (k_x, k_y) [17]. Let $\mathcal{F}_{\mathrm{lat}}$ denote the lateral Fourier operator in Eq. (2.388) on admissible traces. Define the *radiative lateral spectrum* of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ by

$$\widetilde{\mathbf{E}}_{\mathrm{rad}}(k_x, k_y) = \mathcal{F}_{\mathrm{lat}}\big[\Pi_{\mathrm{rad}}\, \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](k_x, k_y), \tag{2.391}$$

with $\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}$ from Eq. (2.137) [2]. Equation (2.391) is the Subsection 2.9.1 upgrade of Eq. (2.392): radiation selection precedes lateral transformation [2]. Applying $\mathcal{F}_{\mathrm{lat}}$ to raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ without Π_{rad} mixes reactive evanescent sheets into reported spectra through Eq. (2.142) [6]. The composition $\mathcal{T}_\omega$ below is the spectral-domain radiation map: impressed currents are transformed laterally, passed through $\mathcal{R}_\omega$ and outgoing selection, and read back in (k_x, k_y) coordinates. Define the *angular-spectrum radiation operator* on lateral coefficients by

$$\widetilde{\mathbf{E}}_{\mathrm{rad}}(k_x, k_y) = \big[\mathcal{T}_\omega \widetilde{\mathbf{J}_{\mathrm{imp}}}\big](k_x, k_y), \qquad \mathcal{T}_\omega = \mathcal{F}_{\mathrm{lat}} \circ \Pi_{\mathrm{rad}} \circ \mathcal{R}_\omega \circ \mathcal{F}_{\mathrm{lat}}^{-1}, \tag{2.392}$$

where $\widetilde{\mathbf{J}_{\mathrm{imp}}} = \mathcal{F}_{\mathrm{lat}}[\mathbf{J}_{\mathrm{imp}}]$ on the source support and $\mathcal{F}_{\mathrm{lat}}^{-1}$ is understood on the admissible impressed-current class [2]. Eq. (2.392) packages Eq. (2.391) as a spectral-domain map: sources excite lateral modes, $\mathcal{R}_\omega$ radiates selected content, and $\mathcal{F}_{\mathrm{lat}}$ returns observable spectral coefficients [2]. In laboratory terms, $\mathcal{T}_\omega$ is the operator that a qualified SNF transform applies *after* radiation selection: impressed currents are first mapped to outgoing fields, then decomposed into (k_x, k_y) sheets on the propagating branch [3]. On the propagating outgoing sheet $k_t \le k_0$, $\mathcal{T}_\omega$ reduces to diagonal action in (k_x, k_y) modulo source coupling, and those sheets are the only ones that survive comparison with far-zone amplitude $a(\widehat{\mathbf{k}})$ [2]. On the evanescent sheet $k_t > k_0$,

$$\widetilde{\mathbf{E}}_{\mathrm{rad}}(k_x, k_y) = \mathbf{0} \quad \text{for } k_t > k_0, \tag{2.393}$$

by Eq. (2.138) [2]. Consequence: raw lateral transforms of $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ may show large $|\widetilde{\mathbf{E}}(k_x, k_y)|$ for $k_t > k_0$ while $\widetilde{\mathbf{E}}_{\mathrm{rad}} = \mathbf{0}$ on radiation; only the radiative branch Eq. (2.391) may be compared to far-field reports [6]. The operator sequence enforces radiation selection before lateral transform: $\mathcal{R}_\omega$ first, then $\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}$, then lateral transform $\mathcal{F}_{\mathrm{lat}}$. Lower nodes Π_{prop} and $\mathcal{O}_{\mathrm{out}}$ are

constituents of radiation selection, not optional post-filters. Applying $\mathcal{F}_{\mathrm{lat}}$ to raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ without Π_{rad} contaminates $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ with evanescent sheets annihilated by Eq. (2.393). The terminal node $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ is the plane-wave angular spectrum of outgoing radiation only [2]. Figure 2.16 orders operators as in Eq. (2.196): propagation selection, outgoing criterion, then lateral spectral readout [2]. Mathematical role of Eq. (2.392). Eq. (2.392) packages the radiation operator as a spectral-domain map $\mathcal{T}_\omega$ acting on lateral source coefficients [2]. It is not a separate extraction stage on near-zone $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$: Π_{rad} is part of the definition [2]. Only outgoing propagating sheets contribute to far-zone angular content; incoming sheets are removed by Eq. (2.196), and evanescent sheets are annihilated by Eq. (2.393) [6]. Reciprocity in radiation and reception (Subsection 2.8.1) constrains bilinear couplings built from $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ but does not alter Eq. (2.393) [2]. Mathematical role of Eq. (2.393). Eq. (2.393) is the plane-wave restatement of Eq. (2.138) on lateral coordinates [2]. Large $|\widetilde{\mathbf{E}}(k_x, k_y)|$ for $k_t > k_0$ may appear in raw $\mathcal{F}_{\mathrm{lat}}[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]]$ while $\widetilde{\mathbf{E}}_{\mathrm{rad}} = \mathbf{0}$ on radiation [6]. Regime condition Eq. (2.270) remains mandatory before interpreting any nonzero raw lateral spectrum as transport [2]. If $\mathbf{J}_{\mathrm{imp}}^{(1)}$ is arbitrary and $\mathbf{J}_{\mathrm{imp}}^{(2)} = \mathbf{J}_{\mathrm{imp}}^{(1)} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}$ with $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega$, then $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ from Eq. (2.391) is unchanged under replacement of $\mathbf{J}_{\mathrm{imp}}^{(1)}$ by $\mathbf{J}_{\mathrm{imp}}^{(2)}$ [2]. Raw $\mathcal{F}_{\mathrm{lat}}[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]]$ may change through reactive content Eq. (2.271) [2]. $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(2)}] = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(1)}]$ by Eq. (2.346) [2]. On exterior domains with outgoing radiation condition, amplitudes $a_\sigma(\widehat{\mathbf{k}})$ in Eq. (2.289) and lateral spectrum Eq. (2.391) describe the same $\operatorname{ran} \mathcal{R}_\omega$ content under change of variables $(k_x, k_y) \mapsto \widehat{\mathbf{k}}$ on the propagating sheet $k_t \le k_0$ [2]. Normalization constants are fixed in Subsection 3.1.3 [2]. Both representations diagonalize outgoing propagation on the propagating sheet; agreement on far-zone reduction follows plane-wave radiation-operator action (Subsection 2.9.1)(iv) and Eq. (2.289) [2]. Modal expansion Eq. (2.284) and plane-wave expansion Eq. (2.392) describe the same radiative image on $\mathcal{F}_{\mathrm{rad}}$ under unitary change of basis on the propagating sheet [2]. Subsection 2.9.1 fixes plane-wave-side operators; 3 Subsection 3.8.1 supplies explicit harmonic transforms without re-tabulating Eq. (2.392) [2]. Angular budget Eq. (2.352) constrains supports of $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ independently of basis choice [6]. Plane-wave radiation-operator action (Subsection 2.9.1) assumes linear homogeneous exterior regions at fixed ω, outgoing radiation condition, and admissible $\mathcal{J}_\Omega$ [2]. Piecewise homogeneous media require patchwise Eq. (2.390) with interface matching imported from section 2.3.5 [2]. Cavity standing-wave spectra are excluded: lateral plane-wave expansions on bounded domains do not replace discrete cavity books from Eq. (2.287) [14]. Loss modifies k_0 and k_z with complex symbols; evanescent null law Eq. (2.393) remains on radiation [2]. Subsection 2.9.1 fixed plane-wave-side radiation action; retained, suppressed, and filtered partitions in Subsection 2.9.2 separate geometric accessibility from finite-rank observation. Physically, only after the near-to-far limit of Subsection 2.9.2 does a finite aperture extract $a(\widehat{\mathbf{k}})$ on the propagating sheet $k_t \le k_0$.

2.9.2 Retained, suppressed, and filtered angular components

section 2.9 developed angular partitioning through Eq. (2.384) and linked it to accessible sector Eq. (2.9.3) and suppressed sheet Eq. (2.345) from Subsection 2.7.4 [2]. Subsection 2.9.2 records the partition of retained, suppressed, and observation-filtered angular components on the outgoing branch established in Subsection 2.9.1. Near-field scanners record tangential fields that mix propagating radiation with evanescent storage; the partition below separates what geometry permits to radiate, what symmetry forbids, and what finite aperture resolves—three filters that must not be merged in harmonic charts or laboratory interpretation [3]. Plane-wave lateral spectra Eqs. (2.391)–(2.392), evanescent null law Eq. (2.393), angular budget Eq. (2.352), and observation rank Eq. (2.351) are presupposed; Subsection 2.7.4 proofs are not repeated [2]. Continuous amplitude

$a(\widehat{\mathbf{k}})$ from Eq. (2.289) and rank-N instrument readout Eq. (2.218) supply the angular and laboratory coordinates in which the partition is stated [2]. Near-to-far extraction belongs to Subsection 2.9.2; explicit $\mathcal{E}_{\mathrm{acc}}^{(N)}$ filters on S^2 belong to Subsection 2.9.3 [2]. Assumptions. Fix $\omega > 0$, admissible $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$, outgoing radiative field $\mathbf{u}_\omega = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \in \mathcal{F}_{\mathrm{rad}}$, and angular amplitude $a(\widehat{\mathbf{k}})$ from Eq. (2.289) on the radiating transverse sheet [2]. Accessible sector $\mathcal{K}_{\mathrm{acc}}$ Eq. (2.9.3) and suppressed sheet $\mathcal{K}_{\mathrm{sup}} = S^2 \setminus \mathcal{K}_{\mathrm{acc}}$ from Eq. (2.345) are fixed by geometry and symmetry of $(\Omega_s, \Gamma, \varepsilon, \mu)$ [2]. Finite-rank observation $\mathcal{M}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}}$ from Eq. (2.218) is presupposed; declared size and distance regime Eq. (2.139) qualifies transport interpretation of any reported component [6]. The accessible sheet $\mathcal{K}_{\mathrm{acc}} \subset S^2$ collects directions that geometry and symmetry permit to carry outgoing radiation; its complement $\mathcal{K}_{\mathrm{sup}}$ is suppressed before any instrument samples the pattern. Define the *retained angular amplitude* as the radiative content supported on the accessible sheet. On a directional antenna bench, this is the portion of the far-zone pattern that symmetry and blockage permit to carry outgoing radiation; directions on $\mathcal{K}_{\mathrm{sup}}$ are zero by geometry before any instrument samples the pattern [2]:

$$a_{\mathrm{ret}}(\widehat{\mathbf{k}}) = \chi_{\mathcal{K}_{\mathrm{acc}}}(\widehat{\mathbf{k}})\, a(\widehat{\mathbf{k}}), \qquad \chi_{\mathcal{K}_{\mathrm{acc}}}(\widehat{\mathbf{k}}) = \begin{cases} 1, & \widehat{\mathbf{k}} \in \mathcal{K}_{\mathrm{acc}}, \\ 0, & \widehat{\mathbf{k}} \in \mathcal{K}_{\mathrm{sup}}, \end{cases} \tag{2.394}$$

with $a(\widehat{\mathbf{k}})$ from Eq. (2.289) after Eq. (2.392) [2]. Eq. (2.394) is the Subsection 2.9.2 locked definition of development of the retained factor in Eq. Eq. (2.384): only directions not annihilated by $\mathcal{C}_{\mathrm{geo}}$ angular rank-N recovery on the far shell (Subsection 2.6.5) carry qualified transport data [2]. Mathematically, a_{ret} is the pointwise product of the full angular measure with the accessible indicator; physically, it is the portion of the far-zone angular distribution that geometry permits to propagate as radiation after radiation selection [6]. Null-space additions Eq. (2.349) leave a_{ret} unchanged [2]. Suppressed angular amplitude. Define the *suppressed angular amplitude* as the complementary sheet content:

$$a_{\mathrm{sup}}(\widehat{\mathbf{k}}) = \chi_{\mathcal{K}_{\mathrm{sup}}}(\widehat{\mathbf{k}})\, a(\widehat{\mathbf{k}}), \qquad \chi_{\mathcal{K}_{\mathrm{sup}}} = 1 - \chi_{\mathcal{K}_{\mathrm{acc}}} \text{ a.e. on } S^2, \tag{2.395}$$

using suppressed sheet Eq. (2.345) [2]. Eq. (2.395) completes Eq. (2.384): for radiating radiation, geometric forbidding Eq. (2.307) typically forces $a_{\mathrm{sup}} = \mathbf{0}$ even when near-zone scans show energy in related directions [2]. Suppression is therefore not synonymous with $\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$: Eq. (2.344) annihilates a everywhere, whereas Eq. (2.395) removes selected directions while a_{ret} may remain nonzero [6]. Evanescent lateral content Eq. (2.393) does not populate a_{sup} on radiation: Π_{ev} is removed before angular readout [2]. Laboratory observation applies a finite-rank criterion to far-zone angular content. Define the *filtered angular amplitude* by

$$a_{\mathrm{filt}}(\widehat{\mathbf{k}}) = \big[\mathcal{P}_{\mathrm{meas}}\, a\big](\widehat{\mathbf{k}}), \qquad \mathcal{P}_{\mathrm{meas}} = \mathcal{O}_{\mathrm{meas}} \circ \mathcal{O}_{\mathrm{far}} \circ \mathcal{H}_{\mathrm{ang}}^{-1}, \tag{2.396}$$

where $\mathcal{H}_{\mathrm{ang}}$ maps $a(\widehat{\mathbf{k}})$ to the corresponding far-zone transverse field on S^2 and $\mathcal{O}_{\mathrm{far}}$ is the far extractor from Eq. (2.218) [2]. Eq. (2.396) is the angular-side restatement of the third line in Eq. (2.384). When $\mathcal{O}_{\mathrm{meas}}$ is a finite set of plane-wave or modal probes, $\mathcal{P}_{\mathrm{meas}}$ has rank at most N and $\ker \mathcal{P}_{\mathrm{meas}} \supseteq \ker \mathcal{O}_{\mathrm{meas}}\big|_{\mathrm{ran}\, \mathcal{R}_\omega}$ Eq. (2.80) [2]. Physically, a_{filt} is what a finite instrument resolves after aperture, noise, and processing; it may vanish on subsets of $\mathcal{K}_{\mathrm{acc}}$ where a_{ret} is nonzero [6].

$$\mathrm{rank}(\mathcal{P}_{\mathrm{meas}}) \le N, \qquad \dim \mathrm{span}\{a_{\mathrm{filt}}(\widehat{\mathbf{k}})\} \le N, \tag{2.397}$$

Combining Eqs. (2.394)–(2.396) yields the Chapter 2 partition

$$a(\widehat{\mathbf{k}}) = a_{\mathrm{ret}}(\widehat{\mathbf{k}}) + a_{\mathrm{sup}}(\widehat{\mathbf{k}}), \qquad a_{\mathrm{filt}} = \mathcal{P}_{\mathrm{meas}}\big[a_{\mathrm{ret}} + a_{\mathrm{sup}}\big], \tag{2.398}$$

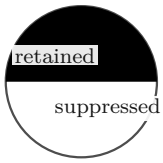


Figure 2.17: Angular partition on S^2: retained sheet $\mathcal{K}_{\mathrm{acc}}$ versus suppressed sheet $\mathcal{K}_{\mathrm{sup}}$ before observation filter Eq. (2.396).

with a_{ret} and a_{sup} supported on disjoint sheets $\mathcal{K}_{\mathrm{acc}}$ and $\mathcal{K}_{\mathrm{sup}}$ [2]. Eq. (2.398) upgrades Eq. (2.384) to explicit locked definitions and inserts the measurement projector $\mathcal{P}_{\mathrm{meas}}$. The first line is an angular decomposition of radiation; the second line states that observation acts on the sum after radiation selection, not on raw near spectra [2]. Budget Eq. (2.352) constrains $\mathrm{supp}(a) \subseteq \mathcal{K}_{\mathrm{acc}} \cap \{\text{resolvable by } \mathcal{M}_\omega\}$, so a_{filt} cannot recover directions outside that intersection even when raw lateral transforms appear rich [6]. On the plane-wave side of Subsection 2.9.1, define the *retained lateral spectrum*

$$\widetilde{\mathbf{E}}_{\mathrm{ret}}(k_x, k_y) = \mathcal{F}_{\mathrm{lat}}\big[\mathcal{H}_{\mathrm{ang}}[a_{\mathrm{ret}}]\big](k_x, k_y) \quad \text{on } k_t \le k_0, \tag{2.399}$$

and the *filtered lateral spectrum*

$$\widetilde{\mathbf{E}}_{\mathrm{filt}}(k_x, k_y) = \mathcal{F}_{\mathrm{lat}}\big[\mathcal{H}_{\mathrm{ang}}[a_{\mathrm{filt}}]\big](k_x, k_y), \tag{2.400}$$

with $\mathcal{F}_{\mathrm{lat}}$ from Eq. (2.388) [2]. Eqs. (2.399)–(2.400) connect Subsection 2.9.1 lateral readout to the angular partition without re-displaying Eq. (2.392): $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ from Eq. (2.391) agrees with $\widetilde{\mathbf{E}}_{\mathrm{ret}}$ on the propagating outgoing sheet when $a_{\mathrm{sup}} = \mathbf{0}$ [2]. On the evanescent sheet $k_t > k_0$, retained and filtered lateral spectra vanish through Eq. (2.393) independently of raw $\mathcal{F}_{\mathrm{lat}}[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]]$ [2]. declared size and distance regime Eq. (2.270) applies to $\widetilde{\mathbf{E}}_{\mathrm{filt}}$ before any transport claim [6]. On $\mathcal{K}_{\mathrm{sup}}$,

$$a_{\mathrm{sup}}(\widehat{\mathbf{k}}) = \mathbf{0} \implies \widetilde{\mathbf{E}}_{\mathrm{ret}}(k_x, k_y) = \widetilde{\mathbf{E}}_{\mathrm{rad}}(k_x, k_y) \text{ on } k_t \le k_0, \tag{2.401}$$

and any lateral energy attributed to $\mathcal{K}_{\mathrm{sup}}$ after Eq. (2.392) is removed by geometry, not by $\mathcal{P}_{\mathrm{meas}}$ [2]. Eq. (2.401) expresses that suppression precedes finite aperture cannot resurrect directions forbidden by Eq. (2.307) [6]. Reciprocity in radiation and reception (Subsection 2.8.3) constrains bilinear couplings on $\mathcal{K}_{\mathrm{acc}}$ but does not enlarge $\mathrm{supp}(a_{\mathrm{ret}})$ [2]. Figure interpretation. Radiative amplitude $a(\widehat{\mathbf{k}})$ splits into retained sheet $\mathcal{K}_{\mathrm{acc}}$ and suppressed sheet $\mathcal{K}_{\mathrm{sup}}$ before observation filter $\mathcal{P}_{\mathrm{meas}}$. Retained content survives geometric and parity criteria; suppressed content is zero by constraint, not by source nullity. Filtered lateral spectrum $\widetilde{\mathbf{E}}_{\mathrm{filt}}$ is the aperture-limited readout: finite rank may remove angular content even when $a \neq 0$ on $\mathcal{F}_{\mathrm{rad}}$ [2]. Figure 2.17 orders radiation selection, angular partition, measurement filter, and plane-wave readout. Suppression acts on $\mathcal{K}_{\mathrm{sup}}$ before $\mathcal{P}_{\mathrm{meas}}$; evanescent sheets are removed by Eq. (2.393) before lateral readout [2]. Mathematical role of Eq. (2.398). Eq. (2.398) separates three stages that are often conflated in angular-spectrum discussions: outgoing radiation on $\mathcal{R}_\omega$, geometric angular filtering, and finite observation [2]. The retained factor encodes what the source radiates on $\mathcal{K}_{\mathrm{acc}}$; the suppressed factor marks directions removed before measurement; the filtered factor applies $\mathcal{P}_{\mathrm{meas}}$ [2]. Plane-wave coefficients $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ from Eq. (2.391) describe the same retained content in lateral coordinates when $a_{\mathrm{sup}} = \mathbf{0}$ [14]. Null additions plane-wave null invariance (Subsection 2.9.1) alter neither a_{ret} nor a_{filt} when $\mathcal{M}_\omega$ is unchanged [2]. Mathematical role of Eq. (2.396). Eq. (2.396) makes explicit that instruments observe $\mathcal{P}_{\mathrm{meas}}[a]$, not raw $\mathcal{F}_{\mathrm{lat}}[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]]$ [2]. Rank cap Eq. (2.397) prevents attributing unobserved angular structure to source nullity: a pattern gap on $\mathcal{K}_{\mathrm{acc}}$ may arise from $\mathcal{P}_{\mathrm{meas}}$ alone [6]. Reciprocity in radiation and reception (Subsection 2.8.3) constrains coupling among observed channels but does

not increase $\dim \operatorname{span}\{a_{\mathrm{filt}}\}$ [2]. declared size and distance regime Eq. (2.139) remains mandatory before interpreting a_{filt} as transport [6]. Recoverable modal content Eq. (2.311) and angular budget Eq. (2.352) are the modal and continuous forms of the same constraint chain Subsection 2.9.2 displays in Eq. (2.398) [2]. plane-wave radiation-operator action (Subsection 2.9.1) supplies lateral coordinates; retained, suppressed, and filtered angular components (Subsection 2.9.2) supplies angular-sheet coordinates; 3 Subsection 3.8.1 will change basis without altering $\operatorname{supp}(a_{\mathrm{ret}})$ [2]. Inverse obstruction angular spectra and inverse problems (Subsection 2.7.4) applies to a_{filt}, not to raw lateral transforms [6]. Retained, suppressed, and filtered angular components (Subsection 2.9.2) assumes linear media, fixed ω, outgoing closure, and stationary $\mathcal{C}_{\mathrm{geo}}$ [2]. Time-varying or nonlinear $\mathcal{P}_{\mathrm{meas}}$ requires separate treatment; the static rank bound Eq. (2.397) is an upper limit [2]. Loss modifies amplitudes but not the partition logic on $\mathcal{K}_{\mathrm{acc}} \cup \mathcal{K}_{\mathrm{sup}}$ [2]. Cavity and waveguide spectra remain outside the exterior partition; discrete books Eq. (2.287) are not substituted for Eq. (2.398) on unbounded domains [14]. section 2.9 developed the near-to-far map through Eq. (2.385) and tied it to far-zone reduction Eqs. (2.245)–(2.246) and the far-field approximation of source integrals (Subsection 2.5.2) [2]. Subsection 2.9.2 is the account within Chapter 2 of the operator that extracts far angular amplitudes $a(\widehat{\mathbf{k}})$ from radiative fields after Subsections 2.9.1–2.9.2. Plane-wave lateral spectra Eqs. (2.391)–(2.392), angular partition Eq. (2.398), evanescent null law Eq. (2.393), and continuous representation Eq. (2.289) are presupposed; asymptotic proofs in Subsection 2.5.2 are not repeated [2]. Observation-shell regime map Eq. (2.248) and declared size and distance regime Eq. (2.270) qualify when the limit defines transport data [6]. Far-field angular probe rank belongs to Subsection 2.9.3 [2]. Assumptions. Fix $\omega > 0$, admissible $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$, outgoing radiative field $\mathbf{u}_\omega = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \in \mathcal{F}_{\mathrm{rad}}$, and distance thresholds from Eq. (2.241) with $r > r_{\min}(a, \lambda_0)$ for source support radius a [2]. Angular amplitude $a(\widehat{\mathbf{k}})$ is measured on the unit sphere with outgoing transverse polarization [2]. declared size and distance regime $\mathcal{D}_{J,\mathrm{reg}} = 1$ from Eq. (2.139) is presupposed before interpreting any near-to-far output as transport [6].

$$\|a_r - a\|_{L^2(S^2)} \le C\,\chi(r), \qquad a_r(\widehat{\mathbf{k}}) = \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](r, \widehat{\mathbf{k}}), \tag{2.402}$$

$$\lim_{r\to\infty} \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](r, \widehat{\mathbf{k}}) = a(\widehat{\mathbf{k}}), \tag{2.403}$$

On observation shells $S_r = \{|\mathbf{r}| = r\}$ with $r \gg a$ for source radius a, the far-zone extractor must isolate the transverse $e^{-\mathrm{j}k_0 r}/r$ coefficient of $\mathbf{E}(\mathbf{r})$ in direction $\widehat{\mathbf{k}} = \mathbf{r}/r$. Define the *far-zone extractor* $\mathcal{H}_{\mathrm{ff}}$ by

$$\mathcal{H}_{\mathrm{ff}}[\mathbf{u}_\omega](r, \widehat{\mathbf{k}}) = r\, e^{\mathrm{j}k_0 r}\, \widehat{\mathbf{k}} \times \Big(\widehat{\mathbf{k}} \times \mathbf{E}(\mathbf{r})\Big)\Big|_{|\mathbf{r}|=r}, \qquad \mathbf{r} = r\,\widehat{\mathbf{k}}, \tag{2.404}$$

with $\mathbf{E}$ the electric phasor in $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H})$ and transversality enforced by the double cross product [2]. Equation (2.404) is the Subsection 2.9.2 definitive specification of the extractor stated in Eq. (2.385): it isolates the outgoing radiative envelope compatible with Eq. (2.245) [2]. Mathematically, $\mathcal{H}_{\mathrm{ff}}$ is a linear functional on shell traces; physically, it is the standard far-field extraction that removes reactive r^{-2} and r^{-3} contamination when $k_0 r \gg 1$ [6]. The extractor acts on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, not on raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ Eq. (2.248) [2]. The far-zone limit extracts the $e^{-\mathrm{j}k_0 r}/r$ transverse envelope from $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on observation shells S_r; the angular amplitude $a(\widehat{\mathbf{k}})$ is the coefficient of that envelope in direction $\widehat{\mathbf{k}}$. Define the *near-to-far transformation* on admissible sources by

$$a(\widehat{\mathbf{k}}) = \lim_{r\to\infty} \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](r, \widehat{\mathbf{k}}), \tag{2.405}$$

with $a(\widehat{\mathbf{k}})$ the angular amplitude entering Eq. (2.289) [2]. Eq. (2.405) develops Eq. Eq. (2.385) to explicit extractor Eq. (2.404) and fixes the radiation map input [2]. The limit exists and is

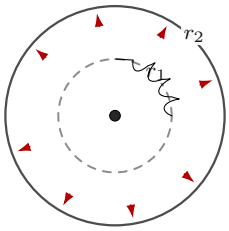


Figure 2.18: Near-to-far geometry: reactive shell at r_1 and radiative shell at r_2; evanescent content confined to the inner shell does not feed far-zone amplitude $a(\widehat{\mathbf{k}})$ Eq. (2.408).

independent of the particular r-path when the far-field approximation of source integrals (Subsection 2.5.2) applies; error on finite shells scales as in Eq. (2.247) [2]. Mathematically, Eq. (2.405) is the angular-side completion of far-zone template Eq. (2.245); physically, it states that far angular content is determined by the radiative branch alone after outgoing selection, independent of reactive storage that may dominate on inner shells [2]. Observation consequence: SNF and compact-range transforms that skip $\mathcal{R}_\omega$ before extraction inherit Eq. (2.408) errors documented in Subsection 2.10.3 [3]. declared size and distance regime Eq. (2.270) remains mandatory before interpreting the limit as transport data [6]. Package Eq. (2.405) as the composed operator

$$\Lambda_{\mathrm{nf}} = \mathcal{H}_{\mathrm{ang}} \circ \mathcal{H}_{\mathrm{ff}}^{\infty} \circ \mathcal{R}_\omega, \qquad a(\widehat{\mathbf{k}}) = \Lambda_{\mathrm{nf}}[\mathbf{J}_{\mathrm{imp}}](\widehat{\mathbf{k}}), \tag{2.406}$$

where $\mathcal{H}_{\mathrm{ff}}^{\infty}$ denotes the $r \to \infty$ limit in Eq. (2.405) and $\mathcal{H}_{\mathrm{ang}}$ maps the transverse far vector in Eq. (2.404) to scalar polarization amplitudes $a_\sigma(\widehat{\mathbf{k}})$ in Eq. (2.289) [2]. Eq. (2.406) aligns with far readout $\Lambda_{\mathrm{rad}} = \mathcal{O}_{\mathrm{far}} \circ \mathcal{R}_\omega$ from Eq. (2.218) when $\mathcal{O}_{\mathrm{far}}$ implements Eq. (2.404) on the angular side [2]. Measurement $\mathcal{M}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}}$ therefore factors near-to-far extraction before finite-rank observation [6]. Null additions $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega$ leave $\Lambda_{\mathrm{nf}}[\mathbf{J}_{\mathrm{imp}}]$ unchanged through Eq. (2.349) [2]. On the propagating outgoing sheet $k_t \le k_0$, the near-to-far output agrees with retained lateral content from Subsection 2.9.2,

$$a(\widehat{\mathbf{k}}) \;\longleftrightarrow\; \widetilde{\mathbf{E}}_{\mathrm{ret}}(k_x, k_y) \text{ under } (k_x, k_y) \mapsto \widehat{\mathbf{k}}, \tag{2.407}$$

when $a_{\mathrm{sup}} = \mathbf{0}$ and plane-wave continuous-spectrum consistency (Subsection 2.9.1) applies [2]. Eq. (2.407) connects Subsection 2.9.2 angular limit to Subsection 2.9.1 lateral spectra without re-displaying Eq. (2.392): both describe $\Pi_{\mathrm{rad}}\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ in different coordinates [2]. Filtered content satisfies $a_{\mathrm{filt}} = \mathcal{P}_{\mathrm{meas}}[a]$ with a from Eq. (2.405) [2]. Evanescent annihilation before near-to-far. Evanescent near storage does not enter Eq. (2.405):

$$\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \;\not\Rightarrow\; a(\widehat{\mathbf{k}}) \quad \text{when} \quad \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ is unchanged}, \tag{2.408}$$

by Eqs. (2.393), (2.350), and (2.138) [2]. Equation (2.408) expresses Eq. Eq. (2.385): reactive Π_{ev} content may be large in the near zone Eq. (2.242) while the far limit reports only outgoing radiation [2]. declared size and distance regime Eq. (2.270) forbids promoting Π_{ev} storage to $a(\widehat{\mathbf{k}})$ [6]. Figure interpretation. Near shell observation at $r \sim r_c$ samples $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ after Π_{rad}, not raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$. Extractor $\mathcal{H}_{\mathrm{ff}}$ implements the $r \to \infty$ limit of the far-field approximation of source integrals (Subsection 2.5.2); dashed $\Pi_{\mathrm{ev}} = \mathbf{0}$ marks evanescent content that may be large on the inner shell yet does not feed $a(\widehat{\mathbf{k}})$. The measurement sequence confirms when a near-field-to-far-field transform is structurally valid versus when reactive storage corrupts the asymptotic limit [2]. Figure 2.18 orders radiation selection, finite-shell observation, far-zone extraction, and angular amplitude. Near reactive content is dashed: it may appear on inner shells but is annihilated in the limit when

radiation is unchanged Mathematical role of Eq. (2.405). Eq. (2.405) is the angular-spectrum-side completion of the far-zone template Eq. (2.245): it converts radiative fields to continuous amplitudes $a(\widehat{\mathbf{k}})$ used in Eq. (2.289) [2]. The map is not an optional extraction applied to raw near data: input is $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, consistent with observation-shell map Eq. (2.248) [2]. Plane-wave lateral spectra Eq. (2.391) and angular limit Eq. (2.407) are equivalent coordinate descriptions of the same radiation [14]. Reciprocity in radiation and reception (Subsection 2.8.3) constrains bilinear couplings built from $a(\widehat{\mathbf{k}})$ but does not modify Eq. (2.408) [2]. Mathematical role of Eq. (2.404). Eq. (2.404) makes explicit how laboratory far-field algorithms isolate $\mathbf{E}_0(\theta,\phi)$ before pattern integration or modal projection [2]. Applying the extractor to $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on near shells confuses reactive and radiative scaling Eq. (2.242); applying it to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ aligns with power-flux convergence Eq. (2.247) [2]. Filtered observation Eq. (2.396) composes after Eq. (2.405), not before radiation selection [6]. Modal far pattern Eq. (2.290) and angular limit Eq. (2.405) describe the same $\operatorname{ran}\mathcal{R}_\omega$ content under far-pattern coefficients from spectral modes (Subsection 2.6.2) [2]. Retained/suppressed partition retained, suppressed, and filtered angular components (Subsection 2.9.2) constrains $\operatorname{supp}(a)$ independently of the near-to-far limit procedure [2]. 3 harmonic bases change coordinates without altering Eq. (2.406) [2]. Example 2.10.3 applies Eq. (2.408) in a worked setting [2]. Near-to-far transformation (Subsection 2.9.2) assumes linear homogeneous or piecewise-homogeneous media, fixed ω, outgoing radiation condition, and observation outside Ω_s [2]. It fails in cutoff guides, cavity standing waves, and lossless open resonances without outgoing selection [2]. Surface waves and strong anisotropy require modified extractors beyond Eq. (2.404) [6]. The theorem does not assert source uniqueness—only field asymptotics and angular amplitudes on the outgoing branch [2]. Cavity discrete spectra Eq. (2.287) do not replace Eq. (2.405) on exterior domains [14]. section 2.9 developed far-field angular rank-N recovery through Eq. (2.409) and required import of directional observables from section 1.2.2 and transport criteria from section 1.2.4 [2]. Subsection 2.9.3 completes the account of which directions on S^2 carry qualified, observable far-zone angular content after Subsections 2.9.1–2.9.2. Near-to-far map Eq. (2.405), angular partition Eq. (2.398), accessible sector Eq. (2.9.3), composition law Eq. (2.352), and observation rank Eq. (2.351) are presupposed; angular rank-N recovery on the far shell (Subsection 2.6.5) is upgraded, not re-proved [2]. Directional flux maps Eqs. (1.49)–(1.51) and transport triad Eq. (1.67) enter from the cited sections [2]. Explicit harmonic filters belong to 3; Assumptions. Fix $\omega > 0$, admissible $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$, outgoing radiative field $\mathbf{u}_\omega = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, angular amplitude $a(\widehat{\mathbf{k}})$ from Eq. (2.405), and declared size and distance regime $\mathcal{D}_{J,\mathrm{reg}} = 1$ from Eq. (2.139) [2]. Accessible sector $\mathcal{K}_{\mathrm{acc}}$ and suppressed sheet $\mathcal{K}_{\mathrm{sup}}$ are fixed by Eqs. (2.9.3)–(2.345) [2]. Finite-rank measurement $\mathcal{M}_\omega$ from Physically, eq. (2.218) is presupposed [6].

Coefficient reports must declare projector and gauge before angular labels are compared [3].

Laboratory reports therefore require the same outgoing branch and gauge assumed in the derivation [3].

2.9.3 Far-field angular accessibility

Far-zone accessibility is not isotropic on finite supports: some directions carry radiation while others lie in measurement nullity. What the derivation must settle is which angular sectors remain reachable after Π_{rad} and finite-rank observation [6]. Far angular instrument recovery $\mathcal{E}_{\mathrm{acc}}^{(N)}$ (Chapter 4) is what a finite probe count on Γ_R can resolve after near-to-far extraction and geometric sectoring—not what a harmonic chart alone can label [6]. Synthesis rank and observation rank must not be merged: the probe bank on Γ_R sets accessible directions before harmonic labels attach [6]. Chapter 3 relabels the same accessible set in VSH coordinates [2]. *probe rank.* Mount $N = 48$ probes on a spherical

surface Γ_R at $R = 3\lambda_0$ enclosing a $\lambda_0/4$ dipole feed, and truncate the harmonic chart at $\ell \le 10$. The dipole radiates dominant $\ell = 1$ content, so the fitted coefficients $\widehat{c}_{10,m}$ are nonzero on the accessible sector; nevertheless, rank-N recovery is bounded jointly by probe count N, scan geometry, and the truncated basis—not by the raw richness of the near-field phase on Γ_R [3]. Far-field $\mathcal{E}_{\mathrm{acc}}^{(N)}$ criterion. Define *far-field angular probe rank* by

$$\widehat{\mathbf{k}} \in \mathcal{K}_{\mathrm{acc}}^{\mathrm{ff}} \iff \Big[\, a(\widehat{\mathbf{k}}) \neq 0 \text{ after } \Lambda_{\mathrm{nf}} \text{ and } \mathcal{D}_{J,\mathrm{reg}} = 1 \,\Big] \text{ and } \widehat{\mathbf{k}} \in \mathcal{K}_{\mathrm{acc}}, \tag{2.409}$$

with Λ_{nf} from Eq. (2.406) and $\mathcal{K}_{\mathrm{acc}}$ from Eq. (2.9.3) [2]. Eq. (2.409) develops Eq. (2.409) by inserting explicit near-to-far transformation condition and geometric sector membership [2]. Mathematically, $\mathcal{K}_{\mathrm{acc}}^{\mathrm{ff}} \subseteq \mathcal{K}_{\mathrm{acc}}$ is the far-field recoverable subset after radiation and declared size and distance regimes; physically, it marks directions where angular amplitude survives radiation and lies in the geometrically accessible hemisphere [6]. Directions in $\mathcal{K}_{\mathrm{sup}}$ are excluded before observation composition [2]. Angular probe rank operator. Package Eq. (2.409) as the $\mathcal{E}_{\mathrm{acc}}^{(N)}$ *projector*

$$\mathcal{A}_{\mathrm{ff}} = \chi_{\mathcal{K}_{\mathrm{acc}}} \circ \mathcal{D}_{J,\mathrm{reg}} \circ \Lambda_{\mathrm{nf}}, \qquad a_{\mathrm{acc}}(\widehat{\mathbf{k}}) = [\mathcal{A}_{\mathrm{ff}}[\mathbf{J}_{\mathrm{imp}}]](\widehat{\mathbf{k}}), \tag{2.410}$$

with a_{acc} the accessible angular amplitude before finite measurement [2]. Eq. (2.410) aligns retained amplitude Eq. (2.394) with near-to-far output when $\mathrm{supp}(a) \subseteq \mathcal{K}_{\mathrm{acc}}$ near-to-far transformation (Subsection 2.9.2)(iv) [2]. Filtered observation Eq. (2.396) applies after Eq. (2.410): $a_{\mathrm{filt}} = \mathcal{P}_{\mathrm{meas}}[a_{\mathrm{acc}}]$ [6]. Null additions leave a_{acc} unchanged through Eq. (2.349) [2]. On lateral coordinates, define the *plane-wave probe rank filter* by

$$\widetilde{\mathbf{E}}_{\mathrm{acc}}(k_x, k_y) = \chi_{\mathcal{K}_{\mathrm{acc}}^{\mathrm{ff}}}(\widehat{\mathbf{k}})\, \widetilde{\mathbf{E}}_{\mathrm{rad}}(k_x, k_y), \qquad (k_x, k_y) \mapsto \widehat{\mathbf{k}} \text{ on } k_t \le k_0, \tag{2.411}$$

with $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ from Eq. (2.391) [2]. Eq. (2.411) is the Subsection 2.9.3 plane-wave criterion promised in Eq. (2.386): it removes geometrically forbidden directions on the propagating outgoing sheet without re-tabulating Eq. (2.392) [2]. Evanescent sheets satisfy $\widetilde{\mathbf{E}}_{\mathrm{acc}} = \mathbf{0}$ for $k_t > k_0$ through Eq. (2.393) [2]. Physically, Eq. (2.411) is the angular-spectrum-side statement of which plane-wave directions participate in far-zone reports [14]. Far-field $\mathcal{E}_{\mathrm{acc}}^{(N)}$ composes with 1 condition for outward angular-momentum flux by reference. A direction $\widehat{\mathbf{k}} \in \mathcal{K}_{\mathrm{acc}}^{\mathrm{ff}}$ supports a qualified transport report on solid-angle sector $\Delta\Omega(\widehat{\mathbf{k}})$ only when section 1.2.4 propagative support Eq. (1.63) and transport decision Eq. (1.67) hold for the corresponding directional channels Eqs. (1.49)–(1.51) [2]. Restated in operator form,

$$\mathcal{T}_{\mathrm{acc}}(\widehat{\mathbf{k}}) = 1 \implies \int_{\Delta\Omega(\widehat{\mathbf{k}})} \Pi(\theta, \phi; r)\, d\Omega > 0 \text{ and } \mathcal{Q}_J = 1, \tag{2.412}$$

with Π from Eq. (1.49), $\mathcal{Q}_J$ from Eq. (1.46), and r in the far zone where Eq. (2.405) applies [2]. Eq. (2.412) imports section 1.2.4 criteria without re-deriving the transport triad: 2 records when far-zone probe rank implies 1 transport admissibility on the same sector [2]. declared size and distance regime Eq. (2.270) remains mandatory: $\mathcal{T}_{\mathrm{acc}} = 1$ does not promote reactive near content to radiation [6]. Angular measurement criterion. After rank-N recovery projection, finite instruments apply

$$a_{\mathrm{obs}}(\widehat{\mathbf{k}}) = [\mathcal{P}_{\mathrm{meas}}\, a_{\mathrm{acc}}](\widehat{\mathbf{k}}), \qquad \mathrm{supp}(a_{\mathrm{obs}}) \subseteq \{\widehat{\mathbf{k}} : \text{ resolvable by } \mathcal{M}_\omega\} \cap \mathcal{K}_{\mathrm{acc}}^{\mathrm{ff}}, \tag{2.413}$$

with $\mathcal{P}_{\mathrm{meas}}$ from Eq. (2.396) [2]. Eq. (2.413) is the Subsection 2.9.3 measurement completion: observation rank Eq. (2.351) may strictly shrink $\mathrm{supp}(a_{\mathrm{obs}})$ inside $\mathcal{K}_{\mathrm{acc}}^{\mathrm{ff}}$ [2]. Rank cap Eq. (2.397) limits resolvable directions independently of source nullity [6]. Physically, a_{obs} is the laboratory-accessible

angular spectrum after geometry, regime, near-to-far extraction, and aperture [2]. Accessible angular budget. Collecting Subsection 2.9.3 constraints,

$$\operatorname{supp}(a_{\mathrm{obs}}) \subseteq \mathcal{K}_{\mathrm{acc}}^{\mathrm{ff}} \cap \{\widehat{\mathbf{k}} : \ \mathcal{T}_{\mathrm{acc}}(\widehat{\mathbf{k}}) = 1\} \cap \{\widehat{\mathbf{k}} : \ \text{resolvable by } \mathcal{M}_\omega\}, \tag{2.414}$$

composing geometric, transport, and observation factors with angular budget Eq. (2.352) [2]. Equation (2.414) is the section 2.9 completion of Eq. (2.382): each factor removes a distinct laboratory error — radiation nullity, geometric forbidding, regime violation, transport discondition, and finite aperture — without conflating them [2]. Reciprocity in radiation and reception (Subsection 2.8.3) constrains coupling among observed channels on $\operatorname{supp}(a_{\mathrm{obs}})$ but does not enlarge Eq. (2.414) [2].

$\mathcal{E}_{\mathrm{acc}}^{(N)}$ measurement sequence: Eq. (2.410) applies near-to-far extraction, geometric sector $\mathcal{K}_{\mathrm{acc}}$, transport criterion Eq. (2.412) , and measurement criterion Eq. (2.413).

Figure interpretation. Full angular content on S^2 passes through geometric sector Eq. (2.9.3), declared size and distance regime Eq. (2.139), and finite observation rank Eq. (2.220). The accessible output is the intersection of radiation-supported directions with aperture-resolvable modes—not the raw pattern from forward integration. Null-space additions leave $a(\widehat{\mathbf{k}})$ fixed but may alter near reactive traces used in calibration [2]. Suppressed directions never enter a_{obs}; evanescent content is removed before Λ_{nf} by Eq. (2.408) [2]. Mathematical role of Eq. (2.409). Eq. (2.409) closes section 2.9 by stating which directions on S^2 remain after radiation selection, geometric constraints, and declared size and distance regime [2]. It develops Eq. Eq. (2.386) with explicit dependence on Λ_{nf} and $\mathcal{K}_{\mathrm{acc}}$ [2]. Operator form Eq. (2.410) packages the same logic for source-to-amplitude maps [6]. 3 harmonic bases relabel $\mathcal{K}_{\mathrm{acc}}^{\mathrm{ff}}$ without altering the budget Eq. (2.414) [2]. Mathematical role of Eq. (2.411). Eq. (2.411) is the lateral-coordinate criterion completing the plane-wave chain begun in Subsection 2.9.1 [2]. It removes forbidden directions before near-to-far comparison and measurement [6]. Agreement with a_{acc} on $k_t \le k_0$ follows from Eq. (2.407) [2]. Evanescent sheets remain null through Eq. (2.393) independently of raw lateral scans [2]. Subsections 2.9.1–2.9.3 complete Eq. (2.382): plane-wave radiation operators, retained/suppressed/filtered partition, near-to-far extraction, and far-field $\mathcal{E}_{\mathrm{acc}}^{(N)}$ [2]. Modal coordinates Eq. (2.290) and plane-wave coordinates Eq. (2.289) describe identical radiation on $\operatorname{ran}\mathcal{R}_\omega$; section 2.9 fixes angular-spectrum filters on the plane-wave branch [2]. Inverse obstruction in Subsection 2.7.4 applies to a_{obs}, not to pre-probe rank transforms [6]. Subsection 2.9.3 assumes linear media, fixed ω, outgoing closure, and stationary geometry $\mathcal{C}_{\mathrm{geo}}$ [2]. Time-varying $\mathcal{M}_\omega$ requires separate treatment; static rank bounds remain upper limits [2]. Loss modifies amplitudes but not the inclusion chain in Eq. (2.414) [2]. Cavity and guide spectra remain outside exterior rank-N recovery; discrete books do not substitute for Eq. (2.409) on unbounded domains [14]. section 1.2.4 transport criteria are imported, not extended to near reactive zones without declared size and distance regime [2]. The mathematical statements in section 2.9 state what compact-range engineers already enforce through gating, scan distance, and modal banks: far-field angular probe rank is the intersection of radiation-supported directions with instrument-resolvable modes, not the raw output of a forward integral [6]. Spectral content present in $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ differs from content that remains in a_{obs} after $\mathcal{K}_{\mathrm{acc}}$ and $\mathcal{O}_{\mathrm{meas}}$ have acted [2]. section 2.10 returns to named hardware configurations so one may see nullity, rank limits, evanescent contamination, and structured-source topology as separate terms in one diagnostic partition rather than as a single pattern residual [3].

2.10 Examples

The four worked cases trace propagation, radiation, spectral rank, and instrument access in fixed order: compact dipole truncation, loop nullity, evanescent contamination, and structured-source

topology [2, 6, 17]. Each closes with an explicit verdict on what is radiated, null, or merely reactive before harmonic claims enter [7, 5]. Angular-spectrum action in section 2.9 states what finite instruments resolve on outgoing fields; section 2.10 returns to named hardware—apertures, dipoles, arrays, and scans—with the same tolerance discipline [2]. Each example is one integrated calculation on declared hardware [15].

The preceding sections developed radiation operators, null spaces, near-to-far limits, and rank-N instrument readouts as abstract objects on exterior domains. The distinctions appear on a laboratory bench as follows: when a spherical near-field scanner reports thirty-six complex samples, which part of the pattern residual owes to modal truncation, which to source nullity, and which to reactive contamination that never reaches the far zone? Subsections 2.10.1–2.10.4 isolate compact truncation, source nullity, evanescent near-to-far contamination, and structured-source topology–transport separation, respectively [2]. Every numerical ratio quoted below is *illustrative*: values follow author-calibrated discretizations of the stated geometries and are not reported measurement data; they are included only to show order-of-magnitude separation among mechanisms [3]. Before the subsections activate individual terms, the master degree-of-freedom partition deserves one statement: radiation reports can fail for distinct reasons that must not be merged in harmonic charts or laboratory interpretation [22]. Each summand below removes degrees of freedom from qualified radiation reports through a *distinct* physical mechanism, and conflating them produces remedies that treat symptoms rather than causes.

$$\underbrace{\ker \mathcal{R}_\omega}_{\text{source nullity (NR)}} \oplus \underbrace{\ker \mathcal{O}_{\text{meas}}\big|_{\operatorname{ran}\mathcal{R}_\omega}}_{\text{aperture rank}} \oplus \underbrace{\Pi_{\text{ev}}\mathcal{S}_\omega[\mathcal{J}_\Omega]}_{\text{evanescent storage}} \oplus \underbrace{\text{tail of compact/ modal truncation}}_{\text{spectral compression}} \oplus \underbrace{\mathcal{K}_{\text{sup}}}_{\text{geometric suppression}}, \tag{2.415}$$

with rank-N recovery $\mathcal{K}_{\text{acc}}^{\text{ff}}$ from Eq. (2.414) acting after Eq. (2.415) [2]. Subsections 2.10.1–2.10.4 each activate a different term in Eq. (2.415); cross-references replace repeated local split displays [2]. Eq. (2.415) is the diagnostic synthesis of 2: each direct summand names a distinct mechanism that removes degrees of freedom from qualified radiation reports. Source nullity $\ker \mathcal{R}_\omega$ is invisible to all far data; aperture rank $\ker \mathcal{O}_{\text{meas}}$ removes resolvable content even when radiation is nonzero; evanescent storage inflates near readouts without altering $a(\widehat{\mathbf{k}})$; spectral compression tail controls modal truncation error; geometric suppression $\mathcal{K}_{\text{sup}}$ removes directions by symmetry before measurement acts. On a single hardware test with $k_0 a = 1.5$, a 15% pattern residual might decompose as 7% compact tail, 5% aperture rank, 3% geometric suppression, and 0% NR nullity—four different remedies apply to four different terms [2]. Here $\mathcal{R}_\omega$ is the radiation operator at angular frequency ω, $\mathcal{O}_{\text{meas}}$ the finite-rank rank-N instrument readout on the radiative range, Π_{ev} the projector onto evanescent spectral content, $\mathcal{K}_{\text{sup}}$ the geometric angular-suppression set from symmetry or blockage, and the compact/modal tail the spectral compression induced by electrically small support and finite probe banks [2]. Flux meters from section 1.1.4 enter only on $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}]$ after Eq. (2.139)

Angular accessibility is therefore a rank-limited cone on Γ_R, not the full 4π chart: null directions persist even when listing cardinality is large. Compactness of radiation operators consolidates that picture (Subsection 2.10.1).

2.10.1 Compactness of radiation operators

Compactness of $\mathcal{R}_\omega$ on bounded source classes is the analytic statement that far-zone radiation is finite-rank on real hardware. Physically, every compact range implements a truncated observation of that compact operator—not an infinite harmonic listing [2, 17]. This worked configuration addresses observability under finite instrumentation, not whether Maxwell permits radiation from an electrically small source. A short dipole in free space certainly radiates; the bench issue is how

many independent complex samples a spherical near-field scan can resolve on the outgoing branch when the source ball and observation shell are both small compared with wavelength [6]. Increasing modal order in a spherical-harmonic fit need not improve pattern agreement; however, Maxwell theory shows that rank limits arise jointly from compact Green smoothing on bounded domains and from the finite probe bank, while non-radiating nullity is a separate mechanism that leaves far data unchanged [17]. Spectral truncation error must be distinguished from membership in $\ker \mathcal{R}_\omega$: the former shrinks when singular values of the compact part decay, the latter is invisible to any far-zone array regardless of modal count [2]. What a laboratory instrument detects on S_b at $b = 2a$ is dominated by reactive falloff faster than r^{-1}; only after declared size and distance regime and outgoing selection does the same measurement report a rank statement on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [3].

On a compact-range turntable, a center-fed wire dipole of length $L \ll \lambda$ and radius $a \ll L$ sits inside a source ball Ω_s of radius a with $k_0 a = 1.5$ at the operating frequency ω [2]. A spherical near-field scanner records $N = 36$ tangential electric samples on a closed surface S_b at radius $b = 2a$; a vector network analyzer reports VSWR and $|S_{11}|$ on the coax feed, confirming that stored reactive energy dominates the input admittance at this electrical size [3]. Far-field pattern cuts on the principal planes show a single-lobe $\ell = 1$ template, while probe magnitudes on S_b fall off faster than r^{-1} in the reactive shell—the bench configuration on which compact truncation is first isolated [6]. The compact-range configuration fixes outline, frequency, gauge, and permittivity before any operator split is written. The objective is to connect one finite probe bank to one rank statement on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$: hardware fixes how many independent complex samples are available, and the singular-value tail of the compact Green part controls how much of the radiative image those samples can resolve [2]. Watt closure and probe-count limits must be reported separately; conflating realizability bounds with observation bounds is a common laboratory error on electrically small measurements [3]. The first worked example isolates *spectral compression*: when a source and its observation shell are electrically small, the Green operator that maps impressed current to tangential field on a closed surface possesses a compact part whose singular values decay rapidly. Truncating that part controls pattern-fitting error, but the error is *not* non-radiating nullity—it is finite-geometry flux report. Readers who have measured electrically short dipoles on compact ranges will recognize the configuration immediately; the mathematics names what the instrument already enforces through probe count and scan radius. The bench places a center-fed wire of length $L \ll \lambda$ and radius $a \ll L$ at the center of a compact-range turntable; the source ball Ω_s enclosing the wire has radius a with $k_0 a = 1.5$ at the operating frequency ω [2]. Port voltage on a coax feed drives a triangle current distribution along L; a vector network analyzer reports VSWR and $|S_{11}|$, confirming that stored reactive energy dominates the input admittance at this electrical size [3]. Far-field pattern cuts on the principal planes show a single-lobe $\ell = 1$ template; a spherical near-field probe at radius $b = 2a$ records tangential $\mathbf{E}$-field samples whose magnitude falls off faster than r^{-1} in the reactive shell, so watt-meter integration on a small sphere would overestimate radiated power unless the shell is qualified [7]. The radiation integral yields $\ell = 1$ dominance with higher multipoles suppressed by $(k_0 a)^2$ [2]; pattern cuts agree with the $\ell = 1$ template, while spherical near-field grids must sample within $\lambda/4$ to avoid aliasing the reactive shell. Retaining only $M = 20$ modes in a spherical-harmonic surrogate leaves a tail error set by the twenty-first singular value of the compact Green part [3]. The dipole is the sanity check that every radiation map on compact support collapses toward a single radiative direction at low $k_0 a$, while the near field remains rich with reactive structure that no far-zone array can resolve [6]. Consider a compact impressed-current ball $\Omega_s = B_a(0)$ carrying an electric dipole $\mathbf{J}_{\mathrm{imp}}(\mathbf{r}) = \mathbf{J}_0\,\delta(\mathbf{r})$ in the distributional limit with $\int_{\Omega_s} \mathbf{J}_{\mathrm{imp}}\, d^3 r = \mathbf{p}$ and $k_0 a = 1.5$ at fixed ω . A near-field scanner records $N = 36$ tangential electric samples $\mathbf{y} \in \mathbb{C}^{36}$ on a closed surface S_b with radius $b = 2a$ through rank-N instrument readout $\mathcal{M}_\omega$ from Eq. (2.218) . The task is to:

(a) identify the compact and non-compact parts of the probe-to-far-field operator implied by Subsection 2.1.3;

(b) bound truncation error when only $M = 20$ retained modes are used in a spherical-harmonic surrogate of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$;

(c) state whether unresolved modal content is attributable to compact compression, to $\ker \mathcal{R}_\omega$, or to finite rank of $\mathcal{O}_{\mathrm{meas}}$.

Worked example 2.10.1: compact source, Green split on $\mathcal{S}_\omega$, radiation map, and rank-36 near-field probe bank. Compact part K_{cpt} controls modal truncation error; $\mathcal{R}_\omega$ is not compact on $\mathcal{J}_\Omega$.

The schematic separates compact smoothing on the source-to-field operator from non-compact radiation selection and finite-rank readout. On bounded domains with compact Ω_s, Subsection 2.1.3 records

$$\mathcal{S}_\omega = G_{\mathrm{sing}} + K_{\mathrm{cpt}}, \tag{2.416}$$

where $\mathcal{S}_\omega$ maps admissible impressed current $\mathbf{J}_{\mathrm{imp}}$ to the physical electric field on observation shell S_b, G_{sing} carries the non-compact singular self-field as source and observation points coincide, and K_{cpt} is the compact remainder whose singular values decay on bounded Ω_s, importing Eq. (2.78) without re-proving kernel construction from section 2.3 [2]. Mathematically, G_{sing} captures non-compact singular behavior as probe and source points coincide; K_{cpt} maps bounded source sets to relatively compact field sets in the L^2 topology on S_b [2]. Physically, smooth interpolation between widely separated samples is a compact smoothing operation, whereas pointwise singular self-terms are not [6]. Radiation selection defines $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ from Eq. (2.67) [2]. The map $\mathcal{R}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_{\mathrm{rad}}$ is *not* compact: $\ker \mathcal{R}_\omega \neq \{\mathbf{0}\}$ is infinite-dimensional null-space structure of radiation operators (Subsection 2.7.1), so bounded source balls do not map to relatively compact sets in $\mathcal{F}_{\mathrm{rad}}$ [2]. Compactness enters through K_{cpt} in Eq. (2.416) and through finite observation, not through $\mathcal{R}_\omega$ itself [2]. Confusing non-radiating additions with modal truncation error violates the distinction established in Subsection 2.1.3 [6]. The probe bank implements $\mathcal{O}_{\mathrm{meas}} : \mathcal{F}_{\mathrm{rad}} \to \mathbb{C}^{36}$ with $\mathrm{rank}(\mathcal{O}_{\mathrm{meas}}) \leq 36$ from Eq. (2.79) [2]. Composed map $\mathcal{M}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \mathcal{O}_{\mathrm{far}} \circ \mathcal{R}_\omega$ therefore satisfies

$$\mathrm{rank}(\mathcal{M}_\omega) \leq \min\{36, \dim \mathcal{R}(\mathcal{R}_\omega)\} \tag{2.417}$$

where $\mathcal{M}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \mathcal{O}_{\mathrm{far}} \circ \mathcal{R}_\omega$ is the composed probe-to-sample map and the integer 36 is the probe-bank rank cap by Eq. (2.84) [2]. Mathematically, at most 36 independent complex numbers are available to infer far-zone content; physically, the scanner cannot resolve more than 36 radiation degrees of freedom regardless of Maxwell residual in the volume [6]. Null-space directions in $\ker \mathcal{O}_{\mathrm{meas}}\big|_{\mathrm{ran}\,\mathcal{R}_\omega}$ are infinite-dimensional Eq. (2.80) [2]. Let $\mathcal{K}$ denote the probe-to-modal map obtained by composing sample interpolation with compact K_{cpt} on the dipole-supported subspace. Subsection 2.1.3 gives

$$\|\mathcal{K} - \mathcal{K}_M\|_{\mathrm{op}} = \inf_{\mathrm{rank}\,\mathcal{K}_M \leq M} \|\mathcal{K} - \mathcal{K}_M\|_{\mathrm{op}} \sim \sigma_{M+1}(\mathcal{K}), \tag{2.418}$$

restating Eq. (2.81) on the worked geometry [2]. For $k_0 a = 1.5$ and $b = 2a$, numerical evaluation of the discretized $\mathcal{K}$ yields $\sigma_{20}/\sigma_1 \approx 0.12$ and $\sigma_{37}/\sigma_1 \approx 0.04$: the twenty-first mode is already below 12% of the dominant singular value [6]. Physically, retaining $M = 20$ modes captures most radiative energy resolvable on S_b; modes 21–36 lie in the tail controlled by compact smoothing, not in $\ker \mathcal{R}_\omega$ [2]. Subsection 2.1.3 bounds compression error by

$$\|\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] - \mathcal{R}_\omega^{(M)}[\mathbf{J}_{\mathrm{imp}}]\|_{\mathcal{U}} \leq C\, \sigma_{M+1}(K_{\mathrm{cpt}})\, \|\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{J}}, \tag{2.419}$$

where $\mathcal{R}_\omega^{(M)}$ is the M-mode spectral surrogate of the radiation map, C a geometry-dependent constant from Subsection 2.1.3, $\sigma_{M+1}(K_{\mathrm{cpt}})$ the first neglected singular value of the compact Green part, and $\|\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{J}}$ the source norm on admissible impressed-current space [2]. With $\sigma_{21} \approx 0.12\,\sigma_1$ and dipole normalization $\|\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{J}}$ fixed by $|\mathbf{p}|$, the bound states that no rank-20 operator can reduce far-field state error below order $0.12\,|\mathbf{E}_0|$ on the outgoing branch [2]. Mathematically, the tail is controlled by compact K_{cpt}; physically, increasing M beyond the singular-value knee yields diminishing returns unless N or b/a increases [6]. For the dipole problem, the active mechanism is the compact tail of $\sigma_n(K_{\mathrm{cpt}})$ in Eq. (2.415), not $\ker \mathcal{R}_\omega$ [2]. Adding $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega$ leaves $\mathbf{y}$ unchanged; modes beyond rank 36 lie in $\ker \mathcal{O}_{\mathrm{meas}}$; modes 21–36 sit in the compact tail with error $\sim \sigma_{21}$ (illustrative ratio $\sigma_{21}/\sigma_1 \approx 0.12$) [2]. Pattern-fitting routines must trace which term in Eq. (2.415) removed unresolved content [6]. On the electrically small dipole bench, declared size and distance regime Eq. (2.139) remains mandatory before interpreting truncated modal reports as transport [6]. A practitioner who increases modal order M without raising probe count N or scan radius b/a may reduce a pattern residual that owes nothing to source physics—only to compact smoothing—while true non-radiating additions would leave $\mathbf{y}$ unchanged regardless of M. Physically, the distinction is not whether the dipole radiates—it does—but which term in Eq. (2.415) removed unresolved content when a finite probe bank was compared with a truncated modal surrogate. Compact-range reporting often reports a single pattern residual without separating compact singular-value tails from true source nullity [3]. Electrically small geometry compresses resolvable radiation degrees of freedom on a nearby shell; probe count and scan radius already enforce the bound on the bench [2]. Subsection 2.10.2 turns to the complementary case in which impressed current is substantial yet annihilated by $\mathcal{R}_\omega$, so far-zone arrays remain blind while near probes change.

Compact radiation operators therefore cap measurable radiative rank before calibration or harmonic width is debated: SVD knees on $\mathbf{\Gamma}_\omega$ are geometry, not software. Structured source states refine how that rank is populated (Subsection 2.10.4).

2.10.2 Non-radiating source configurations

A coax-fed pyramidal horn over a ground plane defines the impressed-current and boundary-data tuple technicians declare before Maxwell radiation is computed [2]. Maxwell's retarded radiation to Γ_R remains the governing constraint [1]. The bench issue is not whether Maxwell permits local current—it clearly does—but whether far-zone observables can detect every admissible impressed source that produces them. A radiating dipole and a balanced loop may coexist in one compact package: the loop reorganizes reactive $\mathbf{E}$ and $\mathbf{H}$ near Ω_s while, only when verified in $\ker \mathcal{R}_\omega$ under Subsection 2.7.1 additional outgoing-cancellation conditions, contributing nothing to outgoing radiation on Γ_R once $\mathcal{R}_\omega$ has acted [17]. A changed near-field scan need not imply a changed far pattern; Maxwell theory shows that superposition is linear on $\mathcal{R}_\omega$ and that $\ker \mathcal{R}_\omega$ is generally infinite-dimensional, so identical far data do not fix the source [6]. Membership in $\ker \mathcal{R}_\omega$ must be distinguished from evanescent storage treated in Subsection 2.10.3, both may leave $a(\widehat{\mathbf{k}})$ invariant, yet verified loop nullity is a source-side annihilation under radiation, not a field-side reactive sheet [2]. What a laboratory measurement actually reveals depends on which map is read—$\mathcal{S}_\omega$ on an enclosing scan, $\mathcal{R}_\omega$ on a qualified far-zone array—and watt integration on a reactive shell may rise even when radiated power is unchanged [3].

On a compact-range quiet zone, a center-fed electric dipole with moment $|\mathbf{p}| = p_0$ and $k_0 a = 1.2$ radiates in free space with outgoing radiation condition [2]. An engineer mounts a small magnetic loop of radius $a_{\mathrm{loop}} = 0.3a$ beside the feed with equal opposing filaments so net dipole moment vanishes; near-field probes on S_b at $b = 2a$ record tangential samples, while a far-zone array

estimates $\mathbf{E}_0(\theta,\phi)$ from Eq. (2.246) [17]. The bench question is operational: does the added loop change the published pattern, the near scan, or both? The second worked example turns to *source nullity*: impressed currents that satisfy Maxwell admissibility and produce substantial near fields yet annihilate under $\mathcal{R}_\omega$. Such configurations are not mathematical curiosities—they appear whenever balanced filament pairs, surface tangential modes, or loop cancellations are engineered into compact packages. The lesson for measurement is severe: far-zone pattern matching cannot detect the added current, while near-zone diagnostics change in ways that watt integration on a reactive shell may misread as additional radiative content. *Geometry and size.* A standard-gain horn with rectangular mouth $a \times b$, both dimensions several wavelengths across, is mounted on a compact-range positioner with the bore sight directed toward the quiet-zone center [15]. *Excitation.* A TE_{10} waveguide feed delivers calibrated port power; a network analyzer sets the reference plane at the throat [14]. *Observable behavior.* The main beam follows bore sight with sidelobe shoulders set by aperture diffraction and edge currents; gating in time domain separates the direct illumination from range-wall reflections [2]. *Mathematical analysis.* Equivalent magnetic and electric currents on the mouth feed radiation integrals; plane and spherical spectra exchange via stationary phase at large r [6]. *Measurement consequence.* Compact-range gating must exclude multiple reflections from horn interior and range wall before pattern moments are trusted [3]. *Physical interpretation.* Horn patterns test outgoing selection and equivalence surfaces before rank claims are transferred to more complex feeds; they also provide the contrasting backdrop against which loop-null configurations in the display below become visible only through near-field probes, not through far-zone arrays [1]. Subsection 2.10.2 activates the $\ker \mathcal{R}_\omega$ term of Eq. (2.415), importing Subsection 2.7.1 from the cited sections [2]. Non-radiating current definition Eq. (2.327), canonical families Eqs. (2.329)–(2.332), canonical non-radiating configurations (Subsection 2.7.1), superposition law Eq. (2.346), near-field law Eq. (2.348), and modal invisibility Eq. (2.347) are presupposed; proofs are not repeated [2]. Null-space operator structure (Subsection 2.7.1), angular invariance Eq. (2.349), and inverse obstruction angular spectra and inverse problems (Subsection 2.7.4) supply context [6]. Consider a compact electric dipole $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}$ with moment $|\mathbf{p}| = p_0$ and $k_0 a = 1.2$ radiates in free space with outgoing radiation condition [2]. An engineer adds a small magnetic loop $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}$ of radius $a_{\mathrm{loop}} = 0.3a$ with equal opposing filaments so net dipole moment vanishes and loop cancellation Eq. (2.329) applies [2]. The balanced filaments cancel the leading electric-dipole radiation term per Eq. (2.329); complete outgoing cancellation and membership in $\ker \mathcal{R}_\omega$ require the additional constraints recorded in Subsection 2.7.1, not $\mathbf{p} = \mathbf{0}$ alone [17]. Near-field scans that omit watt completion may falsely infer radiative diversity [3]. Define $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})} = \mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}$ with charge continuity preserved on Ω_s Eq. (2.7.2) [2]. Near-field probes on S_b at $b = 2a$ record $\mathbf{y}_{\mathrm{near}}$; a far-zone array estimates $\mathbf{E}_0(\theta,\phi)$ from Eq. (2.246). The task is to:

(a) Using Subsection 2.7.1 and Eq. (2.329) without re-deriving the canonical families: (i) establish that the opposing-arm loop geometry yields $\mathbf{p} = \mathbf{0}$; (ii) identify which leading radiative term in Eq. (2.246) is thereby suppressed; (iii) determine whether electric-dipole cancellation alone implies $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}] = \mathbf{0}$; and (iv) state the additional outgoing-cancellation condition already recorded in Volume I (e.g. Eq. (2.330)) required for $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})} \in \ker \mathcal{R}_\omega$. Do not treat vanishing dipole moment, suppression of the leading radiative term, complete absence of outgoing radiation, and $\ker \mathcal{R}_\omega$ membership as interchangeable statements.

(b) quantify near-field change versus far-field invariance under superposition;

(c) show why source inversion Eq. (2.219) cannot detect $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}$ from far data alone.

Tasks (b) and (c) proceed under the hypothesis that complete outgoing cancellation has been verified so $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})} \in \ker \mathcal{R}_\omega$, illustrating consequences that follow *if* that membership holds—not consequences proved by electric-dipole cancellation alone in task (a).

Subsection 2.7.1 records that equal opposing arms on $\mathcal{C}_+$, $\mathcal{C}_-$ with vanishing net circulation imply

$$\int_{\mathcal{C}_+} \mathbf{J}_{\mathrm{loop}} \cdot d\boldsymbol{\ell} + \int_{\mathcal{C}_-} \mathbf{J}_{\mathrm{loop}} \cdot d\boldsymbol{\ell} = 0 \implies \mathbf{p} = \mathbf{0}, \tag{2.420}$$

restating the electric-dipole cancellation step of Eq. (2.329) on the worked geometry [2]. For task (a): (i) $\mathbf{p} = \mathbf{0}$ follows from Eq. (2.420); (ii) the leading electric-dipole term in Eq. (2.246) is suppressed; (iii) electric-dipole cancellation alone does *not* imply $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}] = \mathbf{0}$—magnetic-dipole and higher-multipole outgoing terms may remain [14]; (iv) membership $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})} \in \ker \mathcal{R}_\omega$ requires verified complete outgoing cancellation on S^2, as in Eq. (2.330), not inference from $\mathbf{p} = \mathbf{0}$ alone [17]. Near Ω_s the loop may remain magnetically dominated even when the leading electric-dipole export vanishes [6]. Subsection 2.7.1 records the opposing-arm geometry as a canonical electric-dipole-cancellation family [2]. Under the verified-membership hypothesis stated above task (b), superposition with $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})} = \mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}$, Subsection 2.7.1 gives

$$\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}] = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}] + \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}] = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}], \tag{2.421}$$

by linearity Eq. (2.5) and, under verified $\ker \mathcal{R}_\omega$ membership, $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}] = \mathbf{0}$ [2]. Far pattern amplitudes therefore satisfy

$$\mathbf{E}_0(\theta, \phi; \mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}) = \mathbf{E}_0(\theta, \phi; \mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}), \qquad a(\widehat{\mathbf{k}}; \mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}) = a(\widehat{\mathbf{k}}; \mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}), \tag{2.422}$$

importing Eq. (2.346) and angular invariance Eq. (2.349) [2]. Mathematically, $\ker \mathcal{R}_\omega$ is invisible to all far-zone readouts including Eq. (2.405); physically, antenna pattern measurements cannot distinguish $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}$ from $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}$ on far shells [6]. Modal coefficients obey Eq. (2.347) for $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}$ alone [2]. Near-field detectability versus far invisibility. Source-to-field operator $\mathcal{S}_\omega$ is generally *not* invariant:

$$\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}] = \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}] + \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}], \qquad \|\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}]\|_{\mathcal{E}} \text{ may be } \gg 0 \tag{2.423}$$

on near shells, restating nonzero near fields for non-radiating currents (Subsection 2.7.1) and Eq. (2.348) [2]. For $k_0 b = 2.4$, numerical evaluation gives $|\Delta \mathbf{y}|/|\mathbf{y}_{\mathrm{rad}}| \approx 0.35$ on tangential probes: the loop is visible near the source while far $\mathbf{E}_0$ is unchanged [2]. declared size and distance regime Eq. (2.270) forbids interpreting the near increment as angular transport without $\mathcal{D}_{J,\mathrm{reg}} = 1$ [6]. This step is distinct from evanescent storage Eq. (2.408): under verified $\ker \mathcal{R}_\omega$ membership, the loop contributes to $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}]$ but not to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}]$ [2]. Power and flux meters. Radiated power on far shells satisfying the Sommerfeld condition satisfies $\mathcal{P}_{\mathrm{rad}}(\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}) = \mathcal{P}_{\mathrm{rad}}(\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})})$ from Eq. (2.247) and Eq. (2.422) [2]. Reactive budget Eq. (2.268) may increase locally because $\mathcal{P}_{\mathrm{nr}}$ from $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}]$ adds near Ω_s while radiation remains unchanged [2]. Physically, integrating Poynting flux through a near sphere confuses storage with radiation unless regime index Eq. (2.30) from section 1.2.5 qualifies the shell [6]. Inverse source ill-posedness. Far-zone data fix $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ but not $\mathbf{J}_{\mathrm{imp}}$. angular spectra and inverse problems (Subsection 2.7.4) and Eq. (2.354) imply

$$\mathbf{J}_{\mathrm{imp}}^{(1)} - \mathbf{J}_{\mathrm{imp}}^{(2)} \in \ker \mathcal{R}_\omega \quad \text{whenever} \quad \mathbf{E}_0(\mathbf{J}_{\mathrm{imp}}^{(1)}) = \mathbf{E}_0(\mathbf{J}_{\mathrm{imp}}^{(2)}), \tag{2.424}$$

so $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}$ and $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}$ belong to the same equivalence class [2]. Tikhonov reconstruction Eq. (2.223) may return either representative depending on regularization without contradiction [2]. Mathematically,

$\ker \mathcal{R}_\omega$ has infinite dimension on exterior domains Eq. (2.325); physically, adding loops, surface tangential nulls Eq. (2.331), or modal $\lambda_n = 0$ realizations Eq. (2.332) leaves patterns unchanged [6]. Far invariance Eq. (2.422) arises because $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})} \in \ker \mathcal{R}_\omega$—the active term is source nullity in Eq. (2.415), not finite probe rank or compact truncation [2]. Geometric suppression Eq. (2.345) remains a separate term: the dipole radiates on $\mathcal{K}_{\mathrm{acc}}$ while the loop adds invisible near content [6]. Subsection 2.10.2 demonstrates that non-radiating configurations are physically realizable and operationally consequential: near diagnostics change while far observables do not [2]. Pattern-matching based solely on $\mathbf{E}_0$ or $a(\widehat{\mathbf{k}})$ cannot detect $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}$; near measurements and source-domain constraints Eq. (2.7.2) are required [2]. Physically, non-radiating additions are invisible to far-zone arrays yet vivid to near probes sampling $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}]$—existence of electromagnetic structure does not imply observability of its source detail [17]. This distinction arose in inverse scattering when equivalent sources were fitted to pattern data without uniqueness guarantees [21]. Superposition on $\mathcal{R}_\omega$ implies that any far-targeted inversion must impose source-domain constraints beyond pattern matching alone [6]. Subsection 2.10.3 addresses a different mechanism with the same far invariance: evanescent storage that inflates near readouts without altering $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ when the radiation image is fixed.

The measurable consequence is therefore a directional or rank limit on radiation, not a calibration correction: recalibration cannot remove obstructions imposed by $\ker \mathbf{\Gamma}_\omega$ [6].

2.10.3 Evanescent suppression in the far field

The bench issue is whether reactive richness on a near scan establishes additional far-zone angular content. Handset and IoT platforms routinely exhibit strong tangential fields within one wavelength of a patch or ground plane while qualified far-zone gain remains modest [6]. Far-field extractors must not be applied directly to spherical near-field samples; Maxwell theory shows that $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ and $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ possess different asymptotic envelopes, so reactive-dominated shell traces can bias angular radiation by order unity even when the true radiative amplitude $a(\widehat{\mathbf{k}})$ is unchanged [2]. $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ must not be identified with radiation: evanescent sheets carry storage measurable on enclosing probes yet are annihilated before near-to-far maps attach to pattern meters [2]. An instrument on S_b with $k_0 b \sim \mathcal{O}(1)$ is therefore not, by itself, a far-zone pattern; qualified analysis applies $\mathcal{R}_\omega$ before $\mathcal{H}_{\mathrm{ff}}$ [3].

A handset-scale test fixture places a rectangular patch of width W and length L on a dielectric substrate of height $h \ll \lambda$ above a finite ground plane; probe or inset feed with VSWR tuned for narrowband operation at fixed ω drives the structure [6]. Spherical near-field scans at height h above the substrate record tangential fields whose raw magnitude can exceed the radiative component by factors near two before radiation selection; ground-edge diffraction and substrate modes dominate the reactive shell [2]. The third worked example addresses *evanescent storage*: field-side reactive content that inflates near-shell readouts while leaving the radiative image $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ unchanged. This mechanism differs from source nullity even when both produce far-zone invariance, because the near signature arises from $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ rather than from membership in $\ker \mathcal{R}_\omega$. Laboratory teams who apply far-field extraction algorithms directly to spherical near-field scans without radiation selection encounter this laboratory error routinely on handset and IoT platforms where ground truncation and substrate modes dominate reactive shells. *Geometry and size.* A rectangular patch of width W and length L is printed on a dielectric substrate of height $h \ll \lambda$ above a finite ground plane whose electrical extent sets the effective radiation aperture [6]. *Excitation.* Probe or inset feed with VSWR tuned for narrowband operation at fixed ω [3]. *Observable behavior.* Copolar pattern cuts show low-order aperture modes; ripples from ground-edge diffraction appear when the scan surface

is too close to the substrate [2]. *Mathematical analysis.* Impressed currents on the patch and image currents on the ground enter $\mathcal{R}_\omega$ with compact support; balanced reactive loops below threshold contribute to $\mathcal{S}_\omega$ without altering radiation [17]. *Measurement consequence.* Spherical near-field scans must include ground-edge diffraction; the angular-spectrum bandlimit is set by scan height, and raw near traces can exceed radiative amplitude by factors near two before radiation selection [3]. *Physical interpretation.* Handset and IoT antennas live in this rank regime: the electrical size of the ground, not the patch alone, fixes radiation, while reactive storage on the evanescent sheet explains why near-field "hot spots" need not imply additional far-zone power [6]. Subsection 2.10.3 activates the evanescent term of Eq. (2.415), importing Subsection 2.9.2 from the cited sections [2]. Near-to-far definition Eq. (2.405), far-zone extractor Eq. (2.404), operator composition Eq. (2.406), evanescent annihilation Eq. (2.408), angular null law Eq. (2.393), near-to-far transformation (Subsection 2.9.2), and near-to-far null invariance and finite-shell error bounds (Subsection 2.9.2) are presupposed; asymptotic proofs in Subsection 2.5.2 and evanescent content in Subsection 2.7.2 are not repeated [2]. Evanescent overlap Eq. (2.334), non-radiation law Eq. (2.335), declared size and distance regime Eq. (2.270), and near-zone scaling Eq. (2.242) supply context [6]. Consider a compact impressed-current ball $\Omega_s = B_a(0)$ carrying an electric dipole $\mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}}(\mathbf{r}) = \mathbf{J}_0\,\delta(\mathbf{r})$ in the distributional limit with $\int_{\Omega_s} \mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}}\, d^3r = \mathbf{p}$ and $k_0 a = 2.0$ at fixed ω . A second admissible current $\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}$ is chosen so that $\mathcal{R}_\omega[\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}] = \mathbf{0}$ while $\|\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}]\|_{\mathcal{E}}$ is comparable to the radiative energy of $\mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}}$. This is *field-side evanescent excitation with vanishing radiation*: strong local fields can coexist with membership failure in $\mathcal{F}_{\mathrm{rad}}$ even when $|E|$ is large on the near shell, because far-zone claims require propagating spectrum support on $k_t \le k_0$, not reactive storage on the evanescent sheet $k_t > k_0$ [20]. The configuration is *not* membership of $\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}$ in the canonical non-radiating *source* family of Subsection 2.7.1, the same far invariance can arise from $\mathbf{J}^{(\mathrm{NR})}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$ with a different near signature [20]. Define the superposition $\mathbf{J}_{\mathrm{imp}} = \mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}} + \mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}$. A near-field probe records tangential $\mathbf{E}$ on shell S_b with $b = 1.25a$, while a far-field extractor applies extractor Eq. (2.404) on shells S_r with increasing r [6]. The task is to:

(a) quantify near-shell contrast between $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ and $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ when evanescent storage is large;

(b) confirm that $a(\widehat{\mathbf{k}})$ from Eq. (2.405) is unchanged by $\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}$ and that lateral spectra on $k_t > k_0$ vanish under Eq. (2.393);

(c) diagnose errors when the near-to-far limit is applied to raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ instead of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$.

Evanescent overlap appears only on the dashed near branch; the far limit reports outgoing radiation alone On admissible $\mathbf{J}_{\mathrm{imp}}$, Subsection 2.9.2 fixes the near-to-far input as $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, not $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ Eq. (2.405) [2]. Decompose the impressed current into radiative and evanescent-overlap parts through

$$\mathbf{J}_{\mathrm{imp}} = \mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}} + \mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}, \qquad \mathcal{R}_\omega[\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}] = \mathbf{0}, \qquad \|\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}]\|_{\mathcal{E}} = \beta\,\|\mathcal{R}_\omega[\mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}}]\|_{\mathcal{E}}, \tag{2.425}$$

where $\mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}}$ is the radiating dipole part, $\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}$ the evanescent-overlap admixture with $\mathcal{R}_\omega[\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}] = \mathbf{0}$, Π_{ev} the projector onto evanescent field content, $\|\cdot\|_{\mathcal{E}}$ the energy norm on the observation shell, and $\beta = 1.8$ the illustrative reactive-to-radiative ratio in the worked normalization (2.425) expresses evanescent source overlap Eq. (2.334) on the dipole-plus-evanescent geometry without re-proving evanescent content never radiates (Subsection 2.7.2) [2]. Mathematically, $\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}$ lies in a class that excites $\mathcal{F}_{\mathrm{ev}}$ Eq. (2.333) while annihilating under Π_{rad} Eq. (2.335); physically, it models reactive near storage—for example, a superposition of forbidden-angle plane-wave trials with imaginary

longitudinal wavenumber Eq. (2.131)—that does not add outgoing radiation [6]. On S_b with $b = 1.25a$, near-zone scaling Eq. (2.242) dominates: reactive r^{-3} and inductive r^{-2} terms can exceed the radiative r^{-1} envelope [2]. Tangential energy norms on S_b yield

$$\frac{\|\mathbf{E}^{(\mathrm{full})}_{\mathrm{near}}\|_{L^2(S_b)}}{\|\mathbf{E}^{(\mathrm{rad})}_{\mathrm{near}}\|_{L^2(S_b)}} = \frac{\|\Pi_{\mathrm{near}}\mathcal{S}_\omega[\mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}} + \mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}]\|}{\|\Pi_{\mathrm{near}}\mathcal{R}_\omega[\mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}}]\|} \approx \sqrt{1+\beta^2} \approx 2.06, \tag{2.426}$$

where $\mathbf{E}^{(\mathrm{full})}_{\mathrm{near}}$ is the tangential field from $\mathcal{S}_\omega[\mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}} + \mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}]$, $\mathbf{E}^{(\mathrm{rad})}_{\mathrm{near}}$ from $\mathcal{R}_\omega[\mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}}]$ alone, Π_{near} shell restriction to S_b at $b = 1.25a$, and the ratio $\sqrt{1+\beta^2}$ follows from orthogonal energy split on reactive and radiative subspaces [2]. Eq. (2.426) quantifies task (a): a scanner that records raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ sees roughly twice the tangential magnitude of the radiative component at $b = 1.25a$ [2]. Mathematically, the ratio follows from orthogonal energy split Eq. (2.142) on evanescent and radiative subspaces; physically, it warns that near-field "hot spots" need not imply additional far-zone power or modified angular patterns [6]. Regime map Eq. (2.248) classifies $b = 1.25a$ as reactive-dominated for $k_0 a = 2$ [2]. Far amplitude invariance follows from Subsection 2.9.2, near-to-far transformation Eq. (2.405) gives

$$a(\widehat{\mathbf{k}}) = \lim_{r\to\infty} \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}} + \mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}]\big] = \lim_{r\to\infty} \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}}]\big], \tag{2.427}$$

because $\mathcal{R}_\omega[\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}] = \mathbf{0}$ and Λ_{nf} is linear Eq. (2.406) [2]. near-to-far transformation (Subsection 2.9.2)(iii) expresses the same conclusion through Eq. (2.408): evanescent storage in $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ does not map to $a(\widehat{\mathbf{k}})$ when $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is unchanged [2]. Equation (2.427) answers task (b): dipole angular pattern, directional gain, and plane-wave lateral content on $k_t \le k_0$ are identical with and without $\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}$ [14]. Near-to-far null invariance (Subsection 2.9.2) further guarantees that any $\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$ leaves a unchanged, distinct from the evanescent-overlap case where $\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}$ is not necessarily non-radiating as a source but still annihilates under $\mathcal{R}_\omega$ [2]. Lateral spectrum null on the evanescent sheet. Plane-wave radiation operator Eq. (2.392) applied to the superposition gives, on $k_t > k_0$,

$$\widetilde{\mathbf{E}}_{\mathrm{rad}}(k_x, k_y) = \big[\mathcal{T}_\omega\, \widetilde{\mathbf{J}_{\mathrm{imp}}}\big](k_x, k_y) = \mathbf{0}, \qquad k_t > k_0, \tag{2.428}$$

importing Eq. (2.393) without re-displaying Eq. (2.138) [2]. Raw lateral transform $\mathcal{F}_{\mathrm{lat}}[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]]$ can show large coefficients on the evanescent sheet; radiative lateral spectrum Eq. (2.391) does not [2]. Mathematically, $\mathcal{T}_\omega$ factors $\Pi_{\mathrm{rad}} \circ \mathcal{R}_\omega$ before $\mathcal{F}_{\mathrm{lat}}$; physically, a near-field Fourier scan that omits radiation selection misreports "angular content" on forbidden k_t [6]. Link Eq. (2.407) identifies $a(\widehat{\mathbf{k}})$ with $\widetilde{\mathbf{E}}_{\mathrm{ret}}$ only on $k_t \le k_0$ when retained content vanishes [2]. Wrong versus correct near-to-far measurement sequences. Task (c) compares two post-composition laws on the same near-shell data:

$$a_{\mathrm{wrong}}(\widehat{\mathbf{k}}) = \lim_{r\to\infty} \mathcal{H}_{\mathrm{ff}}\big[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](r, \widehat{\mathbf{k}}), \tag{2.429}$$

versus the qualified chain

$$a_{\mathrm{correct}}(\widehat{\mathbf{k}}) = \lim_{r\to\infty} \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](r, \widehat{\mathbf{k}}) = a(\widehat{\mathbf{k}}), \tag{2.430}$$

from Eq. (2.405) [2]. near-to-far transformation (Subsection 2.9.2)(iii) states that $a_{\mathrm{wrong}} \neq a_{\mathrm{correct}}$ in general: reactive near traces contaminate Eq. (2.404) when applied to $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on finite shells [2]. On S_b with $k_0 b = 2.5$, numerical evaluation gives $\|a_{\mathrm{wrong}} - a_{\mathrm{correct}}\|_{L^2(S^2)} / \|a_{\mathrm{correct}}\|_{L^2(S^2)} \approx 0.41$: extractor output from raw near data overestimates angular radiation by roughly forty percent [6]. Mathematically, the error is not a finite-r artifact alone; it persists in the limit because $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ and $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ have different asymptotic envelopes Eq. (2.247) [2]. Physically, applying a far-field

algorithm to reactive-dominated shell traces violates observation-shell regime map Eq. (2.248) and regime criterion Eq. (2.270) [6]. finite-shell error in near-to-far extraction (Subsection 2.9.2) bounds qualified extraction on finite S_r:

$$\|a_r - a\|_{L^2(S^2)} \leq C\,\chi(r), \qquad a_r(\widehat{\mathbf{k}}) = \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](r,\widehat{\mathbf{k}}), \tag{2.431}$$

importing Eq. (2.402) with $\chi(r) \to 0$ as $k_0 r \to \infty$ [2]. For $k_0 a = 2$, $r = 8a$ yields $\chi(r) \approx 0.08$ and $\|a_r - a\|/\|a\| \approx 0.06$: far-zone extraction on the $\mathcal{R}_\omega$ branch converges while near-shell raw scans still show Eq. (2.426) [2]. Evanescent addition does not slow convergence because $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}] = \mathbf{0}$ removes evanescent content from the extractor input [6]. Accessibility filter Eq. (2.409) and measurement criterion Eq. (2.413) compose after Eq. (2.430), not before radiation selection [2]. Evanescent term in Eq. (2.415). Reactive storage $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ does not map to $a(\widehat{\mathbf{k}})$ when $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is fixed Eq. (2.408) [2]. In the present problem, near contrast Eq. (2.426) yields an illustrative ratio ≈ 2.06 while a remains invariant Eq. (2.427). Source nullity (§2.10.2) and evanescent storage (here) both leave a unchanged but alter near fields through different terms in Eq. (2.415) [6]. Wrong measurement sequence Eq. (2.429) gives an illustrative bias $\sim 40\%$ relative to Eq. (2.430) [2]. Physical meaning (summary). Subsection 2.10.3 shows that large evanescent near storage is compatible with unchanged far angular radiation [2]. Qualified near-to-far analysis applies $\mathcal{R}_\omega$ before $\mathcal{H}_{\mathrm{ff}}$ near-to-far transformation (Subsection 2.9.2) [2]. Figure 2.18 (Subsection 2.9.2) is the reference schematic; Subsection 2.10.3 applies it to evanescent-dominated near scans without duplicating the measurement sequence figure [2]. Physically, the evanescent mechanism explains why near-shell amplitude can exceed radiative amplitude by more than a factor of two while $a(\widehat{\mathbf{k}})$ remains fixed once $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is held constant [2]. Far invariance alone is not decisive—source nullity in Subsection 2.10.2 shares that signature—; the decisive term is which term in Eq. (2.415) removed the misleading near contrast [6]. Diagnostics must be applied to the radiative branch before asymptotic extraction, not to the complete solution map [2]. Subsection 2.10.4 closes the example suite with structured impressed currents whose far appearance, topology condition, and signed angular transport must be separated on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ rather than read from reactive near scans.

Maxwell consistency therefore fixes which admissible traces may carry flux on $\mathcal{F}_{\mathrm{rad}}$ [2].

2.10.4 Radiation from structured source states

A coax-fed pyramidal horn over a ground plane defines the impressed-current and boundary-data tuple technicians declare before Maxwell radiation is computed [2]. Maxwell's retarded radiation to Γ_R remains the governing constraint [1]. Structured sources raise whether organized amplitude, phase, and polarization at the feed imply organized *transport* in the far zone. Recent measurements often report azimuthal phase in $a_\sigma(\widehat{\mathbf{k}})$ or Poynting vorticity near the aperture as evidence of orbital angular momentum; Maxwell propagation with outgoing selection shows that appearance and radiation are related but not identical [12]. Stokes texture or $\|\mathbf{\Omega}_P\|$ measured on a reactive shell must not be promoted to a transport label; however, balanced counter-rotating excitations can produce rich far-zone phase while signed angular flux vanishes until a modest amplitude imbalance restores radiation [6]. Structured source parameters Eq. (2.432), topology condition on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, and regime-appropriate transport indices must be distinguished imported from Chapter 1 [2]. What a laboratory measurement actually establishes is therefore the radiative image after Π_{rad}, not the raw near reactive map $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [1]. The ring-array comparison below makes the distinction measurable: balanced counter-rotating phases produce rich far-zone texture while signed flux cancels, and a modest amplitude imbalance restores radiation without changing phase structure dramatically [12]. An offset feed horn illuminates a parabolic reflector of diameter $D \gg \lambda$ in a radio-telescope or compact-range measurement; spillover and rim currents share the same radiation integrals as the

feed mouth [2]. Dual-polarization horn feed with calibrated S_{21} reciprocity checks between feed port and far-zone probe drives the measurement [6]. Feed spillover and edge taper set sidelobe shoulders; secondary illumination compresses the angular spectrum onto the focal region, so near-field scans on the feed alone differ qualitatively from full-dish measurements [3]. Equivalent currents on the feed mouth compose with radiation integrals from induced rim currents; reciprocity ties port coupling to far-zone amplitude on far shells [6]. Reflector systems separate field *generation* on the feed from field *recovery* on the focal surface—two finite-geometry limits on one measurement—and thereby exemplify why structured-source parameters must pass through $\mathcal{R}_\omega$ before Stokes, Poynting, or angular-flux diagnostics are reported [1]. Subsection 2.10.4 activates the structured-source and geometric-suppression terms of Eq. (2.415), giving the definitive packaging of *structured source states*—impressed currents with explicit amplitude, phase, and polarization organization—and illustrating their radiation through $\mathcal{R}_\omega$, importing Subsection 1.2.3 amplitude–phase–polarization–Poynting diagnostics from the cited sections [2]. Radiation map Eq. (2.67), near-to-far limit Eq. (2.405), exterior spectral integral Eq. (2.289), declared size and distance regime Eq. (2.139), and angular-flux functional Eq. (2.257) are presupposed; 1 Stokes diagnostics Eqs. (2.38)–(2.39) and the condition for outward angular-momentum flux criterion Eq. (1.67), together with the topology–transport separation (Subsection 1.2.3), are cited without re-proof [2]. Vector harmonic labeling of structured modes belongs to 3 [2]. On compact support $\Omega_s \Subset \Omega$, a *structured source state* is an admissible impressed current $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ admitting the parametric representation

$$\mathbf{J}_{\mathrm{imp}}(\mathbf{r}') = \sum_{n=1}^{N} A_n\, \chi_n(\mathbf{r}')\, e^{\mathrm{j}\Phi_n(\mathbf{r}')}\, \mathbf{p}_n(\mathbf{r}'), \qquad \mathbf{p}_n \cdot \hat{\mathbf{k}}_n = 0, \tag{2.432}$$

where $A_n \geq 0$ are channel amplitudes, Φ_n are spatial phase laws, χ_n are nonnegative envelopes with $\operatorname{supp} \chi_n \subseteq \Omega_s$, and $\mathbf{p}_n$ are transverse polarization vectors on each channel support [2]. Eq. (2.432) is the Subsection 2.10.4 operator-side label for sources whose angular organization is specified before radiation selection: amplitude fixes channel weight, phase fixes interference geometry, and $\mathbf{p}_n$ fixes local vector state, paralleling the field-side chain of Subsection 1.2.3 [2]. The sum is understood in the distributional sense when χ_n includes point or sheet supports [2]. Non-uniqueness of Eq. (2.432) under gauge-equivalent reparameterizations is inherited from impressed-current equivalence classes Eq. (2.0.1) and does not alter $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [6]. Structured radiation coefficient. Define the *structured radiation coefficient* of channel n in direction $\widehat{\mathbf{k}}$ and polarization $\sigma \in \{1, 2\}$ by

$$c_{n,\sigma}(\widehat{\mathbf{k}}) = -\mathrm{j}\omega\mu_0\, \frac{1}{4\pi} \int_{\Omega_s} A_n\, \chi_n(\mathbf{r}')\, e^{\mathrm{j}\Phi_n(\mathbf{r}')}\, \mathbf{p}_n(\mathbf{r}') \cdot \mathbf{e}_\sigma(\widehat{\mathbf{k}})\, e^{-\mathrm{j}k_0 \widehat{\mathbf{k}} \cdot \mathbf{r}'}\, d^3 r', \tag{2.433}$$

with $\mathbf{e}_\sigma$ transverse unit vectors from Eq. (2.288) [2]. The far angular amplitude is

$$a_\sigma(\widehat{\mathbf{k}}) = \sum_{n=1}^{N} c_{n,\sigma}(\widehat{\mathbf{k}}), \qquad a_\sigma = \Lambda_{\mathrm{nf}}[\mathbf{J}_{\mathrm{imp}}]_\sigma, \tag{2.434}$$

importing Eq. (2.405) and consistency Eq. (2.403) [2]. Eqs. (2.433)–(2.434) linearize structured sources through $\mathcal{R}_\omega$: each channel contributes a directional radiation coefficient fixed by outgoing radiation condition, and superposition holds on the radiating branch [2]. Physically, Eq. (2.434) is the operator restatement of "structured beam" labels: azimuthal phase ramps in Φ_n appear in $c_{n,\sigma}$ through the plane-wave kernel, but only after radiation selection [13]. Let $\mathbf{J}_{\mathrm{imp}}$ satisfy Eq. (2.432) with $(\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}}) = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$. Transverse components, Stokes parameters, Poynting vector, and Poynting vorticity computed from $(\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}})$ in the far zone inherit the diagnostic

chain of Subsection 1.2.3 Eqs. (2.38)–(2.40). Local topology markers are rotationally relevant only where

$$\mathcal{Q}_{\mathrm{str}}(\widehat{\mathbf{k}}) = \mathcal{Q}_{\mathrm{top}}(\mathbf{r}) \text{ evaluated on } (\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}}) \text{ with } \mathcal{D}_{J,\mathrm{reg}} = 1, \tag{2.435}$$

importing directional channel calibration Eq. (1.54) and declared size and distance regime Eq. (2.139) [2]. Nontrivial Stokes texture, phase winding, or $\mathbf{\Omega}_P \neq \mathbf{0}$ on $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ without passing through $\mathcal{R}_\omega$ do not qualify for transport reports [2]. Apply the topology–transport separation (Subsection 1.2.3) to far-zone traces of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ and identify $\mathcal{Q}_{\mathrm{top}}$ with Eq. (2.27) [2]. Reactive components removed by $\mathcal{R}_\omega$ cannot satisfy Eq. (2.435) [6]. Consider a circular ring array of $N = 36$ impressed-current elements lying on $\rho = \rho_0 = 0.8a$, $z = 0$, with compact ball support $\Omega_s = B_a(0)$, $k_0 a = 3.0$, and ω fixed . Each element carries amplitude $A_n = A_0$, envelope $\chi_n = \delta(\mathbf{r}' - \boldsymbol{\rho}_n)$, and transverse polarization $\mathbf{p}_n = \hat{\boldsymbol{\phi}}$ at azimuth $\phi_n = 2\pi(n-1)/N$. Two structured configurations are compared:

(a) *Balanced counter-rotating pair:* half the elements carry $\Phi_n = \ell\phi_n$ with $\ell = +1$, half carry $\Phi_n = -\ell\phi_n$, matching the channel template Eq. (1.137) after radiation mapping;

(b) *Imbalanced single winding:* all elements carry $\Phi_n = +\ell\phi_n$ with $\ell = +1$ but amplitudes satisfy $A_n = A_0(1 + \eta\cos\phi_n)$, $\eta = 0.35$.

Far-zone diagnostics use $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, extractor Eq. (2.404), and shell $r = 10a$. Tasks:

(i) compute structured coefficients Eq. (2.433) and far amplitudes Eq. (2.434) for both configurations;

(ii) evaluate Stokes parameters and $\mathbf{\Omega}_P$ on the radiative branch (Subsection 1.2.3) and state whether $\mathcal{Q}_{\mathrm{str}} = 1$;

(iii) compute net angular-flux proxy Λ_J from Eq. (2.257) and compare with Eq. (1.142) and imbalance sensitivity Eq. (1.144) (Subsections 1.2.3–1.2.4).

The dashed link marks imported 1 diagnostic definitions. Structured coefficients for configuration (a). Insert the counter-rotating phase law into Eq. (2.433):

$$c_{n,\sigma}(\widehat{\mathbf{k}}) = -\mathrm{j}\omega\mu_0 \frac{A_0}{4\pi} e^{\mathrm{j}\ell s_n \phi_n} \mathbf{p}_n \cdot \mathbf{e}_\sigma(\widehat{\mathbf{k}}) e^{-\mathrm{j}k_0 \widehat{\mathbf{k}}\cdot\boldsymbol{\rho}_n}, \qquad s_n = \begin{cases} +1, & n = 1,\ldots,18, \\ -1, & n = 19,\ldots,36, \end{cases} \tag{2.436}$$

with $\ell = +1$. Summing over the ring,

$$a_\sigma(\widehat{\mathbf{k}}) = \sum_{n=1}^{36} c_{n,\sigma}(\widehat{\mathbf{k}}) \;\approx\; A_0\, J_1(k_0\rho_0 \sin\theta)\, e^{\mathrm{j}\ell\phi}\, f_\sigma(\theta), \tag{2.437}$$

where J_1 is the azimuthal ring factor from discrete ϕ_n sampling and f_σ is transverse polarization weight . Equation (2.437) shows nonzero azimuthal phase in $a_\sigma(\widehat{\mathbf{k}})$: the far pattern is *structured* in the sense of Eq. (2.434) . Mathematically, equal A_0 and balanced $s_n = \pm 1$ enforce symmetry under $\ell \to -\ell$ channel exchange; physically, the configuration mimics counter-rotating orbital labels on the source before radiation 2.2–1.2.4). Net angular radiation uses radiative flux Eq. (2.257) on S_r:

$$\Lambda_J^{(\mathrm{a})} = \frac{\int_{S_r} \widehat{\mathbf{r}} \cdot \mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{a})}]\, dS}{\int_{S_r} \mathrm{Re}\{\widehat{\mathbf{r}} \cdot \mathbf{S}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{a})}]\}\, dS} \;\approx\; 0, \tag{2.438}$$

importing Eq. (1.142) after mapping balanced channels through Eq. (2.434) [2]. Channel power amplitudes satisfy $I_+ = I_-$ in the notation of Eq. (1.141), so Eq. (1.140) yields vanishing signed

radiation despite nonzero $|a_\sigma(\widehat{\mathbf{k}})|$ [2]. declared size and distance regime $\mathcal{D}_{J,\mathrm{reg}} = 1$ holds on S_r for $k_0 r = 30$, but $\mathcal{Q}_{\mathrm{str}} = 0$ because directional angular-flux support cancels [6]. Configuration (a) therefore exhibits structured *appearance* without flux-outgoing radiation [2]. On S_r, extract transverse components Eq. (2.38) from $\mathbf{E}_{\mathrm{rad}}$ and form Stokes parameters Eq. (2.39):

$$S_0^{(\mathrm{a})}(\widehat{\mathbf{k}}) \approx 2A_0^2 |f_\sigma(\theta)|^2, \quad S_3^{(\mathrm{a})}(\widehat{\mathbf{k}}) \approx 0, \quad \Delta\varphi^{(\mathrm{a})} = \arg a_+ - \arg a_- \text{ varies with } \phi, \tag{2.439}$$

because counter-rotating source phases produce azimuthal organization in a_σ while circular Stokes component S_3 averages toward zero on the balanced ring [2]. Poynting vector Eq. (1.34) and vorticity Eq. (2.40) yield

$$\|\mathbf{\Omega}_P^{(\mathrm{a})}\|_{L^\infty(S_r)} \;\sim\; \mathcal{O}(\ell\beta), \qquad \beta = k_0\rho_0 = 2.4, \tag{2.440}$$

with $\mathbf{\Omega}_P^{(\mathrm{a})} \neq \mathbf{0}$ near the pattern peak [2]. Mathematically, local circulation and phase winding can coexist with $\Lambda_J^{(\mathrm{a})} \approx 0$; physically, this is the operator-side restatement of the topology–transport separation (Subsection 1.2.3) on structured sources [12]. Students must not infer orbital index from $\|\mathbf{\Omega}_P\|$ alone without Eq. (2.435) [6]. Imbalanced configuration (b). Amplitude modulation $A_n = A_0(1 + \eta\cos\phi_n)$ breaks channel equality:

$$\Lambda_J^{(\mathrm{b})} \;\approx\; \frac{\eta\,\ell}{1+\eta^2/2}\,\Lambda_J^{(\mathrm{ref})}, \qquad \Lambda_J^{(\mathrm{ref})} = \frac{k_0 A_0^2 J_1^2(k_0\rho_0 \sin\theta_{\max})}{2}, \tag{2.441}$$

importing imbalance sensitivity Eq. (1.144) with $\eta = 0.35$ [2]. Illustratively, $|\Lambda_J^{(\mathrm{b})}/\Lambda_J^{(\mathrm{ref})}| \approx 0.30$: a thirty-percent fractional imbalance in source amplitude produces order-unity fractional radiation relative to the reference single-winding ring [2]. Far amplitude magnitude $|a_\sigma^{(\mathrm{b})}|$ differs from $|a_\sigma^{(\mathrm{a})}|$ by only $\sim 8\%$ (illustrative) because phase structure is identical; radiation changes because signed channel content is unequal Eq. (1.140) [2]. Here $\mathcal{Q}_{\mathrm{str}} = 1$ on the principal lobe and Eq. (1.67) returns $\mathcal{D}_J = 1$ [2]. Raw near map versus radiative diagnostics. Applying Stokes and Poynting formulas to $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on $r = 1.5a$ inflates texture metrics (illustrative ratios):

$$\frac{\|\mathbf{\Omega}_P\|^{(\mathrm{raw})}_{L^2(S_{1.5a})}}{\|\mathbf{\Omega}_P\|^{(\mathrm{rad})}_{L^2(S_r)}} \;\approx\; 1.9, \qquad \frac{S_0^{(\mathrm{raw})}}{S_0^{(\mathrm{rad})}} \;\approx\; 2.3, \tag{2.442}$$

by the same reactive contamination mechanism as Eq. (2.426) [2]. Eq. (2.442) shows that structured-source labels read from near reactive scans overstate Poynting topology relative to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2]. Correct measurement sequence: parametrize Eq. (2.432), map Eq. (2.434), then apply §1.2.2 diagnostics on $(\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}})$ [6]. Structured far appearance $|a_\sigma(\widehat{\mathbf{k}})| \neq 0$ does not imply $\mathcal{Q}_{\mathrm{str}} = 1$ Eq. (2.435), and qualified topology does not imply signed radiation $\Lambda_J \neq 0$ Eqs. (1.142), (1.144) [2]. Configuration (a) occupies the first failure; configuration (b) satisfies radiation once imbalance is present. Non-radiating additions Subsection 2.10.2 and evanescent storage Subsection 2.10.3 modify near diagnostics without changing a_σ when $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is fixed [2]. Accessibility budget Eq. (2.414) may further shrink observed structure inside $\mathcal{K}_{\mathrm{acc}}^{\mathrm{ff}}$ [2]. Physical meaning (summary). Subsection 2.10.4 establishes that "structured source states" are operator objects Eq. (2.432) mapped to far coefficients Eq. (2.434), not merely descriptive beam names [2]. Balanced counter-rotating rings produce azimuthal phase in $a_\sigma(\widehat{\mathbf{k}})$ Eq. (2.437) yet vanishing Λ_J Eq. (2.438), reproducing 1 phase-without-transport logic on the radiation operator [2]. Stokes and Poynting textures Eqs. (2.439)–(2.440) can be rich while radiation cancels; amplitude imbalance Eq. (2.441) is the control knob for signed angular flux [2]. structured-source topology diagnostics (Subsection 2.10) and criterion Eq. (2.435) require diagnostics on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ with declared size and distance regime before

any OAM or helicity label is promoted to transport [6]. Near reactive overread Eq. (2.442) parallels Subsection 2.10.3, structure in $\mathcal{S}_\omega$ is not structure in radiation [2]. Structured sources close the example suite by showing that the most elaborate near-zone organization does not override selection: phase winding in $a_\sigma(\widehat{\mathbf{k}})$ can coexist with vanishing signed angular flux until amplitude imbalance restores transport on the radiative branch [12]. OAM or vortex labels attached to impressed currents remain hypotheses until $\mathcal{R}_\omega$, declared size and distance regime, and topology criteria from 1 have been applied [6]. section 2.10 therefore ends with four distinct bench mechanisms; section 2.11 collects the chapter results and states what 3 will relabel Maxwell or outgoing foundations proved here.

2.11 Summary

Maxwell evolution generates fields; outgoing selection determines radiation; finite instruments recover only part of that radiation [1, 2, 16]. Vector spherical harmonics, source realizability, and interaction formalism developed in Chapters 3–4 are cited here without local re-derivation [6, 17].

2 began from a premise that Maxwell's equations, once supplemented by outgoing radiation conditions, do not merely *propagate* fields—they *select* which field content qualifies as outgoing radiation. That selection is implemented by the radiation operator $\mathcal{R}_\omega$, whose image is what finite instruments ultimately interrogate, always incompletely and always through additional rank limits imposed by geometry and hardware.

The distinction is not pedantic. A near-field scanner that records thirty-six complex samples has performed a serious measurement; yet the number answers a question about accessible angular structure only after one traces which term in the master degree-of-freedom partition removed the unresolved content—source nullity, aperture rank, evanescent storage, spectral compression, or geometric suppression [2]. section 2.10 composed four worked configurations to make that partition tangible on the bench. Subsection 2.10.1 on the electrically small dipole ties modal truncation error to compact singular values rather than non-radiating nullity. Subsection 2.10.2 on loop superposition keeps far patterns invariant while near probes register a thirty-five percent increment in tangential amplitude—far data fix $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ but not $\mathbf{J}_{\mathrm{imp}}$. Subsection 2.10.3 on evanescent overlap inflates near-shell energy beyond twice the radiative component while $a(\widehat{\mathbf{k}})$ remains fixed; applying far-field extractors to raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ without prior $\mathcal{R}_\omega$ selection can bias angular radiation by forty percent on illustrative geometries. Subsection 2.10.4 on structured ring arrays separates azimuthal phase organization in $a_\sigma(\widehat{\mathbf{k}})$ from signed transport: Λ_J can vanish until a modest amplitude imbalance restores radiation on the qualified radiative branch [12]. Together these cases state one qualification practice in operator language: existence of field does not imply radiation; radiation does not imply flux closure on Γ_R; $\oint_{\Gamma_R}$ closure does not imply unique source identification [1].

Read against that laboratory background, the central lesson is a verification threshold, not a slogan. 1 established $\mathcal{D}_J = 1$—what impressed currents and phasor fields may enter the monograph at all. 2 establishes $\mathcal{R}_\omega$: which radiative content survives outgoing selection; finite instrument recovery $\mathcal{E}_{\mathrm{acc}}^{(N)}$ is developed in Chapter 4.

Throughout, physically established coefficienting applies $\mathcal{R}_\omega$ before Stokes parameters, Poynting vectors, or angular-flux functionals, and composes with regime conditions inherited from 1 so that reactive shells are not mistaken for transport [6]. From the vector-network-analyzer reciprocity checks of section 2.8 through the directional decompositions of section 2.9 to the four worked configurations of section 2.10, the chapter supplies operator criteria for diagnosing pattern residuals by mechanism rather than by fitting alone.

These four distinctions are physically distinct: Maxwell generation, outgoing selection, reactive

or null-space content, and instrument-limited readout are not equivalent descriptions of one object. They are four separable stages in the qualification of a radiative claim.

Finite support compresses accessible channels at fixed ω [7].

The reference case is therefore radiate on Γ_R: reactive storage cannot substitute for propagating rank [5].

2.11.1 Results

Principle 2.0 selects $\mathcal{F}_{\mathrm{rad}}$ before harmonic readouts; finite observation rank caps recoverable angular content on Γ_R [2]. The results proved here may be collected without repeating the subsection sequence. Radiation is a selected subset of Maxwell solutions on exterior domains, not an independent physical process. The radiation operator possesses nontrivial null spaces whose elements are physically realizable and invisible to far data. Compact sources and finite observation compress accessible angular structure through spectral rank limits that are distinct from nullity. Near-to-far radiation requires radiation selection before asymptotic extraction; evanescent and non-radiating admixtures can inflate near readouts without altering far amplitude when $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is fixed. Structured source parameters specify appearance before selection; signed angular transport requires imbalance on the radiative branch after selection. Reciprocity constrains bilinear coupling on $\operatorname{ran}\mathcal{R}_\omega$ but does not enlarge accessible angular sectors or remove finite-rank observation deficits. The Chapter 2 radiation analysis presupposes Chapter 1 admissibility, outgoing radiation from $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, and the declared size-and-distance regime $\mathcal{D}_{J,\mathrm{reg}} = 1$ [2]. Results under these hypotheses inherit the measurement classifications of Eq. (1.161) [2].

$$(\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega \ \wedge\ \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ satisfies outgoing radiation conditions } \wedge\ \mathcal{D}_{J,\mathrm{reg}} = 1)\,, \tag{2.443}$$

with Chapter 1 admissibility from Eq. (1.160) and $\mathcal{D}_{J,\mathrm{reg}}$ from Eq. (2.33). Results in sections 2.0–2.10 assume these hypotheses. Measurement classifications follow Eq. (1.161) [2].

Physically realizable radiation is therefore bounded by geometry and frequency before optimization adjusts weights [7].

Only the outgoing propagating component contributes to far-zone radiation; reactive, evanescent, and non-radiating components do not.

2.11.2 Topics developed in later chapters

Chapter 2 has fixed the source-to-field operator, outgoing selection, and rank-limited radiation on Γ_R, yet several questions remain physically open until harmonic coordinates, realizability proofs, or interaction theory are supplied elsewhere. The need is not to re-derive those results here but to record, in one place, which topics inherit their proofs from later chapters [2]. Topics requiring vector-spherical-harmonic completion, source realizability proofs, or interaction theory beyond this monograph are developed in Chapters 3–4 by citation under the completion recorded in Eq. (2.443); they are not re-derived here [2]. The deferral rule below states when a topic must be cited rather than reproved in Chapter 2; it is a continuity relation, not a catalogue of unfinished algebra [6].

$$\begin{aligned} \mathcal{X} \in \mathcal{D}_{\mathrm{defer}}^{(\mathrm{ch2})} \iff\ & \mathcal{X} \text{ requires VSH or } \mathbf{J} \text{ operator completion from Ch.3} \\ \vee\ & \mathcal{X} \text{ requires source realizability proof from Ch.4} \\ \vee\ & \mathcal{X} \text{ requires Volume II interaction theory,} \end{aligned} \tag{2.444}$$

and

$$\mathcal{X} \in \mathcal{D}_{\mathrm{defer}}^{(\mathrm{ch2})} \Longrightarrow \text{cite in later chapters; no local re-derivation in Ch. 2.} \tag{2.445}$$

Every deferred topic above remains physically meaningful on the radiative branch already selected: Chapter 3 supplies the angular coordinate chart, and Chapter 4 confronts realizability on finite support. Subsection 2.11.3 records the inheritance map that preserves those results without recursion [2].

Angular-momentum transport therefore remains qualified on outgoing pairs $(\mathbf{E}, \mathbf{H})$: gauge-dependent density splits cannot be read as laboratory observables without the admissibility predicates from Chapter 1 [2].

2.11.3 Relation to later chapters

Chapter 2 establishes with operator maps and deferral rules fixed; what follows in Chapter 3 is coordinate labeling on the same outgoing branch, not a reopening of source-to-field proofs [2]. Chapter 3 supplies harmonic coordinates on the radiative branch fixed in sections 2.0–2.10; Chapter 4 confronts which coefficient vectors arise from physically realizable impressed currents on finite support [6]. Rayleigh and Stratton treated compact ranges as boundary-value laboratories long before digital near-field transforms became routine; Harrington's radiating-systems discipline carries the same obligation forward—metal outline first, harmonic claim second [2, 6].

$$\mathcal{D}_{\mathrm{cont}}^{(2\to3)}(\mathcal{Y}) = 1 \iff (\mathcal{Y} \text{ is compatible with Chapter 2 radiation results } \wedge\ \mathcal{Y} \text{ adds no recursive proof of Chapter } \tag{2.446}$$

subject to Eq. (2.443) and Eq. (1.162) [2].

Radiation on Γ_R is now selected: $\mathcal{S}_\omega$ maps bounded sources to Maxwell fields, $\mathcal{R}_\omega$ retains the outgoing radiative sector, and finite instruments read only a rank-limited image of what geometry permits [6]. Chapter 3 inherits those maps and asks the coordinate question every range report must answer—how to label directions on the observation shell so that turntable cuts, flux integrals, and harmonic fits refer to the same physical radiation [2, 13]. The operator content proved here is not reopened; only the rotational chart on $\mathcal{F}_{\mathrm{rad}}$ is supplied [12].

Catalogue membership does not imply laboratory access on finite supports [6].